Design after Capitalism

Transforming Design Today for an Equitable Tomorrow



Matthew Wizinsky

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Preface

I never should've walked into that Sunglass Hut. So begins my imaginary memoir of life in a consumer society, where every narrative is shaped by objects, environments, symbols, and desires of consumption. Growing up in the suburban sprawl of metro Detroit in the 1980s and 1990s, I witnessed the power of a single product both to build and to destroy a metropolis. Even as the American economy was swinging toward postindustrialization, everyone I knew had family working for the auto industry, in one way or another. Some friends' parents worked as engineers, administrators, or marketers in the gleaming, low buildings along the freeway in Dearborn; others worked in the few remaining factories or steel mills. In high school, some kids even harbored dreams of dropping out to "work the line." There was still a sense that you could build a career of high wages, good benefits, and early retirement from unionized factory work. That version of the middle class, along with the entire economic system that created it, was vanishing quickly, though.

As a kid, I would regularly visit the North American Auto Show in Detroit. It was a breath-taking spectacle of style and technology. The newest, coolest, sleekest, and fastest. This was my hometown's calling card. It was also design! Yet walking out of Cobo Hall into a city literally crumbling was an exercise in cognitive dissonance. How was it possible that the sexy curves, crisp lines, glitz, and shine of automobiles had both built this city and let it fall apart? Detroit became emblematic of the ruins of industrial capitalism and the punchline to jokes about urban dysfunction. In the film *RoboCop* (1987), Detroit was the setting for a dystopian sci-fi vision of authoritarianism, political corruption, corrosive media, unbridled privatization, technology companies (literally) getting away with murder, and other obscene tropes of capitalism.[1] This is not exactly the image of your hometown you wish to see on screens all over the world. However, Detroit in the 1980s was an excellent place to observe the worst effects of late twentieth-century American economy and culture. Utter stripping of the inner city by decades of lost employment as automotive manufacturing

dwindled, white flight, wholesale divestment of all things public, militarized police pitted against crime (organized and disorganized), and disenfranchisement of citizens: all convened to produce a nearly perfect inversion—a shadow—of the once glittering capital of American industry.

The system that created the largest and most materially well-off middle class in history had run its course. The Great Gobbling-Up Machine gobbled up as much as it could and moved on. A similar fate has been met by countless cities across the country—not to mention by old-growth forests, precious-metal and coal mines, freshwater streams and lakes, and landscapes stripped of their natural value. Of course, much is still left behind: people's lives, elements of a natural ecosystem, and man-made infrastructures, albeit perhaps quite broken. Detroit's population has dwindled to less than half the size it had been when my dad was born there in 1950. Yet along with every other place on the planet that has been gobbled up, what remains are the seeds for what grows next.

My entry to the world of graphic design came via teenage infatuation with punk rock and, inevitably, the art of the album cover. When I started my first band at fifteen years old, the punk subculture I joined had a particular ethos. It was independent and DIY. It was emphatically antiracist and pro-equality for all people, regardless of class, race, gender, or orientation. Within our community, we all were outsiders from the broader social worlds in which we were weirdos, rejects, or just "off." Some flavors of the punk movement were decidedly anticapitalist. My personal politics as a teenager and young adult were too simplistic to grasp the meaning of anticapitalism. It was obvious that there was tremendous disparity in the world—just look at Detroit. But while respecting the value of all people seemed an easy mindset to adopt, reckoning with the problems of the global political economy was just beyond my capacity. As life moved on, I became a designer and then a design educator. An underlying agitation with "business as usual" began to reveal what that teenage angst might have been better directed toward: transcending a historical system of hierarchy that was already in decline. In my life as an academic, my research agenda developed around studying and practicing participatory design methods mixed with speculative and critical design approaches—critiquing the business-as-usual approaches toward which design tends to default. Writing a book against capitalism seemed the most punk rock thing I could do as a design scholar.

In 2015, I began teaching a design studio that challenges undergraduates in industrial, fashion, and communication design to imagine “urban futures” by materializing “critical speculations.” There has been much warranted critique of speculative design. Indeed, it appears the backlash *against* speculative design helped bring critical attention to the colonizing practices of design more generally. Speculative designers have at times been ruthlessly admonished for the elitist, white, heteronormative capitalist assumptions embedded in their scenarios—even those intended to be seen critically.[2] Many speculative design projects can rightly stand accused of being conservative in their political implications due to their emphasis on technology alone.[3] These misgivings aside, it was through speculative design that I first embraced “after capitalism” as a framework to think about possible futures. I came to understand that this framework carried enough substantive research, theory, and empirical grounding *outside of design* to propose a reorientation to *transform design fr*om within.

In 2017, I curated an exhibition of student projects from this course. I sent copies of the exhibition catalog to many scholars whose work was studied by students in the class—most of whom I had never met. One recipient was Erik Olin Wright, professor of sociology at the University of Wisconsin–Madison. Wright’s book *Envisioning Real Utopias* (2010) is an incredible compendium of case studies, empirical evidence, and theories on *real utopias*—people making alternative life systems amid global capitalism. I hardly expected my design exhibition catalog would elicit a response. Yet Wright responded quickly and enthusiastically. This began an incredible yet all too brief correspondence that I experienced as mentorship. Wright was emblematic of the warm, generous, and fully engaged intellectual—a testimony of his own radicalization in the 1960s and a sadly fleeting sight today.

Wright invited me to join his annual “real utopias” retreat in Madison. Unfortunately, he was diagnosed with leukemia and became too ill to attend the conference that year. He Skyped in from the hospital and was seemingly more “alive” than any of us—still personally and politically engaged in the project of building *utopias* wherever he went. He asked a nurse to join the symposium briefly as he spoke about forging cooperation across disciplines and dismantling destructive hierarchical structures in society. The conference was an incredible experience and connected my teaching and

research interests in design to an action-oriented and interdisciplinary community of scholars. Inspired by this encounter, I continued teaching the studio course, now with the subtitle “Design after Capitalism.” I’m proud to recognize how profoundly the students in this course have challenged, inspired, and motivated me on this topic over the past several years.

After the conference in Madison, I continued to invite Erik Olin Wright to the course as a guest lecturer. In the fall of 2018, he wrote a very succinct and clear-eyed note that after an upcoming bone marrow transplant, either he’d be too sick to join the class in discussion, or he’d be dead. Sadly, Erik Olin Wright passed away in early 2019. My relationship with Wright near the end of his life was a major influence. Perhaps this whole project is a love letter to a gone mentor and friend. Wright was brilliant yet incredibly humane, a courageous pioneer, a generous intellect, and a very special organization of “stardust.” For me, a designer starting out an academic career, he was also a healthy antidote to the cynical, self-serving careerism that plagues academia. This book is dedicated to his memory, and he is very much alive in these pages.

Design after Capitalism starts by recognizing that there are no more externalities. We can’t design on the assumption of infinite resources at our disposal: material resources from a dying planet, labor resources from a global labor market with wages racing to the bottom, dangerous working conditions, and so on. It looks to me as if this particular historical model of social and economic organization has run its course. I think of *home*, in the broadest sense, when I hear Anna Lowenhaupt Tsing say, “If we end the story with decay, we abandon all hope—or turn our attention to other sites of promise and ruin, promise and ruin.”[4] What’s next is up to us. The future doesn’t really exist; it’s being produced right now.

Introduction

Design is world making.[1] As grandiose as that sounds, the various fields that compose the territory of design are organized around intentional production of the objects and symbols that make up everyday life. Design practices—institutionalized patterns of knowledge and action—contribute to the material culture of the worlds we all inhabit. In the act of designing, designers identify "users" and "use-case scenarios," creating idealized subjects and *ways of being* in the world. In this way, designers contribute to determinations about who has what kinds of agency and how those agencies are made possible—physically, symbolically, and experientially. At the same time, designers are often employed by others (their clients) to determine what is produced to achieve these ideals and how. This describes both sides of the political dimension of design: mediating the everyday lives of unknown others but typically doing so through the authority and financial support of their clients. Design alone cannot (and certainly does not) replace politics or economics or culture. Yet design is everywhere, and everywhere it is, it represents and enacts politics, economies, and cultures.

There is an inherent tension between disciplinary norms and any attempt to change those norms, to reorient patterns of knowledge and action. This is different from the *expansion* of design disciplines witnessed over the past several decades. Designers have taken on greater responsibilities for determining what to produce and how, particularly in the expanding space of digital products and services. This has brought up myriad new challenges, including ethical concerns. However, my proposal is for a fundamental shift in purpose, which requires a conscientious move toward change at a higher systemic level than before. Whereas many design practices, no matter how advanced, systemic, or complex, assume capitalism as the limit of its transformative actions, this model attempts to extend that limit. Prominent design scholars and historians, even those critical of this context, tend to accept that capitalism is the primary ideological *container* for modern design practices.[2]

This book proposes a rationale, theoretical framework, and practical principles for design to transcend the logics, structures, and subjectivities of capitalism. Most contemporary design practices have evolved out of the historical conditions of industrial capitalism; however, converging forces point to the possibility that capitalism is coming to an end or, at the very least, that it is mutating into something quite different. Capitalism's end will pose a problem for the future of design if designers are not prepared for—or, perhaps, actively engaged in—practicing in ways that anticipate "after capitalism."

Capitalism has long been critiqued for a range of detrimental social, ecological, and political impacts, from the way it alienates people from the objects of their labor to the ways that the logics of competition and commodification seep into everyday social interactions. The forces driving this exploration of *design after capitalism* are focused on design's social adaptations to a changing economy. In other words, the argument is as much economic as it is moral.

First, capitalism requires constant growth. We are now witnessing both the immediate and long-term effects of global warming[†] at every level of human and nonhuman earthly existence. The infinite growth required by capitalism is at odds with the finite resources of our planet, which is literally burning up.

Second, digital goods and services as well as the rapid automation of production, distribution, and design of all kinds of material "things" may simply be incompatible with market mechanisms alone. Capitalism has impressively adapted itself to changing historical conditions; however, it may not be suited to an economy that is dematerialized, postindustrial, and knowledge based.

Addressing both of these points, this book critically assesses design practices operating in an evolving digital information and service economy to propose a transformation that blends design entrepreneurship with social empowerment. To do so, it theorizes and historically contextualizes a modified discipline of design and articulates fundamental principles, strategies, and guidelines for its practice. The argument builds from evidence and theoretical positions well outside the design of material culture—economics, sociology, anthropology, philosophy, digital and media studies, political science, history, environmental and sustainability studies,

critical theory—but returns to design theory and practice with new principles. This assemblage of theoretical positions, historical and contemporary case studies, and practical guidelines intends to provide a foundation for more broadly connected and intentionally implemented approaches to the practice of a postcapitalist design (PCD).

The principles and practical guidelines put forth here can be put to use now, without the need for political revolution or specific policy reforms. The purpose is to inform and influence designers, design students, as well as professionals and scholars in various fields that interface with design to consider the "institutional" nature of design practices and how they might be more equitably and sustainably modeled through a new disciplinary model. Design itself is synonymous with change. It is about materializing preferable states of the world and, in doing so, establishing preferable *ways of being* in that world. This book is a plea for designers to take seriously their own claims about design's potential to incite broad and meaningful social change. My hope is that it will help designers to better *think what they* are doing.

After Capitalism

My argument is not that capitalism is inherently evil or that all capitalist, profit-seeking activity is greedy and manipulative. Capitalism is a social construct. Like any social system, it is embedded with its own contradictions and limits. Most important to remember, it is an economic and social organization that has been imagined, implemented, institutionalized, and normalized *by people toward certain goals*. If those goals are not being achieved or are not agreeable, people can modify or discard the system. Capitalism is also a *historical system* of organization whose origins are rooted in a very different social and economic context than the present. It was within those historical conditions that modern design emerged. Over several centuries, capitalism has proven incredibly productive and at times has successfully distributed wealth and increased material prosperity for countless people. However, for reasons to be discussed, that system appears to have run its course and is no longer credible for claims to the broad distribution of improved livelihoods. As the

system exacerbates negative tendencies that were always inherent, and as positive social contributions diminish, designers are positioned to transform design practices to institutionalize new social and economic organizations within the dying one.

The term *after capitalism* has been used for decades in disciplines engaged in envisioning future states of a more just, sustainable, and democratic political economy. The term appears in sociological texts, philosophical works, and scholarship in strategic foresight to describe the next paradigm of political economy and its implications. It is applied to the concept of a new political economy that is emerging from within capitalism and might further grow to erode and eventually replace capitalism. This slow transformation will make use of the productive technological and innovative gains of capitalism but repurpose them toward more democratic and egalitarian social and economic outcomes. This change will *not* take place through a sudden or violent political revolution, nor will it happen organically on its own. It will require collective effort toward remaking work, culture, and society beyond the competitive, free-market models defined by private ownership of the systems of value production. This process contrasts with revolutionary or ruptural transformations that might bring about change quickly and forcefully—such as the many violent revolutions of the twentieth century. Many scholars of the topic believe that such a revolutionary transformation is highly unlikely in places such as the United States and, even if it were to happen, would not guarantee a more just social and economic system.

Another critical element of *after capitalism* is that it is *not* an endorsement of centralized, state-run socialism. Most proponents of the move to “after capitalism” generally support “free” enterprise (within regulated constraints) and support markets as spaces of social, economic, and intellectual exchange. The major transformations have to do with how production is managed and how to build broader and more democratic participation in investments that yield greater social benefits rather than accumulating massive wealth for a tiny sliver of people.[3]

This proposal for an intentional and collective shift to *design after capitalism* comes at a time when design discourse is consumed with challenges of expanding the purview of design—to operate systematically and intentionally in broader, more diffuse contexts—while also confronting

extant and new ethical challenges. Efforts are currently underway to decenter the Western/European-centric model of modernity that gave rise to the specific, historical disciplinary mindsets, methods, and practices referred to here as *modern design*. These practices arose from the *productivism* of European modernity, an economizing, growth-oriented, and "rational" ideology that provided the ground-plane for modern design's ideal of "functionalism"—an ideal applied to the merger of aesthetics and usefulness in the "artificial world" constructed by humans, from cities to cooking utensils to social networks. This apparatus of knowing and doing was born from a civilizational history built on imperialism, colonialism, plunder, and both physical and symbolic impositions of dominance. Today, as we continue to witness the long-term implications of such practices, various strands of design discourse are examining how to "decolonize" design practices and/or to transition out of models of oppression and erasure.[4] Yet this discourse scantly addresses capitalism head on—even when capitalism is acknowledged as a pillar of the Western/European *hegemonic* form of modernity and modern design. The term *hegemony*[*] refers to the dominance of a system that, even when not chosen, asserts itself over other ways of thinking or doing things. Putting into sharp focus capitalism's constraints, contradictions, and distortions on the logics and structures of design is a necessary step to addressing broader issues of injustice and oppression. Meanwhile, reformist movements such as "conscious" or "values-based" forms of design and production, even those centered on "sustainability," are often lacking critical analysis of capitalism in terms of both its economic and its social/political implications for design.Introduction

The kinds of design that are the subject of this study include the design of things, systems, and symbols and how they produce both value and values for humans. They also include the complex systems of value and values that fold into the designing and making of things, systems, and symbols and how those systems entangle people, material resources, technologies, labor, and desires all around the world. These kinds of design include:

1. The design of material things, such as products, garments, furniture, and so on (e.g., product design, industrial design, fashion design)

2. The design of information and communication products, services, and experiences (user-experience design, interaction design)
3. The design of symbols that produce identity and value (graphic design, branding, experiential design, symbols of association)
4. The economies created around the practice of design itself (employees, ownership, communities of practice, and so on)

This argument engages critical analysis of the social and economic outcomes of design activities within the larger social systems of contemporary capitalism—not individuals' intentions or values. This is the structural nature of the argument. A structural analysis is the rationale for the proposal that design as a discipline, including its various professional fields, needs to be structurally transformed. This book offers critical analysis as a framework for creating an alternative political economy of design whose principles and strategies are described and toward which *design practices can begin today* to create long-term transformation.

Motivation

My motivation to write this book began with my own experiences and observations of design's capacity for world making and with my sense of disappointment at how narrowly that capacity is typically applied. Over the course of two decades in design practice, research, and education, I have been fortunate to work with an enormous variety of smart, skilled, compassionate, and dedicated people—from design collaborators to creative directors to clients and collaborators from other creative practices and academic disciplines to faculty colleagues and students. I don't think any of them were malicious by intent. I don't think any of them sought to destroy the social fabric of communities around the world, flatten the relationships people have with their neighbors into mere transactions, perpetuate historical domination of repressed groups, or destroy the planet. Yet I worry that all of them (and I) were doing those very things. In various projects and contexts, I sought explicitly to work against those outcomes through design. Yet in my own thinking and work, in every attempt to try to imagine design truly remaking the world into something more egalitarian,

democratic, and “sustainable,” I kept running into the same wall: capitalism.

To systematically address the twin contemporary crises of climate change and social justice, we need to identify new models of practice that address both. If we are to talk of design’s capacity for systemic change, then we need to identify the problematic system, diagnose it, and work toward transformation. There is a system directly tied to these twin crises, and it’s the same one that directs and contains a vast quantity of design activity. It’s capitalism. In response, I propose a modified discipline: design after capitalism.

This proposal is for a model of design practice that expands social empowerment through design entrepreneurship while producing the things and the symbols that make up everyday life. I come at this project from the perspective of a designer who has grown increasingly dissatisfied with both the ways in which designing is done (design practice) and the way design is portrayed, discussed, and perceived in practice, in academic contexts, and by the public (design discourse). However, designers are critically optimistic, and this book is an attempt to navigate out of that dissatisfaction. Critical optimism is required to perceive the world as improvable (diagnosis and critique), to identify and specify opportunities for change (alternatives), and then to suggest how to bring those changes into being (transformation).

Today, design is struggling to materialize “preferable states” and humane ways of being in the world while confronting new ethical and ecological challenges. My suggestion is to examine how design is practiced under capitalism, to challenge those logics, structures, and mindsets, and to come up with something better. You’ll be glad to know that I have specific suggestions in mind. This is, however, just the beginning.

My education in graphic design was built largely around a discourse that was humanistic. This discourse situated graphic design in its relationship to the human capacity for language. After all, communication is about the “communal,” what “communities” share in common and what we can uniquely make from or do with that shared “commons” of language. Outside the directives of any particular application, this way of thinking positioned graphic design as the pursuit of novel graphic forms that are both communicative and imaginative—that can “take us somewhere new.”

During the early twenty-first century—at the start of my own career—rapidly growing opportunities for designers to work on digital systems and interfaces transitioned many graphic designers, including me, into the world of interaction and user-experience design. This transition combined with the rise of *design thinking* and widespread adoption of user-centered design methods. In my experience, the dominant discourse began to speak of design in instrumental terms. Design was declared beholden to the "business needs" of its clients, thus making design instrumental to "business needs." As the principal of a design studio, and thus a business owner, I know what it means to eat and pay the bills based on doing design and on making decisions that maximize income over expenditure, minimize my tax expenses within legal boundaries, and push me to operate and think competitively. These are the moves to be played within the rules of the game. However, it's the game that I am challenging here.[5]

As the various disciplines and professions operating under the flag *design* continue to expand and modify in the shifting political, economic, and technical landscape of what will soon be the "mid-twenty-first century," designers are often finding themselves ill-equipped to critically assess design's role in globally connected yet locally distinct societies. Many voices in design discourse have been making that argument for a long time. Some critics point to failures in design education. In particular, there is the claim that design is still being taught from a crafts-based or industrial-apprenticeship model. Yet this claim is often followed by the suggestion that what designers *really* need to understand is business. As the argument goes, design is made possible only by business interests, its surest sign of success is profit for its clients, and the work of global corporations shapes the world.[6] My proposal counters that argument, insisting that design entrepreneurship *can* and *should* enact social empowerment and build community economies, not just competitive advantage for profit.

The work of researching and writing this book has exposed me to whole galaxies of ideas that are probably familiar to many. I have (admittedly slowly) realized that my own conception of capitalism—a conception similar to the one held by many of those with whom I have worked and by those I have taught—has been shaped by an era that is long gone. The mental models in my head about how the economy works and what roles

design plays in that economy started going out of date in the 1970s. I have been slow to catch on. I don't think I'm alone.

I did not initially approach this project with a deep political motivation. My earliest exposure to the ideas of postcapitalism came from studies related to digital media and digital culture more broadly—a territory full of ideas and practices defined by free access, open sources, and/or sharing in kind with a view toward individual productions built from a common pool of shared assets. Although I have studied and made use of some Marxist language and methods of analysis, I do not consider myself a Marxist, communist, or socialist—at least not in any specific identifiable form. My attempt here is to describe a political economy that requires new orientations between matter and culture, information and value, people and place—an "after capitalism" that doesn't fully exist and perhaps can't be fully envisioned. Nonetheless, I see that design can play a fundamental role in the existence and becoming of this new political economy. The perspective and proposal put forth in this book were shaped by my own research, the foundational work of many scholars, plus countless hours of discussion in classrooms and conferences and personal conversations with students and colleagues from around the world. It is truly a privilege to do this work.

Speaking of privilege, I would be remiss in not explicitly addressing my own positionality in relation to this work. In every conceivable way, I am a beneficiary of the Euro-centric model of modernity. The world I inhabit was designed to benefit me at the cost of others based on social status indicators and arbitrary events that have placed me near the top of the current social hierarchies of race, gender, and class. I did not earn these benefits, but I clearly recognize them and hope to leverage them toward a more just and egalitarian design—and world. Despite claims that capitalism is a system intricately linked to other systems of domination, it is seldom directly identified as being central to efforts at creating a more equitable model of design. This is my attempt to shake loose a brick in that wall.

As I was in the middle of writing this book, a pandemic startled the world. Millions of people around the globe have now died from a novel coronavirus, and all countries have been challenged to marshal the best social, political, economic, and medical responses they could muster. In 2020, the United States of America, the wealthiest country in the world, a

nation with a reputation for leading the world in innovation fueled by private competition, fared particularly badly. It set records for hospitalizations and deaths. A nation comprising just 5 percent of the world's population was home to 25 percent of global COVID-19 infections. Meanwhile, economic fallout from the pandemic accelerated many of the key challenges laid out in this book. With public and workplace lockdowns and restrictions in place, remote work, online shopping, and delivery services skyrocketed. Tech companies that own the platforms of information and economic exchange raked in huge profits. Amazon, Apple, Facebook, and Google all reported positive results in the second quarter of 2020 even as overall US economic growth fell by 32.9 percent. Nearly every other industry reported record-level layoffs, furloughs, and budget cuts, leading to the highest levels of unemployment in the United States since the Great Depression. In 2020, the 500 richest people in the world gained $1.8 trillion in net worth. Meanwhile, one in three residents of New York City faced food insecurity, with racial disparities disproportionately affecting Black and Hispanic New Yorkers—more than half of whom faced food hardship.[7]

As machine learning, artificial intelligence, and other expanding forms of automation render more and more jobs obsolete (or soon to be), the rapid transformations witnessed during the COVID-19 pandemic are just a montage glimpse of the future to come if we continue down the path we are currently traveling—massive disparities enabled by the decoupling of market values and profits for platform owners from the lived realities of the vast majority of humanity.

"Everyone designs who devises courses of action aimed at changing existing situations into preferred ones."[8] As we move into the post-COVID era, ways of living, being, and doing will remain affected. However, this is also a moment of sublime opportunity for rethinking, reengaging, and remaking a design that truly delivers on its promise to transform the world into "preferred states" and to do so in ways that are inclusive, sustainable, and just. This transformational process starts by acknowledging the profound responsibilities designers hold to their constituents—not just to clients or funders. This responsibility includes a deeper commitment to the communities, local and global, in which they work and in which their efforts directly affect countless anonymous lives. We are entering a new

global paradigm, so let’s shift the paradigmatic model of contemporary design. Let’s not shirk from the challenge, but embrace the possibilities.

To Think What We Are Doing

Sadly, designers are rather notorious for lacking in their breadth of knowledge. Generally, it’s not their fault. An emphasis on *doing* is a structural—and I believe necessary—element of design as an applied discipline. In fact, I believe the material-experiential expertise of design remains its critical distinction and is absolutely essential to creating a humane world. Yet the pleasure that designers seek from their work is also in part what makes design so dangerous. They can easily fall in love with the moves they make in the game without really paying attention to the rules of the game—or what game they’re playing!

The education of most designers is typically limited to studio and project-based work, substantiated by bits of history and theory, often cherry-picked and decontextualized. Designers seldom study political economy, let alone the history of or alternative possibilities of political economy. Some may have a general understanding of economics or business. Yet these perspectives typically take the structures and logics of capitalism to be a given. In practice, this narrow education means that designers often have limited theoretical frameworks to assess and, therefore, to even think about the implications of their work beyond the immediate or near-term effects of projects. This typically leaves designers limited in their capacity to analyze, discuss, and therefore truly think what it means to engage in “design practice” beyond being attentive to operational concerns or creating a coherent (and marketable) body of work. Their only recourse is to support and perpetuate existing economic and social paradigms, without acknowledging that they are doing so. This book questions those “givens,” in an attempt to transcend their structures through design practice.

I argue that design practice is *institution making*. This means that over the course of years working with various collaborators, affecting countless unknown “users,” and participating within systems of communications, technologies, material resources, labor resources, distribution and supply-chain systems, and so on, design puts into motion huge quantities of labor,

materials, technology, and investment. How designers do this work—how they engage with materials and labor, how they exercise their authority in decision making, how they engage the communities in which they work and live—makes a tremendous impact.

Capitalism is both an abstract idea that shapes design activities and a particular historical model of economic and social organization. This model has a history that has been both lauded and critiqued, a present that is undergoing significant challenges, and a future that we all can participate in shaping. Capitalism's enduring power lies partially in the ways it is cast as the only, the best, or maybe even the most "natural" model of resource allocation. By understanding the true hybridity of "the economy," however, one can recognize there are other ways to live, work, and act. Simply recognizing the existence of social and economic hybridity allows us to imagine different ways of designing.

Designers are doing new and seemingly more complex things almost daily. It wasn't so long ago that *user-experience design* wasn't even a term. Now it's one of the fastest-growing job titles. Designers are taking on new strategic roles within organizations across all domains, from health care to finance to government. These designers are affecting the lives of "users" around the globe, even as their work is often "experiential" or immaterial.

Meanwhile, many designers continue to design material things. Rapidly evolving technologies are quickly changing how material things are designed, produced, and distributed. Designers are strategic partners, conducting research, studying or shaping markets, and in many cases doing as much to determine *what* should be made and *why*, as to determine *how* it should look or be made.

The boundaries of "design" as a discipline and even of subdisciplines such as "fashion design," "graphic design," and "industrial design" are blurring such that these industrial-era terms seem antiquated. Even harder to assess than shifting boundaries is where all these disciplines are going. As a design educator, I am fascinated by the simple fact that when I look out at a class of design students today, I can't possibly imagine what their careers will be like. This is both exciting and terrifying!

As design practices become more diverse and more diffusely integrated into economic, political, and social systems, there grows a need to assess the fundamental epistemologies and ontologies of that work. What these

fancy words mean are how we know things to be true or valid (epistemology) and how we establish the principles or essences of *being in the world* (ontology). Designers might seldom think about it, but designing is, among other things, a way of knowing and making things known, just as the work of designing produces new *ways of being* in the world—for humans and nonhumans alike.

Temperature's Rising

In early 2019, a 16-year-old Swedish activist named Greta Thunberg started a global youth movement. She initiated school strikes around the world in which students walked out of their classrooms to protest the failure of leaders everywhere to make any meaningful changes to remedy global warming. The message was clear: the adults have failed. Children today and children in the future will suffer from the actions and inactions of those who have squandered the resources of this planet—particularly through an addiction to fossil fuels. They are right. We, the adults, have not done enough to ensure basic habitability for them. Young people are fed up, pissed off, and rightly so. We know that the planet is heating up. We experience the effects directly during exceptionally hot summer days. We read about the toll of overconsumption and its ecological impacts in parts of the planet that we might not see directly: melting glaciers, species going extinct, plastic gyres spinning in the ocean, and so on. As I am writing this sentence, the headlines are lit up with record-breaking heat waves, massive flooding, and wildfires. Soon enough, we'll be back in hurricane season. Or perhaps hurricane season will be year-round. What seems to get the most attention is that all these changes are having a major economic impact.

We live in a fully designed, artificial world. Whatever "nature" has been, it is now fully subsumed into the human world making that has produced the artificial world of our daily lives. What scientists have named the *Anthropocene*—the era during which human activity has the dominant influence on the planet's ecology—is more than just a new geological era. It's a new mindset. This mindset tells us that there is no more "away" in which to dump our personal garbage or industrial waste. There is no "out there" where raw materials can be strip-mined or fracked or where people

laboring under atrocious conditions can be safely hidden from sight. Incidentally, our current era has also been dubbed the *Capitalocene*[9] as the timeline of the rapid deterioration of our planet's health is also well aligned with the history of capitalism—something that may be more than coincidental.

In economic terms, an "externality" is "a consequence of an industrial or commercial activity which affects other parties without this being reflected in market prices."[10] In the Anthropocene, there are no more externalities.

We are living in a closed loop of connections. The scale and reach of those connections and their points of intersection may be enormous, but they are finite. The complex relational forces that have brought the entire world of physical materials and human sociotechnical activity into daily life has also brought us to an era in which we are witnessing an incredibly rapid loss of biodiversity as our planet heats up and every aspect of our ecosystem changes.

Design is everywhere. In the awakening to the world of the Anthropocene, in which there is no "away," no more "externalities," no infinite material resources, no cheap, faceless, unnamed labor in massive manufacturing facilities in locations we can't find on a map, no more unlimitedness of any kind—with, perhaps, the exception of the human imagination—what becomes of the work of design?

This awakening comes at a time when designers themselves can often be heard struggling, squeamishly sometimes, with questions of what is "ethical" in various practices of design. As design practices that might once have been focused on the production of stand-alone objects become more complexly intertwined with higher-level systems of production, distribution, and consumption—as distinctions between design and general "innovation" are also blurred—these questions become more difficult to ask, let alone address.

The design students I encounter today are increasingly discontent with the environmental degradation, corporate and political corruption, and yawning global disparity that constitute "business as usual." These students give credence to the notion that designing products, garments, and services requires adapting to and working within existing models of production and consumption. Although they may recognize complicity in the negative impacts of designing *for and in* these unsustainable models, they also have

no cohesive vision of alternatives. These concerns typically manifest with a sense of resignation or even paralysis: *But . . . what* can we do?

Hyperobjects

The British philosopher Timothy Morton has defined global warming as a *hyperobject*. *Hyperobjects* are entities "massively distributed in time and space relative to humans."[11] Their effects are immediately and continuously experienced, yet the hyperobjects themselves are too big and too diffuse—both physically and temporally—to be touched, perceived, or even fully *thought*. Yet they are here with us, and they are real. "Hyperobjects are not simply mental (or otherwise ideal) constructs, they are real entities whose primordial reality is withdrawn from humans."[12]

Global warming is a *hyperobject* par excellence. It is much larger than we can comprehend, yet it is both caused by and experienced by human activity every single day. Although I might experience a weather event, I don't directly experience climate. The physical and time scales of "climate" are just too big. When I turn on my air-conditioning on a record-breaking summer day, I am both experiencing and contributing to global warming. This is called a "positive feedback loop": the system's outcomes feed into the system, making it stronger. I can't feel it now, but that extra carbon dioxide I put into the atmosphere will be there for hundreds of years. Global warming permeates the entire planet, from the depths of the ocean to the outer reaches of the atmosphere and every location in between. Yet we also experience it when a category 5 hurricane destroys a city or when we feel the burning rays of the sun on our skin on an exceptionally hot day. "Hyperobjects are here, right here in my social and experiential space."[13] A hyperobject started long ago, yet it affects our thoughts and actions today and shapes what will be our future. Capitalism is also a *hy*perobject.

> Paradoxically, capitalism has unleashed myriad *objects* upon us, in their manifold horror and sparkling splendor. Two hundred years of idealism, two hundred years of seeing humans at the center of existence, and now the objects take revenge, terrifyingly huge, ancient, long-lived, threateningly minute, invading every cell in our body.

> When we flush the toilet, we imagine that the U-bend takes the waste away into some ontologically alien realm. Ecology is now beginning to tell us about something very different: a flattened world without ontological U-bends. A world in which there is no "away."[14]

Design, too, is a *hyperobject*. It is everywhere and shapes our every interface with the material world. Yet it is also intangible, ineffable, constructed by (often) unknown and (often) faceless others, distributed through space and time. At the scale of material culture, design is the stuff that occupies our physical lives, our work and leisure, and our social lives. Designed things, experiences, and the symbols that we use to perceive, understand, and navigate our lives are much more than just props to our existence. They directly shape how we live. They provide certain opportunities while closing out others. As the fundamental activity of building the "artificial" world of human experience, design has always been with us, has and always will shape us in ways that we can't fully comprehend. Design is too big and too diffuse to fully think.

Political Economy

A term used often in this book is *political economy*. This term is not commonly used these days because the term *economics* has generally eclipsed it. Economics as a discipline has advanced primarily through mathematical modeling and the development of theories to explain mathematical models and realities of economies. The term *economics* usually refers to study of the economy in rational or even neutral (as in "without value judgment") mathematical terms. The term *political economy* refers to an approach that sees the economy as entirely interdependent with social and political institutions, practices, and norms.

> From Adam Smith's *Wealth of Nations* in 1776 (or perhaps the Physiocrats even earlier) until at least John Stuart Mill's *Principles of Political Economy* in 1848, what we now call "economics" was in fact generally referred to as "political economy." This terminology in large part reflected the belief that economics was not really separable from

> politics. This was more than an administrative classification of disciplines; it arose from the widespread view that political factors are crucial in determining economic outcomes. Hence, as a discipline economics historically viewed political forces not only as influencing economic outcomes, but often as a determining influence.[15]

I've decided to use the term *political economy* here because it seems more specific and appropriate to the subject of design. As I hope to argue, design is inherently an activity with political and economic implications. The political nature of design lives in the simple fact that by designing things or experiences that are to be used by others, designers are shaping the choices, actions, outcomes, and desires of unknown "users" out there in the world. This "political" aspect of design concerns how it asserts or creates power relationships among people, objects, and symbols, which is different from the "politics" of government laws, processes, and regulations. My use of the term *political economy* is an attempt to remind us that we're constantly talking about both the *political* and *economic* dimensions of design.

Designing is systems oriented in its processes and procedures, in the way that it synthesizes disparate needs into the constraints that define and shape its action. Design is object or at least outcome oriented in its realization of objects or experiences whose every detailed phenomenon are the purview of the designing process. The political dimensions of design are embedded both in the designing of things and in the political power of *design as labor*, attached as it is to the production of nearly the entire material world.

The Structure

A formative influence on this project is the work of the American sociologist Erik Olin Wright, whose "real utopias" project calls for the intellectual enterprise of an "emancipatory social science." That objective requires three primary tasks: "a systematic diagnosis and critique of the world as it exists; envisioning viable alternatives; and understanding the obstacles, possibilities, and dilemmas of transformation."[16] The goal of my

work here is similar, yet oriented toward design. Based on Wright's model, the book is structured in three parts.

Part I: Diagnosis & Critique

Since its inception, capitalism has been celebrated and critiqued, and whole libraries have been written on its meanings and impacts. The professional fields of modern design were born into industrial capitalism, and the history of modern design is intimately connected to that of capitalism. While capitalism has mutated to adapt to changing conditions, design has also evolved along with capitalism's changing forms. However, particularly since the global financial crisis of 2008, many economists, sociologists, and cultural and political theorists assert that capitalism may soon devour itself. Capitalism is in the midst of two crises: a crisis of productivity and a crisis of legitimacy. Networked digital technologies are now producing free, abundant information goods and services that threaten the basic logics of property, production costs, and value. Meanwhile, escalating global inequality, corporate and political corruption, widespread and rapid destruction of the environment, economic precarity, "bullshit jobs,"[17] and massive accumulation of wealth by the very few have disillusioned a wide spectrum of people. Today, most young people in the United States no longer support capitalism, yet they hold no cohesive vision for an alternative. This means that the future users, clients, and *constituents** of design—and future *designers themselves*—have lost faith in "business as usual." A new political economy and new kinds of design are in the making, so long as they can escape the clutches of a dying system.

Part II: Alternatives

One of capitalism's enduring strengths is its illusion of totality. Any economy is actually *hybrid*, and, as such, many forms of economic activity already break the basic logics of capitalism even within capitalist societies. Various anticapitalist, noncapitalist, and even some fragmentary examples of postcapitalist models of design practice have existed throughout history. We can be informed and inspired by these case studies to rethink basic assumptions about how design works, how it produces value, and how its

fundamental activities can operate outside the seemingly inescapable logics of capitalism. These case studies, along with theoretical positions from a range of disciplines, are followed by a theoretical framework and theory of change for postcapitalist design practices.

Part III: Transformation

The final part begins with an exercise in strategic foresight to put forward two possible futures for design after capitalism—one cautionary and one aspirational. Design practice is argued as being in itself "institution making." Design practices direct massive flows of investment, labor, technology, and material resources, while normalizing cultural ways of knowing, doing, and being. Untethered from goals of profit maximization, design is capable of massive reorientation. The rationale for an egalitarian postcapitalist design sets the course for a defined set of principles, strategies, and practical guidelines that can be used today to begin the work of transformation.

The reader may disagree with some or all of the positions taken here. That is just fine. Political questions arise only when people disagree, and the value of disagreement is that it can generate positive transformation—so long as the discussants act in good faith. In part or whole, my hope is that the ideas shared in this book will empower designers to rethink and transcend some of the current detrimental social and economic models of design practice. Your labor in designing and strategically thinking through the impacts of designing is your power. I hope the book gives the historical precedent, a rationale, and the practical guidelines to work collectively in determining the ways that design itself might come into a "preferred state"—starting now and working into the future.

† Throughout this book, I use the term *global warming* to refer specifically to the heating of Earth's climate system due to human activities, primarily the burning of fossil fuels. I use this both in combination with and contrast to *climate change*, which indicates the broader effects of a changing climate.

* Terms that appear in the glossary at the back of the book are indicated at their first significant mention with an * (asterisk).

1
What Is Design after Capitalism?

New Frontiers

On October 4, 1957, an unprecedented event in the history of humankind took place. An earthly object, designed and constructed by humans, broke the bonds of Earth's gravity and hurled into the greater universe. This object, whose Russian name roughly translates to "fellow traveler of Earth," was a Soviet satellite. Sputnik 1 left the planet from a launch site in central Asia and briefly joined the celestial symphony of sun, moon, and stars; "it dwelt and moved in the proximity of the heavenly bodies as though it had been admitted tentatively to their sublime company."[1]

Sputnik was the first human-made object to leave the earthly grip of gravity and atmosphere. Beyond a first for the species, it was also the first major event of the Space Race between the United States and the Soviet Union, ushering in a new era of political, military, technological, and scientific developments. For most of the second half of the twentieth century, the two global superpowers were not only locked in a nuclear weapons stand-off but also engaged in an ideological showdown between capitalism and communism. They were competing technologically for the military upper hand as well as for symbolic victories—bragging rights—in "world's first" achievements. Although the military advantages of reaching space-flight milestones were seen as necessary for each country's security, the struggle for this competitive advantage was blended with the symbolism of demonstrating which political economy could create the most rapid technological innovation.

The Soviets struck first with the unmanned Sputnik. Shortly after that, in 1961, the Soviet Union sent the first human into space when cosmonaut Yuri Gagarin completed one orbit around Earth in the capsule Vostok 1. This sent American efforts into overdrive. Less than a year after Gagarin's landmark trip, John Glen became the first American astronaut to orbit Earth

in 1962. Later that year, President John F. Kennedy audaciously announced that the United States was planning to send a man to the moon. That bold ambition was achieved in 1969, when American astronaut Neil Armstrong became the first human to walk on the moon. This “giant leap for mankind” was part of a suite of Apollo missions, which included six subsequent missions that landed astronauts on the moon and eventually ended in 1972. The total cost for American taxpayers was $25.4 billion in 1973 dollars (equivalent to about $153 billion in 2019).

In her introduction to *The Human Condition*, a book on the nature of political action in the era of science that was published in 1958, the German American political theorist and philosopher Hannah Arendt reflects on the curious response to the Sputnik launch just a year earlier. What she found remarkable was not so much the combination of awe and fear but the sense of joy. Yet that “joy was not triumphal; it was not pride or awe at the tremendousness of human power and mastery which filled the hearts of men, who now, when they looked up from the earth toward the skies, could behold there a thing of their own making. The immediate reaction, expressed on the spur of the moment, was relief about the first ‘step toward escape from men’s imprisonment to the earth.’”[2] In that moment, in the shadow of global war and on the precipice of a potential nuclear Armageddon, there was the sense of relief that now, finally, humankind might be set free from the mess made on this planet.

Just 70 years after Sputnik, on February 7, 2018, the world’s most powerful operational rocket delivered a very unconventional payload to the universe. The billionaire engineer and entrepreneur Elon Musk’s rocket firm SpaceX sent a 2008 cherry-red Tesla Roadster into heliocentric orbit. The goal was to set the sports car—one item from Musk’s empire of technical products—into an orbit around Mars, but it missed the mark. The vehicle is occupied by Starman, a spacesuit-wearing mannequin who is, ostensibly, a stand-in for anyone. This stunt immediately became a planetary sensation with viral images and videos of Starman coolly cruising the cosmos lighting up terrestrial and social media networks. In a period less than the average lifespan of an American, the launch of an earthly object into space went from an event “second in importance to no other” to a spectacle of brand making.

Although the SpaceX stunt appears less existential than the pivotal achievement of Sputnik, it is telling of our current cultural moment that technical achievements are fully enmeshed within a commercial project. NASA has done its due diligence as a public institution by classifying the Tesla Roadster as a celestial body and will continue to track it *indefinitely* along with all the other objects being monitored in the solar system, such as planets, asteroids, and satellites. It is very possible that Musk's goal of space tourism or even colonization will become a reality in the near future —at least for a handful of very wealthy humans. Space is no longer a frontier. Space is now a luxury "experience" or an opportunity for competitive advantage, defined by lifestyle images such as Starman whose cosmic journey is building a new market of desires very different than those elicited by the science fiction of the previous century. This rapid transformation in which space exploration has moved from a species-level achievement to luxury branding may be the ultimate expression of what has recently been described as *innovation pessimism**: a growing sense that the technological achievements of our era are of decreasing significance. Instead of creating visions for new ways of life on Earth, we flood the world—and the cosmos—with gadgets of minimal social value.

With the dissolution of the Soviet Union and China's economic transformation at the end of the twentieth century, the Cold War rivalry between capitalism and communism came to an end. At the dawn of the twenty-first century, capitalism seemed to prevail as the clear winner of the global ideological conflict of the long twentieth century. With its triumph over communism—the only other enduring model for political and economic organization—it seemed no competitor for an alternative to capitalism could ever stand a chance. This moment was even declared the "end of history,"[3] suggesting that all historical attempts at achieving a perfect model for social and economic organization had been achieved in liberal, democratic capitalism. With this success, particularly amid the globalization of communications, labor, and markets for buying and selling things and ideas, capitalism has assumed a sense of destiny. In the first decade of the twenty-first century, it appeared to be the final dominant global model—a "natural" or at least an unavoidable condition of the contemporary world. In 2003, the American literary and political theorist

Fredric Jameson famously stated that it is now “easier to imagine the end of the world than the end of Capitalism.”[4]

However, since the global financial crisis of 2008, a growing chorus of economists, sociologists, cultural theorists, political theorists, and other researchers and social commentators have asserted that capitalism may soon defeat (or devour) itself. This assessment runs alongside the voices of millions of global activists shouting and sometimes dying for a more just world. Beyond the crisis cycles inherent to capitalism, the rise of networked digital technologies has delivered the capacity to produce free, abundant information goods and services that are threatening the basic logics of “scarcity” and “property.” At the same time, new and emerging technologies suggest that material products may soon approach zero marginal production costs, as many of the forms of labor that go into the material production of global society are trending (or threatening, depending on your perspective) toward near total automation.

Although diverse, these perspectives tend to agree that capitalism is unlikely to end in an abrupt rupture or revolution. Rather, it poses a threat to its own existence. Technological revolutions of value and labor, saturation of markets, and escalating global inequality through wealth accumulation are synching up with social unrest over corporate and political corruption, unprecedented destruction of the environment, economic precarity, “bullshit jobs,”[5] and massive accumulation of wealth by a very small number of people—all while productivity increases yet wages stagnate.

In 2016, a series of signals indicated that the American public’s support of capitalism had waned dramatically in recent years. Perhaps the “end of history” was prematurely declared. Of course, that year will remain memorable in the history of US politics for a presidential election whose front-runners included a billionaire real estate developer with no record of public service, the longest-serving Independent US senator in history, who is also a self-described democratic socialist, and a former secretary of state, who is also a former senator and a former First Lady and whose name has become synonymous with modern American political dynasties. The rhetoric was swirling at whiplash speed. The 24-hour news cycle of the cable television era was liquefied into the constant flow of Twitter and other social media outlets, where “the news” is doused in opinion, conjecture, and

deliberate misinformation and where the fragmentation of information exacerbates political divisions. So much for “the news”!

During the run-up to this wild election, the Harvard Institute of Politics conducted a poll to survey the political views of Americans 18 to 29 years old. The questions were focused primarily on the presidential election and connected policy positions (or at least talking points related to them). Amid the barrage of ideological rhetoric, a small nugget of seemingly significant information seeped out but received little attention. When asked “Which of the following, if any, do you support[: capitalism or socialism]?,” only 42 percent of respondents said they support capitalism. Only 33 percent supported socialism. Two years later, Gallup conducted a poll with similar results, demonstrating that young Americans are steadily souring on capitalism.[6]

It would be easy to read too much into specific polling results, yet, clearly, these numbers indicate a rising sentiment. Zach Lustbader, a Harvard student who was involved in the 2016 poll, said, “The word ‘capitalism’ doesn’t mean what it used to.”[7] Americans in the age range polled likely have little memory of the Cold War, a time when capitalism was a model of freedom in comparison to the command economies of totalitarian regimes. Instead, young adults today grew up during the 2008 global financial crisis and its aftermath. They saw what unfettered and unregulated economic self-interest looked like as reckless risks taken by financial institutions brought down the global economy and its citizens, requiring those same citizens to pay the balance sheet (and then some). They watched their own futures written off.[8]

What these polls reveal is that a majority—slim, but a majority nonetheless—of the American population is turning away from capitalism. This indicates that the emerging generation of entrepreneurs, professionals, politicians, civic activists—and, yes, *designers*—are disillusioned by one of our society’s most fundamental organizing structures as well as by the many institutions that support and reinforce its principles. Many young people today recognize they are reproducing a society they don’t want. Yet there appears to be no cohesive vision among them for a preferred alternative.

No doubt, many diverse factors contribute to this growing malaise. However, two stand out: climate change and the growing social and economic disparities fueled by technological change. First, global warming

tells us that we are living in a finite system. Even space travel and plans for colonization of other planets indicate that we are constantly living inside closed loops of finite resources. For the past 250 to 300 years, capitalism has more or less succeeded in treating this closed loop as an endless succession of new markets and new territories, with various "cores" being fed and sustained by their "peripheries"—zones where material and labor resources can be exploited. The Anthropocene instructs us that there are no more externalities (and, actually, that there never were any). Second, as the financial journalist and author Paul Mason has pointed out, there are growing contradictions between digital products, services, and automated forms of production and an economic model built on the assumption of scarcity. Mason says, "The main contradiction today is between the possibility of free, abundant goods and information and a system of monopolies, banks and governments trying to keep things private, scarce and commercial. Everything comes down to the struggle between the network and the hierarchy, between old forms of society moulded around capitalism and new forms of society that prefigure what comes next." He describes the situation in terms of conflicting models: "*An information economy may not be compatible with a market economy*—or at least not one dominated and regulated by market forces primarily" because a system "based on markets, property ownership and exchange . . . cannot capture the 'value' generated by the new technology." In other words, "it is increasingly evident that information goods conflict fundamentally with market mechanisms."[9]

Two powerful and unpredictable forces are shaping where we go next: climate change and technology. We'll return to these two forces regularly in this book. However, before a substantive discussion can be had on how capitalism shapes and limits both society and design, the subject needs to be defined.

What Is Capitalism?

There are generally two ways to think about what capitalism is. In abstract terms, it is a model that organizes relationships between people, things, and institutions. In historical terms, capitalism is a model for social and

economic organization that emerged from specific historical conditions and has its roots in seventeenth-century England.

Clearly, societies and economies have changed a great deal since the seventeenth century, yet many economists agree that there is continuity from then to now. In fact, one of the most commented upon aspects of capitalism is its ability to adapt to changing conditions while retaining its basic contours. Capitalism has taken on both different general forms—such as free-market capitalism, welfare capitalism, and state capitalism—as well as specific forms unique to the different ways that different countries have adopted it into their political models. Perhaps the most obvious but important thing to note about capitalism as a historical model is that *it didn't always exist*. Its origins are specific to a particular time and place, and this contingency is what makes it almost certainly a historical phenomenon that will come to an end.

To define capitalism in abstract terms, we can identify some of the key characteristics that have carried through its many mutations. According to *Merriam-Webster*, the definition of the term *capitalism** is: "an economic system characterized by private or corporate ownership of capital goods, by investments that are determined by private decision, and by prices, production, and the distribution of goods that are determined mainly by competition in a free market."[10]

Although this short definition is handy, it may be useful to extend it a bit further by identifying key features:

1. Private ownership and control over the means of production and value generated from production of goods and services (land, factories, machinery, business equipment and technology, intellectual property, digital platforms, etc.)
2. Transformation of labor power into a commodity via wages (paid labor)
3. The production of goods or offering of services for profit through an exchange mechanism (markets)
4. Market competition as the motivator for innovation, technological development, and efficiencies in labor and material production

This is in no way an exhaustive list of characteristics, but it provides a foundation so that we can begin to see how design activities intersect with capitalism. The one thing missing here—and it's important—is how this seemingly simple economic arrangement also shapes social organization and social relations. Markets have existed through much of known history, and producers have owned their own means of production in different historical models. To further distinguish capitalism from what existed before and what might come after, we should emphasize how capitalism establishes specific *social relations** of production. One of capitalism's unique features is the particular kind of *class** hierarchy it produces. Under capitalism, there is a class that privately owns the means of production, and there is a class that subsists by selling their labor to the owners. Either you have money to invest in making more money, or you work for money to live. It is worth considering the "middle" role that designers play in this structure. Designers sell their labor power to others—their "clients." However, designers also direct the capital of their clients to bring their own efforts into production. It is this liminal position that offers designers unique opportunities to direct or redirect—to varying extents—social relations in the work of producing things.

Capitalist Logics

An observation made by many critics of capitalism is that the logics that operate economic activity—such as ownership of private property and market competition—have seeped into the everyday fabric of society. For example, the logic of "commodification" can be seen working in the ways people living under capitalism relate to each other. In the abstract terms of *commodities* and *labor*, each *thing* and every *action* can be seen as having *value* on a *market*—real or imagined. In a consumer culture, the connection goes further. Under the logics of capitalism, things begin to stand in for human attributes.

The American anthropologist Elizabeth Chin describes the attachment an infant makes to a favorite blanket. The infant cries to be held and comforted by the warm embrace of a parent. The parent, too tired to keep coddling the infant or needing to attend to some matter more important than embracing a

baby, comforts the child with a blanket. Over time, the blanket becomes a substitute for the person as the infant's subject of love. The infant is thus at a very young age indoctrinated into drawing "comfort from a thing."[11]

Today, designers can be found working on many different forms of robotic companions—robot-looking robots, human-looking robots, cat- and dog-looking robots.[12] They are designed and marketed as friendly and supportive companions for children, the elderly, the depressed, and people going through various forms of rehabilitation. How can we read the capitalist logics built into these projects? Commodity items are substituted for the companionship of parents, children, friends, and even *real* cats or dogs—often with a big price tag.

Beyond physical things that produce comfort in our lives, designed symbolic representations go a long way toward determining or expressing our social identities. The German economic sociologist Wolfgang Streeck has studied how capitalism has adapted, particularly in the decades since the turning point in "mature" capitalist economies in the 1970s, toward what is called "late capitalism." What he has observed can be generally described as the transition from an industrial economy organized around producing value by making things to a postindustrial economy organized around producing value from knowledge, information, and other activities we loosely call "innovation." Streeck points out that "today's markets for consumer goods in the rich capitalist countries are by and large saturated, making it essential for capitalist profitability to get individuals whose *needs* are covered to develop *desires* that give rise to new desires the moment they are fulfilled. Product design and advertisement are instrumental for this, but also low prices as made possible by the sweatshops of today, out-of-sight of final consumers and out-of-reach for collective solidarity."[13]

Streeck is specifically naming key roles that design plays in generating unnecessary and ultimately unfulfillable desires. These efforts serve primarily to produce profits for business owners and shareholders (who are compelled to drive toward ever new ways to make money off existing money in order to survive the competition inherent to the system) rather than to address real "needs." One feature of this arrangement in a global labor market—where the cheapest labor that can be found is the most competitive, regardless of the working conditions—is the proliferation of what he refers to as sweatshops "out-of-sight of final consumers."

The anthropologist Anna Lowenhaupt Tsing describes Nike's transformation from a distributor for the Japanese shoemaker Onitsuka Tiger to a global footwear brand: "When Nike's founders established their independence from their Japanese chain, they added style—in the form of the Nike 'swoosh' and advertisements featuring black American sports heroes. Learning from their Japanese experience, however, it never occurred to them to manufacture shoes. 'We don't know the first thing about manufacturing. We are marketers and designers,' explained one Nike vice-president.'"[14]

Today, the labor needed to manufacture and assemble consumer products, garments, and material goods is sourced globally, producing a downward race for lower wages and less-regulated working conditions. Nike's regular run-ins with criticism and consumer backlash over the use of unregulated labor and child labor, unsafe working conditions, and what we generally call "sweatshops" is a prime example. Although proponents of global labor competition might argue that any job is better than nothing, recurring disasters at garment factories and high rates of suicide in electronics factories are a reminder of the social costs this system is designed to produce. This is to say nothing of the massive environmental toll of both fast fashion and consumer electronics.

In addition to the scaling-up and moving "out of sight" of material products, the value that can be generated through symbolic representations becomes increasingly valuable in a postindustrial economy. Designers work with corporations to create "brands" as a series of symbols. These symbols produce lifestyle types and thus create norms and desires—always less than fully satisfied—by which individuals understand and communicate their values, identities, and positions within a complex and diverse society. However, in this process the "consumers" are not really producing their own values or identities; they are simply picking from those put forth by brands on the market.

What is meant by the term *capitalist logic*, therefore, is the way that capitalism inflects a certain way of thinking about the world into aspects of life that have nothing in particular to do with economic activity. To think about "investing" time in an activity or a relationship ("capital"), to think about the anticipated returns on this investment ("profit"), to think about the material world or people and their talents ("labor") as resources ("raw

materials") for other ends is to engage in capitalist logic. Admittedly, it's really hard to avoid thinking like this because this logic has seeped out of an economic system and into our everyday lives. Particularly under the constant barrage of marketing, all language eventually sounds like marketing language. Yet it may still be possible to think differently.

Technology, Innovation, & Value

Much of the disciplinary knowledge of modern design has evolved primarily out of the historical conditions of industrial capitalism. Design has consistently played dual roles in shaping products (what to make and how) and in creating markets for those products (generating desire through brands, advertising, lifestyle imagery, and so on)—from the mass-produced clothing made from cotton mills at the start of the Industrial Revolution to sophisticated digital products and services today.

Both design and technology are touted as engines of "innovation." What this really means is that they contribute to change. One of the basic tenets of capitalism is that it leverages market-based competition as a motivation for innovation. Everything must change and change constantly to produce the kind of growth that capitalism requires. Both design and technology are catalysts of this constant change, and they have a long history of working together. As the speed of technological innovation increases, so, too, does it seem that design and technology become more intertwined. One of design's common roles in relationship to technology is to "domesticate" new technologies by transforming them into consumer products or services.

Take, for example, the transistor radio. As for many consumer electronic products, the technical research behind the transistor radio's invention is rooted in military origins. During World War II, Bell Labs began working on attempts to produce "crystal" mixer diodes for use in radar units. After the war, a group of American physicists continued this research and in 1947 invented the first transistor. Before that, most radios used vacuum tubes, which were inefficient, fragile, bulky, and heavy compared to transistors. Commercial radios for domestic use were large, furniture-like objects.[15] With the invention of transistors, a much smaller radio receiver could be produced that used far less power and had greater durability and shock

resistance. This meant listening to the radio could now be a mobile activity—something you could do anywhere.

Several companies made plans and prototypes for a commercial transistor radio in the late 1940s and early 1950s. Ultimately, it was the combined efforts of two technology firms, Texas Instruments and Industrial Development Engineering Associates, that produced the world's first commercially available transistor radio, the Regency TR-1. The object was designed by industrial design firm Painter, Teague and Petertil.[16] Legend has it that the design was completed in just six weeks—remarkable for a first in its category—and largely done over the phone and via drawings that were exchanged via mail. The circuitry for TR-1 was housed in a plastic case, initially in the colors black, bone white, mandarin red, and cloud gray, and the new transistor radio was advertised to weigh just 12 ounces.

The TR-1 was an immediate success. Soon after that, companies began competing to make radios that were smaller and cheaper in order to get in on the market for this new sensation. In 1955, the small Japanese company Tokyo Tsushin Kogyo (Tokyo Telecommunications Engineering Corporation) introduced the first transistor radio to utilize all miniature components. The company's name would be unpronounceable to American audiences, so a new brand name that sounded a little like the American slang "sonny boy" was invented: Sony.

In 1955, the company introduced both the first Japanese transistor radio and the first product to carry the Sony name: the TR-55. Working with an unprecedented size, much smaller than the TR-1, the Sony designers sought an entirely new approach to the design and shape. The radio was the first to use punched aluminum for a speaker grill.[17]

Fast on the heels of Sony, the German company Braun got in on the action with the T3 pocket radio. Released in 1958, the T3 was designed by Dieter Rams in collaboration with the Ulm School of Design. The T3 is a stunning example of Rams's design philosophy materialized in the form of a consumer product: it is an unadorned white box; the tuner is a circle; a grid of holes that serve as the speaker cover create a perfect square.

In the documentary *Objectified* (2009), Dieter Rams told director Gary Hustwit, "I always strove for things to be sustainable. By that I mean the development of long-lasting products—products that don't age prematurely,

which won’t become out of style. Products that will remain neutral, that you can live with longer.”

The T3 was more expensive than its competitors. However, the combined dedication to design quality and technical superiority identified it as a premium item. The T3’s minimal design inspired and influenced many subsequent products, including another landmark in the portable audio revolution: the first-generation Apple iPod.

Transistor radios quickly became the most popular electronic communication device in history. Billions of units were manufactured during the 1960s and 1970s. Technological innovation dwindled quickly, and, eventually, new models were essentially just different versions of plastic packaging for the same functional core. This packaging incorporated a huge range of novelty designs, including radios that looked like cartoon characters such as Huckleberry Hound or Tom and Jerry or product promotions in which the radios appeared to be a pack of Marlboros or even a tin of Spam!

In 1983, the Argentinian architect and industrial designer Daniel Weil responded with a novel design that made a statement on the nature of designing radios at the time. He created the “Radio in a Bag.” Designed by Weil and manufactured by APEX in Tokyo, the object demonstrates quite clearly what the design of transistor radios actually entailed. The functional components of the radio are strewn within a clear PVC bag, stating that the design of such objects was at that point essentially an act of designing packaging. According to the V&A Museum, which collected one of the objects: “The working parts of this transistor radio are displayed as ornament. The traditional hard, protective case of the radio, which hides its components, is replaced with one of flexible, see-through PVC. It is constructed as an envelope, so all the parts are easily accessible for repair. 10,000 were made. They were sold mainly in Japan within a year.”[18]

Weil’s project is an example of “discursive design.”[19] Despite the fact that the Radio in a Bag was created for commercial purposes—it sold 10,000 units in about a year!—it also generated discourse. In this case, the object’s discursive nature was its provocation of the role of design as something akin to window-dressing the technical components of a consumer electronic product.

Transistor radios fundamentally transformed the experience of listening to music, further influencing popular culture. Beginning in the 1980s, cheap transistor radios began losing ground to products that allowed further freedom and freedom of choice—the ability to listen not only to what's on the radio but to whatever you like—such as portable cassette players, including the Sony Walkman. Of those billions of transistor radios, it's likely that many are still operational to this day. However, one after another, they lost their value in the market by being subsequently designed or "innovated" out of their desirability.

This narrative can be told many times over. Beyond the work designers do to domesticate new technologies into consumer goods, design also accelerates the pace at which products become obsolete—not only for technical reasons, but also for stylistic reasons alone.

The industrial economy has given way, partially but not fully, to a postindustrial economy in which information, knowledge, and services are fundamental economic drivers. In this transition, design has found new roles in territories such as service design, designing software as product or service (user-experience and user-interface design), and strategic design. Yet designers also still design *things*. The material conditions of designing a chair, for example, may not seem to have changed much. Wood is still wood, right? However, in the era of urgently responding to the immediate and long-term threats of global warming, wood is no longer simply wood. It is no longer one of many natural "resources" to be selected. New material sensibilities are also required for the changing landscape of materiality, such as biotextiles. Meanwhile, new ways of fabricating, designing, sharing, communicating, and distributing also change how materials might be transformed into designed objects. These and other paradigm-shifting trajectories collide with the outdated logics of a capitalist system of production.

Economic Hybridity: Capitalism Is Only the Tip of the Iceberg

One of capitalism's most powerful features is the way it is portrayed as a complete, dominant system. Although capitalism is no doubt hegemonic—exerting its influence beyond economic activity alone—it never operates in a "pure" form.[20] The varieties of capitalism across history and geography are diverse, but the logics and principles that are "alien from capitalistic norms of individualism, competition, the belief in endless economic growth, and accumulation, among others, exist 'here and now' in the interstices of modern capitalist societies." The sustainability researcher Giuseppe Feola claims that "scholars have demonstrated that capitalism rests on 'hidden' (unvalued in economic terms) social reproduction in the household, non-market exchanges, and nonutilitarian rationality as expressed in social solidarity networks and related manifestations."[21]

Feminist economic geographers writing under the pseudonym "J. K. Gibson-Graham" have contributed an important critique of capitalism by demonstrating that any capitalist economy comprises a wide range of economic diversity, including community economies. They have coined the term *capitalocentric*: "When we say that most economic discourse is 'capitalocentric,' we mean that other forms of economy (not to mention noneconomic aspects of social life) are often understood primarily with reference to capitalism: as being fundamentally the same as (or modeled upon) capitalism, or as being deficient or substandard imitations; as being opposite to capitalism; as being the complement of capitalism; as existing in capitalism's space or orbit."[22] By rejecting a *capitalocentric* viewpoint, Gibson-Graham argue for recognition of the much broader range of economic activities, models, relationships, and institutions that coexist.

This rejection of capitalocentrism is significant for designers because it provides a framework to assess other models of design practices—institutions in themselves—that already have, currently do, or might in the future coexist with capitalism. Establishing such models in current conditions sets the stage for building capacities toward more intentionally strategized, broadly implemented, and collaboratively powerful postcapitalist institutions—including design practices.

The End of the End of History?

Two decades into the twenty-first century, the future of capitalism looks uncertain. Perhaps, most obviously, it is impossible to maintain this engine of perpetual growth on a planet of finite resources. “Garbage time is running out.”[23]

The matsutake mushroom is a rare and valuable item associated with traditional Japanese gift rituals. In her incredible study of the human and nonhuman actors that make up the global matsutake trade, the American anthropologist Anna Lowenhaupt Tsing describes the making of new life systems “in Capitalist ruins.” In the past few decades, the matsutake has demonstrated a fascinating trait: it grows remarkably well in a symbiotic relationship with certain species of pine trees that flourish in areas of the Pacific Northwest that had previously been completely stripped of trees by the timber industry. These pines sprout up in the left-behind remains of commercial timber and logging. So, too, does the matsutake. This new growth has drawn a curious and complex network of people into the forests, including many South Asian refugees, to forage, sort, and export the mushrooms.

Tsing describes these activities as “salvage accumulation: the creation of capitalist value from noncapitalist value regimes.” Tsing is cautious about prematurely calling the politics of these activities *postcapitalist*. She sees this network of political economy producing itself on the *periphery* of capitalism and so prefers the term *pericapitalist*. She describes the spaces of “salvage accumulation” as “unlikely platforms for a safe defense and recuperation” from capitalism.[24]

Design operates differently, though. Its politics can be directed at the *core* of capitalism, as well as the *periphery*. Yet to do so requires designers to recognize the institution-making nature of their practices—whether those practices are internal or external to corporate capitalist operations. Recognizing the value that designers bring in both their knowledge and their material expertise, this labor power is substantial in the kind of hybrid economies we see in the postindustrial era.

Erosion

As a research project, *Design after Capitalism* blends empirical and theoretical contributions from contemporary researchers in sociology, philosophy, sustainability and environmental studies, digital and media studies, cultural theory, political science, history, *critical theory*,* and economics with concerns of design practice. In particular, the real utopias project by the American sociologist Erik Olin Wright has been formative. Wright synthesizes nearly 40 years of empirical research on anti- and noncapitalist models of social and political institutions.[25] Translating Wright's work on social institutions, we can begin to formulate similar models and strategies specific to design.

The first step is to position "design practice" as "institution making." Design practices systematically generate value, social functions, and codified *ways of doing* for their employees, clients and constituents, business or social partners, and the communities in which they work. In the particular ways that *designers do designing*, design practice becomes an economic and social institution that operationalizes relations between labor, materials, property, and value.

Wright's description of the "strategic logics of anticapitalism" offers a valuable framework. Wright uses a two-by-two matrix to identify four loci of "anticapitalist" activity, with an axis for the *location* of the strategy (state or civil society) and an axis for the *goal* of the strategy (neutralizing harms or transcending structures).[26] Wright describes different versions of the individual loci: *taming capitalism* by neutralizing the harms of the state; *resisting capitalism* by neutralizing the harms of civil society; *escaping capitalism* by transcending the structures of civil society; and *smashing capitalism* by transcending the structures of the state. What Wright proposes is "a way of thinking about strategies opposed to capitalism that combine taming and escaping and connect both of these to resistance." The strategy he proposes "is to tame capitalism in ways that make it easier to escape and to escape capitalism in ways that make it easier to tame[;] . . . this is a way of approaching the problem which does point beyond capitalism, but it does so without also demanding a rupture."[27]

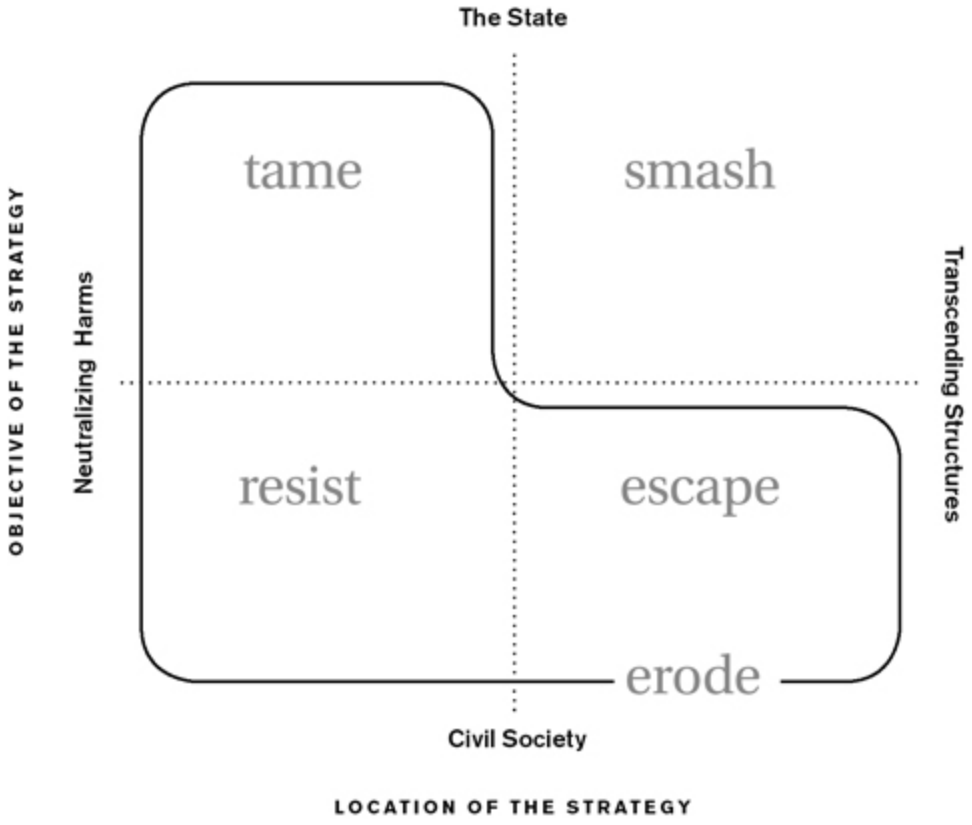


Figure 1.1
Strategic logics of anticapitalism, based on ideas by Erik Olin Wright.

Wright calls this logic the “erosion” of capitalism—transcending structures and neutralizing the harms of capitalism in civil society while also neutralizing the harms inflicted by government—to create new forms of economic and social functioning that can collectively grow and permeate social relations more broadly. Situating design practice into the logic of “erosion” suggests that substantial transformation can occur outside—or before—significant transformations of capitalist government. This can be accomplished by focusing primarily on the capitalist structures of civil society. We will continually return to the concept of *erosion** throughout the book.

Design after Capitalism

The notion “after capitalism” does not suggest a specific form of socialism or communism or a “return to” any precapitalist structure. It means moving

past or beyond the structure we have now, in ways that *build upon and transcend* the logics, structures, and subjectivities that capitalism has created. It has been argued that as capitalism devours itself, a new form of feudalism or neofeudalism, may emerge.[28] Some think this has already happened.[29] Contrasting this pathway, “after capitalism” names moving toward a new and as-yet-undefined system that we can’t yet fully see or think—or maybe even appreciate—because it is in a state of becoming. It is a new system of economic and social organization that will bring with it new forms of social relations and economic relations, new symbols of social and economic life, and new logics for existing together on a crowded and warming planet. While we may not be able to see its entirety, we can sketch the basic contours of this undefined new system and begin creating it today.

Once a designer recognizes that capitalism is only the “tip of the iceberg”—as J. K Gibson-Graham have argued—that it is hegemonic but not totalizing, that “the economy” is and always has been hybrid, that value is constantly created out of noncapitalist forms of social relation, that one of design’s primary functions is to find new and “better” ways to operate within the given parameters of the situation, that design practice is itself institution making and affects both the world and the lives of countless humans and nonhumans well beyond the lived experiences of the involved designers, clients, and users . . . then the opening for design after capitalism is made visible and ready.

The sci-fi author William Gibson has famously stated that “the future is already here—it’s just not very evenly distributed.” Design after capitalism is also already here in various ways—it is just not evenly distributed and is under constant strain from the capitalist structures and logics that envelop it. It has been here in the practices of designers who have intentionally and systematically operated against, outside, or beyond capitalist structures and logics. It is here transcending the structures of civil society and neutralizing the harms of capitalist institutions in an *erosion* of the constraints established by capitalism. However, it has not yet fully flourished. Practices outside capitalist structures suggest design’s potential to align with the historical moment of postcapitalism. Perhaps then it will become the central premise of a transformed discipline: design after capitalism.

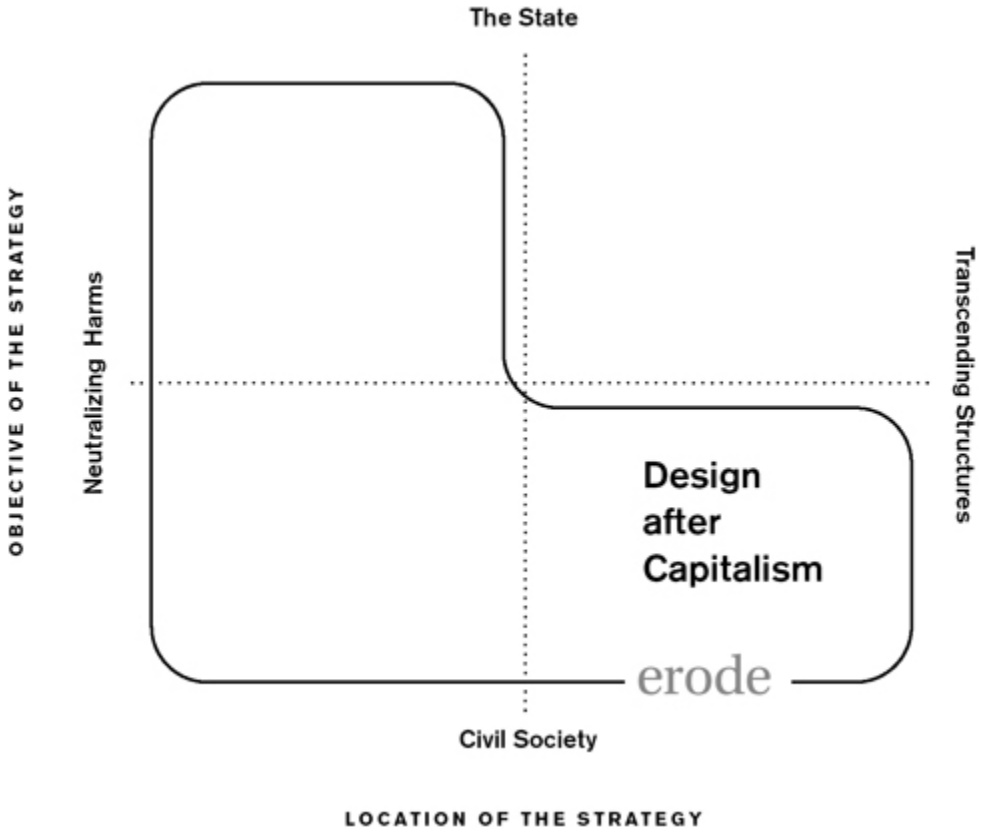


Figure 1.2
Strategic location of design after capitalism, within Erik Olin Wright's strategy of *eroding* capitalism.

I

Diagnosis & Critique

But baby's on fire
And all the instruments agree that
Her temperature's rising
But any idiot would know that

—Brian Eno, "Baby's on Fire"

2

Design, Capitalism, & Economic Theory in the Industrial Era

Parallel Histories

Capitalism is a model of social and economic organization that emerged from specific historical conditions. This chapter and the following give a brief survey of the origins and evolutions of three interconnected subjects: capitalism, modern design, and capitalist economic theory. To do so, these chapters treat each of these three subjects to episodic descriptions that are woven together to follow general stages in the evolution of capitalism's history.

The historical perspective is admittedly Western and specifically American. It is not intended to further perpetuate the colonial mindset that prioritizes European modernity but, rather, is based on a specific thread of historical development that has informed a particular set of modern design practices. When these three histories are juxtaposed in ways not typically explicit, each might be understood differently by direct relation to the others. Ultimately, the goal is to help see modern design's complex historical associations with capitalism and the theoretical ideas that have connected material production and consumption to value and values in Western capitalist societies. The sequence is as follows:

Origins: Feudalism & Mercantilism to Capitalism: 1500s ~ 1700s

Early Industrial (Anarchic) Capitalism: 1750s ~ 1850s

Advanced (Managed) Industrial Capitalism: 1850s–1970s

The (So-Called) Golden Age of Capitalism: 1945–1970s

Postindustrial (Late) Capitalism: 1970s–present (see chapter 3)

Origins

In the early 1600s, several European empires were in competition to colonize the far reaches of the globe. Competing British and Dutch trading companies regularly voyaged to India and Southeast Asia for spices, silk, and textiles to be sold back home. These journeys took months or even years and were fraught with the hazards of shipwreck, piracy, mutiny, and disease. The ships had to be built, repaired, and outfitted with enough food, water, provisions, and weaponry to maintain large crews—sometimes more than 150 sailors—for very long and very dangerous expeditions. Shipyards became major operations, employing local labor and requiring the purchase of huge quantities of supplies.

All of this required massive up-front investment, or "capital."[1] This capital was often obtained from wealthy London or Amsterdam merchants or aristocrats with access to family fortunes who sought access to exotic goods. Each voyage was a high-risk venture. All the capital invested to send a single ship off to distant lands could suddenly disappear with one catastrophic storm or act of piracy. Voyages were initially financed separately, each its own capital investment with the possibility of a hefty return or total loss. Eventually trade companies began an innovative new approach, shifting their finance methods to spread out investments (and risk) over multiple voyages and multiple years. Diversified investments reduced risk and allowed for even those with no direct ties to the company to participate in the gamble of investment into the company's activities. Thus, the "joint-stock" company was born.



Figure 2.1
Logo for the Vereenigde Oostindische Compagnie (VOC), or Dutch East India Company.



Figure 2.2
VOC flag with the letter *A* indicating “Amsterdam,” flying on a replica East Indiaman ship. (Photograph by McKarri.)

Founded in 1602, the Dutch East India Company (Vereenigde Oostindische Compagnie, or VOC) became the world’s first publicly traded corporation. Through its innovations in finance and business practices, VOC laid the groundwork for many features of modern-day global corporations. VOC ships flew under both the Dutch flag, symbolizing that the ship was protected by the royal Dutch navy, and the corporation’s “trademark.”[2] This mark, a monogram of the characters VOC, is generally

considered the first globally recognized corporate logo. It was emblazoned on items denoting company property, including flags, shipping containers, and cannons, and later adopted on merchant goods such as porcelainware. The primary monogram, spelling VOC, appeared below the first letter of the name of the city chamber conducting the operation. For example, an A placed over the VOC indicated that an expedition was coordinated from Amsterdam. The simplicity, legibility, and flexibility of this graphic system of identification make it the first example of a corporate identity system, setting a standard for those that would be produced hundreds of years later.

This story is fascinating in its demonstration of design symbolically integrating new forms of enterprise and ownership at the very moment new economic infrastructures, such as joint-stock corporations, were invented. Monograms, symbols, and even trademarks had already existed a long time. Ancient Greek and Roman pottery were stamped with symbols identifying their makers, and European monarchies relied on emblems and heraldry to identify their sovereignty and property. However, the need for clearly identified ownership attached to global commercial ventures gave birth to the first system of corporate branding. The English word *branding* shares roots with the Dutch term *branden*, which means “to burn,” referring to the burning of a mark into something. This practice dates at least as far back as ancient Egyptians marking ownership of livestock by burning a symbol on their hide and has also historically been used on humans in the marking of criminals as well as slaves.

VOC’s intricate, coordinated, global operations unfolded during a time when European powers operated under the economic system of *mercantilism*. In the mercantile system, a nation’s prosperity was believed to be based on its share of resources within a fixed volume of global trade —in other words, it was a zero-sum view of potential wealth. This economic system operated within the feudal social order in Europe, a rigid hierarchy based on personal ownership of resources by nobility and the loyalty of subjects to their lords. Mercantilism supported the need for huge sums of capital, high-risk ventures, a system of trade for profit, and a variety of market systems that allowed for the exchange of goods.

In the VOC corporate identity, we see a transition from symbols of imperial conquest (the Dutch royal colors) to the design of symbols,

objects, and systems supporting private enterprise amid expanding forms of private ownership. Yet this wasn't quite capitalism.

The New Laws of Motion: Origins of Capitalism

Several interconnected social and economic structures began to coalesce in Europe in the sixteenth and seventeenth centuries that paved the way for capitalism. They included long-established patterns and infrastructures of global trade, new infrastructures for private means of production across different models of land and property ownership, the transition of the peasantry into a workforce relying on wage labor for survival, technological changes leading to industrial machinery and mass production, and state-level legal systems protecting private property that allowed for the accumulation of wealth.

The Austro-Hungarian economic historian Karl Polanyi defines capitalism by its *abstractions,* most notably what he calls the three *fictitious commodities*: land, labor, and money. Prior to markets for the purchase and sale of land, "property" in the form of land was held chiefly by noble families. Landed property was passed down one generation to the next. One couldn't simply *buy* land. For the majority of the population, performing labor was primarily for the purpose of self-sufficiency, subsistence, or otherwise making what was needed to live. Money was a medium of exchange that set values for commodities, which were typically made by small-scale craft industries. All three were "embedded"[3] in social relationships, which required that they be governed by social, cultural, religious, or moral considerations. Land and human beings were thought to have their own sovereignties, separate from commodities bought and sold on markets. However, once land, labor, and money became commodities in and of themselves, a "great transformation" took place that gave way to a "market society."[4]

Another key differentiation between mercantilism and capitalism has to do with the nature of *goods* and their production. Mercantilism operated primarily with the exchange of luxury goods, including exotic items from faraway places that were intended for wealthy households. "There was no mass market for cheap everyday consumer products. Peasant producers

would typically produce not only their own food but other everyday goods like clothing."[5]

> Capitalism is a system in which goods and services, down to the most basic necessities of life, are produced for profitable exchange, where even human labor power is a commodity for sale in the market, and where, because all economic actors are dependent on the market, the requirements of competition and profit maximization are the fundamental rules of life. Because of those rules, it is a system uniquely driven to develop the forces of production and to improve the productivity of labor by technical means. Above all, it is a system in which the bulk of society's work is done by property-less laborers who are obliged to sell their labor power in exchange for a wage in order to gain access to the means of life. In the process of supplying the needs and wants of society, workers at the same time create profits for those who buy their labor power. In fact, the production of goods and services is subordinate to the production of capital and capitalist profit. The basic objective of the capitalist system, in other words, is the production and self-expansion of capital.[6]

Markets for the exchange of goods and services, money as a medium for exchange, and varying institutions of private property had existed for much of history. However, the economic historian Ellen Meiksins Wood challenges the view that capitalism is somehow the "natural" destination of historical progress. She claims that this assumption makes "no acknowledgement of imperatives specific to capitalism, of the specific ways in which the market operates in capitalism, of its specific laws of motion which uniquely compel people to enter the market and compel producers to produce 'efficiently' by improving labor productivity—the laws of competition, profit maximization, and capital accumulation. It follows that adherents of this model saw no need to explain the specific social property relations and the specific mode of exploitation that determine these specific laws of motion."[7]

Many historians point toward flourishing European cities as a key ingredient to the formation of capitalism. According to this argument, relatively independent centers of commerce and production were linked

together by a system that allowed for independent, privately owned producers to specialize in wide-reaching forms of trade. Again, Wood counters that this growing urban commercial activity does not constitute the specific "laws of motion" of capitalism; these European merchants "could function perfectly well within non-capitalist systems" such as mercantilism. Wood instead points toward economic and social transformations in the countryside, where the "specific dynamics of capitalism were already in place in English agriculture before the proletarianization of the work force. In fact, those dynamics were a major factor in bringing about the proletarianization of labor in England."[8]

*Proletarianization** is the social process of people becoming employed as wage laborers rather than working for their own subsistence. According to Wood, "agrarian capitalism was the root of British economic development." She goes even further, stating that "without English capitalism there would probably have been no capitalist system of any kind: it was competitive pressures emanating from England, especially an industrialized England, that, in the first instance, compelled other countries to promote their own economic development in capitalist directions."[9]

Without a highly productive agricultural sector that could sustain a large nonagricultural workforce, industrial capitalism could not have emerged. As England's agricultural productivity grew rapidly, a decrease in the proportion of people needed to grow enough food met with a population boom. Combined, these forces created a mass of people with no land or property seeking new ways to make a living. Without a population dispossessed of their agricultural lives, there would have been no mass consumer market for cheap everyday goods—such as food and textiles—that drove the process of industrialization in England. This new "market" of consumers was historically unique in both its size and its limitations, notably the relative poverty of consumers in need of cheap goods. This market had far more in common with later mass-consumer markets than with the luxury trade under mercantilism.[10]

In this view, capitalism was born within specific historical relationships between landowners, tenants, and peasants. The new mode of economic pro-duction—large-scale production of cheap goods by people working for wage labor—was *embedded* into new social relationships. Owners of the factories and machines represented the "capitalist" class, while those doing

the productive labor represented the “working” class. Thus, British society transformed from a mercantile and feudal society to a capitalist one just at the start of the Industrial Revolution. “Industrialization was, then, the result not the cause of market society, and capitalist laws of motion were the cause not the result of mass proletarianization.”[11]

Meanwhile, the British Empire’s colonial conquests raised demand for cheap goods around the globe. What followed may sound like a local matter, but it had a profound effect on the social order of British peasantry, and a ripple effect followed. Large open areas of land known as the *commons** were the property of land-owning nobility but were accessible for use by peasants. They used the commons to conduct small-scale agriculture, to collect wood for fuel and cooking, and to raise small amounts of livestock for their families. As global demand for wool to be used in clothing and other goods increased, landowners sought more grazing land to increase wool production, so they privatized the commons, fencing them off and thus cutting off access to the peasants. By this process of the *enclosure* of the commons, peasants were cut off from land that had been their primary resource for survival for centuries. Today, there are interesting correlations between the physical commons of English pasturelands and the digital commons of the internet. Both are rich with opportunities for self-subsistence, expression, flourishing; both are wrought with challenges from *enclosure* and *landlords* (e.g., platform owners who charge fees for access or use).

The social transformation brought about by capitalism was dramatic:

> Beginning roughly in the sixteenth century—although with roots that can be traced much further back—a process of change, sometimes gradual, sometimes violent, broke the bonds and customs of the medieval world of Europe and ushered in the market society we know. In England the process bore with particular severity on the peasants who were expelled from their lands through the enclosure of common grazing lands. This enclosure took place to make private pasturage for the lord’s sheep, whose wool had become a profitable commodity. As late as 1820 the Duchess of Sutherland evicted 15,000 tenants from 794,000 acres, replacing them with 131,000 sheep. The tenants, deprived of their traditional access to the fields, drifted into the towns,

where they were forced to sell their service as a factor of production: labor.[12]

Like Wood, the American economists Robert Heilbroner and Lester Thurow refute the notion of capitalism as a "natural" system, claiming that "far from representing an eternal 'human nature,' capitalism comes as a volcanic disruption to time-honored routines of life."[13] An important point emerges when we see capitalism as contingent upon a specific set of historical conditions: the logics, or "laws of motion," that we have come to accept as seemingly natural in a market society are, in fact, social constructions, created out of historical factors. They "are not eternal attributes of a natural order. They are the creations of a process of historic change, a change that divorced labor from social life, that created real estate out of ancestral land, and that made treasure into capital. Capitalism is the outcome of a revolutionary change—a change in laws, attitudes, and social relationships as deep and far-reaching as any in history."[14]

Running parallel to these volcanic social transformations was a revolutionary disruption in the ways goods were produced and how people labored to produce them.

From Linz to Detroit: Origins of Modern Design

Modern design is predicated on the industrial production of goods at a large scale, in contrast to craft- or trade-based production. The terms *graphic design* and *industrial design* were not coined until the early twentieth century; however, seeds of these modern practices can be found in working methods developed in trades, guilds, and cottage industries just before the industrial era. Capitalism came into being, in part, by adapting these modes of production into more efficient industrial forms.

According to the design historian Phillip Meggs, the invention of typogra-phy—printing with independent, moveable, and reusable units of type—ranks "near the creation of writing as one of the most important advances in civilization."[15] The use of paper and ink were well established by the eleventh century, when the first known invention of moveable type—made from clay, wood, and later ceramic—took place in China. However,

further innovation by Johannes von Gutenberg in the fifteenth century in Linz, Germany, transformed printing forever. Gutenberg's synthesis of existing inventions led to a major transformation. In place of the hand cutting of each unit of type to be used for printing, metal type could be more efficiently produced by using molds to cast the letter forms. Combined with a mechanical press to stamp inked type to paper, two orders of efficiency were brought to the world of printing: a mechanized process of printing and an efficient, repeatable process for making the type needed by the press.

The European innovation of movable type relied on several existing factors: the availability of paper, the proliferation of block printing, and an increasing demand for books by a growing literate middle class. Demand for books was due in part to a new institution flourishing in Europe: the university. These movements precipitated the rapid explosion of printing across Europe from the moment of Gutenberg's innovations. Prior to the printing press, a single book was as valuable as an entire farm or vineyard. In the early 1400s, the university library at Cambridge held only 122 books. By 1500, printing presses were operating in more than 140 European towns.[16] The fact that Gutenberg created his first printing press from a modified wine press illustrates the thoroughly agrarian context out of which typographic printing emerged, stoking the flames of Enlightenment ideas on democracy, science, and religious reform.

At the center of this evolution of production was the division of labor. Print shops required a range of specific tasks to be executed repetitively. Medieval scribes had already practiced a form of division of labor for inking and producing illuminated manuscripts. However, the print shop required a more tightly coordinated operation with increased segmentation of tasks operated at much faster speeds. Not only was typographic printing "a radical improvement in achieving efficiency for the reproduction of multiple copies of a text, but the rationalization of labor that was embedded in this innovative shift created a model that would inform practices well into the industrial revolution hundreds of years later." This division of labor in the print shop "would serve as a model ahead for the transformation of other industries into their increasingly modern form, anticipating some of the modularity reinforced by 19th-century industrialization."[17]

The very idea of mass production requires some form of design to specify how an object is to be produced, by what materials, and under what conditions of cultural desires, tastes, and economic capacity. It would be difficult to overstate the impact of the Industrial Revolution on contemporary life and on the contemporary practices of design as we know them. Yet the print shop's preindustrial mode of production also helped set in motion a model that would be radically expanded by the factories to come in places such as Manchester and, later, Detroit.

Already near the very beginning of that process of industrialization, new explanations were being hatched to explain how mass-produced goods were to be produced, sold, and priced and how these dynamics affected not only individuals but also the economies and politics of whole nations.

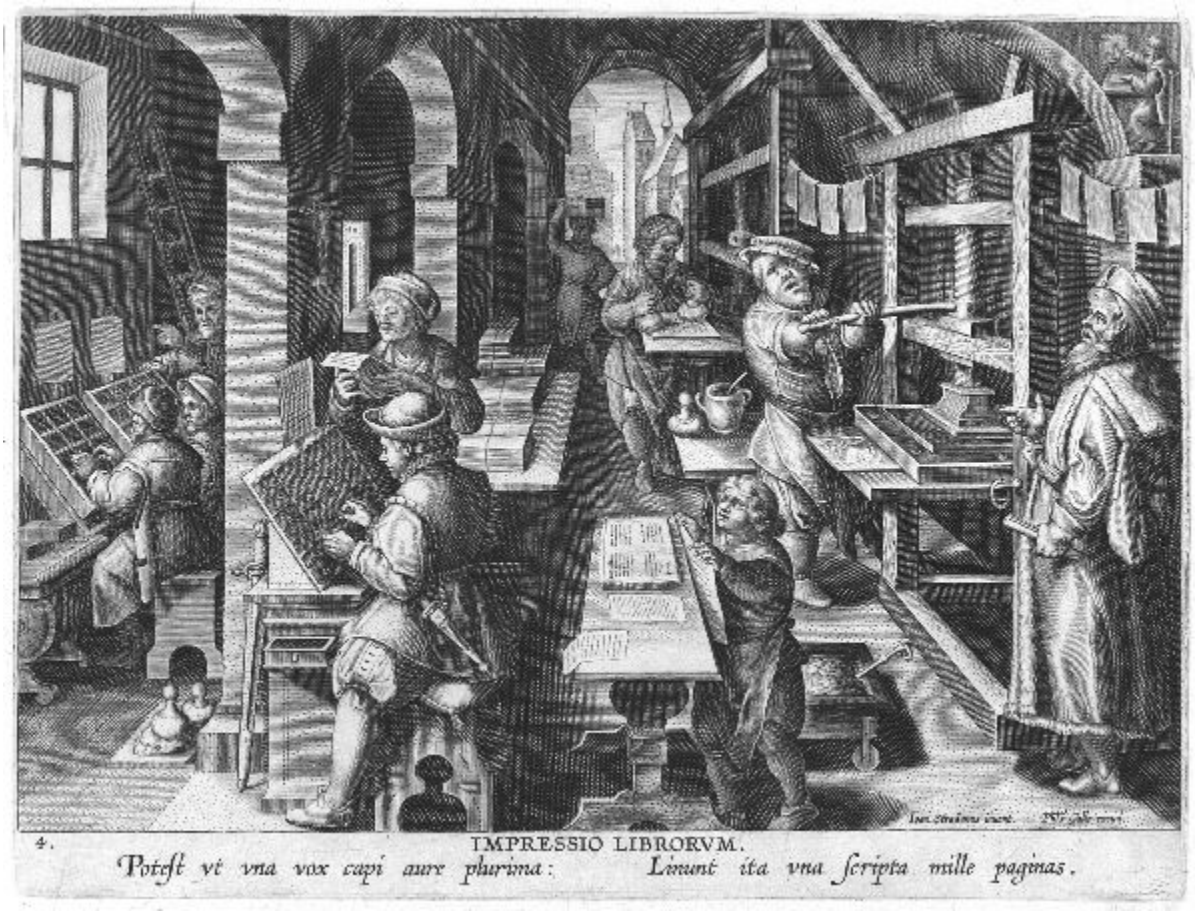


Figure 2.3

Scene from a fifteenth-century print shop showing various specialized tasks of book making. *Impressio librorum* (Book printing) from the *Nova reperta* (New inventions of modern times), c. 1580–1605. Engraving by Theodoor Galle after Jan van der Straet, c. 1550. (Open Access Image from the Davison Art Center, Wesleyan University; photograph by R. Lee.)

Adam Smith & Classical Economic Theory: Supply & Demand in Static Equilibrium

In 1776, during the early days of the Industrial Revolution, the Scottish philosopher and economist Adam Smith published the first thorough description of the "nature and causes" of nations' wealth, a book commonly called *The Wealth of Nations*. To this day, it remains the fundamental work of classical economics. It looks at the intersection of capitalist economic and social organization, such as the division of labor, the measure of productivity, and the functioning of free markets.[18]

Smith introduced the concept of the "invisible hand." The term is used only once in *The Wealth of Nations* and in a very specific context about supporting domestic industry; however, subsequent generations of economists have applied the term and the concepts it entails much more widely. The "invisible hand" has come to signify the idea that a "free" market, unhindered by interference from governments, takes in all information relevant to supply and demand to determine—almost magically—the correct pricing of goods and services and to send signals about what and how much to produce. Smith laid the foundation for today's neoclassical economics, a market-operated and price-guided approach to describing resource allocation that found resurgence in the late twentieth century.

According to the British design historian John Heskett, "a market only exists because of *scarcity*; it allocates goods that are scarce in relation to the number of people who desire them. . . . Equilibrium is the point where supply and demand intersect and determine the price customers are prepared to pay. Equilibrium implies balance and is essentially a static condition."[19] It is precisely this assumption of a "static condition" that has provoked criticism of classical and, later, neoclassical economics. Heskett points out that the fundamental premise of equilibrium leaves little room for competitive dynamism, in which products themselves can shift pricing. He claims that design "is essentially about change, and concepts of equilibrium have limited relevance in explaining change."[20]

Neoclassical economic theory also operates on several other dubious assumptions. Among them, consumers must operate in perfectly rational ways, paying the right price for a product based on supply-and-demand

calculations. Similarly, owners and workers must operate under conditions of perfect competition, meaning no one has the upper hand, for wages to be properly set. For those two assumptions to hold, all actors in the system must always have access to perfect information on all their potential options. Obviously, theories can take us only so far. However, these assumptions limit classical and neoclassical theory, particularly in the capacity to account for the kinds of value design can create. As Heskett puts it, "Just as there is little in Smith's *Wealth of Nations* that enlightens us as to why people find things useful or desirable, so also is there in Neoclassical theory . . . [little] substantial concern with how products might be different. . . . If market decisions are, indeed, based on goods that already exist, there is little left on which to focus beyond price and quantity." Despite the fact that "neoclassical economic theory raises many questions and provides few answers for design theory and practice," Heskett somewhat resignedly states that it is "the dominant theory of the capitalist system" and that criticizing it is "likely to have little effect."[21]

Regardless of the economic theories about what was transpiring in the eighteenth century, one part of the story seems well agreed upon: the Industrial Revolution remade life for untold numbers of people on this planet—and for the planet itself.

Early Industrial (Anarchic) Capitalism

From about 1760 to about 1840, new manufacturing and production processes marked a major turning point in human history. Nearly every aspect of everyday life was transformed by the Industrial Revolution. Traditional and craft-based modes of production were overtaken by the scale, speed, and efficiency of machines. The chief characteristic of the factory system of manufacturing was the complex division of labor, which gave each worker a specialized and repetitive task. All these changes contributed to the mode of production known as industrial capitalism.

Designers became "designers" not by making goods—as would have been the case for craftsmen from time immemorial—but by specifying the material forms to be produced by others. "By others" is also a central feature distinguishing capitalist production from previous models. Workers

are said to be “alienated” from their labor by producing goods that they themselves do not determine or design or use and by selling their efforts for a wage, which is required then to purchase the necessary goods for their own survival (rent, food, clothing, etc.). The difference between what is paid for labor, materials, and tools to produce commodities and the money made from selling the commodities—called *surplus value**—is translated into profits for the firm’s owners. Capitalists use profits as new capital to invest in further ventures, thus putting in motion both the accumulation of wealth and a growing divide between the owners and workers.

According to Polanyi, “not until 1834 was a competitive labor market established in England, hence industrial capitalism as a social system cannot be said to have existed before that date.”[22] Everyone began to need either capital—one’s own seed of investment to generate money—or a wage to survive within this commodity system. This is when the abstraction of labor came to be a *fictitious commodity*, something to be bought and sold on a market. As Arendt describes it, “the modern age has carried with it a theoretical glorification of labor and has resulted in a factual transformation of the whole of society into a laboring society.”[23]

The first era of industrial capitalism, from the mid-eighteenth century to the mid-nineteenth century, has also been referred to as *anarchic capitalism* because the “activities of capitalist entrepreneurs were relatively unchecked either by organized labour or the state.”[24] During this period, many small-scale factories, workshops, and industrial-entrepreneurial enterprises were engaged in intense but small-scale competition with one another—trying to produce more, more quickly, and less expensively. Large peasant populations throughout Europe began flocking from the countryside to quickly industrializing cities. The unregulated nature of this competitive labor market produced terrible working conditions: incredibly long hours, dangerous work with unpredictable machines, little to no protection from safety or health hazards, employment of children, and so on. However, the desperate need for employment that brought thousands of people into the cities also built the roads, railways, canals, and other infrastructures for the movement of goods and people that constitute some of the basics of urban life today. Yet the disparities produced by this new social order were impossible to miss and fueled critique, rebellion, and revolution.

Karl Marx: The Capitalist Mode of Production & Class Struggle

As populations grew and urbanized, new social dynamics developed among the stratified social classes. A “middle class” (or bourgeoisie)—distinguished from both the “owner class” (capitalist owners of production) and the “working class”—emerged and gained economic power. Industrialization played a role in further social and political revolutions as new class dynamics conflicted with each other, as well as with preindustrial social and political hierarchies. Naturally, historians have disputed exactly which forces affected the others. However, one nineteenth-century historian produced a sweeping analysis of the dynamics of this system in terms that set a course for almost all discussions of capitalism to follow.

Although his name alone often elicits a sense of revolution, the German historian, economist, philosopher, and social critic Karl Marx sought mainly to reveal economic patterns shaping society through the capitalist mode of production. He is perhaps the most “remarkable analyst of capitalism’s dynamics who ever lived.”[25] His work contrasted with that of classical political economists such as Adam Smith. What was fundamentally different about Marx’s economic view was his perspective on history. Neoclassical economic thought was generally shaped by a belief in a progressive series of stages of historical development, each improving upon the last, with commercial society as the final stage. However, Marx saw history as an ongoing struggle between the social classes of each era, the outcome of which was based on the “tension and antagonism” of the struggle. In Marx’s view, this meant capitalist society was a historical stage, not a permanent one. As an economist, he contested the classical view of market growth as a force that produced harmonious, price-setting equilibrium. Instead, in his struggle-centered view of history, he saw market-based economic growth as a source of constant crisis and malfunction.

Marx identified the labor process that goes into the production of commodi-ties—things to be sold—as the primary source of value creation. Working in England in the middle of the nineteenth century, Marx observed that it was on the factory floor where raw materials became commodities that created the economic value of capitalist production. This is called the *labor theory of value*.* The difference between the value added by labor

and the wage paid to the worker is a *surplus value*, which transfers to the capitalist owner in the form of profit. This transfer constitutes a major feature of the capitalist "mode of production": surplus value is extracted from labor, which creates both an "exploitation" of the labor force and an "alienation" of the workers from their labor. Both outcomes become the fuel for workers to engage in a struggle against the owner class of capitalists—demanding higher wages, better working conditions, or even partial ownership of the enterprise.

Against the harmonious view of self-directed, market-shaped equilibrium, Marx saw capitalism as inherently unstable: procuring raw materials, hiring a labor force, and selling finished products in a market are all actions laden with risks of failure. The materials can't be had at the necessary low price (less profit). The workers go on strike to demand more pay (less profit). People don't want the finished good, or they want it for a lower price (less profit). Even in the contemporary moment of entrepreneurial fervor, it's good to remember that most new businesses fail within their first five years.[26] The primary reason for failure is lack of market need; the secondary reason is not enough capital.[27]

This failure-prone activity becomes an engine of economic crisis, particularly as the capitalist mode of production seeks continual growth or *accumulation*. With profit exploited from labor, the capitalist invests—or spends more *capital*—on a bigger or new profit-seeking venture. In Marx's view, this process necessarily trends toward big business. At each crisis of economic downturn, smaller firms go bankrupt, and their assets are bought up by the surviving firms. As the cycle continues, a steady increase in the size of business firms results (thus decreasing competition, consolidating wealth, building economic power). The growing scale and economic power of large firms intensifies the class struggle as small-scale business owners are pushed out of competition through a boom-and-bust cycle. Growing numbers of property-less workers have the surplus value of their labor extracted by a smaller number of wealthy capitalist owners. In this process, Marx saw the seeds of a society that would eventually be reduced to two classes—owners and workers.

Marx did not claim that *all* production or *all* social relations are derived from or based in capitalism. He also knew that there are many noncapitalist forces at play within generally capitalist societies. His explicit goal was to

describe the *capitalist* "mode of production" and to reveal the social process within it. The capitalist mode of production is centered on the private ownership of the means of production—factories, land, machinery—and Marx saw this *private property* as the seed of capitalism's structural exploitation. A distinction that seems to be commonly missed in Marx's analysis is that between private property and personal property. Marx was not against individuals owning their own personal property, or items for personal use. Rather, he argued against the private ownership of "capital"—the means of production that put one class in power over others through ownership.

In Marx's view, any *commodity*—something to be exchanged—has a *labor value*, a *use value*, and an *exchange value* (typically expressed as a price in money). His labor theory of value rests on the observation that all forms of labor, despite their varieties, are essentially reducible to a single, measurable phenomenon of *socially necessary labor*—that is, the work required for a society to subsist and/or to prosper.

> Use-value as an aspect of the commodity coincides with the physical palpable existence of the commodity. Wheat, for example, is a distinct use-value differing from the use-values of cotton, glass, paper, etc. A use-value has value only in use, and is realized only in the process of consumption. . . . Use-values serve directly as means of existence. But, on the other hand, these means of existence are themselves the products of social activity, the result of expended human energy, materialized labour. . . . To measure the exchange-value of commodities by the labour-time they contain, the different kinds of labour have to be reduced to uniform, homogeneous, simple labour, in short to labour of uniform quality, whose only difference, therefore, is quantity. . . . This reduction appears to be an abstraction, but it is an abstraction which is made every day in the social process of production. The conversion of all commodities into labour-time is no greater an abstraction, and is no less real, than the resolution of all organic bodies into air.[28]

In Marxism,* as in classical economics, the kinds of valuable or desirable things that labor transforms into commodities seem to just appear out of

thin air. This rational assessment of how values are created and assessed has little to say about how value might be generated out of desires beyond utility, such as irrational consumer demand for items of seemingly little or no *use value*. They are not the deliberative work of designers or others determining what is to be made, for whom, and how. Regardless, Marx sets the groundwork for many ensuing analyses of the social dynamics of the capitalist system of commodity production.

By illuminating social machinations of the capitalist mode of production through his perspective on class struggle, Marx revealed economic life to be fully entwined with political and social life. By placing this model of social organization into historical perspective, he also revealed that capitalism is a temporary situation, sure to give way to future historical changes. The work of Marx and his collaborator Friedrich Engels subsequently influenced major political upheavals, as is well known. These ideas inspired significant labor reforms brought about by worker strikes and a developing *class consciousness* in the later nineteenth century. These ideas have been influential in economics, politics, and social sciences ever since. While Marx focused primarily on the social dynamics of production, others during this time were studying the social dynamics of the impacts of consumption.

(Conspicuous) Consumption

In economic terms, *consumption* refers to the *use* of goods and services, which is generally considered their primary and final purpose. Neoclassical economics generally assumes that consumers base their household spending on rational, informed decision making that considers both current and future circumstances, a process called *rational opt*imization.

> This “rational optimization” assumption is untestable, however, without additional assumptions about why and how consumers care about their level of consumption. . . . [E]conomists usually assume (1) that the urgency of consumption needs will decline as the level of consumption increases (this is known as a declining marginal utility of consumption), (2) that people prefer to face less rather than more risk in their consumption (people are risk-averse), and (3) that unavoidable uncertainty in future income generates some degree of precautionary

saving. In the interest of simplicity, the standard versions of these models also make some less-innocuous assumptions, including assertions that the pleasure yielded by today's consumption does not depend upon on one's past consumption (there are no habits from the past that influence today's consumption) and that current pleasure does not depend upon comparison of one's consumption to the consumption of others (there is no "envy").[29]

Based on contemporary trends of high consumption[30] and low savings (in 2019, 40 percent of Americans were only one paycheck away from poverty[31]), these assumptions seem far-fetched. Regardless, *consumption* is commonly used as a benchmark for measuring a capitalist economy's health.

At the turn of the twentieth century, the economist and sociologist Thorstein Veblen coined the term *conspicuous consumption** in a scathing indictment of the deliberately wasteful display of consumption by the wealthy—in particular the "new rich" who emerged in the transition from the Victorian to the industrial era. He chided the wealthy for their disparagement of the poor as being "lazy," lacking "thriftiness," and splurging what little they had on immediate gratification by succumbing to base emotions. To Veblen, such accusations of irrational economic behavior were both misplaced and hypocritical. The rich assumed that their class carried a kind of virtue that they had earned through their patient and calculated investment, through frugality that demonstrated their civilized self-control. Yet what Veblen saw was just the opposite. The defining behavioral characteristic of the *leisure class*—those above and beyond the need for engaging in industrial or otherwise *productive labor*—is transforming objects of function or beauty into display pieces, or ornaments. Material goods become valuable and desirable not for their use value. Rather, they become symbolic of status. Spending money on such goods is an end, not a means. This process creates its own positive feedback loop, and the purpose of consumption is reoriented while satisfaction from consumption becomes elusive.

However widely, or equally, or "fairly," it may be distributed, no general increase of the community's wealth can make any approach to

> satiating this need, the ground of which is the desire of every one to excel every one else in the accumulation of goods. If, as is sometimes assumed, the incentive to accumulation were the want of subsistence or of physical comfort, then the aggregate economic wants of a community might conceivably be satisfied at some point in the advance of industrial efficiency; but since the struggle is substantially a race for reputability on the basis of an invidious comparison, no approach to a definitive attainment is possible.[32]

Veblen provided a fascinating description of the evolution of societies that produce a *leisure class*. In the early stages of *predatory cultures*, warring tribes would capture and take captive their enemy's women, who would become *trophies* of the successful combatant's prowess. The captive women would then be forcibly coerced into marriage, making the *trophy wife* a "durable result of their exploits." "The ownership of women begins in the lower barbarian stages of culture, apparently with the seizure of female captives. The original reason for the seizure and appropriation of women seems to have been their usefulness as trophies."[33] Over time, this custom translated into other forms of trophy taking: the spoils of war, the commemoration of victories, and a general pursuit of glory through accumulation of goods with no practical utility other than as ceremonial symbols of conquest.

Since Veblen's time, the term *conspicuous consumption* has been more broadly applied to the psychology of a society comprising individuals who see their sociopolitical positions defined largely by their function as consumers. Such a society demands constant growth in the quantity and variety of consumer goods and services on the market, which yields constant production of new desires. This social-psychological phenomenon has aligned well with capitalism's economic need for constant growth and has provided designers with a permanent project of creating new objects and new experiences of consumable desire.

Advanced (Managed) Industrial Capitalism

The anarchic era of early industrial capitalism was a time of polluted cities, overcrowded tenements, and dire, dangerous working conditions, including for children. Small-scale operations competed with each other, and the winning firms' accumulation of capital resources produced growth. Meanwhile, living conditions for urban factory workers and their families got so bad that legal changes were put in place to protect the health and welfare of workers.

In England, the Factory Act of 1833 regulated the employment of children in the textile industry. Children younger than nine were not allowed to work, working hours were limited to 9–12 hours per day (depending on the worker's age), children nine and older had to attend at least a couple of hours of school every day, and officials were hired to ensure these rules were followed. Over the next decades, several more factory acts were passed and expanded to other industries, each act aimed at improving safety, health, and working hours, particularly for women and children. Along with these new legal constraints, the escalating scale and complexity of production presented a new set of conditions in which industrial production became more complicated, more regulated, and required greater managerial control.

During this time, both the management and labor sides of industry became more organized. Labor unions formed to leverage collective bargaining, using the power of numbers to fight for better wages and working conditions. Industrial corporations increased management operations, employing more people to handle the expanding complexities of their growing operations, regulatory controls, and their workforce's demands. Because industrial production required so much coordination and planning, there was a high level of economic risk. Instability and uncertainty were the enemy, not only in terms of labor shortages through strikes or walkouts but also in terms of new competitors who might spring up to steal another's market share or workforce. "The main way in which employers reduced uncertainty was not, however, through association but through concentration. The simplest way of dealing with the competition was to buy it up or merge with it."[34]

As corporations grew and individual units within them also grew, so did the quantity and variety of managerial positions. This has been called a "managerial revolution," which changed the character of industrial

capitalism—from a system controlled by risk-taking industrial buccaneers instilling fear and discipline on the factory floor to one managed by bureaucratic middlemen quietly running business operations from their bland offices. Some people became concerned that

> the growth of management, together with the spread of shared ownership to many small and powerless owners, meant that managers rather than shareholders now controlled corporations. Instead of seeking simply to maximize profits, managers took account of the interests of all with a stake in the company. Plausible as it was, this notion of a "managerial revolution" overstated the power of managers, for owners were still ultimately in control and profitability remained the "bottom line," but industrial production was undoubtedly a much more managed process than it had been before.[35]

With the growing complexity in operations, an expanding universe of commodity types, and the constant risks associated with long-term planning for fickle consumption habits, it became clear that new kinds of professionals were needed to anticipate, specify, and give material forms to consumer desires.

The Birth of Modern Design

It was during this stage of advanced, managed industrial capitalism that the professional fields of industrial design and graphic design truly came into being. The increasing scale and complexity of production required organization, strategic planning, and an integration of the detailed planning of products—from the conception of new product types down to the details of their production. Meanwhile, increased competition among firms for similar product types required stronger methods of product distinction and promotion in the market.

As companies grew and expanded the varieties of their products, consumers witnessed a deluge of new options for both fundamental needs and discretionary spending. From a design standpoint, this surge created opportunities to build audiences, customers, and even loyal return consumers to companies whose offerings might span a huge range of

product types. Designers participated in the process of creating new markets through product design but also through visual communications promoting products and persuading the public into becoming consumers.

In early twentieth-century Europe and America, the exuberance of social transformations defining “modernity” forged a dramatic change in the language and forms of material culture. The leading principle of design was a sleek, bold, and rational *functionalism*,* which broke with traditions of the past, in particular the lavish *ornamentalism* of the Victorian era. The spirt of the machine age inspired a new aesthetic grammar of everyday life.

Industrial Design & Graphic Design

The German architect and designer Peter Behrens has been credited with being the first industrial designer in history. His work spanned from the design of buildings and industrial products such as clocks and tea kettles to the graphic design of corporate identities, promotional materials, and even typefaces. In 1907, Behrens participated in founding the German association of artists, architects, and designers called the Deutscher Werkbund (German Association of Craftsmen). The Arts and Crafts movement of the late nineteenth century had called for the return to a preindustrial model of producing small-scale crafts inspired by nature and imbuing everyday objects with utility and beauty. The Werkbund sought similar goals but sought to achieve them through industrial processes—to impart a sense of coherence and beauty in everyday life of the Industrial Age. The practical embrace of industrial mass production made the Werkbund highly influential among industrial capitalists.

Behrens had a prolific career, but is perhaps best known for his work with the Algemeine Elektrizitäts-Gesellschaft (AEG, General Electrical Company). The AEG initially produced electrical equipment, such as light bulbs and motors, but expanded rapidly in the first half of the twentieth century and is credited with many inventions in electrical engineering.

> Emil Rathenau, director of the AEG, appointed Behrens its artistic advisor. After Rathenau purchased European manufacturing rights to Thomas A. Edison’s patents in 1883, the firm became one of the world’s largest manufacturing concerns. Rathenau was a visionary

> industrialist who sought to give a unified visual character to the company's products, environments, and communications. In 1907 the electrical industry was high technology; electric teakettles were as advanced as digital electronics are today. As design adviser to the AEG, Behrens began to focus on the design needs of industry, with responsibilities ranging from large buildings to stationery and electric fans.[36]

In 1907, Behrens began as an "artistic consultant" for the AEG. Although he was never actually an employee of the company, he became a central figure in its use of design to distinguish its products, thus establishing an industrial-era example of corporate "branding." Behrens designed the entire corporate identity for the AEG, everything from logotype to products to promotion and publicity. "The AEG graphic identity program made consistent use of three linchpin elements that would be present in corporate identity programs as the genre evolved half a century later: a logo, a typeface, and a consistent layout of elements following standardized forms."[37]

Given the range and scope of his work, Behrens might be seen not only as the first industrial designer but also as an early practitioner of the "total-design" programs taken up by large multinational corporations in the middle of the twentieth century.

The Dutch company Philips was founded in Eindhoven in 1891 as a manufacturer of light bulbs. From that humble beginning, Philips grew into a multinational conglomerate and is today one of the world's largest corporations. In 1925, a young architect named Louis Kalff wrote a letter to Anton Philips, president of the company and son of its founder, boldly claiming that "the company's advertising was not modern enough and did not have enough allure."[38] Apparently, Anton Philips agreed, and Kalff's letter paid off. "Just a few days later he was called into the head office to be offered a job."[39] Kalff was initially hired to oversee the "commercial and artistic propaganda departments" (advertising) that would modernize and standardize all publicity materials for the company. Kalff designed the first official Philips wordmark, which became the model for the logo still used

today. His first advertisement, published in 1925, was for the Duplo line of car lights.

From his early work producing visual communications for the company, Kalff moved on to help Philips expand into new product territories. When Philips made its first radios, such as the “Chapel” radio, Kalff helped define the form and materials. “This striking and instantly recognizable radio stands out as key in the move towards the importance of design at Philips. Previous radio designs were boxy and impersonal, but the Chapel, designed by Kalff, was shapely and incorporated the stars and waves of the Philips logo.”[40] Kalff translated the two-dimensional language of the company’s graphic identity into a three-dimensional product form. The Chapel radio evokes the Philips identity with a speaker grill designed as the company’s logo. As product categories continued to expand, Kalff designed pioneering new products such as the Philishave, one of the first electric rotary razors.

The preceding examples show how two specialized design professions began to emerge from a swirl of industry-related creative practices. Yet the disciplinary language for this kind of work was not yet fully identified. The US Patent Office formally introduced the term *industrial design* in 1913 to describe “the distinguishing form of products that have marketable value.”[41] The term was little used for about two decades, but eventually, out of a milieu of architects, individual craftsmen, and “commercial artists,” a new breed of designer emerged to “transform the process of design from an individual creative act that produced one artifact at a time into a complex industrial mass production system that made millions of identical products.”[42] During the Great Depression in the United States, which began in 1929 and stretched into US involvement in World War II, these professional designers demonstrated the economic value that design could contribute to manufacturing. Through visual differentiation, *styling* and *branding*, distinct lines of common product types generated dramatically increased sales.

The term *graphic design* was first coined in 1922 by William Addison (W. A.) Dwiggins, an American type designer, calligrapher, and book designer. He used the term to describe his own various activities in printed visual communications, including advertisements that merged illustration and typography, book design, type design, and calligraphy. The term was not commonly used until after World War II, though.

With Behrens, an architect turned attention to the styling, technology, and production specifications of industrial-era domestic products. From there, a design language developed in material products was transformed into visual design systems for communications, brand identification, and promotion of products. With Kalff, a “commercial artist” used the logic and language of two-dimensional visual communications to inform the development of new product types and new product categories. These skills became increasingly specialized and gave birth to two distinct disciplines, industrial design and graphic design. Both were born of and into the capitalist endeavor. Yet they also mapped out an expanding field of opportunities to remake the material culture of everyday life in a concerted effort, responding to the unique social, cultural, and political opportunities of their historical moment.

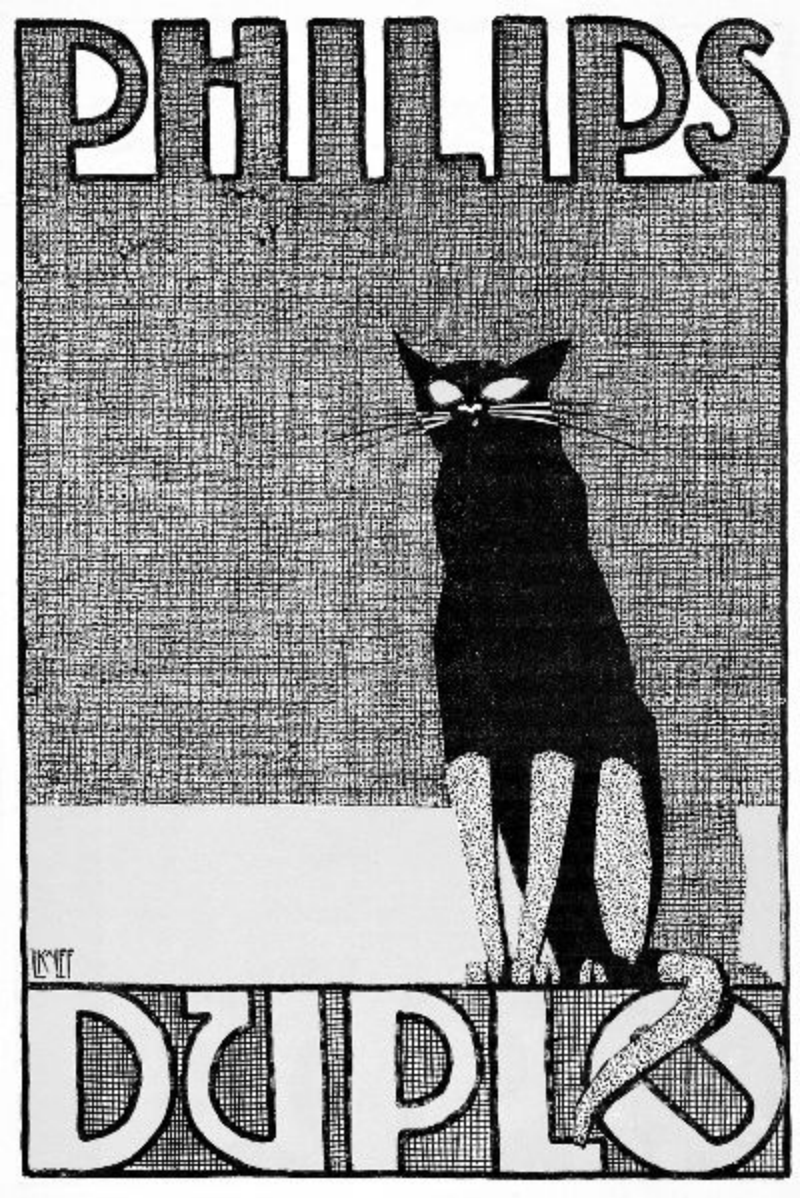


Figure 2.4
Poster for Duplo headlights, designed by Louis Kalff for Philips, 1925. (Philips Company Archives.)



Figure 2.5
The “Little Chapel” radio with Philips logo as speaker grill, designed by Louis Kalff for Philips, 1931. (Philips Company Archives.)

This model of “modern design” emerged in the political context of early twentieth-century Europe and America, a time dominated by the forces that led to a world war, a brief and tense interlude, then another world war. Strong humanistic sentiments defined design as a political project that created common cultural experiences through an optimistic and shared material culture. There were inherent paradoxes in the idea that competing self-interested enterprises could be tasked with creating a common culture

in democratic societies. Yet against the backdrop of both fascism and global war, collective design movements saw the potential for democratic cultural unification. At the same time, though, modern design was recognized as a valuable instrument of business with the capacity to differentiate products, grow markets, and create new ones. One of the more profitable strategies discovered was the way design could differentiate products specifically to increase the quantity and frequency of consumption by a nascent and growing consumer culture.

Planned & Product Obsolescence

During the era of advanced, managed industrial capitalism, workers organized unions to demand improved working conditions and higher pay. In turn, owners began to constantly look for cheaper labor alternatives, often through technological efficiencies in production. Although technological innovation *could* yield longer-lasting products, this posed a profit risk: if consumers purchased less frequently, less consumption equaled lower profit. Two long-lasting strategies to deal with this problem have been *planned obsolescence** and *product obso*lescence.*

Planned obsolescence refers to intentionally producing a product with an artificially limited lifespan. If a consumer needs to replace the product sooner, higher long-term sales can be generated by this increased frequency of purchases. *Product obsolescence*, sometimes called *design obsolescence*, refers to the intentional design of a product so that it becomes less usable, less desirable, or simply appears out of date before reaching the end of its functional lifespan. Generally speaking, planned obsolescence is a technical strategy, whereas product obsolescence is a design strategy. Both strategies share the goal of increased long-term sales, yet designers tend to play a larger role in product obsolescence by making products fashionable and therefore capable of quickly going out of style.

Planned Obsolescence One of the most stunning historical examples of pro-duct obsolescence comes from the history of light bulbs. When homes were first outfitted with electricity in the late nineteenth and early twentieth centuries, they were fully serviced by electric companies. Typically, this service even included free replacement of burned-out light bulbs—someone from the electrical service would actually come to your home and replace

the bulb![43] Based on this model, manufacturers had an incentive to make the bulbs last as long as possible. However, as the service model changed and homeowners became responsible for purchasing and replacing their own bulbs, shorter-lasting bulbs meant more profit.

During the same period that the AEG and Philips were integrating design into their corporate strategies, several European and US corporations, including General Electric and Philips, convened to create a light-bulb-manufacturing cartel. Founded in 1925, the Phoebus cartel formalized a joint commitment to control efficiency standards and the pricing of incandescent light bulbs in order to maximize profits. This agreement held the members accountable to each other to drive up prices and limit product efficiency. At the time, incandescent light bulbs could operate normally with a life expectancy of about 2,500 hours.[44] The cartel set a decreased standard life expectancy of 1,000 hours. Light bulbs produced by cartel members were tested randomly to ensure that no one was breaking the rules by making more-efficient, longer-lasting bulbs. Anyone caught breaking the standards would be heavily fined by the others. The cartel also set prices at a higher rate than competitively necessary. These agreements were made privately so the public wouldn't suspect the possibility of longer-lasting or less-expensive light bulbs. To justify the organization's activities—product testing, meetings, and interactions between competitors—the cartel claimed its primary purpose was to engineer standardization of light bulbs for greater consumer efficiency.

The term *planned obsolescence* wasn't commonly used until the 1950s. However, by then, strategic obsolescence of products had already permeated many industries. The Phoebus cartel reduced competition in the light bulb industry for more than a decade and prevented technological improvements that might further extend light bulb efficiency. Both moves are antithetical to the proclaimed capitalist ideals of economizing efficiency and innovation through competition, yet neither of these ideals stood in the way of an effort to maximize profits.

Product Obsolescence By the early 1920s, the United States was quickly reach-ing what appeared to be *saturation* of the automobile market. In response, the CEO of General Motors set out a new strategy for the company: to divide up the market of consumers and make annual style changes to drive more frequent sales. Henry Ford was opposed to this

approach, preferring simplicity, design integrity, and a commitment to economies of scale. However, GM surpassed Ford in sales as early as 1931 and essentially dominated the industry thereafter.

GM's approach to segmenting the market was successful in producing stratified tiers of *desire*. Central to the new strategy was the concept of a "ladder of consumption": "The person who could afford a Buick was doing better than the one driving a lower-priced Chevrolet—and both envied their better-off neighbor who drove a Cadillac."[45] Everyone knew the rank order of the brands, and this ranking was visible in the design of the cars themselves. In the 1950s and 1960s, tail fins often were a tell-tale sign of rank. Each year, higher-tiered brands introduced a new coveted style. New design cues in this hierarchy of brands started at the top of the "ladder," and the older style from the previous year was moved down to the lower-tiered brands. "The goal of these style changes was to give customers both a positive and negative incentive to upgrade their cars—the desire for an even more attractive vehicle paired with the fear that others would judge them for sticking with last year's model."[46] From today's standpoint, such strategies make it clear how complicit design has been in, among other things, the environmental disaster we currently face.

Automakers from other parts of the world would eventually seize on growing consumer discontent at the center of planned obsolescence. The German-made Volkswagen Beetle was known for its rigid consistency, even its "antistyle." The Beetle had barely changed since its original design in 1938, when it was first produced under Nazi regime–controlled production as a simple, flexible, and inexpensive "people's car" (which is the literal translation of the term *Volks Wagen*). These functional objectives were dictated by Adolf Hitler and carried out by an engineering team lead by Ferdinand Porsche.[47] As early as the 1950s, VW began to tout its low price, small size, and consistency in design as differentiating factors in the American market.[48] A VW advertisement in 1962 cleverly contrasted this consistency to the annual calendar of product obsolescence followed by American automakers. The copy read: "We do not believe in planned obsolescence. We don't change a car for the sake of change."[49] In decades to follow, the "Bug" would prove even more appealing as smaller, more fuel-efficient, and less-expensive imports began to successfully compete with American automobiles.

Today, similar observations can be made of both planned and product obsolescence. In 2018, Italian antitrust regulators fined Apple and Samsung millions of euros over what they described as planned obsolescence of smartphones. Customers were prompted to install new operating systems, which "caused serious malfunctions and significantly reduced performance, thus accelerating phones' substitution." The regulators surmised that "Apple and Samsung implemented dishonest commercial practices," particularly as the companies had not provided adequate information about the software updates or "any means of restoring the original functionality of the products."[50]

Just as each individual capitalist enterprise requires constant growth, so too does an entire capitalist economy. Regulations aside, many political institutions and policies in capitalist economies are designed to encourage spending. There are, of course, competing views about government's role in the economy, yet it is clear that a government's policies and level of spending play a major role in the nature and performance of a nation's economy.

John Maynard Keynes: Aggregate Demand & Government's Role in the Economy

Named after Henry Ford, the term *Fordism** refers to assembly-line methods for increased efficiency along with the division and specialization of labor for mass production. In the early twentieth century, Ford famously paid his employees nearly double the market rate for labor, claiming that every one of his workers should be able to afford a car he helped to produce.[51] This was an expression of the economic concept "aggregate demand," which sought to explain the social relationship between workers and their wages and industrial firms and their revenues. To scale up production (and profits), firms need to increase demand for those products, which requires a consumer market with enough income to make purchases. Whether through altruism, self-interest, or some combination thereof, Ford saw it as the capitalist owner's role to ensure there was a market for his goods. He did this by paying higher wages for workers.

Then things got bad. From October 24 to October 29, 1929, the US stock market experienced the most devastating crash in its history, signaling the

start of the Great Depression. As the economic collapse spread, it wiped out half of the value of American production, rendered one quarter of the nation's workers unemployed, erased millions of savings accounts, and led to the foreclosure of more than a million families' homes.

Both classical theory and Marxist theory knew there were occasions when investors were reluctant to invest, thus slowing, stalling, or even reversing economic growth. However, previous theories generally contended that such occasions were temporary. The market would *self-correct* and pull itself out of the slump. "In Smith's scheme, the rising supply of young workers would keep wages in check. In Marx's conception, each crisis (up to the last) would present the surviving entrepreneurs with fresh opportunities to resume their quest for profits."[52] Within this context, the British economist John Maynard Keynes (rhymes with "brains") developed an economic theory that came to be identified by his own name: *Keynesianism*.* "The great economists were all products of their times: Smith, the voice of optimistic, nascent capitalism; Marx, the spokesman for the victims of its bleakest industrial period; Keynes, the product of a still later time, the Great Depression."[53]

Against the contemporaneous assumptions of classical economic theory, Keynes argued that "*there was no self-correcting property in the market system to keep capitalism growing*."[54] Economic decline under capitalism means social disaster, but the market system alone cannot assure growth. Keynes saw that a market system could reach a state of *underemployment equilibrium*, or low levels of economic activity paired with high levels of unemployment—in other words, very little consumption, very little production, and very little work. That is exactly what started to happen in the Great Depression.

From 1929 to 1933, manufacturing output plummeted by a third, which pushed unemployment from 4 percent to 20 percent. Nearly half the nation's workforce was working part-time or for severely cut wages or both. Because businesses lost capital, they slowed down production and laid off workers whom they could no longer afford to pay. The workers, in turn, had less income to spend, so consumption declined as people bought less. Decreased spending on goods and services further decreased capital for business, further decreasing the need for workers, leading to even greater unemployment in a downward economic spiral toward underemployment

equilibrium. With no insurance on bank deposits, many people lost savings they had entrusted to the banks. With no Social Security or unemployment benefits, the breadth and depth of poverty became overwhelming.

Keynes proposed that government has an active role to play in the economy. Against the “invisible hand” ethos, he argued that governmental spending—even if it requires the government to take on debt—is a necessary component in economic policy, particularly for a depressed capitalist system. This argument clearly runs counter to the notion that “free markets” are best suited to efficiently determine production needs and set prices. However, Keynes saw the only way out of the widespread poverty and unemployment of the Great Depression was for the government to stimulate the economy with its own spending, thus increasing the “aggregate demand,” or total amount of spending in the economy. If the government injected capital, businesses would generate more revenue, which would produce more capital, which would increase business spending. Thus, businesses would hire more workers, workers could again reach levels of income that would permit more spending, and as the economy returned to higher productivity and employment, government spending could wane, and debts could be paid off. Similar economic thinking was seen in US stim-ulus policies enacted after both the 2008 financial crisis and the COVID-19 pandemic.

Starting in the 1930s, President Franklin Roosevelt initiated a series of public-works projects, financial reforms, and new regulatory acts called the “New Deal.” The three primary goals were to provide relief for the unemployed and impoverished, generate economic recovery at the national scale, and put in place financial and business-policy reforms to ensure the situation wouldn’t be repeated. The Works Progress Administration financed projects such as the construction of government buildings, airports, hospitals, schools, roads, bridges, and dams to get unemployed people working again while building up the nation’s infrastructure. More than 8.5 million workers were employed on projects that transformed the American landscape.

By the 1930s, nearly 90 percent of American city dwellers had electricity, but only about 10 percent of people in rural areas did. The Rural Electrification Administration was created in 1935 to bring electricity to rural America. It helped establish member-owned electricity cooperatives,

many of which still operate today. The idea was that affordable electricity could improve rural standards of living while boosting economic competitiveness in America's farm communities. The program also helped manufacturers of light bulbs and electrical products reach a larger market, thus stimulating demand. Designers such as Lester Beall worked with the program to design posters communicating the opportunities and benefits of electrification in simple visual terms, and brought modernist design sensibility to government-sponsored public messaging. Similar to Constructivist approaches to graphic design a few decades earlier in revolutionary Russia, these posters used simple geometric forms to communicate ideas in a way that could be understood even by undereducated or illiterate audiences.

New Deal economic reforms sought to rescue the capitalist system from disaster, primarily by regulating the banking system, the stock market, and corporate activities related to financial reporting and sales of securities. The reforms included the establishment of the Federal Deposit Insurance Corporation, which insures most bank deposits. The Social Security Act was implemented in 1935 to provide universal retirement support, unemployment insurance, and welfare benefits for those not physically able to work and for children in need. Even though many of these financial reforms and relief programs have been weakened or repealed over the decades, they have been crucial for American economic stability and have helped buffer subsequent shocks, such as the 2008 financial crisis, from descending into even worse disasters. Of course, the New Deal wasn't without critics. Some were outraged at the amount of debt taken on by the federal government. Others saw the New Deal as a massive public bailout for a failed economic system—what was really needed was radical change, not just to protect citizens from markets but to improve the lives of workers, women, and minorities. Still others held entirely ideological oppositions to the New Deal and began plotting for the moment to institute a very different, more liberal economic model.

Austrian School: Imperfect Competition & Creative Destruction

In the late nineteenth and early twentieth centuries, ideas from a group of scholars in Vienna came to be known as the “Austrian School” of economic thought. These ideas were based on “methodological individualism,” the principle that individual motivations rather than class or group dynamics can explain social phenomena. Austrian School theories challenged the static nature of equilibrium described by classical theory and generally rejected the value theories of classical and Marxist economics, which were based on costs of production. The Austrian approach instead identified diverse and subjective contexts under which a thing can be said to be a *good*, have *value*, and/or be exchanged at a mutually agreed upon price.

This approach emphatically insisted on a distinction between *value* and *goods* themselves. According to the Austrian economist Carl Menger, who is credited as the founder of the Austrian School: “There is no necessary and direct connection between the value of a good and whether, or in what quantities, labor and other goods of higher order were applied to its production.” *Value* is not an independent property inherent to goods. Value is “a judgment economizing men make about the importance of the goods at their disposal for the maintenance of their lives and well-being. Hence value does not exist outside the consciousness of men.”[55] This theory is referred to as the *subjective theory* of value.*

Part of this theory is the concept of *marginalism*, which attempts to explain how it is possible that goods with higher *use value*—that is, goods whose use by people is of seemingly more importance—could yield lower prices on the market than goods with seemingly lower use value. For example, gold is more valuable than wheat, despite the obvious fact that wheat can sustain human life, a primary utility, whereas gold serves only secondary utilities, such as the display of wealth. Yet although wheat may have a higher *total utility*, gold has a higher *marginal utility* because of its relative scarcity. The concept of marginalism was a rejection of both Adam Smith and Karl Marx, whose theories prioritized costs of production, particularly labor, in the determination of value. In the Austrian view, capital and labor interact in ways that are not predetermined by a model in which the price of any good can be determined simply by balancing supply and demand.

The Austrian School’s enduring contribution to economic theory is the emphasis on *value* and *quality* as dynamic factors in pricing that have

nothing to do with production costs of materials and labor or with a product's use value. Value, including *perception* of quality, is based entirely on relationships among the product, competitors, and potential consumers. "A product is not 'quality' because it is hard to make and costs a lot of money, as manufacturers typically believe. That is incompetence. Customers pay only for what is of use to them and gives them value. Nothing else constitutes 'quality.'"[56]

This economic perspective admits that subjective—emotional, perhaps even irrational—decision making drives consumption. This admission breaks open theoretical space to recognize "opportunities for innovative strategies that generate product innovations, which in turn create conditions of *imperfect* competition."[57] Under *imperfect competition*, a producer can create an advantage not only through lower costs of production or higher demand but also through the creation of "innovative products that function as the principal agents of value creation." Innovative new products can bring about "the destruction of existing products and the dynamic creation of new demand and new markets."[58] The previously described arc of personal audio equipment, from transistor radios to the Walkman to the iPod, is a clear example.

The theoretical role design plays in creating value now becomes clear. Value is in the eye of the beholder—or, in this case, the would-be consumer. The shifting and subjective perceptions of a product's quality, durability, and desirability contribute to its potential for value as an economic *good*. Design contributes to these perceptions in ways that are embodied both in the product itself—its materials, style, physical presence—as well as in the ways a preferred perception of the product's value is promoted through branding, marketing, and promotional messaging.

Many underlying principles of the Austrian School are alive and well in today's mainstream economic thought. The Austrian political economist Joseph Schumpeter contributed the concept of *creative destruction*. According to Schumpeter, "capitalism then, is by nature a form or method of economic change and not only never is but never can be stationary. . . . The fundamental impulse that sets and keeps the capitalist engine in motion comes from new consumer goods, the new methods of production or transportation, the new markets, the new forms of industrial organization that capitalist enterprise creates."[59]

Although critics have argued that the Austrian School is based on an overreliance on individualism, that same core principle has been its enduring influence. Individuals make self-interested—not necessarily rational—decisions that shape markets and subsequently prompt producers to marshal innovative approaches to design and production in order to generate imperfect competition, or *competitive advantage*. This cycle creates new desires that spawn new innovations, which creatively "destroy" existing products and markets in an endless quest for something "new" or "better."

The Austrian School may not have had much direct impact on economic policy during the early twentieth century, but many of its ideas have gradually become quite influential. Ludwig von Mises, one of the school's intellectual leaders, immigrated to the United States in 1940 and has been credited with influencing the "Chicago School" of economics, which pushed America's return to liberal economic policies near the end of the twentieth century. Both schools favor private enterprise, low taxes, minimal government, and free markets. Unconstrained pursuit of self-interest is seen as the engine of progress. However, these principles were a hard sell while Keynesian economic policies appeared to be pulling America out of the Great Depression, and so they would have to wait for their moment to be broadly implemented.

The (So-Called) Golden Age of Capitalism

Wartime Cooperative Production

Although the New Deal employed millions of Americans during the Great Depression, the United States did not reach full employment until entering World War II at the end of 1941. The war effort triggered tremendous government spending, which doubled the gross national product. The US government did not require manufacturers to engage in competitive bidding, which typically leads to lower pricing, and instead provided contracts to cover production expenses plus a modest profit. Barriers that had previously excluded many people from the labor force were removed, bringing women and racial minorities (except in southern states) into the workforce. Factories hired unskilled workers and provided them training,

paid for by the government. Farmers left their land, students dropped out of school, and housewives left their homes to support the massive war effort. This economic state of exception has been called *military Keynesianism*,* a term critical of enormous public spending intended to boost the economy during war but not to support public benefits such as education, health care, transportation, and other public infrastructures during times of peace.[60] Dramatic growth in the US economy during the war was due primarily to a de-emphasis on free enterprise, strict controls on prices and wages, and essentially the elimination of the market's "invisible hand."

As the propaganda machine fired up to promote widespread support of the war effort, graphic designers found new outlets for the burgeoning modernist language of "direct and economical means of simplified text and visual image"[61] in posters promoting citizens to conserve materials, work, volunteer, or serve.

Postwar Boom

The huge ramping up of American and European industrial production for the war created new capacities of scale and production technology that fed into a postwar period of economic growth unusual by all historical standards. In the United States and Europe, this period has been referred to as the "Golden Age" of capitalism because of the sharp increase in material standards of wealth experienced by large segments of the population. However, the American economist Robert Reich has referred to the era as the "Not Quite Golden Age" because of the political economy's incomplete reach in equitable distribution of economic, political, and social well-being.

> Roughly between 1945 and 1975, America struck a remarkable accommodation between capitalism and democracy. It combined a hugely productive economic system with a broadly responsive and widely admired political system. America in those years achieved its highest degree of income equality (since measurements have been available). It generated a larger proportion of good-paying jobs than before or since, and more economic security than ever for more of its people. Perhaps not coincidentally, in those years Americans also expressed high confidence in democracy and trust in government, both

> of which sharply declined in subsequent years. . . . But it was not quite a golden age. Women and minorities still struggled for political equality and economic opportunity. Much of the nation's poverty was hidden away in rural hollows or black ghettos. Foreign policy, ostensibly shaped by the perceived threat of Soviet communism, all too frequently pandered to the needs of large American firms for cheap resources abroad, such as bananas, tin, and oil. Civil liberties were imperiled during Senator Joe McCarthy's anti-communist witch hunt. Much of American life was monotonous, conformist, and deadly dull.[62]

Despite all of capitalism's shortcomings, cooperation between big business, government, and labor unions eased resistance to some of its negative impacts, including cyclical downturns and social inequality. During this time, government played a prominent role in regulating markets and the capitalist system globally. Keynesianism became nearly standard economic policy across the industrialized capitalist economies of the United States, western Europe, and Japan. The successes of this era are credited to the idea that "government, capital, and labor had reached the uneasy agreement that markets produced social ruin when left to their own devices. The state was needed to mitigate inequality, to provide basic services, and —through a combination of monetary and fiscal means—to even out capitalism's boom–bust cycle."[63]

This highly planned and fully regulated economy had shameful racial and gender exclusions but otherwise a generally broad, distribution of economic gains. According to Reich:

> Democratic capitalism of this era involved a complex and continuous set of negotiations—sometimes directly between key players, like big business and big labor; sometimes indirectly, within regulatory agencies and legislatures. There were several important consequences: first, large economies of scale, generating high productivity and hence significant profits; second, tens of millions of steady jobs; third, a wide distribution of the profits (downward to blue-collar workers and outward to smaller communities, farmers, and other constituents); fourth, millions of consumers who used this largesse to purchase the

> goods and services produced in ever larger quantities, thereby stabilizing the economy for large-scale production. The result was a large and growing middle class across the country that stabilized the political system. It was full circle.
>
> This was hardly the textbook model of democratic capitalism, which strictly separates the "economy" from "government" and believes capitalism occurs almost automatically through the workings of an idealized "free market," while democracy happens because voters go to the polls to make their selections as to who should oversee the public sector. In fact, much of the government's unstated role was to *manage* the often untidy process by which these economic and political negotiations took place.[64]

The Keynesian model focuses primarily on *demand* as the driver of economic growth. If more consumers demand more products, they will boost the economy. However, for consumers to generate that demand, they need the money to consume; therefore, employment and wages must be high enough to sustain demand and keep the economy booming. This combination makes for high productivity and high rates of consumption, which contributes to growth and a certain level of social satisfaction (for some). However, it's not particularly good for all of the system's "externalities," such as the negative impacts on the foreign places and their populations where resources are extracted (sometimes destructively and often violently), the (often violent) exclusion of the domestic populations from the system's wealth accumulation and equity, as well as the transformation of air, water, and land into dumping grounds for enormous industrial production.

Postwar Design: Mass Culture & Consumerism

After World War II, steady economic growth in wages and production increased aggregate demand. Increased demand for products that a large *consumer class* could use to symbolize their aspirations put professional design on the map through nearly endless opportunities to materialize the zeitgeist. A combination of rationality, optimism, and striving toward a peaceful and prosperous future—even amid the Cold War's ominous and

nearly constant threat of nuclear annihilation—defined the spirit of this time. There was high demand for inexpensive housing and furnishings as the population depletion of war was followed by a growth in birth rates as young families set out to build new lives. Travel became a new commodity as commercial jet travel became widespread and affordable. The jet age gave way to the space age, both suggesting sleek movement, futuristic materials, and an embrace of technology in the domestic sphere. “New” became its own virtue, an infectious desire to shed tradition and create an optimistic future-oriented culture of everyday life. Inexpensive and accessible, informal and adaptable, functional not decorative: this mindset gave way to a design language that merged new materials and manufacturing technologies with pared-down and often abstract aesthetic sensibilities, all wrapped into affordably priced, mass-produced objects. “There was a tremendous receptivity to the phenomenon of planned obsolescence[,] . . . the calculated efforts of manufacturers to generate demand with a seemingly unending supply of ‘new’ products.”[65]

American companies such as Herman Miller and Knoll manufactured and marketed high-quality, well-designed, and affordable furniture and home furnishings. New materials such as fiberglass and plastic as well as new techniques with molded plywood and veneers gave designers new formal possibilities in ergonomics and sculptural approaches to furniture design. The marketing of these goods adopted similar visual concepts that sought to both appeal to and elevate the aesthetic sensibilities of the postwar consumer market. *Sleek, minimal, uncluttered clean lines*: such terms were the language of the value proposition of design at this time. No tricks here. Just straightforward materials doing things you haven’t seen or experienced before. This was the era of high modernism in design, and it produced some of the most iconic figures of American and European industrial and graphic design, such as Raymond Loewy, Charles and Ray Eames, Massimo Vignelli, Paul Rand, Saul Bass, Eliot Noyes, Armin Hoffman, and others, many of whom remain the idols of design schools around the world even though their work represents a bygone age.[66]

“Good design” fostered capitalist enterprise while developing standards in quality and aesthetics that went beyond mere concerns of taste, instead “embracing politics and the role of design in shaping and strengthening shared cultural values and democratic ideals.”[67] In some cases,

“paternalistic themes of improving and educating the public to appreciate the aesthetic, social, and practical advantages of modern living” were mixed with social themes on the “responsibility of citizens and businesses.”[68] However, whatever good intentions the designers and manufacturers had, these broader social goals were not always well understood “as a universal set of standards that raised general cultural awareness or instilled responsible democratic values.” Rather, these “products and promotions simply offered one among many possibilities for individual fulfillment in an age of increasing affluence and possibilities for leisure. . . . [T]he expanding American middle class in particular responded enthusiastically to constant novelty, fantasy, and instant gratification.”[69]

The role of television was critical to the mass dissemination of dominant cultural ideas, through both programming and advertising. The term *mass culture** describes this historical moment when mass media (glossy magazines, television, and radio) combined with mass production (millions of units of the exact same product) to create a large but fairly homogenized culture—a culture enamored with the “new,” which also began to mean “disposable.” The British design historian and critic Reyner Banham described this emergent phenomenon as *throwaway* culture.[70]

Designers worked all sides of this scenario: industrial designers embraced new materials, new technologies, and new production techniques to create new product lines and entire new categories in all of these areas; graphic designers created stylish and seductive marketing and advertising campaigns to pump up desire for the latest styles and gadgets; design teams worked collaboratively to create new images for contemporary lifestyles through the total-design approach of branding. And, of course, the mass-market engine required constant fuel in the form of growth, growth, growth. This meant people had to buy, buy, buy as much and as quickly as possible. Historians generally agree that “the success of product obsolescence for the mass market could not have taken place without the stimulation of advertising. . . . [T]hrough illustration, photography, or the emerging medium of television, advertisements linked products to individual identity and social acceptance.”[71] The result was a new social, economic, and political order largely dominated by *consumerism*.* The economy needed steady and constantly growing levels of consumption for stability, which made consumption socially and politically necessary.

After World War II, those who had joined the labor force for the first time during the war were for the most part sent back to their prewar social statuses. Women, African Americans, and other racial minorities were promptly displaced by white males from their newly found roles in the economic social system. Amid a housing boom that built up the suburban American landscape, this mass culture exhibited conformity as well as exclusivity. “Conspicuously absent from these new suburban neighborhoods and their television counterparts were Black Americans and other minorities.”[72] Meanwhile, out in the white suburbs, women were subjected to advertising that “outlined the applications of newly-found leisure in a variety of products and appliances” while very clearly encouraging “women to focus their energies upon the home [and] defining their role primarily as consumers while [their] husbands went off to work.”[73]

The Keynesian economic model created the conditions in which the nature of the *commodity* was transformed. There was generally a shift in motivation, “from the commodity itself to the psychological motivation for its purchase, that is, from production to consumption.”[74] The images created through advertising and mass media took on an enormous role in defining social values—symbolized by products as means to satisfy social desires—in ways not previously imaginable. “The relationship between advertising and mass design reminds us that the interaction between products and users was mediated by images as seen in print and on television.”[75]

The Rise of Corporate Identity

With so much growth and conglomeration, big businesses dominated markets. Each industry featured only two to three primary competitors: General Foods, Quaker Oats, and General Mills in food; RJ Reynolds, Liggett & Myers, and American Tobacco in tobacco products; General Motors, Ford, and Chrysler in automobiles; NBC, CBS, and ABC in television broadcast news. This consolidation took place across American industry.

This limited competition posed two new opportunities for designers. First, there was the need to systematically unify an image of continuity and

consistency across the vast and varied offerings of these massive companies. Second, because each of the major media or manufacturing enterprises wished to capture the biggest possible market share, differentiation became critical in generating a *sense* of choice—even when that choice really boiled down to two or three producers selling a huge range of different products. “Particularly in the United States, brand differentiation among products manufactured by a single company was another effective corporate strategy, giving the impression of variety and choice. . . . Such strategies involved a close relation between design, market segmentation, and massive advertising investment.”[76]

During this time, a move toward nondescriptive logos and trademarks was a move away from “things” and toward a symbolic system of differentiators: Coca-Cola versus Pepsi. Out went the descriptive indicators of things; in came far more abstract ideas about what consumption could achieve. Specific expressions of simplicity, efficiency, security, happiness, or sex appeal were the things one consumed, not an automobile, a kitchen appliance, or a soft drink.[77] This philosophy extended into domains where product lines were already mystifying, such as computational machines.

In 1956, IBM hired American architect and designer Eliot Noyes to “rethink the company’s design as a whole, from products to communication right up to the architecture of the buildings.”[78] Noyes brought in the American graphic designer Paul Rand to define the entire visual system for the company’s diverse communication needs and thus to contribute to a massive project emblematic of corporate communications of the time: the IBM Graphic Design Program. What was IBM actually selling? Computers and machines, yes, but, more important, confidence that its machines and its services would improve your business’s capacity to do whatever you wanted it to do. Rand’s design work put a playful face on a somewhat utilitarian business proposition that didn’t have much of a shiny exterior to show. Combining simplicity and abstraction, corporate graphic design produced entire constellations of images to portray the personalities and the many offerings of increasingly complex business practices.

Conclusion

This historical period also came to an end. Midcentury modernist design coupled with a brief period of mutual benefit between capitalism and democracy began to give way to major transformations near the end of the twentieth century. The implications of those transformations are still reverberating in the contemporary economy and design, even as they are not always fully recognized. The icons of design and design practice during the midcentury modernist period are often still regarded as a high watermark, perhaps indicating that such practices are still worth achieving or possible to achieve. However, while history may rhyme, it typically does not actually repeat itself.

3

Design, Capitalism, & Economic Theory in the Postindustrial Era

The New Coworker

For sixty years, West Trenton, New Jersey, was home to a General Motors facility that produced body moldings, door handles, and interior components. The Inland Fisher Guide Plant employed a crew of 1,500 when it opened in 1938. During World War II, it was temporarily converted to build torpedo bombers for the US Navy. In 1998, the plant closed permanently. The 900 remaining workers, who had survived several waves of previous layoffs and downturns, were put out of work. What happened?

In the middle of this timeline, the auto plant witnessed an event of historic magnitude. In 1961, it became home to the first programmable industrial robot in American manufacturing. The robot, named “Unimate,” was used to carry auto components that had just been die cast from molten metal to cooling pools. This was dangerous work for humans because of the safety risks associated with handling extremely hot metal. In one shift, Unimate could do the work previously performed by three shifts of employees. So began the automation of the automobile industry, emblematic of the dramatic recomposition of social and economic life in late twentieth-century America.



Figure 3.1
Built in 1925, Detroit's glamorous Michigan Theatre was designed in the French Renaissance style and became emblematic of the American capital of industrialism. It was built on the same site where there once stood a small garage in which Henry Ford had built the quadricycle, his first automobile. In 1977, the 4,038-seat theater was transformed into a parking garage. (*Source*: Dan Austin, "Michigan Theatre," Historic Detroit, accessed January 29, 2021, https://historicdetroit.org/buildings/michigan-theatre. Image courtesy of Tom Perkins.)

In sociological terms, a society reaches the "postindustrial" stage when the economy shifts from predominantly producing *goods* in the form of objects and physical products to primarily providing *services*. For example, in an industrial economy the majority of workers are employed in manufacturing at factories, textile mills, and chemical plants or in construction. In a postindustrial economy, the majority of workers are employed in professional sectors such as law, medicine, education, and media; in designing, developing, and maintaining the various technical systems that support these professional sectors; and in service industries such as retail, health services, and hospitality. Technology, information, and service delivery become more valuable to the economy than manufactured goods. In the United States, the transition generally took place between the

1950s and the mid-1960s. However, the impact of this transition would be felt most distinctly in the 1970s.

Postindustrial (Late) Capitalism

Writing in the early 1970s, the American sociologist Daniel Bell popularized the term *postindustrial* and used it to articulate shifting social values and perceptions, including massive technological and organizational expansions that grew out of the postwar economic boom.

> The private enterprise system has been the primary institution of Western society not because of its coercive power but because its values—economizing and increasing output of material goods—were congruent with the major consumer values of the society. With all its obvious imperfections the system "worked." Today, however, those values are themselves questioned, not in the way socialists and radicals questioned them a generation ago—that they were achieved at the cost of exploiting the worker—but at the very core, the creation of more private goods at the expense of other social values. . . . [N]o one, meeting collectively, "voted in" our market economy. But now votes are being taken. . . . No social or economic order has a writ of immortality, and the consumer-oriented free-enterprise society no longer satisfies the citizenry, as once it did. So it will have to change, in order that something we still recognize as a liberal society might survive.[1]

What happened in the 1970s to mark such a pivot in the history of Western industrial societies? Historians and economists are, expectedly, divided on cause versus effect yet generally agree on the broad trends. "Since the late 1970s, a fundamental change has occurred in democratic capitalism in America, and that change has rippled outward to the rest of the world. Capitalism has triumphed, and not simply as an ideology. The structure of [the] American—and much of the world's—economy has shifted toward far more competitive markets. Power has shifted to consumers and investors."[2] A few key events were part of this shift.

There was an oil shock. In October 1973, the Organization of Arab Petroleum Exporting Countries proclaimed an oil embargo targeted at nations supporting Israel in the Yom Kippur War. Arab exporters refused to sell or trade oil with Canada, Japan, the Netherlands, the United Kingdom, and the United States. Oil was so central to complex supply chains of production and transportation that the embargo had an immediate and profound impact on the global economy. It lasted nearly two years, and by its end in March 1974 the price of oil had risen nearly 400 percent, with prices significantly higher in the United States. The embargo caused a global shortage, or "oil shock," which in turn triggered many short- and long-term effects on the global economy and global politics. These effects plunged much of the Western world into a recession from roughly 1973 to 1975, creating economic stagnation (frozen wages and low production) along with soaring inflation. This combination induced *stagflation**—inflation during a recession—something that economists and policy makers have studied with the hope to prevent it from occurring again.

Some analysts point to the economic shock of the Arab oil embargo as the trigger of a "crisis sequence"[3] that set off a number of policy changes. These changes would effectively transition advanced capitalist economies away from the Keynesian model of the postwar period toward a liberalization of markets.

> Capitalism has been on a crisis trajectory since the 1970s, the historical turning point being when the post-war settlement was abandoned by capital in response to a global profit squeeze. To be precise, three crises followed one another: the global inflation of the 1970s, the explosion of public debt in the 1980s, and rapidly rising private indebtedness in the subsequent decade, resulting in the collapse of financial markets in 2008. . . . This sequence was by and large the same for all major capitalist countries, whose economies have never been in equilibrium since the end of post-war growth at the end of the 1960s. . . . Unfolding alongside the crisis sequence and the transformation of the tax state into a consolidation state were three long-term trends, all starting more or less at the end of the post-war era and running in parallel, again, through the entire family of rich

capitalist democracies: declining growth, growing inequality, and rising debt—public, private, and overall.[4]

The American economist Robert Reich sees a more specific culprit at the center of the rippling transformations: “The real explanation involves the way technologies have empowered consumers and investors to get better and better deals—and how these deals, in turn, have sucked relative quality and stability, as well as other social values, out of the system.”[5] According to Reich, new tech-nologies enabled entirely new ways of doing business, sourcing labor, and moving materials and goods around the world. Increased global competition pushed the costs of goods and labor wages dramatically downward. This downward spiral created a domino effect: investors’ increased demand for quarterly returns influenced corporate strategies, such as restructuring and downsizing to eliminate “bloat” in the labor force, which led to increased unemployment, which further decreased wages. This process ratcheted up cutthroat competitiveness among businesses leaders, in particular CEOs who knew they might be out of a job if they couldn’t return a profit every single quarter even amid tight global competition. In turn, eye-popping executive salaries and bonuses were used to incentivize ruthless cost-cutting strategies, leading to soaring disparity between top and bottom pay scales. Corporations used their wealth to lobby governments in an effort to squash barriers to profit, such as labor and environmental regulations. Increased lobbying by wealthy corporations and greater financial support of political candidates transitioned a great deal of political power away from citizens and toward the interests of global corporations.

Market Saturation & Product Diversification

The economic challenges of this period were met with new approaches to commodity production. The 1970s are often referenced in design history books, perhaps overgenerously, as an era of *pluralism*. “Mature markets” were already saturated with basic goods. As mass culture splintered, production ran into demand problems for products that were all the same. Segments of the population were taking on new social and cultural identities, and commodity production began to diversify. Production chains

were in the process of becoming more automated and more global, which created opportunities for smaller production runs. Product design responded with a focus on differentiation, customization, and even “limited” runs. This was far different from the mass production of undifferentiated products during the era of *Fordism*. Aligning with the pace of fashion, commodities were produced in smaller quantities with minor and often only aesthetic changes. A dazzling carnival of consumption opportunities was put on offer to consumers who had been profiled by market segmentation. “Late capitalism signifies investment directed toward increasingly segmented audiences (rather than collective or a single shared taste).”[6] This was true for both small, inexpensive products as well as the major, expensive ones, to the point that by 1980 “no two cars built on the same day at the Volkswagen plant in Wolfsburg were completely identical.”[7]

> The re-engineering of product ranges now went far beyond the yearly changes in hubcaps and tail fins that American automakers had invented to accelerate product obsolescence. . . . By the 1980s, accelerated product design and more flexible production equipment and labour made it possible to customize the commodities of the Fordist world to an unprecedented extent, subdividing the large and uniform product runs of industrial mass production into ever-smaller series of differentiated sub-products, in an effort to get closer to the idiosyncratic preferences of ever-smaller groups of potential customers. . . . Product diversification attends to the wants of consumers which, under mass production, had remained commercially untapped; now they could be activated and made profitable. . . . Marketing discovers, but typically also develops consumer preferences; it asks consumers what they would like, but it also proposes to them things they might be prepared to like, including things they never imagined could have existed. Good marketing, in this sense, co-opts consumers as co-designers, in an effort to haul more of their as-yet commercially idle wants, or potential wants, into market relations.[8]

Postmodernism

The period of “late capitalism” finds its cultural form in *postmodernism*.[9] Although vexing in its many murky definitions, the term *postmodernism** “seems to have been able to welcome in the appropriate areas of daily life or the quotidian; its cultural resonance, appropriately vaster than the merely aesthetic or artistic, distracts suitably from the economic while allowing newer economic materials and innovations (in marketing and advertising, for example, but also in business organization) to be recatalogued under the new heading.”[10]

The end of the modern era seemed inevitable by the 1970s in wake of the widespread social and political unrest of the 1960s and the economic challenges of the early 1970s. The norms that had defined Western society and the institutions that authoritatively defended them were rattled by challengers who saw a system built on exploitation with particularly pernicious effects along lines of class, race, ethnicity, gender, and sexuality. The core idea of modernity as a universally applicable project of rational, scientifically supported, industrial production and consumption—wherein the economic interests of private businesses were supported by the state under a proclaimed shared mission of historical progress—was under attack.

In design, postmodernism materialized as a variety of direct challenges to the dominance of modernism as it had been understood since the start of the twentieth century. The defining theory of modernist design was *functionalism*—even though the term itself has often been rejected. The premise that “form follows function” was a rational, economizing ideal from the start. Whether admitted or not, in these words or others the guiding ethos boiled down to the following: “Authentic modern design will reject the coexistence of multiple styles and design according to the demands of instrumental purpose.”[11] This ethos unraveled in the 1970s, along with so much else. The order and clarity found in modernist design, in particular the corporate design of multinational corporations, came to be seen not only as untrustworthy, slick, empty veneer but also as potentially corrupt from the start, based on the original sins of a society built from patriarchy, racial hierarchy, theft, and plunder.

Along with the challenge to rationality came a perceptual transformation of objects or, as seen through the lens of capitalism, commodities. In the modernist conception, all objects of everyday life—chairs, toothbrushes,

train schedules, housing blocks, skyscrapers—were brought into the same domain of aesthetic assessment. This worldview insisted that beauty and utility were one. However, from the Bauhaus to the International Style, this aestheticization of everyday life repositioned the "object" not as a physical item of *use value* (in Marxist terms) but rather as a *sign* (in semiotic terms). The object stood in for an ideological position, an idea, which brought all designed objects into the world of semiotics. In light of postmodernism, all forms of material culture—chairs, kitchen utensils, and buildings every bit as much as novels, advertisements, and films—began to be critically *read* as *texts* and subjected to semiotic analyses, open to multiple interpretations.

The French sociologist and cultural theorist Jean Baudrillard is perhaps best known for his work in *post-structuralism*, a literary and philosophical school that argued that meaning and signification can be understood only by the ways that particular words or "signs" interrelate. For the post-structuralists, there is no existing, ideal, or immutable system or structure of meaning in language. Every word is an arbitrary sign whose meaning can be determined only by its difference from and relation to other words. This is also true for the cultural forms produced by societies constantly in search of meaning—a search that typically ends in the delusion of a total system of knowledge. Turning his attention to consumer society, Baudrillard critiqued the Marxist concept of *use value* and stressed instead that "needs" are constructed by capitalism. He argued against the notion that capitalism can satisfy human needs—a notion sometimes followed by the suggestion that once needs are met, people will moderate their *desires*. Yet capitalism doesn't *satisfy* needs and thus remove them. Capitalism can exist only with constant growth, so it constantly *creates* needs. Capitalism may produce many things, but it cannot achieve a comfortable abundance; instead, it must continuously produce discontent in the form of endless unsatisfied desire.

Baudrillard saw commodities as similar to words. Something is always *signified* by consumption; consumption signifies something socially. Baudrillard effectively extended Veblen's notion of *conspicuous consumption* to all commodity consumption. "You never consume the object in itself (in its use value); you are always manipulating objects (in the broadest sense) as signs which distinguish you either by affiliating you to your own group taken as an ideal reference or by marking you off from

your group by reference to a group of higher status."[12] Objects disappeared into systems.

If postmodernist theory sounds like a lot of mumbo jumbo, consider this passage from the *Harvard Business Review* in 1980: "Products are almost always combinations of the tangible and the intangible. An automobile is not simply a machine for movement visibly or measurably differentiated by design, size, color, options, horse-power, or miles per gallon. It is also a complex symbol denoting status, taste, rank, achievement, aspiration, and (these days) being 'smart'—that is, buying fuel economy rather than display."[13] Perhaps the *Harvard Business Review* was also consumed with post-structuralist theory.

What this means for the concept of *brand* should be immediately apparent. In late capitalism, design's value becomes largely symbolic, creating differential signs. That there are memes dedicated to ridiculing users of Android smartphones as opposed to the socially superior iPhone speaks volumes.

In hindsight, it's not surprising that designers used postmodernist concepts of historical reference and pastiche as cultural currencies in the transforming political economy of the 1970s. "Historical references, decoration, and the vernacular were disdained by modernists, while postmodern designers drew upon these resources to expand the range of design possibilities."[14] As the industrial machine ran out of steam for new cultural forms, the sample, the mashup, and the remix became methods of cultural authorship, blending new forms out of the cultural detritus left behind by 200 years of industrial life. The invention of hip-hop is emblematic. Samples from instrumental "breaks" on existing songs became the material for new sound collages over which an MC would speak, sing, and eventually rap.

The explosively popular, trendy, and high-selling Swatch watch of the 1980s is quintessentially postmodern. The Swatch was "the result of inexpensive plastic materials and production, and an especially fashion-oriented approach to wristwatches to complement clothing." In contrast to the long history of incredibly precise, durable, and reliable clocks and watches produced in Switzerland, "the ephemeral, disposable Swatch . . . targeted a young audience less likely to be influenced by the emotional attachments sometimes associated with personal objects and heirlooms."[15]

The inexpensive watches came in a seemingly endless variety of decorative colors and patterns. Low cost and variety made them an item that was often stylishly worn—and purchased—in multiples.

While the product itself fit the postmodernist moment of fashionable ephemera, the marketing of the watch aligned with postmodernist themes of historical reference and pastiche. In 1984, the American graphic designer Paula Scher created a Swatch advertisement that uses photomontage to depict a woman wearing two Swatches on her wrist amid an Alpine skiing scene, including the white-on-red Swiss cross—all representative of Switzerland. The poster itself is a remix. Scher "appropriates elements of a Herbert Matter Swiss tourism poster from 1934, whose photomontage technique and juxtaposition suggest a broad range of associations."[16] The appeal of Swatch's variety and the encouragement to buy multiple are indicated by the simple fact that the woman in the image is wearing two different watches on her wrist. Graphic design is doing the work of promoting a disposable fashion commodity using a post-structuralist logic of historical images as free-floating signs like language—arbitrary and meaningless on their own and existing only in relation to other signs.

From Objects to Experiences: User-Centered & User-Experience Design

Amid the crisis of history that postmodernism seemed to propose, something substantively new in the human experience also began to take hold. Computational machines, or "computers," began to expand their reach into the everyday lives of regular people. Until personal computers emerged in the late 1970s, the only people who would typically encounter computers were scientific researchers or specialized professionals. The challenges and opportunities posed by everyday people working with these unprecedented machines gave birth to a litany of new disciplines and specializations for professionals and researchers. Designers participated in, became dependent on, or were otherwise caught up in these changes in multiple ways. New and sometimes competing strands of design practice emerged.

As business and personal computing took off in the 1980s, the need to make computer use accessible to nonexperts created new design challenges. The timing coincided with an emerging program of scientific research

called *cognitive science*, which brought together elements of psychology, linguistics, and anthropology. During the early days of designing software, this synthesis of social and physical sciences created practical challenges for designers working in various ways and at various points in the disciplinary mix. The broad field known as human–computer interaction (HCI) emerged in the early 1980s as an interdisciplinary endeavor, which has consistently expanded ever since. Emergent categories of *users* and *usability* created frameworks for designers to understand the new space of design: material–immaterial interdependencies between screens and inputs and bodies and minds. Previously defined designer categories, including "industrial designer" and "graphic designer," were folded into these efforts. Users wanted more than mere functionality or even ease of use. They wanted novel, exciting, delightful, sometimes surprising, or at other times informative experiences that could be understood, repeated, and shared and that would end up shaping much of contemporary life.[17]

Out of these challenges emerged *user-centered design*, a term coined in 1986. It was born with a strong psychological perspective through which designed objects were evaluated as "good" or "bad" based on the positive or negative associations generated in their *experience* rather than on any establishment of cultural, humanist, or aesthetic concerns. A psychological perspective, whether it was supported by actual psychological study or not, began shaping the methods designers used to conceptualize "user needs." The following year, in 1987, another new term emerged: *usability*. "Usability exists in the experience of the person. If the person experiences a system as usable, it is. A commitment to designing for people means that, at base, we must accept their judgement as the final criterion for usability."[18]

User-centered design, usability, and user experience (UX) convened to generate new mindsets and methods for design. Proving valuable in the design of software and other emerging immaterial experiences, a user-centered approach became commonly applied to the design of products, communications, and built environments as well. This fundamental shift was as much economic as ideological. It wasn't just that designers decided that they weren't the all-knowing form givers to cultural totems. Clients wanted to know that *potential users* would become *actual consumers*. User-centered methods of studying desires and evaluative methods of testing prototypes gave assurances and built confidence that products would sell. In

an economy of increasing competition and declining profit margins for markets with very similar product types—such as software or personal computers—user-centered and UX design gave designers tools to provide evidence of both *usability* and *desirability* for new products. This shift ushered in a wave of methods, skills, and specializations in user research that saw designers leave the era of the "creative genius" and enter the era of evidence-based, results-driven rationality.

Meanwhile, user-centered design not only strove to ensure usability across a spectrum of experiences but also sought to extend the tentacles of a brand's imprint on ownable experiences. Thus, *brand* and *experience* became the principle concerns of design. Designers were no longer styling and marketing products; they were shaping the language of every potential communication, determining the friendliness of the employee's uniform, debating which flavor of complimentary coffee would be most "on brand," specifying the smell of the floor polish and the music to play, and in all of these moves they attempted to make major corporations much more like individual people: your friends and companions in a consumer society! Design jobs increasingly became management jobs, many akin to relationship-management jobs.

Cooperative to Participatory Design

Meanwhile, a different model for designing technical systems emerged from Scandinavia. Starting in Norway in the 1970s, the Scandinavian method of "cooperative design" was a democratic means of engaging nonexperts in the process of designing software systems. Cooperative methods were used when new technologies such as digital workplace-management systems would affect not only those directly using the software (managers) but also the people whose daily lives were managed by the software (workers). Unlike the interdisciplinary expert teams assembled in the work of HCI, anyone whose life might be affected by the implementation of a new technical system would be invited to weigh in on how that system would work, how it would interface with their daily lives, and the implications it might have for individual and social life.

A seminal cooperative project was conceived by the Norwegian computer scientist Kristen Nygaard, coinventor of object-oriented

programming, when he conducted computer science research on planning, control, and data processing through the lens of organized labor. In 1972, he began a project with the Norwegian Iron and Metal Workers Union to engage factory workers in the design of workplace technical systems. Instead of an expert-oriented approach, he worked directly with the workers and their unions, giving them active roles in technological innovation supported by their employers. This approach was nearly the opposite to that of the robot showing up next to you on the factory line, which was a sure sign your days were numbered. Nygaard's considerable contributions to the field of computer science are coupled with his socially oriented approach to research, and his work has influenced "all succeeding research and development of user participation in systems development leading into cooperative (or participatory) design."[19] In this approach, democratic control is extended to the private workspace, establishing opportunities for increased knowledge and political power.

In the United States, workers and managers were so segregated in their roles and stratified in their social positions that their attempts at cooperative design did not initially involve cooperation or even direct dialogue between these social classes. There was even friction over the word *cooperative*. The term *participatory* was instead applied and subsequently set the course for what became known as "codesign" methods. The variety of direct-engagement methods that have emerged into practices now commonly known as "participatory design" situate designers not as expert material-problem solvers but more as facilitators to a democratic process of shared authority. User-centered design methods typically study "users," individuals who may or may not end up using the designed experience but who serve to create *personas*—distinct "types" of potential users or consumers. Participatory design in the original cooperative approach takes on a distinctly political dimension by working with everyone from experts to workers to local residents or others who will *actually* be affected by the system being designed. This contrasts with codesign methods when they are used primarily to extract behavioral information from workers or would-be consumers. User-centered design is typically oriented toward individuals (often individual consumers), whereas participatory design is typically socially oriented.

Service Design

In the 1960s, companies began circulating phone numbers exclusively dedicated to answering customers' questions and resolving problems with purchases, replacements, warranties, troubleshooting, and repairs. This new offering required coordinating the behind-the-scenes workforce, offices, and infrastructure to respond to customer calls. Thus, the call center was born. In 1967, American telecom giant AT&T invented the toll-free number. Customers could now call in with their questions—and complaints—for free. This removed the cost barrier to using service lines. As call-based service centers became more widespread, they also became a competitive necessity for businesses. Toll-free service support gave a sense of comfort that could offset consumer concerns about making a bad purchase (concerns described by behavioral theory as "loss aversion"). As service lines and call centers expanded their reach, they spread to industries such as health care for billing, collections, and scheduling; hotels and airlines for questions, reservations, and complaints; banking; and many government services at local, regional, and national levels. The first 911 call was made in the United States in 1968.[20]

Operating high-volume call centers became expensive. The capitalist approach to efficiency typically involves applying technological innovation to replace expensive human labor with machines, outsourcing labor to parts of the world with lower wages, or some combination of both. Fully automated voice-service systems, service-chat operators, and chatbots evolved along with globally outsourced call centers. The history of the call center—including the steps by which technologies were applied to decrease costs, create efficiencies, and maintain (it was hoped) positive points of contact with consumers—contains the seeds of a new design discipline that eventually emerged.

In late capitalism—that is, postindustrial societies with more jobs delivering services than manufacturing things—it became clear there needed to be systematically organized methods for ensuring quality, consistency, and efficiency in the delivery of services. The call to arms came in 1984 from Lynn Shostack, a senior vice president for several financial institutions, in an article she authored for the *Harvard Business Review*. The editorial introduction to Shostack's essay states, "Considering the extent to which the service industry contributes to our GNP, very little

has been said or written about the application of rational management techniques to the development of new services."[21]

The essay starts by recounting myriad familiar failures of services, which were attributed to a lack of systematic methods for producing consistent, predictable, and customer-pleasing service delivery. "Services are unusual in that they have impact, but no form. Like light, they can't be physically stored or possessed and their consumption is often simultaneous with their production." The interpersonal exchange of the "service" becomes the commodity itself. The exchange may be fleeting, but its primary functions to the business are to ensure "satisfaction," to encourage repeat or future sales, and to ensure that the "brand" maintains a positive impression in the market. The on-demand nature of service requires a massive infrastructure waiting in the wings, prepared to deliver on the arrival of consumer needs.

Traditional, industrial manufacturing, of course, could never do this. Large production runs making use of complicated machinery and a large labor force required long-term planning to put commodities into production for future sales and, potentially, long-term consumption. Services require long-term planning and a robust infrastructure, including labor, but the value is immediately captured by the consumer.

In her call for rational management techniques to be applied to services, Shostack says: "Better service design provides the key to market success, and more important, to growth."[22] And with that, a new design discipline was named: *servi*ce design.

Service design is explicitly concerned not only with what the customer directly experiences but also with what is needed behind the scenes to make the right experience happen—employees and their roles, informational or infrastructural resources, and so on. The formalization of the field of service design and its methods created a new territory of design specialization. This territory would be populated not just by operations management and organizational experts but also by creative and innovative strategic thinkers, including designers. Before the *service turn* of the 1980s, most companies were product and logistics focused. The emphasis was on developing and delivering competitive products efficiently—faster, cheaper, more. As the power of "brand" took on greater value in the postindustrial era, though, service design focused on the entire experience of users interacting with companies.

The design scholars Lucy Kimbell and Jeanette Blomberg identify three different perspectives on the object of service design, each requiring different kinds of expertise and drawing attention to different political dynamics. The first perspective focuses on the *service encounter* itself. This lens examines "people's interactions with digital and material touchpoints that shape the experiences of service users," foregrounding "interactions with other people or in relation to artefacts, places and technological systems." This perspective is most "congruent with the usual focus of design practice where designers remain close to people and their interactions with artefacts and other people which are seen to shape experiences of a service. It requires expertise in sketching, modelling and prototyping touchpoints familiar from user interface design, informed by research into people's ways of doing things, capabilities and needs."[23]

Second, a *value co-creating system* perspective would examine "exchange relations between actors in a service system" constituting the "parties within a service being able to negotiate and account for the achievement of outcomes." Developing services from this perspective requires stronger business and analytical expertise, "such as organizational strategy, business modelling, operational process design and data analysis. It establishes what kinds of resources are available to combine into offerings (possibly as platform businesses), and how to gather and assess data to see if outcomes have been achieved."[24]

Third, a *sociomaterial configuration* perspective draws on anthropological approaches to consider "the constituents of a service and . . . how they are assembled dynamically through practice, emphasizing the sociality and messiness of the worlds in which services exist." This lens requires "expertise in sociological and anthropological analysis that inquires into, rather than taking as given, the conditions which shape particular ways of doing things, knowing and being." Through this lens, political questions emerge about "what makes something 'desirable' and for who, and the consequences of particular designs and ways of doing design. For example, it highlights the way that platform businesses result in people —who might otherwise do similar tasks with access to the legal and economic benefits that come with employment—being pushed into being self-sufficient entrepreneurs within a neo-liberal economic model."[25]

Given this range of concerns, Kimbell and Blomberg assert that designers "must work out to whom or what a design is accountable and the nature of these accountabilities" while also "locating themselves in relation to the design activities." "Will they stay close to the service encounter and the experiences the mixture of people and objects affords? Or will they also zoom out and engage stakeholders in thinking about the conditions that inform and shape particular kinds of encounters?" Designers are most familiar with methods required to develop services from the *service encounter* perspective, but this approach "engages a reduced set of accountabilities, emphasizing users and others directly benefitting from, enabling or providing a service experience." This approach is narrow in its political focus, allowing designers to "bracket out as externalities things that are not directly concerned with the delivery of people's experiences," such as the social impact of "invisible workers" and their "emotional labour" or "whether such exchanges are equitable or reasonable."[26] Various class-action lawsuits filed against Uber by its "gig" drivers testify how such narrow concerns on the service delivery deflects attention away from the more complex social relations constructed by service design.

With a user-centered mindset, service design typically engages *potential* users or customers in the design process to increase the usability and/or desirability of services. This language of "collaboration" creates a patina of shared or communal interests in the designing of services, regardless of how narrowly that collaboration is defined.

In 2004, the Indian American business strategy scholars C. K. Prahalad and Venkat Ramaswamy defined the concept of *co-creation* as "the joint creation of value" by a company and its customers by allowing customers "to co-construct the service experience to suit their context."[27] This proposition was rooted in the need for competitive advantage. "High-quality interactions that enable an individual customer to co-create unique experiences with the company are the key to unlocking new sources of competitive advantage. Value will have to be jointly created by both the firm and the consumer."[28] Over the following decade, the idea spread and became a common practice.

The term *codesign* is generally used to describe design processes that openly and actively engage various stakeholders (on the business side and customer side) to ensure their needs are met. "A key tenet of co-design is

that users, as ‘experts’ of their own experience, become central to the design process. The role of facilitation (usually undertaken or coordinated by designers) is an essential component of a successful co-design project. Facilitators provide ways for people to engage with each other as well as providing ways to communicate, be creative, share insights and test out new ideas.”[29] Although often cited as connected to or having emerged from participatory design, the term *codesign* is applied to creative practices in the commercial world as well as in the public sector—moving beyond the explicitly democratic agenda of its origins in the Scandinavian method.

Service design and methods of codesign have mixed with other emergent practices and research methods to generate the broader model of user-centered design, as it is recognized across various fields and domains today. The nearly ubiquitous job title “UX designer” may indicate a broad range of skill sets and working approaches, yet it signals a context awareness of designing things, interfaces, and experiences that include research, prototyping, and testing with potential users. The *experience* of designed things or services is foregrounded. In the context of a competitive global market in which a *brand* typically carries more value than any objective functions of products and services, designed *experiences* operate on a “form follows brand” logic.

Capitalism & Design in the Twenty-First Century

The late twentieth century saw a resurgence of free-market ideals. The term *neoliberalism** is commonly used to describe the policies shaping an economic, political, and social system that dislodged the dominance of Keynesianism in many advanced capitalist economies starting in the 1970s. This model of economic thinking foregrounds private ownership and individually focused competition as the driving forces for social and political organization. Proponents stand against government regulation in almost any form, other than to protect property rights.

Neoliberalism

As the name suggests, the fundamental idea of neoliberalism is a return to the liberal ideology of classical and neoclassical economics. However, neoliberalism modifies these previous free-market forms of capitalism most acutely in its expansion into the public sector. With the prioritization of the free market, nearly all factors of economic control—including those of the public sector—are brought into market competition. The belief is that the market can do things better and more efficiently than any bureaucratically controlled or noncompetitive enterprise. “Since the 1970s, a ‘neoliberal consensus’ has emerged (or been imposed) in which the state withdraws from obligations for public provision in fields as diverse as housing, health care, education, transportation, and public utilities (water, energy, even infrastructures). It does so in the interests of opening up these arenas to private capital accumulation and exchange value considerations.”[30] Proponents claim that by placing once-public services into private ownership, they become more efficient, save tax dollars, create better outcomes, and offer more freedom of choice.

These ideas also have roots in the Austrian School. Early twentieth-century Austrian concepts such as *imperfect competition* and *creative destruction* placed emphasis on the positive benefits of individual, private competition. However, Keynesian policies dominated the booming postwar economies of Europe and the United States. In the context of high employment and economic prosperity, generally perceived as creditable to Keynesianism, it seemed unlikely that there would be much public support for a change of course. In 1947, the Mont Pelerin Society was formed. It was the brainchild of the Austrian economic theorist Friedrich von Hayek, supported by European and American economists and businessmen who sought a new “liberal” order. The Mont Pelerin Society slowly and quietly built a diffuse global network of proponents across academic, industry, and hybrid sectors. During the 1950s, a new kind of policy-producing organization emerged in the form of the “think tank.” The Mont Pelerin Society spread its ideology through a network of think tanks across Europe and in the United States. The American economist Milton Friedman was a member of this society and a founding member of the “Chicago School” of economics, so-named for liberal-minded economic intellectuals teaching and studying at the University of Chicago in the mid-twentieth century. In 1970, Friedman wrote an op-ed for the *New York Times Magazine* titled “The Social Responsibility of Business Is to Increase Its Profits,” in which

he bashed the notion of a corporation's "social responsibility" to things such as "providing employment, eliminating discrimination, avoiding pollution and whatever else may be the catchwords of the contemporary crop of reformers." He asserted that the only responsibility of business was to maximize profits.[31] Naturally, the business world was inclined to agree, and the essay became "arguably the most consequential economic idea of the latter half of the 20th century."[32]

Throughout the period of postwar growth, neoliberal ideas percolated, waiting for a moment to take hold. That moment came in the form of *stagflation* in the 1970s. Politicians and policy makers were desperate for a solution to the economic crisis of the 1970s, and neoliberalism was ready to seize this opportunity. Neoliberal policy came to prominence in the United States and the United Kingdom during the 1980s, particularly during the conservative governments of US president Ronald Reagan and UK prime minister Margaret Thatcher. Claiming a free market is the only economic system that works, Thatcher adopted the slogan "There is no alternative," often abbreviated as TINA.

Whereas Keynesianism focused on *demand* as the key economic driver, neoliberalism operates on a *supply-side* model of economic theory. This is the belief that producers and investors (those with capital) determine the scale and pace of economic growth, including job growth. A neoliberal mindset has been at the center of major economic transformations since the 1970s, including deregulation of industries to allow for competition without restrictions (such as peeling back labor protections and environmental restrictions) and the globalization of labor and supply chains (pushing down wages and moving labor to less-regulated places with much lower labor and production costs). These factors also opened new territories for designers to expand their roles in relation to global business and society.

Design Science & Design Thinking

In the postwar era of globalizing communication, labor, and supply chains, designers became interested in global-scale problems. Elite expert teams of designers, engineers, and scientists came together to improve housing conditions, water supply, sewage, and sanitation, and to tackle other material needs on a global scale. The concept of *design science*, first

introduced by the American architect, theorist, and inventor R. Buckminster Fuller, positioned design as a systematic and research-driven form of designing with measurable facts related to ergonomics, material properties, visual perception, and cost analysis.[33] This work was often aimed at "developing" countries, shorthand for nations that were not industrialized and therefore not fully participating within the global market economy.

By engaging large, interdisciplinary projects, designers became seen as innovative general problem solvers. The design theorists Horst Rittel, from Germany, and Melvin Webber, from the United States, attempted to theorize and codify how designers worked by foregrounding the human dimensions of experience and perception. In 1973, they introduced the term *wicked problems* to refer to complex, open-ended challenges in which designers could intervene only with partial and imperfect knowledge, creating "better" or "worse" interventions when framed from different scales and perspectives.[34] The growing catalog of research-oriented design methods provided many new domains for designers to engage problems through *design solutions*. Design came to be aligned with "creative problem solving," and its methods were seen as generalizable skill sets.

The historical strands of participatory design, service design, and design science trace distinct approaches that began to coalesce at a historical moment of expanding global markets for services, exploding possibilities in digital products, sharply increased competition, globalized production and labor, and deregulated policies to stimulate business competition. Under these conditions, ultrafast "innovation" was seen as the best way to create competitive advantage with first-to-market products, competitively efficient production and management, and a global reach for communications. The growing need for rapid, creative, testable development of new ideas created the conditions for design to export its skillsets for generalized "innovation." That export was called "design thinking."

Design thinking has been applied as a blanket term for "multi-disciplinary, human-centered projects that involve research and rapid ideation (idea generation)."[35] The core attribute of design thinking is, in fact, that there is a *process* for engaging a "problem" by asking, studying, responding with testable hypotheses, testing those hypotheses repeatedly (rapid prototyping), improving upon them, and ultimately implementing

solutions. Specific methods may vary, but the procedural formula claims to be viable for wildly divergent contexts and scales.

In the early 2000s, design thinking emerged from academic research labs and journals to hit the glossy pages of business magazines and conversations in executive board rooms. Writing for *Business Week*, the journalist Bruce Nussbaum was an early proponent. He suggested that design thinking aided and expanded designers' impact by making design "system-conscious."

> By formalizing the tacit values and behaviors of design, Design Thinking was able to move designers and the power of design from a focus on artifact and aesthetics within a narrow consumerist marketplace to the much wider social space of systems and society. We face huge forces of disruption, the rise and fall of generations, the spread of social media technologies, the urbanization of the planet, the rise and fall of nations, global warming, and overpopulation. Together these forces are eroding our economic, social, and political systems in a once-in-a-century kind of way. Design Thinking made design system-conscious at a key moment in time.[36]

He believed the allure for business was that design thinking offers a process for creativity. However, although the process was useful for studying the context of a "problem" and possible "solutions," the real messiness and many failures of true creativity proved to be less palatable to business interests. "From the beginning, the process of Design Thinking was a scaffolding for the real deliverable: creativity. But in order to appeal to the business culture of process, it was denuded of the mess, the conflict, failure, emotions, and looping circularity that is part and parcel of the creative process. In a few companies, CEOs and managers accepted that mess along with the process and real innovation took place. In most others, it did not. As practitioners of design thinking in consultancies now acknowledge, the success rate for the process was low, very low."[37] Nussbaum was one of the earliest and loudest proponents of design thinking within the business world, but his love affair with it was ultimately short-lived. In this same article surveying the history of design thinking published in 2011, Nussbaum also wrote: "Design Thinking has given the design

profession and society at large all the benefits it has to offer and is beginning to ossify and actually do harm."[38]

Regardless, the meteoric rise of the word *design* was evident. Although its actual forms and contributions remained somewhat ambiguous, design was beginning to be understood in business, government, and mainstream contexts as something much more than styling or making things fashionable. Design began to be understood as a rigorous, iterative, contextual, creative process of framing *problems* and intervening to create solutions.

These generalizations of design's capabilities—particularly related to innovation for business interests—paralleled an economic transformation from the production of things to the production of digital systems, networks, and entirely new forms of value.

Digital Economy: Data & Platforms

Amid the swirl of postindustrializing economic transformations and the neoliberal commercialization of public resources, a 1990s "tech boom" built much of the infrastructure that supports the digitally connected world we occupy in the twenty-first century.

> The boom in the 1990s amounted effectively to the fateful commercialization of what had been, until that point, a largely non-commercial internet. . . . As US manufacturing began to stall . . . the telecommunications sector became the favoured outlet of financial capital in the late 1990s. . . . All told, more than 50,000 companies were formed to commercialize the internet and more than $256 billion was provided to them. Investors chased hopes for future profitability and companies adopted a "growth before profits" model. While many of these businesses lacked any profits, the hope was that through rapid growth they would be able to grab market share and eventually dominate what was assumed to be a major new industry. In what would come to characterize the internet-based sector to this day, it appeared a requirement that companies aim for monopolistic dominance. In the cut-throat early stages investors enthusiastically joined, in hopes of picking the eventual winner.[39]

During this time, the logics of an evolving neoliberal capitalism began to take hold through new forms of “ownership” and “labor” made possible by the internet. Companies based on existing business models rose and fell, but a new kind of material promising new forms of value began to emerge: data.

Data could be extracted from consumer behaviors, digital-content consumption, search-engine queries, and activity on social media and were spilling out of production processes across global supply chains. All these data created opportunities for marketable products and services, targeted forms of advertising, and improved efficiencies. Data are like raw material in that they must be extracted, refined, and organized into standard forms to capture value. While data are the raw material (e.g., minerals), the behavioral, economic, and social activities of humans are the source (e.g., the mine).

Data have always been valuable to businesses, from production planning and distribution logistics to consumer preferences and product design, but the dominance of digital communications and connectivity made data easy to capture and inexpensive to store. Industries took notice and began developing new methods to extract data to “optimize production processes, give insight into consumer preferences, control workers, provide the foundation for new products and services (e.g. Google Maps, self-driving cars, Siri).” Yet existing business models were not designed to fully maximize this potential. The “new business model that eventually emerged is a powerful new type of firm: the platform. Often arising out of internal needs to handle data, platforms became an efficient way to monopolise, extract, analyse, and use the increasingly large amounts of data that were being recorded. Now this model has come to expand across the economy, as numerous companies incorporate platforms: powerful technology companies (Google, Facebook, and Amazon), dynamic start-ups (Uber, Airbnb), industrial leaders (GE, Siemens), and agricultural powerhouses (John Deere, Monsanto), to name just a few.”[40]

A platform is a digital infrastructure that provides the ground, the rules, and the interfaces for two or more parties to interact. In that sense, the parties produce their own political economies in which the exchanges that make up their economic and social relations are mediated (and potentially owned) by the owners of the platform itself. One of the primary features of

platforms is their ability to produce *network effects*: the more people use them, the more value they can provide. For example, the fact that more than 2 billion people use Facebook is one of the primary reasons it is so valuable as a platform for social media. Everyone you know is already there, and that density of users is what makes it valuable.

Platforms have come to define entirely new business models, many of which evolved in surprising directions from their initial functions. Google was supported by large amounts of venture capital in the tech boom of the 1990s, which produced an incredibly efficient search algorithm. Data were collected from users' search queries to refine the algorithms and create better search results, and thus a better service. As the Canadian philosopher and digital economy scholar Nick Srnicek points out, "This was an example of the classic use of data with capitalism . . . to improve one's services for customers and users." However, this improvement in service provided no leftover value, no *surplus* to generate more revenue, thus no return to investors. All that changed in 2000 when Google released AdWords. Under intense pressure from investors, the company needed a new model to generate higher returns. This new model was based on translating the "data exhaust" of users' interactions with the platform into saleable ad space that is highly targeted and purchased through an auction system. Data already being extracted were transformed from "a way to improve services to . . . a way to collect advertising revenues." The new model paid off. Immensely. Today, "Google and Facebook remain almost entirely dependent on [ad revenue]: in the first quarter of 2016, 89.0 per cent of Google's and 96.6 per cent of Facebook's revenues came from advertisers."[41] The advertising platform is just one of five platform types identified by Srnicek:

> The first type is that of *advertising platforms* (e.g. Google, Facebook), which extract information on users, undertake a labour of analysis, and then use the products of that process to sell ad space. The second type is that of *cloud platforms* (e.g. AWS [Amazon Web Services], Salesforce), which own the hardware and are renting them out as needed. The third type is that of *industrial platforms* (e.g. GE, Siemens), which build the hardware and software necessary to transform traditional manufacturing into internet-connected processes that lower the costs of production and transform goods into services.

> The fourth type is that of *product platforms* (e.g. Rolls Royce, Spotify), which generate revenue by using other platforms to transform a traditional good into a service and by collecting rent or subscription fees on them. Finally, the fifth type is that of *lean platforms* (e.g. Uber, Airbnb), which attempt to reduce their ownership of assets to a minimum and to profit by reducing costs as much as possible.
>
> These analytical divisions can, and often do, run together within any one firm. Amazon, for example is often seen as an e-commerce company, yet it rapidly expanded into a logistics company. Today it is spreading into the on-demand market and, perhaps most valuably, Amazon Web Services as a cloud-based service. Amazon therefore spans all of the above categories.[42]

What may be surprising is just how valuable data can be to platform and multiplatform business models. Amazon is a prime example. As an early e-commerce business, Amazon marshalled control of vast supply-chain logistics and used competitive measures to force costs down. In the process of building up its massive network, Amazon developed its own internal platform to handle incredibly complex logistics. This was the first purpose of Amazon Web Services (AWS). All the hardware and software Amazon dedicated to this project was quickly recognized to be of value to almost any business operating in the digital world (meaning just about every business on earth). AWS "rents out cloud computing services, which include on-demand services for servers, storage and computing power, software development tools and operating systems, and ready-made applications." Businesses subscribing to these services save an immense amount of time and money by not having to build their own hardware and software infrastructures. With the introduction of Prime memberships that include free two-day and sometimes one-day shipping, Amazon has created a nearly unbeatable economy of scale. In 2018, Amazon spent $27.7 billion on shipping costs—that is more than the gross domestic product of 100 countries in the world.[43] Amazon actually loses money on international retail delivery services, but the data collected are lucrative enough for other lines of business that Amazon considers the loss worth it. AWS is much more profitable than retail sales and delivery. It is estimated to be worth

more than $400 billion as a stand-alone company and is the most profitable and fastest-growing division of Amazon.[44]

Data and platforms compose the foundation of what is now called the "digital economy": the increased reliance by all businesses on information technology, data, and digital communications for their basic functioning. The digital economy applies to all sectors of the economy, not just those focused on offering digital products or services. Data, information technology, and connectivity are required today for everything from hospitals to agriculture to small-scale independent retailers of craft goods. According to Srnicek: "The digital economy appears to be a leading light in an otherwise rather stagnant economic context." As it becomes increasingly important and interconnected to all economic activity, it also becomes "an increasingly pervasive infrastructure for the contemporary economy . . . [as well as] a hegemonic model: cities are to become smart, businesses must be disruptive, workers are to become flexible, and governments must be lean and intelligent."[45]

Throughout their interconnected histories, both design and capitalism have continually evolved to find new ways of producing economic value. Since the early twentieth century, it has been generally accepted that design contributes to the creation of value through the roles it plays in the conception and production of goods and services as well as through its ability to create new markets or expand existing ones—even if these claims have been more "common sense" than directly evidenced. In the twenty-first century, both the political economy and design continue to evolve, adapting themselves to the new value forms exploding out of a data-centric and networked platform economy.

Design's Creation of Value

In the quest for an economic theory that acknowledges design's creation of value, John Heskett studied economic movements since Adam Smith.[46] Based on his analysis, very little economic thinking can identify or account for the varieties or quantities of value design can contribute, even when design's impact on value creation is generally recognized. Classical, Marxist, and neoclassical theories are too static and too rational to account

for the dynamism of change or irrationality of desire that design might produce. Austrian theory is more dynamic with its "subjective" view of value and "imperfect competition" in which design can contribute value through preference for products and/or through innovation in performance or production. Yet given the era of their development, these ideas were still oriented largely toward the production of goods in an industrial and material sense. Economic theory of the late twentieth and early twenty-first centuries has granted a stronger account of technological innovation and the broad distribution of technical knowledge within societies. Simultaneously, this account begins to give a clearer picture of the value created by an expanding discipline of design.

Endogenous Growth Theory

Emerging primarily from the work of the American economist Paul Romer, *endogenous growth theory** (also sometimes called *new growth theory*) refers to a body of economic research and theories that suggest economic growth is *primarily* the result of forces internal (endogenous) to the economic system than of forces external (exogenous) to the system. In particular, it is knowledge of and about technology that allows for the expansive application of technologies in novel and innovative ways. "Romer's work focuses on the missing element of technology and incorporates it directly into models of economic growth by explaining how knowledge is created and spread through the economy."[47] Romer describes this way of thinking as an "economics of ideas" in contrast to an "economics of objects."[48]

The theory holds that knowledge of and experience with technology as well as the distribution of that knowledge to support innovation are significant contributors to economic growth in an economy based on knowledge. This means that investment in human capital is an engine of economic prosperity.

Accelerators of production are typically those innovations that require a great deal of up-front investment in research and experimentation, yet when they are successful, they can create new models and technologies of production that yield processes that can push productivity costs down. Although there are variations in thinking about how this acceleration is best

achieved and to what ends, the basic premise is that "an increase in the share of people working in the knowledge sector will increase economic growth."[49] The more educated and engaged in knowledge work a whole society becomes, the more innovation will flow.

Ideas are not scarce. This assertion runs contrary to the "scarcity" paradigm of most market-oriented economics but aligns with the abundance paradigm of digitally transmitted information. The endogenous growth theory is novel for two main reasons. First, it holds that rising populations can be a source of economic prosperity so long as investment is made in education and direct experience with technical processes and innovation. Second, it argues that ideas are "nonrival," meaning that my use of an idea for my purposes doesn't prevent your use of it for yours. Some claim that intellectual-property law is required to generate the incentive for technological innovation, but it may also be necessary to "restrict competition in the knowledge sector in order to stimulate growth."[50] The theory promotes the value of investing more in technological innovation and distribution of technical knowledge to a greater number of people. This model for noncompetitive innovation through human investment aligns with other contemporary movements such as open source and alternative models of free education. Unfortunately, broad and indiscriminate investment in human capital runs counter to prevailing neoliberal policies, which strive to defund public education programs to privatize them with the claim that this will make them more "competitive."

Heskett's Value Creation Theory

Based on his rigorous examination of historical economic theories and the dynamic ways in which endogenous growth theory accounts for the value of innovation and knowledge, Heskett has proposed his own value theory to account for design's creation of value. According to Heskett's "value creation theory," design is integral to creating opportunities for innovative strategies that can generate three kinds of innovation: *process innovations*, which can lead to improved quality of products and services; *transaction innovations*, which can lead to transaction efficiencies; and *product innovation*, which can lead to new product or service versions (updates) or to entirely new product or service categories. In Heskett's model, all these

innovations lead to *imperfect competition*, or competitive advantage in the marketplace, which results in the destruction of existing products while dynamically creating both new demand and new markets.[51]

In the twenty-first century, the production of *brands* and the production of *experiences* have become design's primary forms of value creation. We experience brands when we follow them on social media, walk into a store, handle packaging, shop online, chat with customer service, respond to a survey, call for technical support, and, to some limited degree, "use" products. We experience brands through services as much as we do through products; in fact, the categories *product* and *service* are typically not entirely clear (e.g., Facebook is a product with multiple products and services inside of it).

Designing and design thinking are change-making activities and therefore drivers of "innovation." Designers have recently moved more broadly into domains such as the nonprofit sector, finance and financial technologies (fintech), health care, and insurance, bringing a robust set of methods for framing, researching, and "solving" complex "problems." Facebook's famous motto "move fast and break things" has become a mantra for those in a hurry to remake the world through new products and services aimed at "improved" experiences. Designers are an integral part of teams doing that work.

The history of modern design is fully intertwined with the history of capitalism and its attendant economic theories. In the 1970s, the pivot toward postindustrial, digital, knowledge-based capitalist economies challenged design to adapt its methods, and designers responded. Particularly in the early twenty-first century, these moves have positioned designers as authorities, strategists, and key decision makers in global-scale business operations. Meanwhile, design has gained traction as a legitimate field of academic study, even as its intentions, methods, and bodies of knowledge have become far more diverse—sometimes representing competing interests.

Conclusion

In specifying a product or service offering, designers operationalize flows of material resources (extraction, refinement, commodification), labor resources (production methods, tools, labor activities), and desires (markets for new goods and services or in new locations). Yet design in action is a method. Under the sway of an unrecognized ideology, it can be a blunt instrument. Design's instrumental role in the production of material culture is intimately tied to a historical system that has developed a networked collection of institutions, laws, rules, customs, and logics that foreground private property, private enterprise, and competition in both production and consumption. That many design practices have operated under the direction of the ideology of capitalism is a historical fact. However, it's also just that: historical.

4
What's Wrong with Capitalism?

Two Kinds of Critique

The word *economy* originates from the ancient Greek term *oikonomia*, which translates to "household management" but also implies "thrift" or a judicious use of resources. To talk of *economy*, then, means to discuss the management of resources with an eye toward both the care of the household and a sense of resourcefulness—making the best of the situation for the best of all involved. To speak of *political economy* adds the consideration of hierarchies of power, generally asking whether the management of resources is just. To critique capitalism as a political economy is simply to ask whether it establishes an efficient, just, and desirable management of the household, which is simultaneously our physical home, the local community in which we live, and the planet.

Over its roughly 300-year history, capitalism has been credited with improving the material conditions of life for countless people around the globe. In fact, according to Erik Olin Wright, "capitalism has generated sufficient material wealth in the world today so that even with no further economic growth no person would have to be poor in the developed capitalist countries, and basic needs could be met for everyone even in poor third-world countries."[1] That sounds like a success story. However, this wealth is very unevenly distributed, with enormous poverty adjacent to extreme wealth. When institutions such as property laws, economic policies, and political and cultural patterns disadvantage targeted groups of people and prevent the reasonable pursuit of self-interest, then the system is not delivering the results it should. Having reached a point of diminishing returns—when the system that fueled intense productivity is no longer doing so—the system has no internal mechanism to slow down or work toward leveling out an equal distribution of its gains. Constant growth is required to avoid internal crises of the system. There is no way for

capitalism or any enterprise working within it to reach a point of “enough.” For individuals and enterprises in capitalist societies, constant growth means that constant competition and constant reinvention are necessary to stay solvent, which equates to staying alive.

Specific claims of capitalism’s success include a dramatic increase in material productivity, merit-based distribution of resources, and technological innovation that continually increases productivity while improving living conditions, which translates to increased health and life expectancies. Yet it seems that all these indicators have reached their tipping points in advanced capitalist societies, in particular the United States, and they are now in decline. What’s more: the issue is not just that these measures of progress have reached their limits of rising returns but that the capitalist system itself is actually pushing them into regression.

Today, capitalism is facing two simultaneous crises. The first is a crisis of *productivity*, signaled by consistently low economic growth in “mature” capitalist countries and a flattening of real wages since the 1970s. These countries, including their citizens and households, have taken on enormous levels of debt to keep things running despite clear signs that the system is not self-sufficient. The great boom of the industrial era ended long ago. A *digital economy* has emerged, generating new forms of productivity, services, and information that could create more value for more people under different conditions. However, these new forms are often artificially constrained by the *scarcity* mindset and property logics of an outdated political economy. Along with this first crisis has come global warming, which has arrived with a vengeance, asserting real limits on the “externalities” that capitalist production has never fully accounted for in its claims of rational efficiency. These two trends make up part of the *economic* critique of capitalism.

The second is a crisis of *legitimacy*, signaled by a dramatic increase in economic, social, and political inequality since the 1970s. Deregulating, defunding public programs, and cutting corporate taxes have propelled rising economic disparity and a massive upward redistribution of wealth. In turn, as the economic power of global corporations allows them to wring governments of tax breaks and subsidies, political leaders in democratic governments have become increasingly beholden to corporate interests, and political competition has come to require enormous fundraising efforts. In

the global financial crisis of 2008 “the promise of self-regulating markets attaining equilibrium on their own was discredited . . . without a plausible new formula for political-economic governance coming into view.”[2] Economic power has subsumed political power, weakening democratic control over the most significant decisions that shape society and wresting opportunities for autonomy and livelihood away from individuals.

Furthermore, capitalism has historically exploited specific populations in ways that stretch beyond the injustices of uneven economic distribution. The social oppression of racial minorities and women, for example, has been both integral to and exacerbated by capitalism. Most of the Western world’s economy was built up by the labor of racialized slavery, and the labeling of labor traditionally performed by women as *unproductive* is still sometimes used to justify lower wages for women in most “advanced” capitalist economies. These trends make up part of the *moral* critique of capitalism.

Economic Critique

Contrary to the conception of capitalism as a rational, efficient model for resource allocation, critics have pointed out its inherent inefficiencies. A system driven by perpetual accumulation leads to irrational outcomes as jobs, resources, and entire industries chase consumer desires rather than human needs understood as anything other than commodities. The very nature of competition leads to endless versions of functionally similar products in the spectacle of “choice,” which in turn lead to tremendous wastes of raw materials, labor time, and cognitive capacity that could be directed elsewhere. The design theorist and philosopher Tony Fry points out that the term *consumption* inaccurately describes what takes place in capitalist economies. Instead, he says, there is mostly *waste*.[3] Meanwhile, the market has long demonstrated a boom-and-bust “business cycle” that periodically wipes out businesses, eliminating jobs, incomes, pensions, and livelihoods along with them. *Externalities* are the territories—physical environments, common resources, cultural assets, populations of people—where gains or losses are not calculated as part of the private cost of production. Private interests push *actual costs* onto others by marking

spaces off as “external.” Pollution and the degradation of the natural environment are prime examples of externalities. Capitalist producers might argue that no product can be produced at a reasonable market cost if all such externalities were the burden of private production. This is also a common argument against enacting government regulations to protect workers, consumers, and the environment and imposing corporate taxes to support public goods such as education and health care. Capitalist producers might argue that these market interferences make costs go up and make their businesses less competitive. So much for managing the household.

Collectively, these arguments indicate that although capitalism “may be rational for individuals, on a systemic level it is highly irrational.”[4] Even beyond its inherent inefficiencies, capitalism is in the midst of a crisis of *productivity* that has reached a point where it is now *undoing* its own positive gains.[5] Capitalism appears to be devouring itself.

Decline & Disequilibrium

Perhaps the most direct criticism of capitalism on its own terms is the fact that total economic growth and real wages for workers have been in steady decline since the 1970s. The broad strokes of the situation look something like this: Due to a combination of automation, neoliberal economic policies, and technologies that made foreign production competitive, Western countries saw a decrease in their manufacturing productivity. To make production as lean and low cost as possible, manufacturing was outsourced to Asia and parts of the “Global South.” This trend began in the 1970s, picked up during the 1980s—when companies were frequently merging, consolidating resources, and purging payrolls to seek greater efficiencies—and then accelerated in the 1990s when various trade deals and the exploding manufacturing capacity of China unleashed globalization. For example, in the 1990s “Nike became an emblem of the lean firm: branding and design were managed in the high-income economies, while manufacturing and assembly were outsourced to sweatshops in the low-income economies.”[6]

Workers displaced from manufacturing often took on service jobs, typically with lower pay and less job security. More women joined the workforce out of a combination of social changes spurred by the feminist

movement and their families' economic needs. This influx increased the supply of labor and in turn decreased wages. The net effect was a decrease in both economic growth and real wages, sometimes running negative to inflation. As competition increased in China, relative wage increases pushed manufacturing into lower-wage regions of the world, such as Southeast Asia. This was followed by growth in service jobs within a global "developing middle class." Massive expansion of the global workforce was led by the growth of "the group earning between $4 and $13 a day . . . from 600 million [in 1991] to 1.4 billion [in 2011]. (Though the demographers call them the 'developing middle class,' the $13-a-day mark corresponds roughly to the poverty line in the USA.)" Three-quarters of the jobs in that new influx of labor were in service industries. "The growth of service sector jobs in the developing world reflects both the natural evolution of the job mix under modern capitalism and a second round of offshoring, focused on call centers, IT departments and back-office functions."[7] Growing numbers of outsourced service jobs and ongoing automation continue to add pressure to wages. The descent continues.

When manufacturing slowed, investment options decreased. Instead of investing in new factories and plants, capital looked to the telecommunications industry. The tech boom of the 1990s burst in the early 2000s, but it ushered in a new era in finance supported by digital tools and communications that created faster and often riskier trade. Speculation in the real estate market led to the housing bubble, which burst in 2008 and triggered a global financial meltdown that "wiped 13 per cent off global production and 20 per cent off global trade. It took global growth negative —on a scale where anything below +3 per cent is counted as a recession."[8] The policy response was a bailout for banks and major industries but austerity measures for the rest. The US Federal Reserve pumped $700 billion into the banking industry, which had largely caused the collapse through reckless investments that took advantage of financial deregulations. Through bailouts and tax cuts, "governments undertook the burden of increasing their deficits in order to ward off the worst of the crisis. As a result, the high levels of private debt before the crisis were transformed into high levels of public debt after the crisis."[9] Capitalist economies are increasingly propped up with financial instruments and monetary policies that are racking up public debt at levels nearly equal or equal to their gross domestic product. With neoliberal policies pushing for

decreased corporate taxes and multinational corporations hoarding cash or hiding it in tax havens, the challenge of recuperating this debt begins to look insurmountable. The great engine of productivity has stalled, and it's taking ever more complicated shell games to keep it afloat.

Meanwhile, not only have wages been stagnant for decades, but work itself has also become far less stable. Wolfgang Streeck observes, "Labour markets in leading capitalist countries are changing to precarious employment, zero hour jobs, freelancing and standby work, not just in small local but also and often in large global firms. An extreme case in point is Uber, a giant of the so-called 'sharing economy,' which with the help of new communication technologies functions almost entirely without a workforce of its own."[10] These trends are a problem for a system whose core function is *supposed to be* enhancing productivity in ways that provide broad opportunities for upward economic mobility. Just the opposite is the case for young workers in "advanced" capitalist countries. In the United States, it has long been a cultural axiom that each generation should do better economically than the previous one. This may have been true in the twentieth century, but it is no longer the case. About 90 percent of children born in the United States in 1940 exceeded their parent's incomes. Americans born after 1980 are much less likely to achieve upward social mobility, with only about 50 percent earning more than their parents.[11]

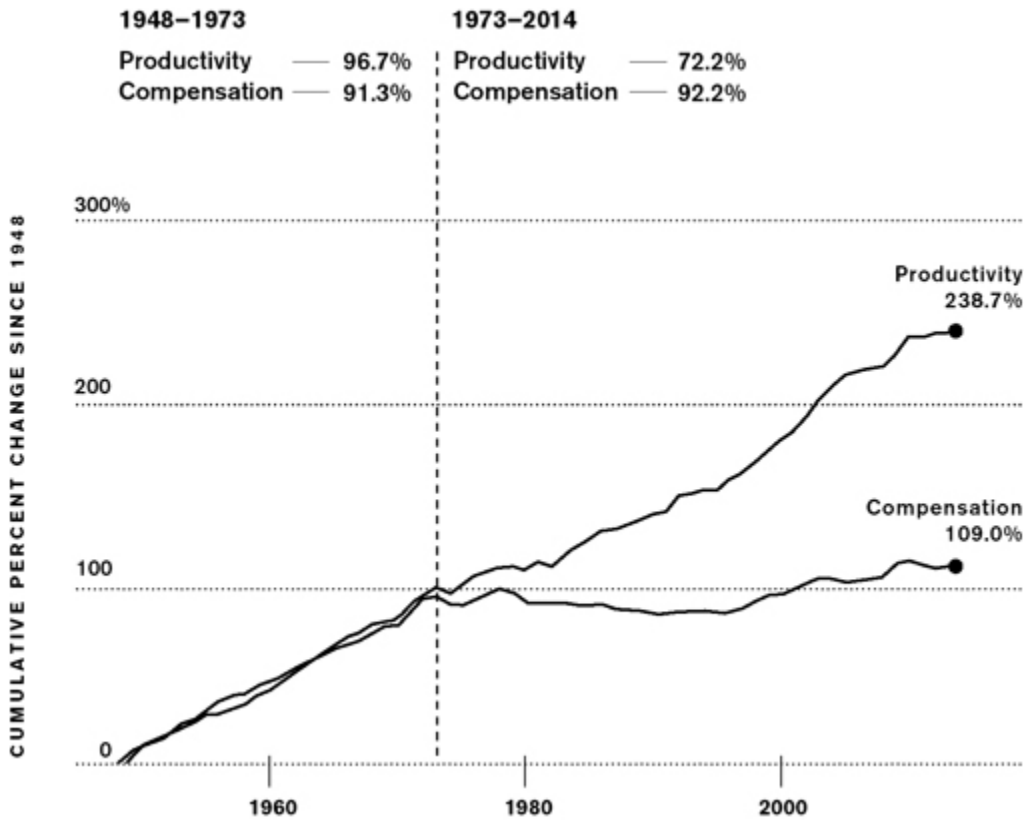


Figure 4.1

Growing disconnect between productivity and a typical US worker's hourly compensation, 1948–2014. Data are for average hourly compensation of production/nonsupervisory workers in the private sector and net productivity of the total economy. (*Source*: Based on an Economic Policy Institute analysis of data from the US Bureau of Economic Analysis and Bureau of Labor Statistics in Josh Bivens and Lawrence Mishel, "Understanding the Historic Divergence between Productivity and a Typical Worker's Pay," Economic Policy Institute, September 2, 2015, https://www.epi.org/publication/understanding-the-historic-divergence-between-productivity-and-a-typical-workers-pay-why-it-matters-and-why-its-real/.)

Economic Inequality

Proponents of free markets point out that capitalism has helped to lift billions of people out of poverty by providing broad distribution of the gains made from increased productivity. Wage-labor opportunities have sprung up where they didn't previously exist; new markets have been created for goods and services not seen before; and the material quality of life has increased for countless people around the globe.[12] Yet this model of allocation has always been asymmetrical, and it has built-in tendencies to increase inequality over time. Wealth accumulation and monopoly

tendencies are central contradictions within capitalism's operation: when capital stops circulating—when companies hoard resources rather than invest them, when workers can't afford to consume things—decline ensues.

> Since 1970, the US economy has grown significantly larger. Real US GDP per capita, a measure of the country's average income, has more than doubled. However, the median household income, a measure of what the typical American family earns, has not kept up. Across the US, median household income has increased by only about 20% over the same period. Such a large disparity between the average and median incomes is possible because the economic gains have not been shared evenly. They have been realized predominantly by high-income households, with only a small share going to the lower majority of the income spectrum. The result is an income distribution that looks like it's been "squashed."[13]

The argument is not that all people should achieve equal results regardless of their efforts. Rather, capitalism is not working to allocate resources with equal opportunities and broad distribution. Is this issue of inequality exaggerated? Not so, according to Jeffrey Winters, professor of political science at Northwestern University. Winters has devised what he calls the Material Power Index (MPI), which estimates "how much material power is concentrated in the hands of a tiny minority." He assigns a base value of 1 to an average American's wealth at the bottom 90 percent. The richest Americans are then measured by their multiples of this base value to provide a "snapshot of MPIs for the United States based on recent income and wealth data." According to Winters's calculations, wealthy people "at the very top of American society have an MPI just over 10,000, which happens to approximate the MPI of Roman senators relative to their society of slaves and farmers."[14] In other words, in the wealthiest democratic and capitalist country of the twenty-first century, there is socioeconomic inequality on par with the inequality between slaves and the slave-owning senators of ancient Rome.

Still not convinced? Consider data compiled from the Federal Reserve, which estimates the distribution of wealth in America: "Between 1989 and 2018, the top 1 percent increased its total net worth by $21 trillion. The

bottom 50 percent actually saw its net worth decrease by $900 billion over the same period."[15]

This remarkable transference of wealth—from the enormous bottom of the economic-power pyramid toward the tiny top—has political consequences that directly threaten democracy. Material power is converted into political power when "it enables the super-rich to buy both political majorities and social legitimacy, the former through campaign contributions of all sorts and the latter by acts of philanthropy partly filling the very gaps in public provision that result from oligarchic fortunes having become safe from taxation in the course of globalization."[16] According to a report published in 2015, "fewer than four hundred families" were "responsible for almost half the money raised in the 2016 presidential campaign, a concentration of political donors that is unprecedented in the modern era."[17]

This massive and growing inequality can also be measured within individual businesses. Describing the advent of the postindustrial society in 1973, the sociologist Daniel Bell seemed shocked by the fact that top executives of American companies were earning up to 25 or even 30 times as much as their lowest-paid employees.[18] By the early 2000s, after decades of skyrocketing incomes, stock options, and other bonus packages, CEOs were typically earning 350 times the wages of the lowest-paid employees in their own companies. In 2005, Wal-Mart's CEO made $17.5 million, or *900 times* "the pay and benefits of the typical Wal-Mart worker."[19]

This vicious cycle of sharply rising financial and political inequality seriously damages the credibility of a political economy that is actively eroding democratic governance.

Technological Innovation

Because capitalism demands constant change from businesses to outperform existing competitors and new upstarts, it has been cast as an engine of technological innovation. In the industrial era, innovation was centered largely on efficiencies of productivity, which gave birth to myriad forms of automation, including Unimate and all its descendants. In the postindustrial era, innovation is often focused on new forms of profit generation from services. The transition between these two eras was marked

by market saturation of goods that were designed to fit consumer “needs.” Refrigerators, washing machines, automobiles—even to some extent transistor radios—all contributed to dramatic changes in the material conditions of everyday life during the postwar boom. By contrast, the technological innovation of the past 40 years—although no doubt filling our lives with impressive and entertaining new gadgets—has not fundamentally altered life in dramatic or for that matter positive ways. The personal computer, the internet, and smartphones have clearly changed aspects of everyday life. However, they have also played a strong role in blurring distinctions between work and personal life while further entangling our daily lives with commercial activity—including the pervasive extraction of personal data that fuels much of today’s digital economy. Connectivity has created new and constantly open portals between private life and work as well as between private life and the market.

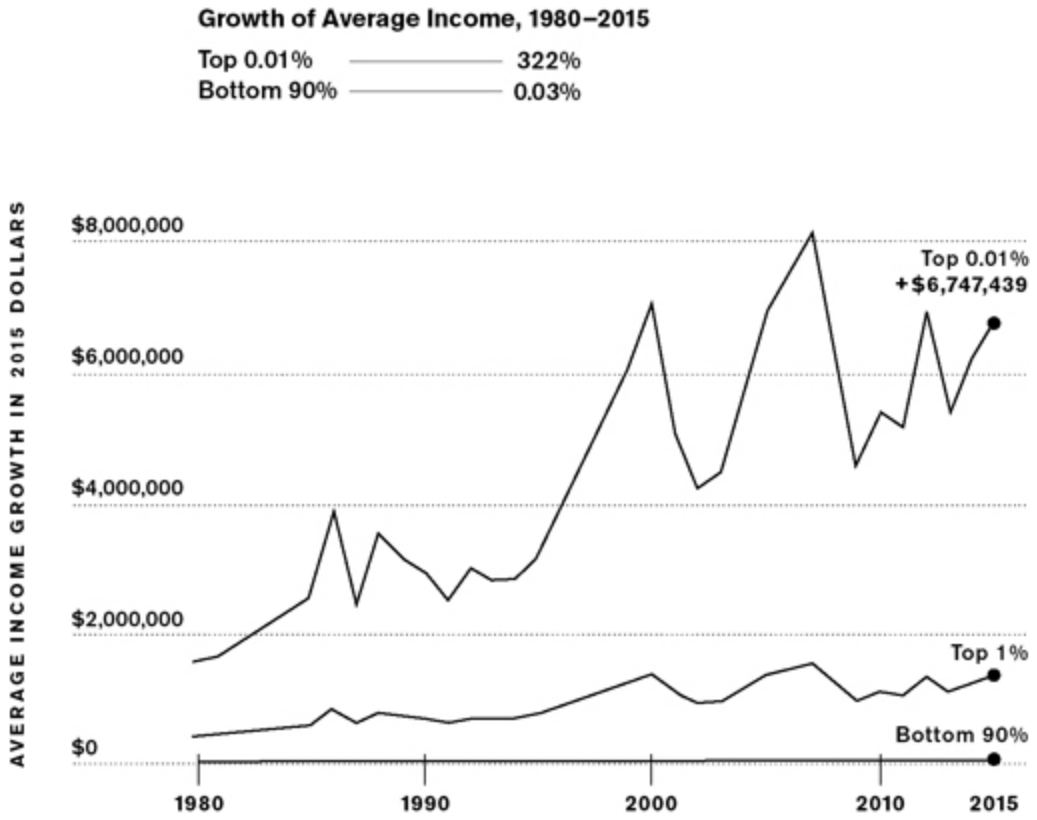


Figure 4.2

Real average US income by percentile, 1980–2015. (*Source*: Based on data from the World Wealth and Income Database as given in Dave Gilson and Edwin Rios, “11 Charts That Show Income Inequality Isn’t Getting Better Anytime Soon,” *Mother Jones*, December 22, 2016, https://www.motherjones.com/politics/2016/12/america-income-inequality-wealth-net-worth-charts/).

The idea that technological innovation has peaked in its capacity for positive social impact is described by the term *innovation pessimism*. You might think it is a term bandied about only by Luddites and tech critics, yet it has been popularized by the German American billionaire venture capitalist Peter Thiel, a founder of PayPal and the first outside investor in Facebook. Thiel believes American innovation is "somewhere between dire straits and dead."[20]

Whether you share this pessimism or not, capitalism's claim to technological "progress" tends to sweep aside the major impact of public spending for research and development that spurs invention and innovation. Many of the most significant, socially beneficial innovations of the past century have been fueled by publicly funded research. This process has fed private enterprise as new technologies based on these innovations became commercialized into new products and services.

> Under a popular narrative, inter-capitalist competition is seen as driving technological changes in the production process, while consumer capitalism demands an increasingly differentiated set of products. But at the same time, capitalism has placed substantial obstacles in the way of technological development. While the carefully curated image of capitalism is one of dynamic risk-taking and technological innovation, this image in fact obscures the real sources of dynamism in the economy. Developments like railways, the internet, computing, supersonic flight, space travel, satellites, pharmaceuticals, voice-recognition software, nanotechnology, touch-screens and clean energy have all been nurtured and guided by states, not corporations. During the golden postwar era of research and development, two-thirds of research and development was publicly funded. Yet recent decades have seen corporate investment in high-risk technologies drastically decline. And with neoliberalism's cutback in state expenditure, it is therefore unsurprising that technological change has diminished since the 1970s. In other words, it has been collective investment, not private investment, that has been the primary driver of technological development. High-risk inventions and new technologies are too risky for private capitalists to invest in; figures such as Steve Jobs and Elon Musk slyly obscure their parasitical reliance on state-led

> developments. Likewise, multi-billion-dollar megascale projects are ultimately driven by non-economic goals that exceed any cost–benefit analysis. Projects of this scale and ambition are in fact hindered by market-based constraints, since a sober analysis of their viability in capitalist terms reveals them to be profoundly underwhelming. In addition; some social benefits (those offered by an Ebola vaccine, for example) are left unexplored because they have little profit potential, while in some areas (such as solar power and electric cars) capitalists can be seen actively impeding progress, lobbying governments to end green-energy subsidies and implementing laws that obstruct further development. . . . Capitalism therefore misattributes the sources of technological development, places creativity in a straitjacket of capitalist accumulation, constrains the social imagination within the parameters of cost–benefit analyses and attacks profit-destroying innovations. To unleash technological advancement, we must move beyond capitalism and liberate creativity from its current strictures.[21]

Public research-and-development money is ultimately captured as private profit when technical innovations are designed into commodities—products and increasingly lucrative service ecosystems. Designers often work to "domesticate" new technologies, making them useful, usable, and desirable so that would-be users can become consumers, thus spawning new markets.

Today, major players who own the platforms of the digital economy are competing for competitive positions within *the stack*[22] of software and hardware that operates so much of contemporary life. For example, Facebook has invested heavily in *chatbots*, low-level artificial intelligence that would allow users to interface with the social media platform by speaking conversationally. "The bet is that these chatbots will become the preferred way for users to interact with the internet."[23] However, what's really driving this heavy investment is Facebook's attempt to work around Google's dominance in internet search and Amazon's dominance in e-commerce. The long bet is that Facebook's investments will allow it to "dominate the e-commerce platform by dominating the interface."[24] The driving force for technological investment and research in this intercapitalist competition has little or nothing to do with humans' actual needs (or even desires). Instead, platform giants are innovating their ways

around each other's current positions of platform dominance and creating new paradigms in interface and interactivity based almost purely on contingent economic conditions. Given the scale of these platforms, it is almost certain these moves will produce new standards or protocols, chart new directions for the digital economy, create new social norms and conflicts, and bring the world along for the ride—all to jockey for dominance in a high-stakes financial chess game.

Life Expectancy

In 1800, the global average life expectancy was just 29 years. Shocking as that may be, even more staggering is that no individual country on the planet had an average life expectancy greater than 40 years. Child mortality was extremely high, and almost everyone lived in conditions of extreme poverty. Over the next two centuries, factors such as global reduction of infant mortality rates, improved nutrition made possible by increased agricultural productivity, and improved medical and health treatments contributed to dramatic extensions of life expectancy. According to the United Nations, "worldwide, life expectancy at birth—a measure of the age to which a newborn would live if subject to current mortality patterns for its lifetime—reached 73 years in 2019."[25]

Unfortunately, the scales seem to have tipped in the other direction recently, particularly in some *wealthy* capitalist societies. According to the US Centers for Disease Control, the United States hit its peak life expectancy in 2014 at 78.9 years. Since then, the number has continued to drop. This decline is not distributed evenly across the population, though, but appears to have strong economic correlations. According to a study published in 2019, the states with the greatest relative increases in death rates among young and middle-aged adults were New Hampshire, Maine, Vermont, West Virginia, and Ohio.[26] In a *New York Times* article, the study's lead author stated, "What's not lost on us is what is going on in those states. . . . The history of when this health trend started happens to coincide with when these economic shifts began—the loss of manufacturing jobs and closure of steel mills and auto plants."[27] For many, the loss of jobs has also meant the loss of access to health care.

The health care system in the United States operates with a strong "free-market" logic, differing from many other wealthy countries. Although the United States spends the most per capita on health care in the world, that spending does not translate to healthier results. "The whole country is at a health disadvantage compared to other wealthy nations. . . . We are losing people in the most productive period of their lives. Children are losing parents. Employers have a sicker work force."[28] Meanwhile, health insurers have seen profits soar in an industry that appears to prioritize shareholders and investors over patients and where "executives are compensated for how well they perform that financial function."[29] In 2020, amid the COVID-19 pandemic, many of the largest American health insurers reported earnings *double* that of the previous year.

As the health industry expands and becomes more diffuse, designers are helping shape a huge range of communication services, self-tracking and self-diagnostic tools, informational and experiential media, telehealth systems, wearable technologies for in and out of clinic, and so on. Much of the discourse follows an almost-too-expected rhetoric in which "the patient becomes a consumer,"[30] deploying user-centered design's instrumental ability to deliver on consumer desires. Unfortunately, this ability embeds a logic of consumption and transaction into both *health* and *care*. A predetermined "consumer" mindset seems likely only to perpetuate the problems of this system.

Ecology

Compare the term *economy* to the term *ecology*. They share the Greek root *oikos*, "house, dwelling, or habitation." *Ecology* combines *oikos* with *logia*, "the study of." Thus, the two words are bound by a sense of place and a grounding in the communal notion of shared experiences of home or habitat. Unfortunately, for the past several centuries the habitat that is our planetary home has not been well considered in capitalism's "household management."

As the Industrial Revolution unfolded, factories began spewing pollutants into the air, water, and land. People flocked to cities in search of work, creating overcrowded living conditions and poor sanitation. In England

during the 1800s, regulatory reforms and governing bodies were established primarily to improve public health and living conditions for the people.

Much later, during the 1960s and 1970s major public outcries were heard over the devastation that industrial production was wreaking on the natural environment. The US National Environmental Policy Act of 1969 was one of the first laws to set environmental policies and goals, followed by other such environmental acts in the early 1970s. They have provided some protections for air, land, water, and plant and animal life; however, they have also faced considerable challenges, corruptions, and systematic deregulation over intervening decades. Meanwhile, the level of industrial production globally has shot up, and many of the largest industrial producers today (e.g., China) have far fewer environmental protections.

Environmental concerns have grown only worse as the once-looming threats of climate change are now well upon us. The explosion of consumer electronic goods has been particularly distressing. These products require global supply chains for extraction of minerals, such as the dangerous and often forced labor of extracting rare-earth metals in the war-torn nation of the Republic of Congo. After a brief life of use, these products are discarded and fill entire landscapes with hazardous disposal, such as the electronic-waste dumps in Ghana.

There are significant moral problems with capitalism's unchecked degradation of the planet, but there is also an economic argument to be made. The financial costs of global warming prove that "externalities" don't exist. A recent Morgan Stanley report estimates that halting global warming and reducing carbon emissions to zero by 2050 would cost $50 trillion.[31] It's a big number, but it's also a defined cost, not an externality.

The design philosopher Tony Fry has described the work of most modern design as *defuturing*: producing material and social "worlds" that actually rob the future of possibilities.[32] The destruction of natural habitats within the very narrow *critical zone*[33] of climate that supports human, animal, and plant life on this planet indicates that future humans will have far fewer choices about how to live because of the choices we are making today. Worse off will be the millions of species of plants and animals that have or are in the process of going extinct.[34] As previously described, both global warming and capitalism are *hyperobjects*, too big to think. However, it's

difficult to dismiss as coincidence the incredibly synchronous historical timelines of capitalism and ecological destruction.

The Industrial Revolution, which generally coincided with and helped propel a fledgling capitalism, was fueled largely by fossil fuels. Coal could be found in large supply near the surface of the earth in industrializing England. This coal was used to power the early machines of the Industrial Revolution, which quickly used up this easy-to-reach coal at the surface and then required more machines to dig deeper into the earth, pump water out of mines, or cut off mountaintops for more coal to power more machines (echoing the "more for the sake of more" logic of accumulation). The climate record shows that the long trend of global warming seems to coincide with the emergence of both capitalism and the dependence on fossil fuels—a dependence that has only accelerated since then.

According to researchers at NASA's Goddard Institute, the average global temperature has gone up more than 1°C (2°F) since 1880—when the Industrial Revolution had finished industrializing much of Europe and North America.[35] Even more troubling is that two-thirds of the warming have occurred *just since 1975*—right as globalization was pushing manufacturing into every corner of the world and flooding postindustrial societies with the disposable goods of a "throwaway culture." These trends are accelerating.[36]

Capitalism has nothing built into its system design to do anything but continue down this path—and more quickly! The Anthropocene, the geological era in which human activity shapes the world's ecosystems, is also sometimes called the "Capitalocene": an ecological era defined by an economic ordering that has "steamrolled this planet and its organisms, gouging out mountains, overexploiting fish stocks, and burning fossil fuels to power the maniacal pursuit of growth and enrich a fraction of humanity."[37]

These criticisms evaluate capitalism's rational logics and capacity to distribute the gains of productivity, even while admitting that productivity has provided many benefits over the past three centuries. However, capitalism has also always been a system based on social hierarchies, most of them arbitrary, and on underlying mechanisms of domination.

Moral Critique

The claim that capitalism is a rational system leads to a sense of moral valorization for those who succeed. In a system that is supposed to reward merit, if you work hard, take risks, and beat out your competitors, you will succeed. Yet the reality is not so clear-cut. The distinct asymmetries of the system on global and local scales attest to this. “Rather than being robust mechanisms for rewarding ‘merit,’ markets often function much more like brutal lotteries.”[38] While the economic criticisms highlight capitalism’s crisis of *productivity*, these are joined by moral criticisms that indicate capitalism’s crisis of *l*egitimacy.

Capitalist Subjectivity

We all are subjects of a world that is designed, but different contexts produce different *subjectivities*.* Subjectivities are those external phenomena that influence, inform, and bias people’s judgments about truth or reality and shape both personal and cultural understandings of the self and the world. Belief in rationality and technological progress, the promise of “freedom,” the valor of success guaranteed by meritocracy, a sense of stable economic and democratic institutions, and a kind of socialization that is based on consumption wherein identity is created through comparison and purchase—these attributes generally shape the *subjects* of capitalism. In democratic, capitalistic, consumer society, a certain kind of citizen-consumer subjectivity is produced in which individualism and choice are foregrounded as ideals.[39]

In his analysis of market economies originating in seventeenth-century England, Karl Polanyi stresses the *embeddedness* of an economy within its social world. He claims that the ideological tug-of-war between state and market—the twentieth-century contest between state-run socialism and capitalism—is an oversimplified view of economic and social control. Instead, the market economy and the modern nation-state form part of a singular human invention he calls the *marke*t society.

What’s so unique about the market society is that it makes possible the evaluation of all “things”—including immaterial and very abstract “things”—within the context of a market of exchanges. Nature is not

nature; it is raw material. People are not *people*; they are labor power and/or consumers. Polanyi calls this market-society way of thinking a "great transformation" toward a rational and transactional worldview of relations between people, things, and the world around them.

Beyond the abstract implications of a transactional worldview, the cost of human life has been explicitly calculated in product strategy. Probably the most well-known example is that of the Ford Pinto. In 1968, the Pinto was designed as an inexpensive American competitor to smaller and cheaper European cars penetrating the US market. The design included a fuel tank placed behind the rear axis to economize space. During safety tests, it was discovered that even a moderate rear collision could puncture the fuel tank and lead to explosion. Ford made every attempt to minimize costs while rushing the car to market. The entire production process, from initial design concept to market availability, was reduced to just 24 months—down from an average of 43 months at the time. When the fuel tank problem was discovered, it was determined that it would cost an additional $11 per unit to change the design. Yet Ford didn't do that. The company knew it would be sued for people's deaths or serious injuries, but it determined that the costs made the risk worthwhile. A cost–benefit analysis calculated each human fatality from the unsafe design would cost the company $200,000 in litigation and settlements, and each major burn injury would cost an average of $67,000. The analysis assumed there would be 2,100 accidents out of all the vehicles produced, resulting in 180 serious burn injuries and 180 deaths. When these costs were spread out over the number of units produced, it was determined the total costs of death and injury to the company would be about $50 million, far less than the $137 million that would be required to improve the known safety risks by redesigning the car. More recently, the Boeing corporation has been accused of similar cost–benefit analysis in a known engineering problem with the Boeing 737 Max fleet, a problem that has contributed to plane crashes killing 346 people.

In the 1930s, a school of social theory and political thought known as the Frankfurt School influenced the development of *critical theory*. In critical theory, culture and society are critiqued through the lens of power structures based on the premise that social relations are shaped by institutions, economic and political structures, and cultural assumptions. Frankfurt School theorists were among the first to study how media influences thought, behavior, and politics, with a particular critique of the role media

play in producing consumer demand as a political strategy. In the twentieth century, this critique was pointed at the *mass-culture* society of mass media and mass consumption, illustrated by consumerism in America and western Europe.[40]

In critiquing the social structures of consumerism, Frankfurt School theorists used the term *culture industry* for the factory-like *production* of cultural *products* that pacify consumer-citizens into a state of political apathy. Products, marketing, advertising, and mass-produced arts such as cinema influence political views and an understanding of social structures. Citizens thus need only *consume* politics like they do other products. The psychological toll of consumerism lies in human identities becoming wrapped up in the things people buy. “The people recognize themselves in their commodities; they find their soul in their automobile, hi-fi set, split-level home, kitchen equipment. The very mechanism which ties the individual to his society has changed, and social control is anchored in the new needs which it has produced.”[41] Against this seemingly inescapable flattening of social and political life, Frankfurt theorists suggest that the only adequate opposition is to *refuse*. This opposition requires not only refusing to adapt to any of the identities put on display for purchase but also refusing to be subjected to the logics and subjectivities of consumer society. This critique acknowledges the incredible influence of culture as a commodity.

Whether named explicitly or not, design is central to this criticism. Through product and service design, branding, and marketing of curated “lifestyles,” designers inscribe social and political views within cultural products. In this way, design normalizes social and political views through commodities, defining the range of “normal” needs, desires, and ways of being. Design creates subjectivities. Under capitalism, design tends to create consumer-subjects.

Class & Social Inequality

In the seventeenth century, as feudalism gave way to capitalism, property ownership changed completely along with social relations. Under feudalism, the structure of nobility, passed down through bloodlines, defined the social hierarchy of society. Under capitalism, however, private

ownership of the means to “produce”—capital, land, machinery, labor—became the distinguishing feature around which new social relations organized. As a social order, capitalism creates hierarchies centered on ownership, which can also be viewed as a form of exploitation.

> The central form of exploitation is based on property rights in the means of production. These property rights generate three basic classes: capitalists (exploiters), who own the means of production and hire workers; workers (exploited), who do not own the means of production and sell their “labor power” (i.e. their capacity to work) to capitalists; and petty bourgeois (neither exploiter nor exploited), who own and use the means of production without hiring others. . . . Because of the power relation between capitalists and workers, capitalists are able to force workers to produce more than is needed to provide them with this subsistence. As a result, workers produce a surplus which is owned by the capitalist and takes the form of profits. Profits, the amount of the social product that is left over after the costs of producing and reproducing all of the inputs (both labor power inputs and physical inputs) have been deducted, constitute an appropriation of the fruits of labor of workers.[42]

This exploitation is the central dynamic of class relations in a capitalist society. In its simplest terms, the Marxist view of capitalist class structure pits the owners against the workers, and the tensions of that economic contest become the grounds of political and social struggle—even history itself. The owners do what they can politically to ensure that what’s in their best interest is protected through laws, policies, political representation, and social norms that present this dynamic system as a game of merit—a game at which they happen to be quite good due to their entrepreneurial risk taking and bravado. Whoever takes the risks should reap the rewards, no matter how big or disproportionate those rewards might be with respect to the efforts of those who labored to create the value that generated the profit. Workers, for their part, struggle against the owners’ attempts to maximize exploitation by organizing labor unions, demanding higher wages, insisting on more benefits or shorter working hours, calling for better or more interesting work, fighting for legal protections, portraying productive labor

as the engine of society, and sometimes slandering owners as parasites who suck out their life and value in the form of profits. As early industrial capitalism became a more complicated affair, there grew a need for more people to organize and manage the relations between the owners and workers. A *middle class* was born with complicated class interests. Over time, the middle class has fluctuated in scale. In the United States, it grew exponentially in the postwar boom and now is in decline—and has changed in nature.

This distinction of a class whose interests are divided "is the problem of the 'middle class'—people who do not own their own means of production, who sell their labor power on a labor market, and yet do not seem part of the 'working class.'"[43] According to Wright, "Highly educated professionals, managers, and many self-employed people, for example, occupy what I have called *contradictory class locations within class relations* and have quite complex and often inconsistent interests with respect to capitalism."[44]

One way the middle class is divided is along the axis of *authority*. To ensure that management can yield appropriate performance from its workers, "capitalist production always involves an apparatus of domination involving surveillance, positive and negative sanctions and varying forms of hierarchy."[45] To work this apparatus, managers and supervisors are hired to exercise "capitalist class powers in so far as they engage in the practices of domination."[46] Employees with authority over other employees are dominated and exploited by capitalist owners in the sense that their labor must also produce a surplus to generate profit, yet they also dominate and exploit other employees in the interests of the owners. Relatively high incomes and pathways to career advancement—the "corporate ladder"—provide opportunities to increase authority, and ongoing incentives (promotions, pay raises, accolades) encourage the achievement of the highest possible productivity through "lean" and "tough" management.

Another axis along which the middle class is divided is that of *skills and expertise*. Employees with high levels of skill or specific expertise are in a privileged position "within exploitation relations." This position allows them to bargain for higher wages and better working conditions. Because skills and expertise "are frequently scarce in labor markets," and because "the control over knowledge and skills frequently renders the labor effort of

skilled workers difficult to monitor and control," highly skilled employees are treated differently than less-skilled workers. For example, "employers must rely to some extent on loyalty-enhancing mechanisms in order to achieve desired levels of cooperation and effort from employees with high levels of skills and expertise, just as they have to do in the case of managers. Employees with high levels of expertise, therefore, are able to appropriate surplus both because of their strategic location within the organization of production (as controllers of knowledge), and because of their strategic location in the organization of labor markets (as controllers of a scarce form of labor power)."[47]

Designers occupy different positions across this expanded class structure. On one end of the spectrum, independent design agencies or studios, including large multinational agencies, operate most similarly to capitalists. They own the means of production, including the labor they hire. However, even large independent agencies typically work for other capitalist owners —their clients—which makes design owners exploited by others who make a profit off their work. Meanwhile, the highest-ranking designers in a corporation, such as chief design officers or chief creative officers, may receive heavy incentives for corporate success, but they are still typically beholden to a profit-producing model. In both cases, owners or executive designers will want to spend as little as necessary (in wages and benefits) to achieve the maximum profit via client fees or departmental budgets.

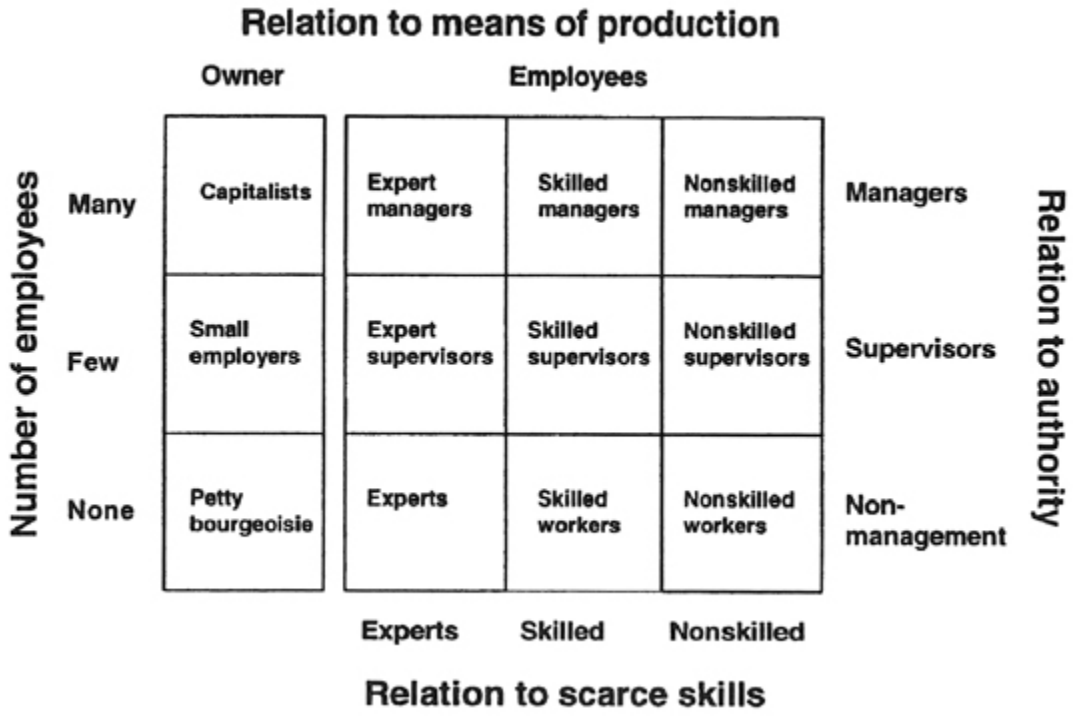


Figure 4.3

Elaborated class typology by Erik Olin Wright: "Adding positions within authority hierarchies and possession of scarce skills and expertise to the fundamental dimension of capitalist property relations" generates this expanded map of class structure. (*Source*: Erik Olin Wright, *Class Counts: Student Edition* [Cambridge: Cambridge University Press, 1997], 20; image courtesy of Cambridge University Press.)

Design typically involves product, service, artifact, or space-making specifications that will be produced by others, such as printers, fabricators, software developers, engineers, manufacturers, and so on. Directing this work, whether on behalf of their clients or for their corporate interests, designers are motivated to get as much of the best labor as they can for as low a cost as possible. Because designers tend to be in a position to understand better than anyone else the parameters of what needs to be produced, they are often directly involved in production negotiations between corporate budget managers and third parties who will do the actual production.

On the other end of the spectrum, few designers or design-related workers are entirely *unskilled* as worker, supervisor, or manager. Junior designers are most likely to belong in the category of "skilled workers" with little relation to authority. Over time, they may move upward in trajectories of authority and expertise. Particularly in large corporate

organizations with in-house design teams, the levels of management and supervisory roles for designers may vary greatly depending on the scale and complexity of the business. Chief creative officers may sit at the level of *expert managers*, and there may be multiple creative directors at the level of *expert supervisors*. Also in the *expert* column would be design researchers, such as UX researchers, who may supervise or manage teams. Designers whose work takes them in the direction of project management may also climb the *relation-to-authority* ladder. Seen in this context, the entire field of design clearly occupies *contradictory class locations*.* Designers are both exploited and exploiters.

Gender Inequality: Sexism & Capitalism

Beyond class relations, boundaries of exclusion provide the basis for further exploitation. In sociological terms, an *ascriptive status** is a mechanism by which people are stratified into social classes based on qualities not in their control. Theoretically, the hierarchies created by ascriptive statuses should be antithetical or, at the minimum, irrelevant to a liberal, rational economic system. Yet, as Polanyi points out, capitalism has always been *embedded* in hierarchies of gender and race—two of the most recognizable ascriptive statuses. Although the economic, political, and legal rules may vary, ideologies of "ascriptive difference help to stabilize a social order by legitimizing its hierarchies of wealth, power, and privilege, including its social divisions of labor, as the natural order of things."[48]

Gender inequality has been a hallmark of Western civilization from its start. Women were excluded from political power in the "free" societies of ancient Greek city-states and the Roman Republic. However, the inequalities attached to the ascriptive status of gender have been compounded by a political economy that views people as units of labor (and consumption) and treats social relations as transactions. Prior to the Industrial Revolution and the parallel rise of industrial capitalism, women were often tasked with domestic labor that included material production: making or repairing clothing for the family, managing the farm, and so on. In a capitalist system, similar domestic work such as cooking and cleaning, keeping a house in order, and caring for and raising children is considered *unproductive labor*. Even though this labor constitutes *social reproduction*

—the labor that provides a labor force—it does not contribute directly to generating profit. “Where as in other systems men and women did specified (but normally equal) tasks, under historical capitalism the adult male wage-earner was classified as the ‘breadwinner,’ and the adult female home-worker as the ‘housewife.’ Thus when national statistics began to be compiled, itself a product of a capitalist system, all breadwinners were considered members of the economically active labour-force, but no housewives were. Thus sexism was institutionalized.”[49]

As the Industrial Revolution dispossessed rural peasants of subsistence-based living and sent them flocking to urban centers in search for work, traditional roles for women were also transformed. Clothing was mass-produced in factories and no longer made at home. Women entered the workforce in textile mills and coal mines. In these new working roles, women (as well as children) were paid much less than men in part because they were believed to be less productive in their physical labor. They were subjected to horrible and dangerous working conditions.

The struggle for better working conditions and wages that animated early labor unions, workers’ rights groups, labor strikes, and socialist political movements during the industrial era also produced the first feminists. The political gains of the early feminist movement were most clearly captured when women achieved *suffrage*, the right to vote, in democratic societies.[50] During the 1950s and 1960s, a *second wave* of feminism emerged that sought equality beyond basic political representation. This movement sought greater economic parity for women, access to employment opportunities and better wages, as well as reproductive rights. This struggle brought millions of women into the workforce. Ironically, these social gains were to the financial benefit of capital. Since the 1970s, an increased labor supply has run adjacent to decreasing wages, lower job security, more working hours for those on the low-end of the wage scale, “and a rise in poverty, increasingly concentrated in female-headed households.”[51]

Feminism now takes expansive forms: “Feminism is not just about eliminating gender-based discrimination. It’s about fighting for and creating equality and a good life for everyone, regardless of their sex, gender, race, ethnicity, education, income, religion, or where they live.”[52] These goals, however, are seen as inimical to capitalism. “Sexism and racism have been a core part of strategies of accumulation in capitalism. Sexism makes

women's unpaid labor in the home, which is essential to society, appear natural, a labor of love. Sexism and racism also continue to be extremely handy tools in the business owners' tool kit to divide and oppress workers, to discourage demands for better pay and benefits, or to block efforts to form a union."[53] In the landscape of globalized capitalist labor, the entire world has become a sweatshop, and the labor is done mostly by young women of color.

The economic success of individual women has sometimes been held up as an indication of progress. Critics of this claim, however, see it as the appropriation of feminism by capitalism. "Feminist ideas that once formed part of a radical worldview are increasingly expressed in individualist terms. Where feminists once criticised a society that promoted careerism, they now advise women to 'lean in.' A movement that once prioritised social solidarity now celebrates female entrepreneurs. A perspective that once valorised 'care' and interdependence now encourages individual advancement and meritocracy."[54] The fight for equality beyond gender-based discrimination has become bound up in the struggle against both sexism and racism. "The most extraordinary fact of capitalist civilization . . . is that the intensity of belief in these two themes [sexism and racism], and the degree to which they have been implemented in social practice, have grown side by side, in tandem."[55]

Racial Inequality: Racism & Capitalism

The first modern democracy—the United States—was based on the ancient Greek model, in which political rights were based on property ownership. When the US Constitution took effect in 1789, "only 6 percent of the population—white, male property owners—was eligible to vote."[56] The Three-Fifths Compromise of the US Constitution states that three out of every five slaves were to be counted as "Persons" for the purposes of taxation and political representation.[57] Immediately, we can recognize a political institution that prioritizes property (thus taxes) as a criterion for participation, while establishing laws of exclusion. Exclusion of ownership or access to wealth equates to exclusion from control over the material production of one's life. The institutions of capitalism are designed to protect and promote private ownership. This poses a particular problem for

groups of people who are considered “property” (as in slavery), who are denied access to property, or who are born into a lineage of denied access to wealth accumulation (as in the long history of racially discriminatory laws). In this system, race transcends social status to become a function of economic relations, “a relation of hierarchy rooted in the capitalist division of labor—turned into an object.”[58]

Although slavery has existed since the earliest known civilizations, the institution of *racialized* slavery is contemporaneous with European imperial colonization and the birth of capitalism. “Race emerged historically along with the institution of slavery in the New World.”[59] The construction of a system of distinct races—“white” or “black” as representative of “European” or “African,” respectively—was a radical flattening of ethnic diversity. Claims of racial differences provided *legitimacy* for the dehumanization of some races for the social and economic benefits of others. They were also a distinguishing foundation upon which many modern republics were built. The African American minister and activist Malcolm X once said, “You can’t have capitalism without racism.”[60]

Racialized slavery and the genocide of indigenous peoples in the Americas were the essential ingredients of the economic growth of the early American republic. In both cases, entire ethnic populations were treated as *externalities* to a system of production that required land and labor to grow its productive capacities and accumulate wealth for owners of farms, ranches, plantations, and industrial enterprises that thrived on the supply of food and materials—all of which were produced out of bondage, pillage, and murder. The “capital” that provided the foundation for Wall Street was enslaved humans, quite literally: New York’s slave market was at the end of the street.[61] Many early industrial technological innovations stemmed from the production of cotton textiles, making cotton from the American South one of the primary materials supporting the Industrial Revolution. The long, gruesome history of slavery and the legally enshrined racial hierarchy of Jim Crow that followed are well documented and establish the social context for the first three centuries of American capitalism. That’s a lot of historical momentum.

In the second half of the nineteenth century, after the abolition of slavery, the expansion of the American railroad system was made possible by large numbers of imported laborers from China. Once again, ideologies of racial

hierarchy played a significant role in this massive economic endeavor. “Railroad operators and other importers of Chinese labor imagined that Chinese workers’ distinctive racial characteristics made them more tractable and capable of living on less than white Americans,” which served as the justification for the low pay and difficult working conditions put upon Chinese workers. After the abolition of slavery, many Chinese and later Sicilian laborers were imported by southern plantations to compete with Black sharecroppers working in cotton and sugarcane fields.[62] In this and many other ways, racial hierarchy was effectively deployed for the class interests of capitalist owners.



Figure 4.4

New York slave market, c. 1730. Located at the end of Wall Street, where it reaches the East River, the slave market was established in 1711 as a place where enslaved Blacks and Native Americans could be hired or purchased. (Courtesy of the New York Public Library.)

The ascriptive statuses of race and gender are still often socially and economically disadvantageous even if they are not legally recognized. The legacy of this constructed hierarchy has contributed to real and staggering economic inequality. In 2016, nearly 150 years after America's abolition of slavery and more than 50 years after the Civil Rights Acts of 1964 and 1968, the average net worth of white households in the United States was about ten times that of the median net worth of Black households. *Ten times!*[63] There is no economically rational justification for a system that results in this scenario. More than 50 years after racial discrimination in housing, employment, voting, and schooling was made illegal, "so much of what makes black lives hard, what takes black lives earlier, what causes black Americans to be vulnerable to . . . surveillance and policing . . . what steals opportunities, is the lack of wealth that has been a defining feature of black life since the end of slavery."[64] This denial of wealth, too, was by design. "As soon as laws began to ban racial discrimination against black Americans, white Americans created so-called race-neutral means of maintaining political and economic power." When the Federal Housing Administration insured loans to help Americans buy homes and thus build wealth, 98 percent of the loans granted from 1934 to 1962 went to white Americans. This program has been credited with building up the American postwar middle class, but it excluded those who were not white. Meanwhile, policies such as *redlining*—discriminatory lending policies that prohibited people of color from buying or renting in many neighborhoods—effectively restricted African Americans to narrowly confined neighborhoods, thus generating concentrations of racialized poverty. These practices have had multigenerational impacts, particularly because home ownership has been central to wealth accumulation for the middle and working classes. "Wealth begets wealth, and white Americans have had centuries of government assistance to accumulate wealth, while the government has for the vast history of this country worked against black Americans doing the same."[65] In leading a nonviolent struggle against American racism, the African American minister and activist Martin Luther King Jr. saw the struggle for racial justice as entirely entangled in the struggle for economic justice. In a speech to the Southern Christian Leadership Conference board in 1967, he said: "The evils of capitalism are as real as the evils of militarism and evils of racism."[66]

Racial economic inequality is not in decline. In fact, it's growing. From 1983 to 2013, white households increased their wealth by 14 percent, while Black households declined in wealth by 75 percent. Median household wealth for Hispanics declined by 50 percent.[67] "This racial wealth gap exists even among blacks who are highly educated and come from two-parent homes. Black families with graduate or professional degrees have $200,000 less in wealth than similarly educated whites. These black or Latino college graduates don't even have as much wealth as white high school dropouts."[68] Since 2008, Black Americans "lost the largest share of their wealth of all racial groups as a result of the last recession and have struggled the most to recover. They are the only group whose household income is less than it was in 2000."[69]

Beyond this abysmal history of oppression and disadvantage, another glaring blot on American capitalism's self-accounting is the incarceration of people of color. America is unique in that it makes up less than 5 percent of the world population yet has 25 percent of the total number of prisoners in the entire world. This enormous prison population is also highly racially disproportionate: while "black inmates make up nearly 40% of the prison population, they only make up 13.40% of the total US population."[70]

Starting in the 1970s, the "war on drugs" pushed many nonviolent offenders into long sentences. Growing prison populations presented economic challenges for federal and state government. Because neoliberal policies operate on the belief that public amenities, including prisons, are best handled by private interests and the market, privatization has transformed American criminal corrections into a lucrative "prison industry." This includes for-profit prisons as well as big contracts for food, construction, employment, and provisions for their operation. The revenue, employment, and profit to be made from incarceration have created economic demand for prisoners. In addition, *penal labor* is a multi-billion-dollar industry. Prisoners are paid incredibly low wages, sometimes 50 cents an hour, to do compulsory work as varied as "building office furniture and making military equipment, to staffing call centers and doing 3D modeling."[71] Companies profiting off prisoner labor include design icons such as Starbucks, whose subcontractor Signature Packaging Solutions has hired prison labor for packaging holiday coffees, and Victoria's Secret, which has hired female prisoners to sew its products.[72] America has the

largest economy in the world, and prison labor has come to play an integral role in that economy.

At no point in the history of American capitalism has the economy functioned (successfully or at all) without the oppressive subjection of women and people of color to its economic needs: slavery, exclusion from ownership, prison labor, and abysmal wage gaps have provided the economic "efficiencies" needed for a "profitable" system. This is hardly a rational, let alone just, form of democratic capitalism. The African American designer and activist Christopher Rudd makes the point clearly: "In order to be anti-racist, you have to be anti-capitalist."[73]

Design plays a role in all these cultural and economic forms of oppression in part by the way it enacts *symbolic violence*. The term *symbolic violence* "describes how priorities, values and even sensibilities are reproduced through cultural practices, processes and institutions. Through symbolic violence, individuals learn to consider unjust conditions as natural and even come to value customs and ideas that are oppressive."[74] Because design embeds ideas in objects, communications, services, and spaces "in subtle and psychologically powerful ways," it often creates or reproduces the sexism, racism, classism, and colonialism that are typical of capitalist economies.

Freedom & Democracy

In some ways, "freedom to choose" is the central moral virtue of capitalism. You can choose the work you do, choose the things you wish to buy, and choose what kind of life you want to lead within a system of free markets for labor, products, services, and ideas. "If there is one value that defenders of capitalism claim it achieves to the highest possible extent, it is individual freedom and autonomy."[75] What extends from this foundational claim of virtue is that capitalism is essential for democracy and the freedom to choose political representation. Capitalism does not guarantee democracy—see, for example, China's model of single-party political power and capitalist economic activity. Yet capitalism is still argued to be a necessary condition for democracy. Let's examine both of these claims.

Freedom of Choice Although market and property-based opportunities for choice are central to capitalism, in reality this hardly amounts to real

freedom. Most people are born with little to no property—no accumulated wealth as capital or land to practice subsistence farming. So they must work. Central to the employment contract for wage labor is that laborers will do as they're told. For most workers in capitalist workplaces, even those occupying exciting and challenging creative careers, "individual freedom and self-direction are quite curtailed."[76] The lack of freedom and self-direction within the world of work is central to the critical concept of *alienation*. Under capitalism, most workers are dominated by a situation in which they are employed for the purpose of turning a profit for the owners.

Yet how can voluntary employment be considered *domination*? The freedom to choose means you may quit any unsavory job to take another. Wright responds that "the freedom of individuals to quit their jobs, however, provides only an illusory escape from such domination since without ownership of means of production or access to basic necessities of life, workers must seek work in capitalist firms or state organizations, and in all of these they must surrender autonomy."[77] Under neoliberal logic, the thrill of competitive, self-directed work is valorized by the beacon of entrepreneurship, coddled by mottos such as "Rise and Grind" and "Do What You Love" (presuming what you "love" is 18-hour days building a tech start-up). However seductive these opportunities may appear, a capitalist world cannot exist without wage earners laboring to create surplus value to make profits for the owners. The profit system requires a higher ratio of laborers to owners, and most people in capitalist society have very little choice in the broad strokes of their lives.

Capitalism & Democracy Public decision making has little or no control over how private properties are used, or how private production is handled.[78] This puts capitalism and democracy at odds, even as *economy* and *politics* are treated as though they are separate spheres. If capitalist firms were to make decisions that have no impact on the public or private lives of others, perhaps all would be well. However, when corporate decisions lead to things such as factory collapses, chemical explosions, oil spills, long-term lead contamination, and so on, "private" interests dramatically affect others. "The key issue, then, is that the private decisions made by the owners of capitalist firms often have massive collective consequences both for employees and for people not directly employed in

the firm, and thus the exclusion of such decisions from public deliberation and control reduces democracy."[79]

During the nineteenth and early twentieth centuries in the United States, capitalism and democracy were often at odds. Capitalist owners—of factories, canneries, textile mills—were regularly afraid that their workers would organize politically and that democratic majorities would abolish private property. Indeed, many workers' organizations struggled toward just these goals. Their efforts, including some major successes, were a continual threat to the capitalist order. Meanwhile, workers and labor organizations were continuously concerned that capitalist owners would use their wealth and political influence to abolish democracy and transform the country into an authoritarian state. "Only in the Cold War world did capitalism and democracy seem to become aligned with one another, as economic progress made it possible for working-class majorities to accept a free-market, private-property regime, in turn making it appear that democratic freedom was inseparable from, and indeed depended on, the freedom of markets and profit-making."[80]

Since the 1970s, the compatibility of capitalism and democracy has again run up against the rocks. The legitimacy of capitalism during the postwar era was predicated on the government's right—even *obligation*—to "intervene in markets and correct their outcomes in the interest of citizens."[81] Big companies worked directly with large, powerful labor unions and community representatives through government and regulatory agencies to create a stable, albeit imperfect, democratic capitalism (with rules of exclusion applied to minorities and women). The "social contract" of this era was built upon reliance between capitalist industry and the nation for mutual prosperity. Stability was critical. It was in the best interest of corporate leaders to maintain a healthy and content labor force, supported through their working life and beyond with living wages and pensions. Well-funded public amenities contributed to a robust, public social sphere and high levels of employment (for white men). Progressive corporate taxes were incredibly high then compared to today, and that tax revenue supported public education, research, medical care, human and health services, unemployment protections, and the safety net of public services that served to keep the working population in good health and spirits.

During this era, industry leaders saw themselves as *corporate statesmen*. In 1953, the president of General Motors, Charles Erwin Wilson, was nominated to become secretary of defense. In his Senate confirmation hearing, Wilson was asked if he could act in the nation's interest over the corporate interest of GM. He could not even imagine how the two could be at odds. He said the "difference did not exist. Our company is too big. It goes with the welfare of the country."[82] GM was indeed considered "too big to fail" in 2008, when the company was bailed out with $13.4 billion in tax revenue just before entering bankruptcy.[83]

Since the 1970s, all that has changed. The Cuban American businessman Roberto C. Goizueta, former CEO of Coca-Cola, once said, "We have one job: to generate a fair return for our owners. . . . We must remain focused on our core duty: creating value over time."[84] With his statement, he made the singular profit motive crystal clear. The iconic American company may sell "happiness" as its core brand offering, but turning a profit on "happiness" is its core value. Today, one need not look far to find stories of corporate foreign investment, tax loopholes, offshore banking, and various means to escape taxation, to locate the cheapest labor and production anywhere on the planet, and to avoid liability to democratic governance as much as possible. Capitalism, which was briefly seen as the sidekick of democracy, is now pushing many societies toward authoritarianism and oligopoly.

It is highly doubtful that American capitalism would have been nearly as successful without its history of genocide and slavery followed by ongoing oppression and economic plunder based on class, race, and gender. Capitalism in the real world appears inseparable from tactics of defining boundaries of exclusion for the sake of exploitation. Therefore, it appears fundamentally flawed as a rational economic system promoting the liberal pursuit of self-interest.

Neoliberalism & Its Discontents

Chapter 3 concluded with a brief history of the rise of neoliberal economic thinking that closed out the twentieth century and rang in the twenty-first. The primary premise of neoliberalism is that the "free market" is the best

model for allocation of resources on a planetary scale. The *rational* idea that all humans are independent economic actors (*homo economicus*) is married to the *liberal* idea that a planet full of individual actors engaged in self-interested competition will work itself out to produce the best economic incentives, opportunities, and outcomes for all who are willing to strive for success (or survival). Neoliberalism considers most forms of government intervention into market activities a *distortion*, something to be minimized or eradicated. The exceptions to this rule are, of course, instances where the government is expected to protect—by any means necessary—the property rights and free reign of economic producers. In other words, the economy becomes the dominant organization under which social and political life are subservient.

According to the British author and activist George Monbiot, neoliberalism "sees competition as the defining characteristic of human relations. It redefines citizens as consumers, whose democratic choices are best exercised by buying and selling, a process that rewards merit and punishes inefficiency." In the neoliberal view, government, taxes, and regulations are "treated as inimical to liberty. Tax and regulation should be minimised, public services should be privatised. The organisation of labour and collective bargaining by trade unions are portrayed as market distortions that impede the formation of a natural hierarchy of winners and losers. Inequality is recast as virtuous: a reward for utility and a generator of wealth, which trickles down to enrich everyone. Efforts to create a more equal society are both counterproductive and morally corrosive. The market ensures that everyone gets what they deserve."[85]

In the United States, the availability of long-term, secure employment with a living wage, health care, and retirement benefits—often made possible by strong labor unions that could collectively bargain for workers—was one of the key ingredients to the economic prosperity of America's middle class for roughly three decades after World War II. Since the 1970s, however, that stability has been shattered, and the labor landscape has become far more precarious. More recently, the radical *disruptive innovation* practiced by companies such as Uber has ushered in a new kind of labor market known as the *gig economy*. The global financial crisis of 2008 doubled unemployment. Nearly a decade later, Nick Srnicek stated "all of the post-2008 employment gains in America have come from workers in non-traditional employment, such as contractors and on-call

workers."[86] On-demand labor, nonnegotiable wages, no long-term employment commitment, limited or no benefits, and little in the way of workplace protections are the new terms of work for many.

An economy that puts massive quantities of workers in competition for increasingly precarious forms of employment to earn decreasing wages and benefits exacerbates the raw social logic of competition for survival. Pervasive data collection in service work leads to a state of omnipresent evaluation of employee performance—surveys, reviews, and customer-feedback channels that can be weaponized by employers. To be employed in these conditions is to live and work under constant surveillance and in perpetual fear of discipline or termination.[87] Meanwhile, the imperative for service employees to be constantly cheerful, friendly, and eager to please their customers has created new social challenges. Recent research has shown that employees pretending to be friendly for their jobs was connected to "feeling anxious or distressed, and also increased the likelihood of feeling emotionally exhausted at the end of the day. Some even reported heightened family conflict at home." This research illustrates that *emotional labor*—although different from manual or factory labor—is nonetheless "difficult, draining work."[88]

In a society defined by fierce individual competition—competition for employment, which correlates to health care, education, and access to material and social survival—neoliberal economic policies have insisted on the primacy of work, productivity, and competitive achievement even in our social lives. "In other words, neoliberalism creates subjects. Paradigmatically, we are constructed as competitive subjects—a role that encompasses and surpasses industrial capitalism's productive subject. The imperatives of neoliberalism drive these subjects to constant self-improvement in every aspect of their lives. Perpetual education, the omnipresent requirement to be employable, and the constant need for self-reinvention are all of a piece with this neoliberal subjectivity. . . . One's personal life is as bound to competition as one's work life."[89] Designers reading this statement may recognize the basic formulation for what is commonly called *hustle* culture.[90]

The highly social creatures we are, "whose brains are wired to respond to other people, are being peeled apart. Economic and technological change play a major role, but so does ideology. Though our wellbeing is

inextricably linked to the lives of others, everywhere we are told that we will prosper through competitive self-interest and extreme individualism."[91] Today, you can take online courses in "personal branding" to capitalize on the most powerful marketing tool at your disposal: you! You are not simply a human being with human relationships, needs, and failings. *You ar*e a brand!

"Even if you do not buy into the ideology [of competitive self-interest], its effects nevertheless force you into increasingly precarious situations and increasingly entrepreneurial inclinations. We need money to survive, so we market ourselves, do multiple jobs, stress and worry about how to pay rent, pinch pennies at the grocery store, and turn socializing into networking."[92] Stress, anxiety, depression, eating disorders, self-harm, loneliness, and other related mental health problems are at catastrophically high levels. We are compelled to play the game whether we approve of it or not, regardless of how stressed out or exhausted it makes us, even if we don't like the rules, and even if the companies we work for and spend our money to support regularly break the rules.

Contemporary Critique

Today, there are new critiques specific to capitalism in its twenty-first-century form, in which *data* are a new material and *platforms* represent a dominant business model. Google, for example, has a long track record of dodging corporate oversight when launching a new product or service that may impinge on privacy protections.[93] Twitter and Facebook have been roundly condemned for profiting off disinformation that has spread political divisiveness, violent and threatening rhetoric, and conspiracy theories that may have swung an American presidential election.[94] Apple, the wealthiest company on the planet and first to reach valuation of $1 trillion, and Amazon, whose CEO Jeffrey Bezos is on track to be the world's first trillionaire, combine foreign investment and heavy corporate lobbying to avoid paying taxes in the United States.[95] The economic and political power these companies wield makes the democratic notion that "every vote counts" pale in comparison.

The contemporary version of capitalism goes by many names, each an attempt to specify its most salient dimension: neoliberal capitalism, postindustrial capitalism, post-Fordist capitalism, supercapitalism, hypercapitalism, finance capitalism, casino capitalism, platform capitalism, cognitive capitalism, and surveillance capitalism, to list just a few.

Cognitive-Cultural Capitalism & the Information-Network Economy

The economy that emerged from the postindustrial shifts of the 1970s is characterized in part by dependence on digital technologies and *cultural labor*. High value is placed on tech industries, media, business and financial services, personal services, and the broad spectrum of cultural and creative industries. “In cognitive capitalism, the nature of work is transformed. Manual labour and industry don’t stop, but their place in the landscape changes. Because profit increasingly comes from capturing the free value generated by consumer behavior, and because a society focused on mass consumption has to be constantly fed coffee, smiled at, serviced by call centers, the ‘factory’ in cognitive capitalism is the whole of society.”[96] However, the proliferation of digital, informational, and cultural *products* has posed new problems for private ownership and property rights and is posing new challenges to an economic system that is predicated on a *scarcity* of goods.

> Since the mid-1990s, a revolution in the way we process, store and communicate information has created the beginnings of a network economy. This has started to corrode the traditional property relations of capitalism in the following ways.
>
> It corrodes the price mechanism for digital goods, as understood by mainstream economics, by pushing the cost of reproducing information goods towards zero.
>
> It adds a high information content to physical goods, sucking them into the same zero-price vortex as pure information goods—and often, as with the [example of Nike] trainers [athletic shoes], making their

value dependent more on socially created ideas (the brand) rather than the physical cost of reproducing information goods towards zero.

It makes financialization necessary, creating two streams of profit flowing to capital from the general population: as workers producing goods, services, and knowledge; and as borrowers generating interest payments.[97]

Think about the neoclassical model of supply and demand meeting in the market to determine the price for a physical good. Now think of a digital product, such as the digital file of a song or a book. The *marginal production cost* of the product is nearly zero—meaning, there is no additional cost for producing an additional "unit" of the product. Because no amount of copying the file will reduce its availability (in economic terms, it is *nonrivalrous*), the supply is potentially infinite. In fact, a product like this is called *antirivalrous*: the more people who have access to it, the more they share knowledge of it, the more they can discuss or critique it, then the more valuable it becomes—like *viral* content in networked media. "Only intellectual property law and a small piece of code in the iTunes track prevent everybody on earth from owning every piece of music ever made. Apple's mission statement, properly expressed, is to prevent the abundance of music."[98] The value of information forces new models of protection to maintain artificial scarcity for an outdated economic system. Under this logic of immaterial private ownership, intellectual-property laws protect "ideas" ranging from truly new and novel inventions to plant species and human genomes.[99]

In such an economy, it is commonly claimed that intellectual-property law is necessary to generate incentive for innovation. The basic idea is that if the large up-front investment required to generate innovation can't be *owned* as a monopoly on the economic advantage it offers, then no one will make the effort or investment to innovate. This claim is common, but is it true?

According to research from the Centre for Intellectual Property Policy at McGill University in Montreal, this is not exactly the case. In a study that "evaluated the strength of IP [intellectual-property] protection in 124 developing countries for the years 1995 to 2011," the researchers sought evidence of positive impact of intellectual-property rights on economic

growth. What they found was that “IP may have few direct effects on growth and that any causality is a result of belief rather than actual deployment of IP.” In other words, “growth has a greater effect on IP than IP on growth, further supporting the idea that politics and belief, rather than direct economic effect, explain the virtuous cycle between IP and growth.” Based on this, the researchers put forth a “placebo theory” that “politics and belief account for higher levels of formal IP and growth based on the faith that IP drives growth.”[100] This stunning discovery of a faith-based economics points to the reality that free-market “common sense” is an ideology. It may be productive, but it is far from a rational system.

“Today, the main contradiction in modern capitalism is between the possibility of free, abundant socially produced goods, and a system of monopolies, banks and governments struggling to maintain control over power and information. That is, everything is pervaded by a fight between network and hierarchy.”[101] Because of these contradictions, Paul Mason sees two basic possibilities ahead: “Either a new form of cognitive capitalism does emerge and stabilize—based on a new mix of firms, markets and networked collaboration—and the remnants of the industrial system find an orderly place within this third capitalism. Or the network erodes both the function and legitimacy of the market system. If so, a conflict will take place that results in the abolition of the market system and its replacement by post capitalism.”[102] Amazingly, Mason uses the same metaphor as Wright to describe how networked information and collective action have the potential to bring about postcapitalism: erosion.

Surveillance Capitalism

If data and platforms are the drivers of today’s economy, then what are the business methods used to create value from them? The American social psychologist Shoshana Zuboff describes the emergence of what she has dubbed “surveillance capitalism.” Of primary interest to Zuboff is the way digital platforms have used a variety of manipulative techniques to collect and commodify data, which creates a new economic order. The data of human experiences are used as “free raw material” for hidden commercial practices of “extraction, prediction, and sales.”[103] The sheer power that digital platforms wield has shifted the balance such that the actual

production of goods and services has become subordinate to the global architecture of digital platforms. In other words, the production of material goods and the provision of services are utterly dependent on digital platforms.

To harvest increasingly targeted data into financial return, this system feeds on "behavioral modification" of its users to create a "parasitic economic logic." Digital platforms and networks manipulate desires and behaviors, then sell off those desires and the granular personal data of their users to product and service providers. Because this system is predicated on surveillance, manipulation, and predictive certainty, Zuboff sees it as a "new instrumentarian power" that challenges market democracy, is "an expropriation of critical human rights," and represents an "overthrow of the people's sovereignty."[104]

These new business models not only profit off murky methods of data collection but also require that participants in the market continually break into new territories that are quasi-legal or in some cases outright illegal. Zuboff describes a framework for this form of extraction that she calls the "Dispossession Cycle," which unfolds in four stages: incursion, habituation, adaptation, redirection. She tells the story of Google Street View to animate the practice.[105]

The incursion stage is "initiated by unilateral incursion into undefended space: your laptop, your phone, a web page, the street where you live, an e-mail to your friend, your walk in the park, browsing online for a birthday gift, sharing photos of your kids, your interests and tastes, your digestion, your tears, your attention, your feelings, your face. The incursion is when dispossession operations rely on their virtual capabilities to kidnap behavioral surplus from the nonmarket spaces of everyday life where it lives." When—or if—there is resistance to the incursion, corporate success will depend on an ability to outpace or outspend opponents, including with "tactics that aim to deplete its adversaries' time, money, and grit." When Google's camera-equipped cars began roaming the planet to capture images that would produce the total map of Street View, there was much legal opposition and social protest—including an incredible story of English villagers who physically blocked a Google car from entering their village![106] Google paid a few fines, but the project continued at such an

intense pace that by the time any serious legal opposition could be made, the incursion was largely complete.

Next there is habituation. Whether due to excitement, resignation to the idea that lack of privacy is the cost of connectedness, or a sense of inevitability, people become users of the new experiences provided by the incursion. Street View became widely used, and as new "dependencies develop . . . it becomes more difficult for individuals and groups to complain." Following habituation, there is adaptation. If there is enough legal or regulatory blowback, some minor tweaks are made to produce "superficial but tactically effective adaptations that satisfy the immediate demands of government authorities, court rulings, and public opinion." Facing pushback on privacy violations from many countries, Google Street View was modified by "blurring faces, instituting opt-out measures, and reducing camera heights."[107]

Finally, once any serious opposition to the seizure of new territory is successfully rebuffed, once dependencies are established and minor adaptations are made along the way, there is redirection: "The corporation regroups to cultivate new rhetoric, methods, and design elements that redirect contested supply operations just enough so that they appear to be compliant with social and legal demands."[108] In the case of Google Maps, the redirection process was partly a campaign of reputation management: a few apologetic blog posts about privacy breeches along with some new disclaimers. More importantly for the company, the ground-plane for future incursions had already been established. In the case of Street View, future incursions are likely to include invaluable commercial opportunities within a yet-to-exist unified platform of virtual and mixed reality (i.e., extended reality) experiences that overlay the fully mapped physical world. Google blended public geographic resources with its enormous reservoir of locational and behavioral data: "In other words, data compiled through public investments . . . are augmented with data taken from a unilateral transfer of surplus behavior and decision rights. The composite results are then classified as private assets."[109] This is neoliberalism at its best.

In this context, UX design and user-interface design appear to invert the underlying logic of user-centered design: centering the users for their own good. Coaxing a user into a digital experience simply to extract behavioral data seems a lot like manipulation. The classical concept of a free market

assumes both parties know what is being exchanged and at what price. This exchange model evaporates in the Dispossession Cycle. UX design that does not make its terms of engagement clear and explicit is stripped of any claims of having "empathy" or of improving the human experience.

Incursions into privacy and designing against users' best interests are not aberrations of the ways that platforms do business in the digital economy: "it becomes clear that demanding privacy from surveillance capitalists or lobbying for an end to commercial surveillance on the Internet is like asking Henry Ford to make each Model T by hand. . . . How can we expect companies whose economic existence depends upon behavioral surplus to cease capturing behavioral data voluntarily?," asks Zuboff.[110] All of these issues pose enormous challenges for democratic oversight. In many ways, data-collecting and platform businesses are simply living up to an ethos of *disruption*. Unfortunately, doing so includes disruption of laws, public trust, and, perhaps more arguably, some basic ethical principles of public and private life.

Capital Is Dead

In this data and platform economy, many see a quickly evolving new mutation of capitalism. McKenzie Wark goes further, provocatively asking, "What if capitalism is already dead?" What if we're just hanging onto the idea that capitalism has changed but is still here simply because its logics are so deeply ingrained in our worldview that even critics of capitalism can't imagine it being over. If capitalism is over, then what is *this*? We may not yet have a name for what we are experiencing now, but Wark is clear on one point: "it is something worse."[111]

Wark suggests a new conception of class structure in the "information political economy" made possible by the global architecture of digital networks and platforms. Those who own, operate, and control the networks are the *vectoralist class** because they control the *vectors* of power and influence. The *vector* is an abstract concept for "the infrastructure on which information is routed, whether through time or space."[112]

Working in tandem with but also exploited by the vectoralist class is the *hacker class*.* "The hacker class produces new information[, which is] whatever intellectual property law recognizes as new." Wark stresses how

this creative labor is different from other kinds of labor, particularly in how the hacker utilizes time. Turning old information into something new can sometimes happen very quickly, while at other times it is tedious. “It’s a strange kind of production. . . . It happens when it happens, including time spent napping or pulling all-nighters.”[113] The hacker class—designers, writers, digital media producers—jams out creative and cultural novelties, producing “differences” that require the vector for distribution, exchange, and revenue.

As in Zuboff’s conception of a surveillance economy, in Wark’s information political economy the capitalist class of producers is subordinate to the new vectoralist class. “The vectoralist class doesn’t need to own the other forces of production anymore. Apple and Google don’t actually make their own products. A sizable chunk of those they directly employ are not workers but hackers, people who come up with new information, whether of a technical or cultural kind, to be incorporated into products whose manufacture can be tendered out to a subordinate class of capitalists.”[114]

Based on Wark’s analysis, we can say that the specialized forms of labor that fall under the category of design are clearly done from the hacker class position. Although this form of labor may sound more pleasurable and stimulating than factory work, neither the hackers nor the workers own their means of production or their output. The hacker, inevitably, sells the rights to what was produced, which makes for employment scenarios in which targeted objectives rather than time commitment are the measure of contribution and, to varying degrees, compensation. It also makes for different models of employment, in which encouraging or coaxing out the valuable *hacks* becomes one tactic for exploitation. This may explain why design studios so often have ping-pong tables and other toys or paraphernalia for play or relaxation. “Hackers can’t be managed like farmers or workers; they are not the same as either class. There’s no relation between the units of labor time and the units of value produced. Something cooked up on the spur of the moment might have enormous value. Long hours of slog might end up being for nothing. Being exempt from routine work is not really all that glamorous in either story, as it just brings uncertainty, frustration, pressure, and (for some) madness.”[115]

Wark hits on the precarity of work in today's creative professions that stems from the oscillating and unpredictable pace of successful (profitable) creative output. The hacker's personal connection to competitive creativity makes for a new kind of class subjectivity. The precarity of this work is entwined with the hustle culture that permeates design practice (*Do what you love!*). Being busy becomes a virtue. Outward expressions of one's passion for work and dedication to the corporate cause—expressions of *toil glamour**—further valorize the designer's lifestyle of labor, making it hard to discern what (or whose) self-interests are being pursued.[116]

Conclusion

Contemporary critics of capitalism point toward two competing forces that indicate that the great gobbling-up machine may soon *gobble up* itself: climate change and the automation of labor. Authors and researchers from a range of disciplines see these two forces as the primary pressures on all social, political, and economic activities in the twenty-first century. The question remains whether capitalism can be reformed (and democratic political institutions updated) or the system will be so disrupted that change will be forced upon it—maybe even to the point of collapse. Between climate change and automation, humans may become expendable to the planet and its owners.

In a personal conversation, the late Erik Olin Wright once shared with me his bumper sticker ethos: "You can never relax." In a brilliant way, Wright's pithy statement reflects both sides of an ideological struggle. On the one hand, capitalism, particularly in its globally competitive, digitally paced contemporary version, assures all participants, owner and manager and worker, that they must keep working and innovating—always faster—or the competition will eat their lunch. On the other hand, for those working toward egalitarian and emancipatory social change, the struggle for transformation is also constant. "After capitalism" is both a horizon and a journey. It requires active work and operates in a constant state of becoming—the becoming of a different world. Fortunately, design is world making.

5
Design’s End of History

Utopia, or . . . ?

It should be said clearly that none of the critiques of capitalism laid out in the previous chapter would claim that capitalism alone is the sole root of historical or contemporary social problems. What these criticisms *do* claim is that capitalism is a powerful, hegemonic structuring of society—an ideology translated into political and social institutions that reinforce its economic model and reverberate through cultures. The ideology is perpetuated both directly and indirectly through institutionalized patterns of knowledge and practice, social logics, and human subjectivities in ways that extend beyond the supposedly separate realms of “economy” and “politics” or even “culture” and “private life.” As such, capitalism affects the structures, logics, and subjectivities of all disciplines, including design.

Medical doctors in a capitalist society are not wholly defined by capitalism. They are motivated, in part, by the prestige that comes from being able to heal, the desire to be of service to their communities, and a genuine desire to help others. However, practicing medicine in a capitalist society, particularly in a health care system shaped by free market policies, is forged within social relations between patients, care providers, and insurance companies that operate on time and cost calculations for treatment or preventative measures and are affected by the fact that doctors often earn high salaries that grant them social power (although the growing costs of medical education may make that salary calculation less desirable).

Similarly, design has existed, does exist, and can exist outside of capitalist logics alone. With or without a political economy defined by profit, accumulation, and growth, designers may desire to produce material and experiential forms that can make everyday life more functionally effortless, more pleasant, more personally distinct, or simply more interesting and unique. Designers might still work for the rewards of peer

recognition, for the challenges and satisfaction of working in applied creativity, and, of course, for vanity's sake! Practicing design in a capitalist society is affected by the logics of competitive advantage, generating difference for the sake of ownership, generating novel, new (*newer!*) functions and images of lifestyles and doing so in heated competition with other designers.

Design finds itself at the very heart of critiques of contemporary capitalism. The ongoing erosion of informed public and democratic discourse is abetted by the distractions of social media, attention-hungry gadgets, and the flood of media-tainment that generally perpetuate social norms of the hierarchical and transactional logics of capitalism, including classism, racism, sexism, and the normalization of poverty in a society that believes itself based on merit. This false notion of separation—economy here, politics there, culture somewhere else—shapes the malaise of advanced Western capitalist societies. A resigned helplessness pervades, in which it appears there are no alternatives other than those we can click, like, buy, consume.

Today, design as innovation works to generate ever new, marginally different product and service systems. Design as "branding" marks ownership of the "experience" of consumption. Designers have seized on the value they can contribute to these combined activities, hacking out endless new possibilities of branded product-service experiences, both material and digital.[1] "The new object is called 'experience,' in the world's most modern sense. Experience is directly attainable. It is definite and cumulative, where happiness is ambiguous and pleasure evanescent."[2]

Yet the contemporary focus on innovation is generally absent any concerted ideological objective other than more, newer, faster, and sometimes cheaper (sometimes not). "Simple and intuitive" has become the user-centered design mantra for everything from robotic vacuum cleaners to banking apps—typically without much question about the larger political-economic impacts of these things. Without acknowledging or even recognizing their ideological origins, many design practices continuously reproduce and strengthen a system whose negative consequences are not symptoms of a temporary disorder but signs of the system's regular and predictable outcomes. Free-market capitalism is ostensibly all about rationality, efficiency, and choice; "it needs no ideological justification," the

argument goes, “because its success is itself sufficient justification.” Yet the “very notion of capitalism as a neutral social mechanism is ideology (even utopian ideology) at its purest.”[3] The question, then, is whether we buy into the ideology or not.

Design practices that don’t actively work against the structures and norms shaped by this unspoken ideology only perpetuate them, making the asymmetries stronger. This is “business-as-usual design” (BaUD). Unfortunately, as highlighted in the critiques of capitalism discussed earlier, business as usual simply isn’t working. It’s not working for the vast majority of people on this planet, and it’s not working for the planet. If design is capable of inciting systems-level change, if design is capable of initiating social change in ways both large and small, it has to think and act ideologically. It must acknowledge the situation, determine alternatives, and make pathways to achieve them. Designers can transcend the structures that limit design’s change-making actions, even as the limits of social possibilities remain constrained by state and legal apparatuses. Like any design challenge, doing so will require framing the problem carefully to intervene effectively.

Building on the criticisms of capitalism explored in the previous chapter, this chapter is a diagnosis and critique of capitalism’s sway over the structures, logics, and subjectivities of design. Critique is not simply a negative operation, however; it can create new possibilities for positive thinking and action.

Practices of Capitalist Design

The practices of a discipline animate its institutionalized patterns of knowledge and power.[4] Design is shaped largely by its practice. Obviously, design practices vary widely, from the different industries in which designers operate to the national or regional differences affected by cultural, historical, or legal variances. This chapter uses the structural and logical entanglements of design and capitalism as a diagnosis and critique of design practices in both the industrial, object-centric paradigm of modern design and the contemporary digital, experiential, cultural economy of twenty-first-century design—both of which are in operation today.

Critique of Modern Design (in the Industrial Era)

As ecological concerns about the destructive forces of consumerism were influencing a generation to work against its unchecked expanse, the Austrian American designer Victor Papanek's undressing of the design professions in *Design for the Real World* in 1971 is still a perennial favorite with design educators and students.

> There are professions more harmful than industrial design, but only a very few of them. And possibly only one profession is phonier. Advertising design, in persuading people to buy things they don't need, with money they don't have, in order to impress others who don't care, is probably the phoniest field in existence today. . . . By designing criminally unsafe automobiles that kill or maim nearly one million people around the world each year, by creating whole new species of permanent garbage to clutter up the landscape, and by choosing materials and processes that pollute the air we breathe, designers have become a dangerous breed. And the skills needed in these activities are taught carefully to young people. . . . [D]esign has become the most powerful tool with which man shapes his tools and environments (and, by extension, society and himself). This demands high social and moral responsibility from the designer.[5]

What Papanek advocated was an approach to design that would work within local conditions and resources, make use of local materials, traditions, and techniques, and involve local populations in the design and production process—a radical departure from the top-down industrial production model. Nearly everything he criticized about design at the time is sadly still true—in many cases, worse. Papanek is generally credited with making "sustainability" a disciplinary concern for design. The sustainability movement within design has made some important strides. New and more sustainable materials, efficiencies in production processes, and the "cradle to grave" life-cycle assessment of products—a method for assessing environmental impacts associated with all stages a commercial product's life cycle—have given designers inroads to shrinking the environmental damage of products through design. Yet we have also witnessed the boomerang effect—when policies, programs, and efforts aimed at the

mitigation of climate change actually result in the opposite. We can't simply "shop" for an end to climate change. We need to transform *consumption* on all levels.

We are still witnessing what appears to be wholesale plundering and destruction of planetary ecosystems—a trajectory that could bring everything, not just the economy, crashing down. The "design sciences" movement of the 1960s and 1970s, which approached design as a technical problem with measurable facts related to ergonomics, material sciences, visual perception, and cost analysis, seemed to promise a pathway for design to help undo this *defuturing* at its core. Unfortunately, this movement generally folded into the privatized, commercial technology sector, essentially thwarting the efforts toward a truly global and humanitarian-focused design. Instead, human-centered design and, later, design thinking became largely co-opted for consumer-oriented design. To truly address the ecological crisis, design must go much further than helping to implement technological improvements. There must be concerted effort to transform subjectivities and ontologies (ways of being) to create any kind of ecological sustainability among "rich" societies that are fully adapted to unsustainable consumption models.

Papanek's critique was not only about ecology, and, of course, he was but one critic. Many criticisms, both general and specific, have been made against design practices for approximately 100 years, since the time that terms such as *graphic design* and *industrial design* were coined.

Here, we'll focus on three main points. Each point concentrates on ways that design practices, operating under the logics, structures, and subjectivities of capitalism, deplete three kinds of "commons":[6]

1. *Natural commons*: the shared planetary resources of all living things on Earth
2. *Commons of internal nature*: the interior lives of human subjectivities and their conditioning by social structures; the body itself, including physical and mental health
3. *Commons of humanity*: the shared experience of local and global commonality and community; respect of cultural distinctions and differences

The criticisms discussed here particularly link design to the critique of capitalism.

Consumerism & the Depletion of the Natural Commons

Design creates production and consumption patterns that exacerbate environmental destruction and dismal, dangerous labor conditions.

This may be the most common and obvious criticism of design. Designers spend their careers giving rise to endless new "forms"—both literal and symbolic products and lifestyle identities—persuading people to buy them, buy new ones sooner, and throw away the old ones faster. Designers make choices about materials that can contribute to how the extraction of raw materials creates environmental problems in both the depletion of natural resources and the enormous piles of discarded stuff littering the planet, choking the atmosphere, clogging the oceans, and contributing to mass extinction of species and rapid loss of biodiversity. This is the depletion of the "natural commons" of our shared world.

Designers also contribute to material and design decisions that prevent products from being durable and/or recyclable. Apple products are a prime example. As laptops have become smaller and lighter, many custom components are standardized for only a particular product or model, making repair nearly impossible. Apple iPhones and iPads are deliberately designed with no screws or hinges and thus with everything sealed inside. This tight control over the component assemblies is a design feature—not a bug—that contributes to a rapid cycle of replacement over repair. This cycle applies to software as well as hardware. Apple is regularly targeted and sometimes successfully sued, fined, or otherwise regulated over "product obsolescence" triggered when the company releases operating-system updates that deliberately reduce performance on older versions of their devices, "thus accelerating phones' substitution."[7] Designers working on apps, games, digital media, and other software that require the latest operating system are similarly accelerating the functional lifetime of devices. Demanding software upgrades eventually demands hardware upgrades.

Globalization of the labor market has sent many manufacturing jobs to Asia to seize on lower wages and less regulation of both environmental contamination and worker safety. In addition to ecological problems,

designers working with or for companies engaged in these practices are contributing to the degradation of labor conditions in the global sweatshop. For example, both Nike and Apple have faced significant criticism for the labor conditions of workers in their factories in China and Southeast Asia.[8] A report on the garment industry issued in 2019 surveyed the wages of workers around the world to determine if they were receiving living wages according to the costs of living in their own countries. The survey included companies such as Nike, Adidas, Puma, and Amazon. "Of the 20 companies surveyed, 19 received the lowest possible grade[;] . . . they produced no evidence that any worker making their clothes was paid a living wage anywhere in the world."[9]

Global indentured poverty is also a gendered issue. Some reports indicate that approximately 85 percent of the world's sweatshop workers are women. And they are predominantly young, most of them 15 to 25 years old.[10] Beyond the grueling conditions and 20-hour days, some of these women are forced to take birth control and routine pregnancy tests to avoid the cost of maternity leave and face routine gender-based violence in physical workplaces, during commutes, and in employer-provided housing. In a study conducted in 2018, women working in Southeast Asia for the global garment industry reported "acts that inflicted sexual harm and suffering" as well as "physical violence, verbal abuse, coercion, threats and retaliation, and routine deprivations of liberty including forced overtime."[11]

At the risk of indelicacy, it should be noted that the designers working for these companies are probably in most cases living fairly comfortably, thus personally profiting from outsourced sweatshop labor.[12] As terrible as the data are, it would be misguided to point blame specifically at any one company, CEO, or designer for being "greedy." All of them are playing moves in the same game—as are consumers. The game itself must be criticized.[13]

Manipulation of the Internal Commons

Design manipulates people, creating desire for things they don't need and don't want or that may be bad for them.

Papanek calls out “advertising,” yet the entire apparatus of product, service, and experience branding can be similarly criticized. Let’s not forget that consumption is not about use value or even symbolic value alone. Focus has shifted to the *experience* of consumption: “we primarily buy commodities neither on account of their utility nor as status symbols; we buy them to get the experience provided by them, we consume them in order to render our lives pleasurable and meaningful.”[14] The term *retail therapy* says it all. The activity of shopping, the experience of making a purchase, the process of evaluating “choices” and making determinations that seem to fit with our internal perceptions of who we are, who we wish to be, and how we wish to portray ourselves—this is the consumption “experience,” perhaps more than actual use of the product. “Happiness has wound up in an ideology of the need for experiences.”[15]

Neoliberal logic would tell us that people should know better; humans are economically rational beings making the best economic choices based on their own calculations of self-interest. Whether you believe that or not, design has proven its capacity to shape consumer desire, and this capacity is often praised as one of its primary business values. The entire category called “lifestyle brands” (which today seems limitless) is a testimony to the affect design generates in motivating people to adopt a prescribed embodiment of certain values, aspirations, interests, or attitudes through a branded image of living. It puts an ideology on sale.[16]

The heavily branded system of objects that constitutes not just “the market” but also nearly every facet of modern life creates a perpetual series of artificially crafted and commercially motivated “choices.” Each of these is supported by intense design efforts to create new, previously unknown desires in the attempt to transform a basic human desire for individual identity into a narrow selection of prescriptions.

Some have argued that “the thesis that consumers are the unwitting pawns of corporate sales strategies” is misleading. Instead, “consumer behavior is motivated by the desire for emulation and differentiation, and that consumers want to create visible distinctions between large social groups or classes, and, within such groups, more subtle distinctions of individuality.”[17] In other words, identity-targeted branding is driven by consumer demand, not the other way around. No doubt, desire for differentiation and individuality stems from humans interacting in complex

networks of sociality. But at the point that this desire for individuation becomes a commercial prospect (*opportunity!*) to be capitalized upon, the seizing actors—corporations and their designers—take it upon themselves to translate a human desire into a consumer desire. This encroachment on the "internal commons" of human subjectivity—as commonplace and banal as it may seem within capitalism—is a politically and ethically charged activity. Yet this activity is plainly the everyday function of much design.

Exploitation & the Commons of Humanity

Design exploits and therefore perpetuates social hierarchies.

Design exploits, often unintentionally, by working as part of a system of class exploitation. Design exploits, sometimes intentionally but other times through implicit bias, for the gain of its clients in the case of perpetuating and profiting from social hierarchies.

Operating from *contradictory class locations*, design is a specialized form of labor that benefits from and perpetuates class hierarchy. Through its relationship to authority, design shares in the power to direct the forms and conditions of labor by others. For example, designers of consumer electronics feed the system of hierarchical class relations and labor exploitation that make up global supply chains. Because it is in the owners' class interests—the authority partially shared with design—to achieve the lowest possible labor and production costs, design is generally obligated to go along with this system of class relations. At times, designers may even contribute to "innovations" in production efficiency that exacerbate poor labor conditions for others. Of course, there are also legions of other forms of laborers with whom design collaborates or directs—copywriters, web developers, printers and fabricators, project managers, office and administrative assistants, support staff for the running of a design studio—who can also find themselves stratified by class hierarchies, directed and supported by design's authority.

Design has also perpetuated social hierarchies, including those of ascriptive statuses, by normalizing hierarchies both symbolically and functionally. Sometimes this is done intentionally. In an oft-cited essay from 1980, the American political theorist Langdon Winner analyzes the racial discrimination inscribed in the design of the infrastructure of New York

City. As the city's planning commissioner during the postwar boom years, Robert Moses has been hailed as the "master builder" of New York City in the twentieth century. When designing the parkways that provide access to beaches on Long Island—a public amenity for city dwellers to beat the summer heat—the bridges were designed too low to allow access by city buses. This was a deliberate maneuver to keep low-income residents (primarily racial minorities) away from the beaches and provide a barrier of exclusion for car-owning (thus wealthier and primarily white) citizens.[18]

Sometimes, and perhaps more often, the social exploitation committed by design is done unwittingly in the application of implicit bias. Designers create subjectivity through symbols and in the use of things, which can perpetuate stereotypes, stigmas, racism, sexism, homophobia, as well as able-ist, bigoted, or other socially detrimental viewpoints. An editorial in *Print* magazine in 2015 reads: "Racism in the United States, especially aimed at African Americans, was so institutionalized prior to the 1970s that the most commonplace commodities, including coffee and children's books, blithely used stereotypes on packaging and book covers. It is more than likely these supposedly incidental depictions, neither deliberately cruel nor malicious yet decidedly rooted in derisive images from the slave era, colored the perceptions of many white adults and children."[19]

Similar criticisms have been directed at the ways designers of products, visual communications, and fashion have promoted restrictive or binary gender roles, sexual objectification, and misogynistic perspectives or have designed functional restrictions into the material culture of everyday life. One need only look at the children's clothing or toy industries to understand how readily gender stereotypes are commodified, even though research suggests that strongly gender-typed toys are less conducive to "children's physical, cognitive, academic, musical, and artistic skills."[20] The "branding" of gendered products for adults also perpetuates binary gender roles and hierarchies or explicitly disadvantages women. For example, the "Pink Tax" is the phenomenon of explicitly marketing or designing goods and services toward female gender types and charging more for them than for those designed or marketed toward male or neutral gender types. Various studies have revealed that "razors, dry cleaning, clothes, haircuts, toys, and a myriad of other goods and services marketed by gender routinely cost more for females than males."[21] Add to this extra "tax" the

fact that women earn approximately only 80 percent of the pay of their male counterparts for the same work,[22] and it's clear that repressive gendered social norms take on real economic weight. Design professions have also been roundly criticized for their lack of racial diversity, gender disparity in leadership and pay, and toxic masculinity in the workplace.[23]

Critique of Twenty-First-Century Design

> Already a literal and uninflected description of daily life sounds like nothing so much as the conspiracy theory of a paranoid schizophrenic: we're surrounded by invisible but powerful forces, monitoring us from devices scattered throughout our homes, even placed on our bodies, and those forces are busily compiling detailed dossiers on every last one of us. They pass the contents of these dossiers onto shadowy, unaccountable intermediaries, who use everything they learn to determine the structure of the opportunities extended to us—or, what may be worse, not extended. We'll be offered jobs, or not: loans, or not: loves, or not: cures, or not. And the worst of it is that until the day we die, we'll never know which action or inaction of our own led to any of these outcomes.
>
> —Adam Greenfield, *Radical Te*chnologies

Today, these ongoing critiques are joined by new criticisms of design's operations in the digital information economy under the logics of *n*eoliberal-*cognitive-cultural-surveillance* capitalism. Many new criticisms are pointed toward UX design because it has become something of a metadiscipline, encompassing and setting agendas for nearly all other practices of design. The field once known as graphic design has expanded to include digital products, services, tools, and other interfaces that privatize or commodify information and communications. For more than 500 years of Western civilization, graphic design gave visual form to language, information, ideas, and the emotive dimension of visual communications, creating many forms of subjectivity along the way. In the early twenty-first century, it appeared that the humanist concerns of graphic design were out of step with design's focused integration into forming technical systems with an ideology of competitive advantage. Many graphic or communication design practices now operate firmly in the space of product and service design.

The fields of industrial design and fashion design often shape material objects that are operated on, by, or through data networks—typically

linking products to services, while collecting behavioral data for purposes beyond performance improvements for the user (Zuboff's "behavioral surplus"). These shifts blur distinctions between industrial-era, object-centric disciplinary divisions (e.g., graphic, industrial, or fashion design). The combination of increased commodification of information and the data-driven mutability of material objects has contributed to a prevailing user-centered mindset, which has been instrumentalized to squeeze value out of branded experiences, transforming experiences into commodities. Design within a postindustrial digital economy finds its highest forms of value in the "innovation" of "brands" and "experiences."

Enclosure: Platforms as Consumption Ecosystems It would be difficult to live and work in the contemporary world without the devices and services that make up constant connectivity. Whether you're working in the gig economy as an Uber driver, running a multinational business, working as a designer for hire, or even (somehow) avoiding capitalist work altogether but still maintaining communications with distant friends or relatives, you are likely networked within vast systems of corporate-owned and controlled communication systems, interfaces, and devices. Once you're entangled in this web of digital networks, you are part of the platform economy—including its various surveillance mandates.

> In the end, the tendency of major platforms to grow to immense size thanks to network effects, combined with the tendency to converge towards a similar form, as market pressures dictate, leads them to use enclosure as a key means of competing against their rivals. If this analysis is right, then capitalist competition is driving the internet to fragment. There is no necessity to this outcome, as political efforts can stall or reverse it; but within a capitalist mode of production there are strong competitive pressures towards this end.[24]

Carefully designed experiences that seamlessly unify products and services reward users who connect in multiple ways to one platform—for example, connecting your calendar in your phone to your thermostat. To get the most out of these systems, you must be all in. Designers help create the product-and-service ecosystems that lure users, sometimes through free services, into the enclosed space of the platform. The digital products and

services sector may be incredibly lucrative—more than physical products alone are; however, physical products are often the vehicles for a new kind of “enclosure.” This enclosure into intricately connected product–service ecosystems limits choices and makes opting out nearly impossible.

Different economic strategies result from the same underlying model. For example, Apple makes more money off its services than off any product category. However, it still receives considerable revenue from its luxury hardware: computers, tablets, smartphones, in-home devices, and wearables. For Apple, the tight integration of products and services makes it difficult to leave Apple’s ecosystem. You’re using the company’s proprietary operating system, iOS, so you’re bound in certain ways to its limited operability with other systems. This restriction is by design, of course. Google, in contrast, makes the lion’s share of its revenue through advertising. Operating on the logic of an advertising platform model, its ecosystem of digital services is centered on extracting as much data as possible from its users. In the company’s offer of different—mostly “free”—digital services that seamlessly interconnect with various aspects of your everyday life, its goal is to make leaving Google’s ecosystem difficult. In both cases, you are bound up in a highly limited set of choices for the products or services you can use or purchase. Enclosure also takes place when public services require the use of a smartphone, specific operating systems (your “choice” comes down to Apple or Android), a third-party app (that charges a service fee), and even proprietary platforms such as Facebook (which extracts and sells your personal data). During the COVID-19 pandemic, it was troubling to witness public-school districts relying on Google’s platform of tools (Meet, Classroom, Gmail, etc.) for remote learning, thus entangling children in the data-surveillance paradigm. What might become of data captured from these children’s virtual-classroom interactions or their academic performance? Will these data become future “behavioral surplus”?

Once you’re enclosed in the ecosystem, it’s difficult to escape.[25] Designers who have made for these seamless connections are helping to ensure that future opportunities for data extraction can be made through the establishment of this enclosure.

Extraction: Microtransactions, Behavioral Data, & Attention In transitioning from an economy of objects to an economy of experiences,

designers have joined forces with interdisciplinary product teams that might include engineers, psychologists, statisticians, anthropologists, and other social scientists. The psychological dimension of designing for the attention of would-be "users" has become paramount as the economic incentives for individualized products and services—including highly targeted forms of advertising—have become enormous. These products and services are enabled by location-sensitive, data-tracking devices at home, at work, in the car, on the body, and that perfectly natural abstraction for a massive infrastructure of data collection, storage, and control, *the cloud*. The metrics of these "interactions" include duration of time spent, clicks, shares, reposts, or responses. In its digitalized form, product design has mapped the factory logics of *Taylorism** onto the psychological space of attention, interaction, and experience—keeping extraction of surplus as the primary goal. Those working in the "cognitive-cultural" labor market are processing data, so they need to be constantly plugged in to ensure their work can capture value.

Although their methods, domains, and interfaces vary, these systems have a common intention: extraction. The digital world is loaded with microtransactions, from "service fees" for using public-parking apps to in-app purchases to expand the experiences of everything from games to grocery deals to Uber upgrades. This diffusion of transactions throughout the digital world is the logical development of neoliberal capitalism's approach to extracting ever more fine and nuanced bits of profit from the nooks and crannies of everyday experience. In her elaboration of surveillance capitalism, Zuboff makes clear that the "deals" we make when engaging with digital service providers are no longer in the same ballpark as the standard exchanges of goods and services of our previous product-oriented market economy. "Behavioral surplus" is an enormous amount of data and metadata that is collected, often without our knowledge or consent, and transformed into a new kind of "futures" commodity—essentially predictions regarding our future actions. Perhaps most familiar is the model Srnicek calls the "advertising platform," which harvests data to sell for hypertargeted and often very reliably convertible forms of advertising. Of the nearly $40 billion in Google's earnings in 2019, more than 80 percent of it ($32.6 billion) came from advertising.[26]

This is ethically slippery territory for designers. They can either work to make these exchanges a clear, explicit contractual offer—"Do you want

this?”—or cloak the extraction in a seamless flow of interactions, where it is difficult to find (or comprehend) privacy notices or to opt out of (rather than opt in to) models for data collection. These practices indicate scenarios in which designers may be working in direct contradiction to the basic ethos of user-centered design, which would focus on the user’s best interests. The extraction of behavioral data from users is commonly used for profitable enterprises other than for product improvements that would benefit the user. Extraction techniques, enabled through UX design, are utilized to maximum use time, attention, and data sharing as revenue potential—with or without the explicit permission of consumers and whether it is believed to be in their best interest or not. This practice has opened a minefield of ethical challenges for designers because the logics of neoliberal capitalism are baked into the microinteractions of our everyday experiences with communication and information systems—the things that make contemporary life what it is.[27]

The extreme revenue streams supporting targeted advertising have created a widely adopted business model that is more than ever laser focused on grabbing attention.[28] And the companies adopting this model are good at getting us hooked. This economics of attention is proving to be bad for us. Anxiety, stress, depression, and even suicide are on the rise, and studies indicate that heavy technology usage may be responsible.[29] Paradoxically (yet somehow unironically), a growing “anxiety economy” has emerged full of new products to help us cope: fidget spinners, weighted blankets, and apps to guide relaxation or meditative practices.[30] What a pipeline!

In many ways, the smartphone, first introduced to the world with the iPhone in 2007, is the perfect object for the neoliberal age. It is a portal for constant extraction of both data and money through endless opportunities for microtransactions, including access by third parties through their own apps. It is a seamless transaction machine—so seamless we often have no idea what’s happening in our own hands. It offers one-button purchases, in-app purchases, and hidden agreements and purchases. The smartphone ensures we are always connected, always available, and always contributing to the endless apparatus of producing social content, generating surplus behavioral data, and opening new avenues for financial extraction. This constant connectivity also makes the smartphone the ideal tool for labor

flexibility by essentially eliminating the distinction between work, personal time, and play. Production and consumption are fused.

In her analysis of surveillance capitalism, Zuboff sees in Apple the seeds of potential for an "advocacy-oriented digital capitalism" that would eschew profiting from "behavioral surplus"—all the extra data that are easy to capture, store, calculate, and transfer within digital interactions. Because of Apple's commitment to "privacy," she sees this possible "third modernity" as a form of capitalism to counter the current surveillance capitalism practiced by Google, Facebook, Amazon, and other megacompanies.[31] In his review of Zuboff's book *Surveillance Capitalism*, the Belarusian American technology scholar Evgeny Morozov takes issue with this distinction, however: "It's an odd argument to make. To contend that the absence of behavioral surplus means that the relationship between Apple and its customers is free from the dynamics of unequal exchange is to ignore all the ways in which Apple regularly pushes its customers around, even preventing them from using third-party repair services. Doesn't that prosaic model of market dominance undermine autonomy as well?" Morozov further questions this highly limited distinction as a false dichotomy: "Would a healthcare system furnished by Apple—expensive but data-secure—be better than a healthcare system furnished by Google—free but data-leaky? Perhaps. But must we really choose between the two? Or can our institutional imagination chart other alternatives?"[32]

Because digital products cost virtually nothing to reproduce, the economic problem facing suppliers lies in how to generate value. Designers are part of this formula, adding seductive, engaging, sometimes humorous or cheeky, or otherwise attention-grabbing visual content to digital experiences simply for the sake of holding our attention just a "micro-moment" longer.[33]

Manipulation: Addiction & Behavioral Modification "In September 2007, 75 students walked into a classroom at Stanford. Ten weeks later, they had collectively amassed 16 million users, $1 million dollars in advertising revenue, and a formula that would captivate a generation."[34] This class was taught by the American social scientist B. J. Fogg and made use of what has become his signature framework. Fogg's behavioral model focuses on how humans are wont to act when three specific forces converge: motivation, trigger, and ability. With the integration of behavioral

psychology into the domain of “persuasive computing,” new precedents for the design of apps and services were put into the hands of an emerging generation of designers, technologists, and entrepreneurs.

Alumni of Fogg’s class have designed products for Uber, Facebook, and Google. Many have started their own companies, often partnering with classmates who would understand and follow the same logics. The class—now often known simply as “the Facebook class” because many of the techniques have proven financially rewarding through third-party Facebook apps—has become the stuff of legends. Many UX and interaction design faculty, students, and practitioners have attempted to emulate its model.

The integration of psychology and design has leveraged an understanding of habits of addiction to create lucrative digital products. The American psychiatrist and neuroscientist Judson Brewer states: “From a psychological standpoint, Twitter taps into our natural reward-based learning processes: trigger, behavior, reward. . . . The harmful side of Twitter comes in the same form. We feel angry at someone’s tweet, our brain screams ‘do something!’ and we instantly send a rage-filled tweet @ that person.”[35] Twitter, like Google, is an “advertising platform.” The company makes more than 86 percent of its revenue from advertising (about $3 billion in 2019), which it can collect based on the network effects of its scale. In other words, Twitter makes more money when people are more engaged on the platform—tweeting, retweeting, commenting, liking—regardless of whether they are sharing valuable information, rage-tweeting, or spewing hateful bigotry. This attention-addiction model of economics clearly has negative social downsides.

The Israeli American author and investor Nir Eyal, a graduate of Fogg’s class, has become a critic of these methods when not practiced within an ethical framework: “Let’s admit it, we in the consumer web industry are in the manipulation business. We build products meant to persuade people to do what we want them to do. We call these people ‘users’ and even if we don’t say it aloud, we secretly wish every one of them would become fiendishly addicted. . . . Unfortunately, our moral compass has not caught-up with what technology now makes possible.”[36]

Eyal developed the Manipulation Matrix, which prompts designers to ask two simple questions to understand what role they are playing when developing new digital products and services: “Will the product or service

materially improve the user's life?" and "Would you use it yourself?" Depending on the answers to those two yes/no questions, designers can identify themselves as being a facilitator (the digital product or service improves the user's life, and you would use it yourself), a peddler (it improves the user's life, but you wouldn't use it yourself), an entertainer (it doesn't improve the user's life, but you would use it yourself), or a dealer (it doesn't improve the user's life, and you wouldn't use it yourself).

The American venture capitalist Roger McNamee was an early investor in Facebook and mentor to Mark Zuckerberg. Despite his involvement in the Silicon Valley story, he sees the negative impacts of an industry he helped create. According to McNamee, "if you want to fix this [mass addiction problem] as quickly as possible, the best way would be for founders of these companies to change their business model away from advertising. . . . We have to eliminate the economic incentive to create addiction in the first place."[37] This change seems unlikely to come about on its own.

One of the most blatantly unethical approaches of manipulation in contemporary UX design is the practice of *dark patterns*. The term was coined by the British designer Harry Brignull in 2010 to bring attention to this troubling development that boomed in the early days of web-based e-commerce.[38] A dark pattern is a misleading or otherwise deceptive decision made by UX and user-interface designers to intentionally trick users to do things they don't really want to do—often by exploiting known aspects of human psychology. Brignull has compiled a library of specific dark-pattern techniques to "name and shame" the practice.[39] Although most designers are most likely against such practices—in theory—examples can be found all over the web, in mobile apps, and in digital interfaces built into products. In Brignull's collection, he has identified many big names using these tricks, including Microsoft, Skype, LinkedIn, TicketMaster, Adobe, *Wired* magazine, and, perhaps less surprisingly, Facebook and Amazon. By deliberately misleading and exploiting users, at its core this practice is antithetical to user-centered design's credo of centering the user's best interest. Nonetheless, it fits squarely within the logic of neoliberal capitalism and has thus added an explicit method of manipulation to the contemporary designer's toolbox.

The still-growing market for wearable technologies and the Internet of Things (IoT) opens new frontiers for design, in which product-and-service ecosystems become diffused and comingled with the frontier of the human body itself. Yet this emerging horizon doesn't shift the agenda so much as reveal new territory for practices of enclosure and extraction as well as new opportunities for behavioral manipulation. According to Srnicek,

> Wearable technologies are another major element of consumer IoT. Nike, for instance, is using wearables and fitness technology to bring users onto its platform and extract their data. While all these devices may have some use value for consumers, the field has not been driven by consumers clamouring for them. Instead, consumer IoT is only fully intelligible as a platform-driven extension of data recording into everyday activities. With consumer IoT, our everyday behaviors start to be recorded; how we drive, how many steps we take, how active we are, what we say, where we go, and so on. This is simply an expression of an innate tendency within platforms.[40]

There's little reason to think that the embodied orientation of UX design will adopt different practical approaches than those deployed through screens.

In today's tech-manic and venture-capital-fueled landscape, "there is virtually no idea so pointless, stupid or offensive that some startup somewhere will not invest resources in trying to develop it as a commercial product."[41] Designers have shared authority in deciding not only what gets produced but also how, so responsibility falls on designers for the implications and outcomes of these decisions. Yet despite increased calls for designers to act "ethically," the complexity and sheer variety of new design scenarios make most ethical standards impractical within the structure of capitalist logic. Google's motto "don't be evil" must be seriously questioned when it is paired with a business model based on tracking users as universally as possible.[42]

The American computer scientist Tristan Harris is also an alum of Fogg's "Facebook class" and has become one of the most vocal insider critics of technology companies' manipulative design practices. He is formerly Google's chief design ethicist, a position he created. Harris claims that

never before in history "has a handful of people working at a handful of tech companies been able to steer the thoughts and feelings of a billion people." He makes these companies' scale of influence clear when he points out that there are "more users on Facebook than followers of Christianity. There are more people on YouTube than followers of Islam. I don't know a more urgent problem than this."[43]

Even though the economic power of a handful of the biggest tech giants is incompatible with the tenets of democracy—among other problems these giants may pose for the rest of the world—they are where much of the action is in the contemporary design economy. Tech companies offer some of the most lucrative and desirable design jobs in the world today.[44] Their successes have paved the way for expanding the roles of UX, product, interaction, and service design in finance, the insurance industry, and health care. And for good reason: designers play the mediating role between what technologies can do and what people can or might wish to do with them. The tech giants may have notoriously small design teams, but they are also the patrons for a much larger cohort of agencies and studios that make their earnings off work directly or indirectly related to the scale and wealth of these companies. Perhaps more significantly, these companies' immense wealth and platform scale gives them near monopoly dominance, which means their every move will have a ripple effect on related disciplines, including design.

Discourse of Capitalist Design: The Logics of BaUD

A discipline's *discourse** determines how people who study or practice that subject use ideas and language to think and to communicate about what they are doing. Discourse shapes and is reciprocally shaped by practice. To criticize design's discourse is to challenge and potentially shift the foundations of its practice. When the logics, structures, and subjectivities of capitalism seep into the logics, structures, and subjectivities of design, design is ideologically relocated from a universal human activity of "world making" to an instrumental set of activities whose purpose and value are defined by their ability to produce competitive advantage. Even without

reference to the hegemony of the larger context from which these logics derive—namely, capitalism—these logics seem self-evident. By explicitly naming those logics, we can question their merits and consider alternatives.

Competitive Advantage & Disruptive Innovation

Business-as-usual design, or BaUD, operates predominantly in the matrix of material-immaterial value creation, specifically for the purpose of producing competitive advantage for business. As defined by Heskett's "value creation theory," design is integral to creating opportunities for innovative strategies that can generate three kinds of innovation: *process innovations*, which can lead to improved quality (or desirability) of products and services; *transaction innovations*, which can lead to transaction efficiencies; and *product innovation*, which can lead to new versions of a product or service (updates) or to entirely new product or service categories. In Heskett's model, all these innovations lead to imperfect competition, or competitive advantage in the marketplace, which results in the destruction of existing products while dynamically creating both new demand and new markets.[45]

Significant innovation—the kind that remakes industries—has been dubbed "disruptive innovation." The term is resonant of the Austrian School's concept of "creative destruction"—popularized by Joseph Schumpeter—to describe the way that new businesses destroy existing ones. Disruption of an entire industry typically also places the new business as the very first in a new territory, which creates coveted monopoly conditions. Lack of competition provides the opportunity to set high prices. Disruptive innovation has become a kind of generalized goal and mantra for many businesses as they seek not only to compete within existing categories but also to create new ones endlessly.

The combined ethos of disruptive innovation and entrepreneurship have helped shape the almost religious fanaticism of start-up culture. Through the start-up process, clever, creative, ambitious, and hard-working young entrepreneurs can "shake up" tired old industries and summon new markets. This can generate tremendous value particularly as return on investment to early supporters (most of whom are probably already quite wealthy given their ability to gamble on high-risk ventures). In the process, the distinction

between real innovation, design, and hype to lure investments gets blurry. The term *exit strategy* is itself striking. Nobody is in this game for long-term commitment or value creation, nor does the game seem to exist for any broader benefits other than the "successful exit."

Although the returns that society receives on the (often failed) economic competition happening among start-ups may be minimal, the gospel of innovation has transformed the image of "work" for many young people today. The allure of entrepreneurship (*Do what you love!*) has drawn many designers into the start-up space. Wealth already accumulated through the process is redistributed by venture capitalists, whose financial power gives them the leverage to shape the ideas they think the world deserves. Along the way, new winners of the start-up game get rich and "give back" by becoming venture capitalists themselves. Many also coach the young and hungry newcomers. This "virtuous circle" ensures that each new cohort of business leaders are fully indoctrinated in the logic of cutthroat competition—a closed loop in which venture capital continually perpetuates itself.

In a scathing indictment of its mythos, the American historian Jill Lepore describes disruptive innovation as "a theory about why businesses fail. It's not more than that. It doesn't explain change." The ideology of manic competition is perpetuated by the venture capitalism that supports it: "Forget rules, obligations, your conscience, loyalty, a sense of the commonweal. If you start a business and it succeeds . . . sell it and take the cash. Don't look back. Never pause. Disrupt or be disrupted."[46]

These are the context and mindset of some of the most celebrated design practices today: *disrupt or be disrupted!* By placing itself at the behest of targeted business needs, by becoming an über-competitive disruptive force, design has remade its relevance while also positioning designers as competitive in a labor market of otherwise stagnant wages. Yet this process has also shown how design is instrumentalized by raw capitalist logics. For designers to exert their efforts toward narrowly defined competitive advantages, putting short-term profit ahead of all else, they are—sometimes knowingly but also perhaps unwittingly—perpetuating an inherently conservative (maybe even retrograde) social agenda. This agenda further reduces social relations to economic or even transactional relations, justifying success or failure as though the world were a level playing field of opportunities. Audiences become users, users become consumers, and

consumers become streams of marginal revenue through microtransactions and extracted behavioral data.

By placing the individual user at the center of the universe, designers perpetuate an ethos of individualized competition, in which each user's unique needs are addressed at nearly any cost. This mentality is at least in part to blame for the erosion of the social contract that has defined democratic capitalist society.

Designerly Ways of Thinking in Neoliberal Logic

> Americans love design most when we're afraid. Just as the Depression enabled industrial design to present itself as the solution to US manufacturing woes, the 2000 to 2002 dot-com crash and 2008 recession, with their long tails, have enabled the rise of a new embrace of design and a new broadening of design's imagined jurisdiction. This time the specific fear is that the knowledge economy is coming for everyone. Bewildered and anxious leaders, public and private, have responded by throwing in their lots with the seemingly magical knowledge-work that is design.
>
> —Maggie Gram, "On Design Thinking"

Design discourse has long been a space of remarkable platitudes (the present volume included). The discourse of *design thinking* takes these platitudes to a whole new level. Design thinking situates design's capacities for research, ideation, management, and the interplay of material, social, and cultural production within the broad context of innovation. For good and ill, designers (and the business world) have recognized the impact of design well beyond aesthetic concerns. By claiming a "seat at the table," presumably in a corporate boardroom, design has positioned itself centrally in the fast-paced and highly competitive contemporary landscape of capitalist enterprise. Several high-profile corporations even now feature chief design officers, indicating the level of corporate responsibility design can play in industries as diverse as finance, technology, and food-and-beverage products.

Without denying the benefits of this situation, which potentially gives responsible designers access to very large levers of global power, we must also honestly assess how design's growing presence in corporate activities within a global capitalist order also represents complicity in all the criticisms just lobbed at capitalism itself. The design-thinking movement, in particular, gave rise to a rapid expansion of design's credibility in business

circles. It is responsible not only for a great expansion of design's domains of work yet also for a great reduction in its purpose. "Disruption" and "innovation" are almost always fully enclosed by the logics, systems, and goals of capitalism in its postindustrial, neoliberal, digital/platform version of political economy.[47] The *problems* framed by design thinking are nearly always reduced to *business problems*. So although design thinking has opened a broader territory for design, that broader territory has paradoxically come with a reduced field of operations.

Design activities direct massive flows of labor, materials, capital, and technological development. Yet to make significant and lasting change within our global political economy, designers need to move beyond the cult of "disruption" and "innovation." Design that operates in the latter way continues to create negative "externalities" both upstream and downstream. When *design problems* are fully inscribed within *business problems*, then *design solutions* can only be *business* solutions.

Conclusion

Has design reached its own end of history? Does the design of experiences, as hybrids of physical and digital interactions with predominantly private, branded enterprises, represent the pinnacle of a century of modern disciplinary practice and theory? Or is design already dead, and what we have now something else—perhaps, as in McKenzie Wark's assessment of the death of capitalism, something worse?

Judging by the enthusiastic and celebratory mood of proponents of UX design and the passionate fervor with which young designers throw themselves at the creation and maintenance of novel brands, it seems possible that design has reached the end of its own history. What else could design do? Perhaps disruptive innovation for the sole purpose of competitive advantage has become design's final state. Perhaps the only work that remains is perfecting and continually applying the methods of competition to whatever untouched spaces remain, so they too may be captured by the structures, logics, and subjectivities of capitalist design—so that all human experiences are owned, crafted, and monitored for new value potential.

To say that design's primary function is to operate in service to business, that its primary value is measured in profits, that its primary purpose is to create new and novel experiences for private gain is to swallow whole the ideology of capitalism as the dominant and *only* organizing principle of society, not just one economic model. Not so long ago, designers bemoaned that design was an essential social activity, so much more than making beautiful or pleasing things. What began as a rejection of design's limited role in the "styling" of objects has returned full circle. The designing of experiences similarly blends function with style, in no particular order. Perhaps the primary difference is that the "function" of the current designed experience is split between the functionality gained by the user and the functional extraction of value that the experience makes possible.

And yet: the shrinking returns of capitalism's productivity in terms of wage stagnation, public debt, and private debt; the innovation pessimism that sees contemporary innovation limited in terms of real social impact, limited by its exclusion of many, and resulting in paltry gadgets that are nearly instant garbage; the radical and growing inequality within wealthy nations and among global regions; the declining life expectancy and mental health disasters in the wealthiest nations; the way that megascaled private wealth and corporate interests have upended meaningful decision making, public and personal, in capitalist democracies—these are features of a political economy aided by a design that transforms opportunities into goods, services, and experiences while turning humans into users and consumers. If modern design's promise was to domesticate technologies and materials into new cultural forms for better, smarter living for all—well, it seems that dream has been foreclosed.

In 2020, a novel coronavirus pandemic infected the world, killing millions. All countries struggled to contain the virus, with varying results in the health and death rates of their citizens. The United States of America, the wealthiest country in the world, beacon of liberal capitalism, and the nation that spends by far the most on health care, seemed to fare the worst. Much has been said about the lack of leadership from an inept federal government—probably all of which is true. Yet, more generally, the pandemic laid bare the structural fallacies of a free market.

Capitalism can't cure a pandemic. Capitalism can't fix poverty. Capitalism can't solve global warming and climate change. Capitalism can't

undo features central to its operation: exploitation, extraction, manipulation, and the detrimental social pathologies of classism, racism, and sexism. The natural trend for a socioeconomic order built on competitive individualism is the dissolution of community and communal well-being. “Communities are being actively dis-imagined by their own people. As a result, Western capitalism is starting to resemble something like a bomb shelter.”[48] Designers rank chief among those communities whose work, whose best intentions, and whose creative energies are active in that *disimagining* and *defuturing*. The post-COVID world may look like the widespread diffusion of “bomb shelters”: home offices, private learning pods, and socially distanced living for those who can afford it.

However, perhaps similar to the declaration of capitalism’s “end of history,” this vision of the future is also premature. Design is implicated in every domain of human productive activity. The easiest argument to make in response to claims that the system can be changed is that if companies don’t like the designers they have (either in house or as consultants), they will simply go get another group of designers to do their bidding instead of working to change the system. This easy resolve in favor of the status quo signals the need for a disciplinary sea change. Design can’t remain instrumentalized and without a prevailing credo as its reason for being. Design must find in its own knowledge, methods, tools, and philosophy the courage to chart new territory—not when it’s convenient or when the money’s good, but now and always.

Although deeply destructive in many ways, the historical era of capitalism has produced a world with enough abundance to sustain the planet’s inhabitants. Now that we have achieved this abundance, we must “wonder whether society’s wealth is best spent financing an advertising arms race.”[49]

What if design and designers worked on something else? What if design, as a discipline and an interconnected set of professional practices, set about to intentionally and collectively create alternative ways of being in this world? What if design were integrated in the making of a more just, equitable, postcapitalist world?

II

Alternatives

When you realize that capitalism, once, did not exist—either as an economy or a value system—a more shocking thought arises: it might not last forever.

—Paul Mason, *Post*capitalism

6
Alternative Designs

Design Hybridity: Capitalist Design Is Only the Tip of the Iceberg

Across history, there have been design movements, practices, and individual designers that have been directly *anticapitalist*, antagonizing the capitalist system, or *noncapitalist*, operating outside the basic logics of capitalism. These "alternative designs" have created ongoing tensions and dialogue with capitalist or commercial forms of design. There are also historical and contemporary precedents of design practices that demonstrate elements of what *postcapitalist design* might be.

This chapter offers an alternative telling of the relationship between design and political economy from the middle of the nineteenth century to the present. We'll look at alternative designs from these categories: anticapitalist, noncapitalist, and postcapitalist. Although limited, this brief survey of case studies helps identify features of this taxonomy, providing perspectives (and I hope inspiration) on the many ways designing can be done beyond capitalist structures and logics—even within capitalist societies. These alternatives demonstrate that capitalist design is also just the tip of the iceberg.

Many of these examples come from European and American contexts, or the "Global North." Some come from Latin America, India, or the "Global South." The preponderance of the former is due in part to the fact that Europe and the United States have long been *core* zones of capitalist production, including the various forms of exploitation and symbolic violence they have exerted on other parts of the world. Case studies from the Global North demonstrate forms of design that have worked or are working against the internal social and economic structures aligned with capitalism. Case studies from indigenous and/or local practices of the Global South demonstrate forms of design in tension or conflict with

capitalist modes of production imposed upon people, such as the exploitations of colonialism or colonialism's contemporary equivalent, international "development." These alternative ways of doing things are often attached to a specific social or political movement. We can see in these models the ways that designing can express diverse motivations for social change.

Anticapitalist Design

Anticapitalist design practices represent those that actively work against the capitalist mode of production and/or openly attack it. This attack begins with naming capitalism and capitalist structures and logics as the target of antagonistic activity.

Nineteenth-Century Socialism: British Arts and Crafts

Practically the moment industrial capitalism began, it was met with resistance. The Industrial Revolution brought radical economic and social change to England and shortly thereafter to much of the rest of the world. As factories spewed toxic pollutants into the world and workers were subject to alienating, dangerous, and endless work while living in poverty, owners became wealthy and set rules for the social and political institutions that would protect their interests. As a response to the expanding hegemony of the capitalist system, socialism emerged as an alternative. Socialists emphasized public (rather than private) ownership of means of production and social organization built on cooperation (rather than competition). Early nineteenth-century socialists such as the Welsh textile manufacturer Robert Owen and the French philosopher Charles Fourier argued for a system of social and economic organization based on small collective communities rather than on a market or centralized state. Karl Marx undoubtedly influenced socialist theory and helped make its ideas widely known. In particular, Marx's conception of society as being produced through the dynamics of different social classes struggling for power has generated new ideas, new identities, and in some cases, new political realities.

Perhaps the most familiar anticapitalist episode from design history is that of the British Arts and Crafts movement. Nominally led by the British designer and poet William Morris and the British philosopher John Ruskin, these nineteenth-century designers rejected what they saw as the destructive effects on nature, beauty, and human well-being by the use of machines for producing the objects of everyday life. The Arts and Crafts movement led a return to the traditions of craft and small-scale production, citing a kind of spirituality that connected humans to the act of making. Arts and Crafts designers created books, furniture, textiles, lamps, and many other home objects through laborious processes such as hand-stamping wallpaper and block-printing books. William Morris's Kelmscott Press followed this logic of glorified labor to design and produce highly ornate, limited editions of medieval texts, such as the works of Geoffrey Chaucer. Politics was at the front and center of this movement. Morris was active in various organized socialist activities in England, working in opposition to the economic inequality and degradation of the human condition brought about by industrial capitalism. Describing his position, Morris said, "What I mean by Socialism is a condition of society in which there should be neither rich nor poor, neither master nor master's man, neither idle nor overworked, neither brain-sick brain workers nor heart-sick hand workers."[1] However, "while Morris's ideology was opposed to industrial capitalism . . . he could not account for or effectively fight the increasing attraction of consumption (and shopping) across all classes of modern society."[2]

Anticonsumerism: Critical Theory & the Situationists

As briefly discussed in chapter 4, the Frankfurt School of social sciences emerged in the 1930s and is connected to the development of critical theory. Members of the Frankfurt School studied and critiqued *consumer culture* and the *culture industry* that creates it. Consumer culture "demonstrates the power of capitalism to reproduce the parameters within which citizens of a consumer society live their everyday lives. Consumer culture gives us the tools to express who it is we are, but while doing so it simultaneously reinforces an economic system in which the individual's ability to be free or to choose is, ironically, constrained."[3] These criticisms were directed at a

mass culture made possible by mass media and mass production at a time of expanding consumption and the invention of new forms of desire.

Although the Frankfurt School was an intellectual movement comprising primarily social scientists, their work was influential for creative practitioners, including designers, seeking ways to work against the cultural norms of consumerism. Perhaps the most well-known and bombastic example is the Situationists. An alliance of avant-garde artists, writers, and poets, the Situationists International formed in Paris in 1957 amid the growing consumer culture of the postwar period. They remained an official organization until 1972. The Situationists worked to address the alienation, exploitation, and fetishization of commodities that capitalist society was producing. However, they did so in a way that prioritized the "real" activities of everyday life, conceding that everyday life can be a continuous process of experimentation and iteration rather than the mere following of a particular ideology or script for a lifestyle. Fundamentally, they aimed to eliminate capitalism through the "revolution of everyday life." Their method was the *situation*—a tool for the liberation of everyday life by negating the pervasive alienation produced by and accompanied by the *spectacle* of a culture of consumerism.

The French theorist Guy Debord was a cofounder of Situationists International and one of the most prominent figures of the movement, in part due to his writing and publishing, which included books and journals dedicated to Situationist ideas. One of the most influential contributions by the Situationists is a particular set of methods for creative practices intended to subvert consumer culture. Three of these methods are *détournement*, *dérive*, and *psychogeography*. *Détournement* refers to turning expressions of a capitalist media culture against itself—for example, turning marketing slogans or corporate logos against advertisers or political representatives of the capitalist system. *Dérive* is "a mode of experimental behavior linked to the conditions of urban society: a technique of rapid passage through varied ambiances."[4] The term is literally translated into English as "drift," and the Situationists would use this method of "drifting" through the urban environment in unexpected ways in an attempt to subvert the functional, commercial logics of the city and its extensions into media spaces. *Psychogeography* is "the study of the precise laws and specific effects of the geographical environment, consciously organized or not, on the

emotions and behavior of individuals."[5] These methods have been adopted in various ways by successive generations of anticapitalist creative practitioners.

One notable example is *Adbusters* magazine: "[The] Adbusters [Foundation] is a global collective of poets, punks and philosophers implementing radical design and media strategies to shake up complacent consumerist culture. We're aiming at the stale systems suffocating society: the power hungry forces that leave people and the environment in disarray; the toxic capitalism that creeps into our bodies and imaginations."[6] The magazine has been in publication since 1989 and is produced by the Adbusters Media Foundation, a Canadian-based not-for-profit. Its publishers have produced a range of related projects and events, including an event that kicked off one of the major global protest movements of the early twenty-first century.

In 2011, Adbusters designed, produced, and plastered a poster all over New York City featuring an image of a young woman striking a ballet pose on the top of a bronze sculpture of a bull, the symbol of Merrill-Lynch and a prominent landmark on Wall Street. In the background, barely visible in a cloud of smoke or chemical gas, there emerges a crowd wearing riot gear and gas masks. The bottom of the poster reads simply: "#OccupyWallStreet, September 17th, Bring Tent." Thus the Occupy movement began. It started in New York's Zucotti Park, spread to more than 951 cities across 82 countries, and claimed the slogan "We are the 99%." The Occupy movement brought a new generation into political consciousness and broadcast globally its critique of the intense economic inequality of late capitalist society.

Antiracist Anticapitalism: The Black Panthers & Symbols of Revolution

The long history of racial oppression in the United States met with the struggle for political representation and power in the civil rights movement of the 1950s and 1960s. In 1966, two young men studying at Merritt College—a community college in Oakland, California—cofounded a revolutionary socialist political organization that they called the Black Panther Party for Self Defense. Huey Newton and Bobby Seale based the

intellectual orientation of the organization on a Marxist perspective that saw the American Black community as a colony that was being perpetually exploited by white businessmen, the police, and the US government. Their political focus on class struggle and solidarity led them to build alliances with other radical groups across racial divisions—radical whites and Chicanos in the United States—and with revolutionary groups around the world. The Black Panther Party was founded on a 10-point program that "addressed the major issues facing urban black communities with demands for full employment and adequate housing."[7] The final demand summarized the party's objectives: "We want land, bread, housing, education, clothing, justice, and peace."[8]

Economic injustice was central to the Black Panther Party's mission. The Panthers clearly recognized that even as legal forms of racial discrimination were beginning to fade, the economic oppression of Black communities would keep them in bondage. Point three of the party's plan initially read: "We want an end to the robbery by the white men of our Black Community." This was later changed to "We want an end to the robbery by the capitalists of our black and oppressed communities."

The Black Panther Party's activities consisted of two main programs. One was community defense through armed patrols that monitored the police in Black communities. They wore immediately identifiable uniforms on these patrols. Leather jackets and berets represented military discipline while also creating a new identity for Black power. The second was directed at community service, particularly in making Black communities self-sufficient. By 1969, the Black Panthers were serving free breakfasts to 20,000 school-age children in 19 cities around the country every school day. They also operated medical clinics and after-school programs, creating community economies outside the limits and control of a racist capitalist system.

The Panthers needed to communicate with Black communities in the US and radical groups globally to maintain a flow of information on political and social events, build support for the party's activities, and encourage participation in their programs. The party's official news channel began on April 25, 1967, with the first issue of what was initially a four-page newsletter. During a meeting at the apartment of activist Eldridge Cleaver where Bobby Seale was working on the inaugural issue, party member

Emory Douglas offered to help. Douglas had taken graphic design classes at San Francisco City College and had access to the print shop there. He redesigned *The Black Panther* and began to produce it on a web press, which gave him the opportunity to print color images. The front and back covers were used for dynamic graphic images and collages that aligned with the party's messaging. A new issue's front and back covers were also designed to operate almost like posters with imagery that could be readily identified and understood in Black communities. Douglas "realized The Black Panther needed potent images to cut through the high illiteracy rates in poor communities." From 1968 to 1971, *The Black Panther* became the most widely read Black periodical in the United States and was distributed internationally, with weekly circulation reaching 400,000 copies. It continued publication until 1979.[9]

Emory Douglas was the art director, designer, and main illustrator for *The Black Panther*, and he also served as the "revolutionary artist" and minister of culture for the Black Panther Party. In addition to the newspaper, he designed posters and other printed matter for recruitment and to spread the Black Panther ideology to oppressed and militant groups, using imagery to build solidarity: images of bloodied pigs (a dig at the police), "people in stances of active armed resistance, men draped in bandoliers, women holding infants and toting rifles." Through these images, Douglas "branded the militant-chic Panther image decades before the concept became commonplace. He used the newspaper's popularity . . . to incite the disenfranchised to action, portraying the poor with genuine empathy, not as victims but as outraged, unapologetic and ready for a fight."[10]

Through the newspaper, posters, and other printed materials, Douglas generated and circulated hundreds of illustrations, collages, and other designed images that chronicled the Panthers' programs and struggles and created new subjectivities for the Black power movement globally. Meanwhile, the newspaper operated to unify otherwise embattled and (quite literally) segregated communities around the world through a medium of visual communication, steeped in the political motivations of a community imagining and fighting for its liberation.

Autonomous Politics & Anti-design

The postwar period in Italy was a time of rapid industrialization in a country of deep and sometimes ancient traditions. It was a period of political dissensus with many parties, even those holding generally similar ideological positions, struggling for voice, support, and power. During the 1960s, unionized factory workers for Fiat lost confidence in the centralized model of their trade unions. They felt the union leaders were too narrowly focused and not properly addressing the full range of issues facing factory workers. Emerging from this potent milieu came a variety of decentralized, antiauthoritarian, and generally anticapitalist political and social movements identified as *operaismo* (workerism) and *autonomism*. The autonomous movement began, in part, as a result of the failure of communist and other leftist movements in Italy (*autonomia*) but later spread to Germany (*Autonomen*) and France. Today, autonomism exists "as a global network of alliances between occupied social centers and media activists in Europe, Zapatistas and Piqueteros in Latin America, Black Blockers in North America, cyber hacktivists in Japan, and autonomous workers, unemployed youth, students, dispossessed peasants, and urban squatter movements in South Korea, South Africa, and India."[11]



Figure 6.1

Thousands of protestors, some carrying flags with the Black Panther Party logo, at a march in Detroit on September 24, 1971. The crowd was protesting the deaths of two Black teenage boys at the hands of the Detroit police. (Photograph by Richard Sheinwald, courtesy of the Associated Press.)

Autonomous social movements "involve people directly in decisions affecting their everyday lives. They seek to expand democracy and to help individuals break free of political structures and behavior patterns imposed from outside. . . . [T]hey seek to decolonize everyday life." The concept of *autonomy* contrasts "the centralized decisions and hierarchical authority structures of modern institutions." In Italy during the postwar period, autonomous movements built solidarity among factory workers, university students, and activists in the feminist movement—all of whom were dissatisfied with the centralized politics and consumerist culture on offer. The people in these groups wanted "to live lives of their own collective making, not ones determined by decisions in corporate boardrooms and

government ministries. As one observer put it: ‘More than a struggle for a new contract, this has been a rage against work.’”[12]

Although there have been many political, theoretical, and practical variations of autonomism, they nonetheless share “a number of principles that provide coherence.” Members of these movements “see their ideals as a revolutionary alternative to both authoritarian socialism (Soviet-style societies) and pseudodemocratic capitalism. . . . [They] seek to change governments as well as everyday life, to overthrow capitalism and patriarchy.”[13]

Italian Anti-design & Critical Design Contemporaneous with autonomous movements in Italy, a radical period of design blossomed, some of which has been identified as *anti-design*. In Europe in the 1960s, several design collectives formed “out of the disillusionment with modernist ideals that had dominated since the early 1900s.” These collectives viewed design, in particular industrial design, as “having aggravated social and environmental problems.”[14] They established a new kind of political perspective rooted in critiquing the world of consumption, which is largely shaped by design, and they did so through design. Anti-design can be seen as a form of “immanent critique,” a method of cultural discourse that articulates contradictions in the structure of society to suggest possibilities for emancipatory social change (a critical approach associated with the German social theorist Theodor Adorno of the Frankfurt School, among others). This mode of critique values the role ideas can play in shaping society and generating future change and embeds those ideas in designed objects.

Worker strikes, protests, student demonstrations, sit-ins, and feminist demonstrations were prevalent in Italy at the end of the 1960s and in the early 1970s. Universities and factories alike were sites of political speech and action. Several Italian designers “began to view their role in the creation of sophisticated domestic objects as part of a repressive collaboration between government and corporate management interests that reinforced class distinction and fueled commodity consumption for the wealthy.”[15] Originating in Italy, anti-design collectives such as Superstudio, Archizoom Associati, Gruppo Strum, and Gruppo 9999 shared a critical approach to the world of consumption through a combination of publications, films, images, installations, and prototypes that were never intended for mass production. “Rather than design positioned as being in

service to problems, design was used to facilitate active participation through happenings, interventions," and other forms of participation used "as a means of communication and political instrument geared to provoke debate on and around the work." In this facilitation, anti-design practices established specific political positions and "aimed to open an intellectual discourse through design."[16]

This politically charged design movement took place during an economic downturn in which there was little work available for designers. In some ways, the lack of economic opportunity triggered a critique of these social conditions out of sheer necessity. These designers couldn't find work, so they "occupied themselves in other ways leading to experimental practices critical of prevailing disciplinary and commercial orthodoxy. The broader air of protest and political activism prominent during the 1960s also had an influence on creative practice and design culture."[17]

At the heart of Italy's critical design movement, Superstudio was a design collective founded in Florence in 1966. In 1971, Adolfo Natalini, architect and cofounder of Superstudio, announced at a lecture in London that the group would no longer practice design. His now famous declaration included these words: "If design is merely an inducement to consume, then we must reject design; if architecture is merely the codifying of the bourgeois models of ownership and society, then we must reject architecture; if architecture and town planning [are] merely the formalization of present unjust social divisions, then we must reject town planning and its cities . . . until all design activities are aimed towards meeting primary needs. Until then, design must disappear. We can live without architecture."[18]

Instead of producing buildings or designed objects, Superstudio worked in a variety of other media, including film, collage, publications, and exhibitions. Its members' work critiqued institutionalized patterns of knowledge and practice in the designed environment—sometimes savagely so. Superstudio's critical practice can also be seen as a "refusal to work" in that these young architects decided not to provide their professional training as labor toward what they viewed as corrupt capitalist interests (although they did translate some of their discursive ideas into furniture that went on to be commercially produced). Informed by autonomism, if we consider "Superstudio's unorthodox decision to refrain from building, one is forced

to think closely about the role of the autonomous designer in the profession itself." Their practice became an instrument for examining both design's impact on social life and the politics of design. "Superstudio's declared right to refuse the labor assigned to them had the utopian potential to upset the efficient operation of the design field."[19]

Designers involved in this critical design movement, such as the members of Superstudio, Ettore Sottsass, and Andera Branzi, also formed Global Tools, a "free school for the development of individual creativity that offered opportunities for discussion." Global Tools operated as a "diffuse system of laboratories" in Florence, Milan, and Naples. Organizers and participants worked with the design media, such as the long-running Italian product design magazine *Casabella*, "to establish an alternative relationship with the Italian design industry."[20] Through published media, designers were able to propagate new ideas about what design is or could be to a public audience even without producing objects for sale.

In 1972, a landmark exhibition at the Museum of Modern Art (MoMA) introduced the work of radical Italian design to a broader audience. *Italy: The New Domestic Landscape* showcased examples of Bel Design, "postwar modern Italian design in furniture, lighting, and product housings featuring molded plastics and abstract sculptural forms." But it also featured a category of "objects selected for their socio-cultural implications," including work by Superstudio, Archizoom, Gruppo 9999, Gruppo Sturm, Ettore Sottsass, and Joe Colombo. The exhibition was curated by the Argentinian American architect and designer Emilio Ambasz, just 25 years old at the time. To realize the ideas of these designers, Ambasz commissioned installations as prototypes of new kinds of environments to reflect on changing domestic living patterns in society. Ambasz: "As I felt that the industry was not able to offer an opportunity to make such environments, because they wouldn't be able to sell them, I invited a number of designers to take part in the exhibition and many of them came up with pretty extraordinary ideas." In the exhibition's catalog, Ambasz describes these designers as "motivated by a profound concern for the designer's role in a society that fosters consumption as one means of inducing individual happiness, thereby ensuring social stability. Torn by the dilemma of having been trained as creators of objects, and yet being incapable of controlling either the significance or the ultimate uses of these

objects, they find themselves unable to reconcile the conflicts between their social concerns and their professional practices. They have thus developed a rhetorical mode to cope with these contradictions."[21]

Beyond being an introduction to new directions in Italian design, the exhibition highlighted the confrontational positions of many contributing designers to recognize design's political significance. Gruppo Strum took advantage of MoMA's institutional platform to produce several alternative manifestos rather than a conventional installation. Its *fotoromanzi*, a popular Italian form of comics using photos with speech balloons, took aim at a housing crisis in which southern Italian workers moving to northern towns for industrial work had no place to live (not unlike the situation in the Great Migration in the United States, when African Americans moved away from the oppressive Jim Crow South only to confront new racial and economic challenges in the North, such as redlining practices that segregated neighborhoods and cities). These manifestos echoed the Situationists' *détournement* of popular media forms and were freely distributed at the museum. "The explicit Marxist message contained within the pages caused unease among the museum's board of directors and outright anger among several members of the public."[22] Outside the context of commercial production, these designers became adept at operating through cultural channels, such as exhibitions and the media, to circulate political and social ideas about and through design.

The global financial crisis of 2008 created a new chapter in design history when little in the way of traditional professional work was available to young designers due to the economic downturn. A surge of new studios popped up, primarily formed by recent design school grads, many of whom created alternative and critical models of practice. Whether due to a lack of opportunity or the designers' political commitment to refuse to work for commercial clients amid a global economic crisis caused by hypercapitalist corruption, the net effect was essentially the same. Similar to the Italian movement almost 50 years earlier, this new generation of critical designers relied on a network of design media, including new digital platforms such as the architecture and design magazine *Dezeeen*, galleries, museums, and exhibitions, to communicate their (sometimes radical) visions for alternative societies through design.

The Universitas Project: Design Education for a Post-technological Society

To coincide with the MoMA exhibition in 1972, Ambasz organized a two-day symposium called “Institutions for a Post-technological Society—the Universitas Project.” The discussion included many significant intellectuals from outside the typical world of design. The stated purpose was to establish “a new type of University concerned with the evaluation and design of our man-made milieu.”[23]

In preparation for the event and to provide material for discussion, Ambasz invited select participants to produce and share essays to be read before the symposium. These essays were compiled in a pamphlet called the “black book.” In an essay published in 2018, revisiting the event through this document, the design historian Matthew Holt articulates several still unmet claims by design practice and theory. The contradictions between design’s aspirations toward intervening in the concrete realities of everyday social life and the commodification of aesthetics were called out in the black book—ostensibly in reaction to some proposals in the MoMA exhibition. The German author and activist Hans Magnus Enzensberger, a member of the literary group Gruppe 47, wrote: “Until today, the history of design has remained a series of defeats, suffered by the high-flying aspirations of the designers in their battle against utilization by *Das K*apital.”[24]

Another contributor was the French philosopher, sociologist, and literary theorist Jean-François Lyotard. Lyotard took issue not only with the nature of power in postindustrial economies but also with how the education system had been co-opted by those same interests:

> Postindustrialism, furthermore, is a fundamentally global phenomenon. Political economy is no longer tied to the nation-state, and the research produced by universities is to assist and expedite multinational expansion and transactions. In turn the individual a university produces is a knowledge-worker not a citizen. The university today, therefore, is primarily skills-based, not ideals-based. . . . The legitimation of knowledge is no longer sought in a story of emancipation (from superstition, from authority), but in the success of the ongoing operation of the system. The question driving education is

> no longer “Is it true?” but “What use is it?” “Is it saleable?” and “Is it efficient?”[25]

This view of design’s continual failure to overcome the structures and logics of a capitalist political economy may not be eternal, however. Holt believes the central ideas of this proposal are only now becoming understood:

> In the socialist tradition, whether the Arts and Crafts movement or the Bauhaus and beyond, design has always been taken to be an intervention in the shaping of artefacts for industry or refining the manner of their consumption. William Morris, for example, believed properly designed artefacts ameliorate and improve existing conditions, but would not have considered design as counterfactual communication or as planning ephemeral and intangible information and communication services and systems. In a very real sense, he could not have. That information characteristic of political econ-omy (post-handicraft, post-product, even post-object)—“the political economy of the sign” as Jean Baudrillard, a contributor and participant of the 1972 conference, calls it—only effectively transpires in postindustrial societies. . . . Design understood in its diverse meanings and above all not restricted to artefact design is a contemporary phenomenon. If alternatives to capitalism—or alternative forms of capitalism—are to be considered, let alone embraced, an acknowledgment and understanding of this environmental sense of design is essential.[26]

According to Holt, what this “environmental” sense of design meant to Ambasz was not only the ecological concerns of human cohabitation within a diverse but finite biosystem but also the making of an entire artificial world (echoing Arendt’s language): “Instead of seeing design as the production of commodities, it needs to be understood as inventing, creating and sustaining artificial environments.”[27] This is the original space of political discourse: how to make a shared human world.

Building on the work of Italian radical designers and their refusal to participate in a system they saw as corrupt, these institutional blueprints for

a new kind of design university were prescient. It is unfortunate that, half a century after the black book and its corresponding symposium, this "environmental" model of design education embedded in true social transformation still seems so far off—at least outside of a few somewhat isolated examples.

Enzo Mari's Autoprogettazione One of the most prolific individuals to emerge from Italy's radical period was Enzo Mari. In an interview in 2014, before even being asked a question, he began with: "Would you like me to reveal a secret straightaway? I realize it's a suicidal thing to say, but I'm going to do it anyway: I'm a communist."[28] Mari came out of the gate with this statement to set the tone for his entire philosophy of design as an inherently political activity. In an earlier interview, he stated, "Political activity consists of bringing others round to one's own position. Isn't that perhaps the shared goal of design?"[29]

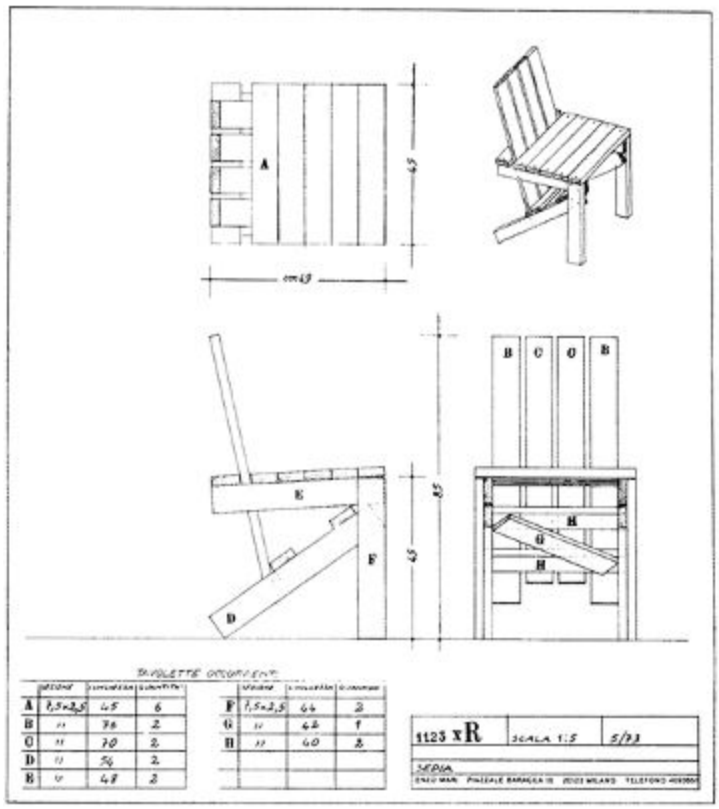




Figure 6.2

Drawings and instructions for a *sedia* (chair), from Enzo Mari, *Autoprogettazione* (1974; reprint, Mantua, Italy: Corraini Edizioni, 2002), 46. (© 1974 Enzo Mari. All rights reserved to Maurizio Corraini s.r.l.)

Frustrated with the growing culture of consumerism in the early 1970s, Mari embraced an alternative approach to the design of everyday objects. His manual *Autoprogettazione* offered easy-to-construct models of tables, chairs, beds, and other household furniture from ready-made cuts of lumber. A precursor to the open-source movement, the published drawings and instructions in the manual meant to empower people to become their own makers and perhaps in the process to modify his designs and become their own designers.[30] This approach to design is both pedagogical and political. The objects people would make from his designs were intended to be more valuable because of their own participation, including the knowledge and skills that were to be gained. Meanwhile, this activity creates a vision of design decoupled from centralized control over production, the generation of false desires, and the exploitation of profit central to a consumer-capitalist system. Mari: "The first problem facing a designer is to define his own model of an ideal world, and not to create an aesthetic. . . . The designer cannot fail to have his own ideology of the world. If he has none, he is a fool who only gives shape to other people's ideas."[31]

Anticonformity: Pluralism & Punk

Contemporaneous with the overtly political Black Panther Party and civil rights movement in the United States and the autonomous movements in Europe, new subcultures began to emerge, which took on symbolic and aesthetic forms as a rejection of consumer culture. When the American poet and activist John Sinclair asked the Black Panthers what white people could do to support their cause, the response was: start your own antiracist political party for white people. Sinclair did just that.

In 1968, Sinclair cofounded the White Panther Party as a militant, antiracist, socialist group composed primarily of young white radicals working toward goals similar to the Black Panthers'. At the time, Sinclair was also managing the Detroit rock band the MC5 (Motor City Five), who were becoming known for their incredibly loud and high-energy performances. The band's sound has been described as directly influenced by the deafening, repetitive noises they heard blasting from the factory lines of Detroit's automotive industry. Meanwhile, the band's lyrics, imagery, and performances were molded in militant politics. The band's general image

included the icon of the White Panther Party, an adaptation of the Black Panther Party's logo featuring a white panther in place of the black one. The White Panthers' agenda included "fighting for a clean planet and the freeing of political prisoners" along with "rock 'n roll, dope, sex in the streets and the abolishing of capitalism."[32]

In an episode infamous in the annals of rock history, the MC5 was the only band to perform at the Democratic Convention in Chicago in 1968. When police violently broke up protests at the convention, many of the scheduled musical acts backed out. The MC5 carried on, though, its members allegedly plugging their amplifiers into a generator used by a hot dog concession stand after they were cut off from the primary power supply. Silent footage of the performance exists, made by FBI field agents who were tasked with following the group in attempts to thwart its political agenda.[33]

Widespread American social uprisings in the 1960s were generally squelched by state violence: protestors beaten by police, political groups spied on and infiltrated by the FBI and CIA, and political leaders assassinated. However, cultural resistance continued in various forms.

> The 1960s did not spark the social or political revolutions forecasted by campus riots in the United States and in Europe or the disruption of the 1968 Democratic Party National Convention in Chicago. . . . [However], the 1960s also signaled the critical recognition of several coexisting cultural expressions in the areas of art and design, a situation sometimes referred to Pluralism, in which no single approach to modernity dominated, and where the exchange value of all designed goods obscures former distinctions between "good," mass, and popular design. The situation accommodated . . . diversity, but only within an acknowledged common culture of consumption, economic growth, and globalization.[34]

During this period of pluralism, mass culture splintered. Out of the social and political challenges of the 1960s, many distinct subcultures emerged, each taking on visual forms in attempts to embody different identities and ideas. In the 1970s, the *punk* movement in the United States and Europe generated one cultural response within this splintering. The movement was

rooted in a combination of music, fashion, performance, and graphic design, which together orchestrated an alternative cultural space that diverged from the conformities of consumer culture. While participants in the movement, including many designers, desired to break out of the dominant modes of cultural production, there was still "the recognition that living within that culture forces compromises."[35]

One technique of punk design was the appropriation of symbols of power, authority, and tradition. Cultural appropriation is the adoption of elements of one culture by members of another culture. This is often criticized when members of a dominant culture appropriate from a minority or socially disadvantaged culture. However, the appropriation of dominant cultural forms, such as authority symbols or corporate logos and advertising, fits the Situationist concept of *détournement* in that it can turn expressions of the capitalist system against itself. This hijacking can be seen clearly in the iconic design work produced for two of the most recognizable punk bands of the era, one on each side of the Atlantic.

During the mid- to late 1970s, the Mexican American graphic designer Arturo Vega held the title "creative director" for the New York punk band the Ramones. The logo he designed for the band directly appropriated the image of the Great Seal of the President of the United States, designed by Charles Thomson in 1782. According to Vega, the Ramones

> reflected the American character in general—an almost childish innocent aggression. . . . I thought, "The Great Seal of the President of the United States" would be perfect for the Ramones, with the eagle holding arrows—to symbolize strength and the aggression that would be used against whomever dares to attack us—and an olive branch, offered to those who want to be friendly. But we decided to change it a little bit. Instead of the olive branch, we had an apple tree branch, since the Ramones were American as apple pie. And since Johnny was such a baseball fanatic, we had the eagle hold a baseball bat instead of the [Great Seal]'s arrows.[36]

The logo was reproduced on posters, album art, and T-shirts—and in various forms it is still being produced today.

The British artist and fashion designer Malcolm McLaren had an appreciation for the Situationists yet felt that a new aesthetic of protest and social rebellion was needed for the 1970s. McLaren and his girlfriend, fashion designer Vivienne Westwood, managed a clothing boutique in London's Chelsea neighborhood, which they treated as home base for testing their social ideas. In 1975, McLaren organized some of the shop's employees and regular customers into a band that he would also manage. He named them Sex Pistols. The British graphic designer Jamie Reid was brought on to design album covers and promotional art for the group. For their single titled "God Save the Queen," Reid took an explicitly *détournement* tactic. He lifted an image of the queen of England familiar from currency and postage in the United Kingdom and "defaced [it] with letters and words torn from newspapers in the manner of a ransom note . . . subverting the usual meaning of both an image of respect as well as the title of Britain's national anthem."[37]

What may have started as serious and concerted political motivation became increasingly commodified as yet one more set of lifestyle choices to be bought and sold on the market of culture. A similar fate befell other subcultural movements of the era, including hip-hop and many others since then. Just as punk designers appropriated symbols of power, the capitalist system in turn co-opted punk's aesthetic for commercial purposes. Although punk may have opened the door for greater inclusiveness of different or new forms of cultural expression, it nevertheless folded back into the capitalist logic. Even Malcolm McLaren admitted, "Punk was just a way to sell trousers."[38]

The eventual trajectory of punk pinpoints one of the major challenges for anticapitalist cultural efforts—overcoming the capitalist potential to co-opt resistance and turn resistance into new cultural commodities. "In many ways the punk phenomenon can be placed within the framework of Pluralism and postmodernism, complex in its relationship to popular culture, appropriating rather than rejecting the multiple signs and meanings of our consumer culture."[39] Similar tendencies can be seen in the cultural and commercial appropriation of images of resistance, from portraits of Che Guevara to the cultural aesthetics of Black power.

Indigenous Resistance & Communal Realization: Autonomous Design

Arturo Escobar is a Colombian American anthropologist and a prominent critic of global capitalist development. His "ontological" view of design characterizes it as a primary function of the relation between human beings and their life worlds. Escobar draws connections between "visions of transitions" and design, particularly situated in a Latin American context of peoples working in defense against global capitalist exploitation of territory, resources, labor, as well as ancestral ways of knowing, doing, and being. His analysis brings together projects of the Global North, such as those aimed at *degrowth** and *commoning*,* with movements and discourses of the Global South, including Buen Vivir, postdevelopment, postextractivism, and local resistance to international development projects. These projects are wildly diverse, and that is, in fact, the point. Escobar sees in them a common project of substantially challenging the "current dominant form of capitalist modernity," including all the manners in which it limits ways of knowing, doing, and being in the world. The practices promoted by these projects are place based and communal "in ways that enable humans to relearn to dwell on the planet with nonhumans in mutually enhancing manners."[40] Escobar names and defines commonalities across these practices as *autonomo*us design.

Escobar defines *autonomous design* as "a design praxis with communities that has the goal of contributing to their realization as the kinds of entities they are."[41] This kind of designing follows some basic presuppositions: every community practices the design of itself; every design activity begins with recognition that people are practitioners of their own knowledge; what the community designs is an inquiry or learning system about itself; every design process involves a statement of problems and possibilities, and the exercise may take the form of building a model of the system that generates the problem of communal concern.

Escobar cites the concept of Buen Vivir, particularly as it contrasts to the market-led growth agenda of development in Latin America. The expression *buen vivir* loosely translates as "good living," the Spanish form of *sumak kawsay*, as this concept is called by the Quechua peoples of the Andes. It is a social philosophy that sees humans only as stewards of earthly resources, a project that is to be undertaken communally. "Buen

Vivir subordinates economic objectives to the criteria of human dignity, social justice, and ecology. The most substantive versions of Buen Vivir in the Andes reject the linear idea of progress, displace the centrality of Western knowledge by privileging the diversity of knowledges, recognize the intrinsic value of nonhumans (biocentrism), and adopt a relational conception of all life."[42] This concept has even inspired wording in the preamble of Ecuador's recent constitution (2008): "We . . . hereby decide to build a new form of public coexistence, in diversity and in harmony with nature, to achieve the good way of living."[43]

According to the Uruguyan social ecologist Eduardo Gudynas, "the subject of wellbeing is not [about the] individual, but [about the] the individual in the social context of their community and in a unique environmental situation." Although Buen Vivir and similar indigenous worldviews are inspiring social movements across Latin America, Gudynas states that these movements are "equally influenced by western critiques [of capitalism] over the last 30 years, especially from the field of feminist thought and environmentalism. . . . It certainly doesn't require a return to some sort of indigenous, pre-Colombian past."[44] The Buen Vivir way of thinking may not be a direct method for transformation, but it can create fresh perspectives on how society, economy, politics, and ecology intersect in ways that undo the hierarchies instilled by capitalism. "It helps us see the limits of current development models and it allows us to dream of alternatives that until now have been difficult to fulfil."[45]

In what Escobar calls a "transition imagination exercise" for autonomous design, he applies this expansive model of designing for transitions to an imagined project in the Cauca Valley of Colombia.[46] He stitches together transformational models of design from the Global North—including speculative design—with models based on indigenous worldviews. The project would require deep commitment from a team with "epistemic, social (in terms of race/ethnicity, gender, generation, class, and territorial bias), and cultural (ontological) diversity," in which authority and expertise are diffused.[47] The many activities this team would undertake include making visible the "defuturing practices" of the current scenario, understanding the various and perhaps competing "life projects" of the communities in the region, designing through cooperative methods of exchange and participation to demonstrate that "other economies are possible," building

scenarios to envision future states of the region through speculative design, designing methods for collaboratively producing local histories, assessing potential impacts of climate change, and creating new media communications and digital platforms to shift the discourse and identity of the region toward the new collective imaginary. Escobar admits, "I am perfectly aware of the overly ambitious nature of the proposal. Let us say that it was intended largely as a theoretical exercise . . . to buttress the idea that 'another design is possible.'"

Although Escobar's focus is on indigenous defense of territories and knowledges through designing communal and place-based ways of being, his project informs "those of us who live in the delocalized and intensely liberal worlds of middle-class urban modernity" to take heed of the imperative for "recommunalizing and reterritorializing."[48] This is a project for an expanded model of design—one that connects a model of design as territorial resistance to community-centered design in urban spaces, as modeled by design for social innovation.[49] Another design that is possible.

Noncapitalist Design

The previous examples were explicitly anticapitalist in their objectives. Diverse *noncapitalist* design practices also exist, operating outside the perceived dominance of market constraints yet not necessarily antagonizing them.

Four Fields Framework

In 2009, American design scholars Bruce Tharp and Stephanie Tharp created the Four Fields Framework for understanding various intentions of design: "Without a compelling, indeed, taxonomic, way of organizing design activity, we are selling ourselves short; we not only have difficulty understanding the profession ourselves, but also in communicating to the world our potency, range, and potential impact."[50] The fields they propose are: *commercial design*, *responsible design*, *experimental design*, and *discursi*ve design.

Commercial design is the most common and probably best-understood field of product design, making up “the overwhelming majority” of professional industrial design activity. The iPhone is an easily understood example. “The primary (though not only) driver of Commercial Design is to make money.”[51]

Responsible design aims to serve people whose needs are not being met by commercial products. Tharp and Tharp cite as examples the One Laptop per Child project and Ableware cutlery, which is designed to help people with limited dexterity to live more independently. Responsible products may be available on the market, but their primary intention is one of service. “The primary (though not only) driver of Responsible Design is to help those in need.”[52]

Experimental design explores new materials, new techniques, or new processes for doing the work of designing. This type of work provides new knowledge about ways of doing things. The Animals Project by Front Design is an example. The designers translated material manipulations by animals into goods for humans: “a lamp cast from a rabbit’s burrow, wall hooks that were formed [to resemble] constricting snakes, a lampshade created after recording a fly’s path around a light bulb, a vase created by casting the impression of a dog’s leg in deep snow,” and so on. None of the resulting products was mass-produced or sold, but experiments were shared as investigations offering new ideas. “The primary (though not only) driver of the Experimental Design is to explore.”[53]

Finally, there is discursive design. It uses the experience of engaging with designed objects to engage in thinking about how designed objects affect everyday lives, including perceptions about the world and about ourselves. “Discursive design as a field encompasses a wide range of ‘conceptual,’ ‘alternative,’ ‘expansive,’ or ‘critical’ practices . . . unified in their concern for intellectual impact.”[54] In their book *Discursive Design*, Tharp and Tharp identify hundreds of projects and practices they qualify as “discursive”: The primary (though not only) intent of discursive design is to express ideas.

These fields are not mutually exclusive; any designed object may merge two or more of the fields. Tharp and Tharp emphasize that the Four Fields Framework is based on “intention,” rather than on outcome. They identify discursive design as a genus with many species, including critical design, speculative design, and design fiction.

Critical, Speculative, & Speculative Critical Design

It was during the 1990s that the personal computer truly became domesticated, in part due to the birth of the internet. Rather suddenly, using computers and making networked digital media became an everyday phenomenon. This shift prompted a variety of responses for how design practices could bring together the design of digital processes and experiences, interface design, industrial design, and communication design—all typically joined by the broader set of interdisciplinary concerns of HCI. Institutions such as MIT's Media Lab and Xerox's Palo Alto Research Center (PARC) as well as research-and-development teams at the Philips corporation became beacons of research and experimentation in their attempts to understand, test, and at times codify the features, possibilities, and processes of this new territory of designing. Many of these efforts were organized around transforming technological possibilities into products.

At the Royal College of Arts in London, the Computer Related Design (CRD) studio "set out an agenda for product and interaction design, positioning it as a vehicle for critical reflection on the role design and technology in society."[55] Based on these interests, a "critical design" unit was set up within the CRD studio and led by research fellows Anthony Dunne and Fiona Raby. It's fascinating that the term *critical design* was coined in the late 1990s, which situates it directly within the new paradigms of designing with, through, for, and about computation as a general-purpose technology—including all the unintended possible outcomes. Dunne first used the term in his book *Hertzian Tales* (1999), and Dunne and Raby further developed the definition of it in their book *Design Noir: The Secret Life of Electronic Objects* (2001). Writing later on this topic, Dunne and Raby state: "Design that asks carefully crafted questions and makes us think, is just as important as design that solves problems or finds answers. . . . [L]et's call it critical design, that questions the cultural, social, and ethical implications of emerging technologies. A form of design that can help us to define the most desirable futures and avoid the least desirable."[56]

Through their teaching, research, and practice, Dunne and Raby have come to be recognized as potent progenitors of a wave of critical design practices to follow, eventually giving form (and a name) to speculative design. According to Dunne and Raby: "Critical Design uses speculative

design proposals to challenge narrow assumptions, preconceptions and givens about the role products play in everyday life."[57]

Matt Malpass further distinguishes speculative design by its specific focus "on science and technology, establishing and projecting scenarios of use; it makes visible what is emerging."[58] The genesis of speculative design within the larger discourse of critical design combined a desire to "find ways of operating outside the tight constraints of servicing industry,"[59] a new intellectual project for design, and a response to the perceived lack of criticality when design engages with new and emerging technologies. While not necessarily anticapitalist, the resulting inquiries through research and design indicate the limitations placed on market-driven technological innovation, suggesting that socially beneficial technologies require a different ideological orientation. For this reason, *speculative critical design*—a mash-up term that indicates the intersection of speculation and critique—is quite often noncapitalist in its pursuit of posing "alternative" scenarios of everyday life through designed objects and experiences.

Critical Jugaad

In Hindi, the term *jugaad* refers to "making do," creating a "workaround," or developing creative solutions to making something useful out of what is at hand. As a noun, it can also refer to a specific workaround, a repair, or even a hack, such as building a vehicle for public transit out of parts from several discarded vehicles. *Jugaad* is not just a term but an entire sensibility of resourceful and creative approaches to working within limited resources. "Although Jugaad is an Indian concept, such tendencies are found in other non-European contexts as well, such as *Jua Kali* in Kenya, *Rasquachismo* in Mexico and *Gambiarra* in Brazil. It could also be understood in terms of practices such as DIY, hacking, kludging, etc. in Western contexts."[60] *Jugaad* also represents the cultural context of a semiskilled labor-market repairing, maintaining, or modifying technical products such as smartphones, speakers, DVD players, TVs, and so on. According to the Indian American design scholar Deepa Butoliya, this practice of "making do" is not a single case in point, but rather a common sight in many Indian cities.

Butoliya translates *jugaad* and similar cultural practices of the Global South as enacting a critical perspective on the "maker culture" and "maker spaces" of the Global North. She introduces the term *critical jugaad* to suggest such "homemade or low-cost artifacts" operate as a "critique to the consumer culture" and are political practices that "permeate boundaries of race, class, affluence, nationality, and gender."[61] As an example, Butoliya cites informal, local repair economies that have long existed in India and China, some of them now particularly focused around the prevalence of smartphones. Low-cost repairs and modifications maintain these devices (primarily "imported" products, which nonetheless might have been assembled in nearby factories) for longer-than-expected functional lives. Sometimes such repairs include modifications that are illegal or violate end-user agreements by manufacturers, such as "jailbreaking" iPhones to remove software restrictions. By offering these kinds of services, informal economies assert a form of local control even when working with the material, legal, social, and political structures of global corporations. Similarly, Butoliya cites *shanzhai*, the Chinese practice of making low-cost copies of popular branded products. This term applies not only to the making of "knock-off" name-brand products such as smartphones and handbags but also to a subculture that parodies products with humorous names and themes or includes additional features not found on the original brand-name products.

These and similar approaches to "making" are culturally unique and at odds with global corporate interests. People use "ingenious making practices like Jugaad as a tool for existence, subversion and criticality against colonial powers of oppression."[62] Butoliya sees them as examples of innovation and ingenuity that should be considered in Western approaches to technological development and maintenance in order to divert Westerners away from a highly unsustainable throwaway culture.

Afrofuturism

The African American author and filmmaker Ytasha Womack defines *Afrofuturism* as "an intersection of imagination, technology, the future, and liberation."[63] Afrofuturist creative works produce cultural, aesthetic, political, and philosophical perspectives on the intersections between the

African diaspora and science and technology. These works move fluidly among music, film, literature, fashion, art, and design. Works by Afrofuturist creators, such as Janelle Monae, Wanuri Kahiu, Kerry James Marshall, Cyrus Kabiru, Manzel Bowman, and Tahir Carl Kamali, among others, *decolonize* Black identities from imperial and exploitative origins. Afrofuturism provides a vehicle for liberating narratives and imagining ways of being distinct from Euro-centric narratives of technology and domination. “Afrofuturism allows black people to see our lives more fully than the present allows—emotionally, technologically, temporally and politically.”[64] As an open and expansive cultural sensibility, Afrofuturism creates space for new subjectivities outside the dominant economic logics of slavery, the subjection to oppressive property laws, the exclusionary political and social tactics, and the ongoing economic inequalities experienced by African people living in a postcolonial world.

Elizabeth Chin’s project AfroGoPro reimagines the portable digital camera GoPro through the lens of African aesthetics. Chin sees the GoPro as a technology that is “heavily raced and gendered in form, format, marketing and use.” Yet there’s nothing about a wearable camera or self-documentation that inherently says, “This oughta be for a guy!” or “This ought to be for white people!” Chin embeds the GoPro in layers of African cultural references. For one model, she “hand-stitched a cover for the GoPro then covered it in Swarovski crystals, mount[ed] it on a headband, and then us[ed] African Dutch Wax fabric . . . bought in Uganda to create the headwrap.”[65] Another AfroGoPro is based on the design of a Maasai wedding necklace. Chin’s project challenges stereotypes about both Africa and personal technologies, in particular the cultural frames in which they are conceived or used.

Afrofuturism offers a framework for envisioning models of technological innovation and invention outside the hegemonic view of a singular, expert-driven narrative of technology driven predominantly by a European worldview of modernity. This idea can be adopted by any community seeking to create its own culturally specific models of technological innovation, whether specific to the diaspora of African people or not.

Participatory Speculation

In the pioneering work of Scandinavian cooperative design in the 1970s, automation controlled by the design of software systems represented a major sociopolitical concern because it had the potential to alter the nature and pace of work. Automated work processes also presented new terrain for contestation among owners, managers, and unionized workers over ownership and control of the workplace. Participatory design practices that adhere to the political function of designing can still be found in nearly every domain of design practice around the world—albeit in varying proportions to other models of practice.

> One of the guiding principles in participatory design is to expect controversy rather than consensus within the process. . . . Like critical design practices, participatory design reflects critically upon the relationship between the designer and the communities that are being designed for, and, or with. These practices operate beyond conditions set by fiscal gain or technological development and are established as intellectual and politically motivated practices, and increasingly extend designs agency as they are used to inform policy in address to societal concerns.[66]

The politics of design remain a central concern for participatory design researchers and practitioners, particularly when it comes to power dynamics that emerge within workplaces and communities. Although much of the work generated by discursive, critical, and speculative designers has taken the form of rhetorical propositions—prototypes, installations, publications, and so on—some practitioners have successfully situated their work in positions of political power.

Strange Telemetry is a London-based research and design consultancy whose work with organizations, including government agencies, is used to "illuminate the contexts in which technologies are imagined, built, used, and controlled."[67] The company is itself a cooperative, owned and managed by its worker-members. According to the Strange Telemetry website, the process involves "developing speculative prototypes and designs that can be taken to clients and publics as a way of stimulating debate and critical thinking. This is part of our larger aim to materialise and utilise speculative and critical design in new contexts and applications."[68]

In 2015, Strange Telemetry engaged in a project with the UK Government Office for Science in what was the first application of speculative design for formulating UK government policy. The purpose of the project was to explore challenges and opportunities of an aging society to determine how the government could best steer resources for appropriate technological development. Through a number of workshops with residents in multiple cities, the designers facilitated discussion and debate around "possible developments in employment, services, and transport over a 25-year time horizon."[69] By building images and prototypes of possible work environments, public spaces, and urban transit systems, the group used design to prompt discussion and debate, which could then be used to shape policy—a true merger of speculative and participatory design practices for public benefit. This approach to setting agendas for technological development is diametrically different from the typical capitalist model, in which decisions for technological innovation are made internal to corporate interests, and the innovation is usually later sold or licensed to public organizations as customers.

Cooperative Design: Worker Ownership & Democratic Control

Worker-owned cooperatives offer an alternative to the capitalist model of profit extracted from wage labor and to hierarchical, nondemocratic models of governance. They are "democratic businesses owned and operated equally by a specific membership. Some co-ops have memberships composed of only workers, some of consumers, some of producers."[70] There are many examples in design.

Sassafras Tech Collective is an American worker-owned co-op focused on web and app design and development. Sassafras offers comprehensive design and development services, including user-research and UX design, interaction and graphic design, usability testing, and a wide range of custom development services as well as hosting and maintenance. The cofounders centered their research and practice on issues of technology and social change, working for social justice organizations, nonprofits, researchers, and academics. Design Action Collective is an independent design and visual communications organization in Oakland, California, that provides graphic design, web development, and new media services—primarily for

partner organizations working toward economic and social justice.[71] Design Action Collective began in 2002 as a spin-off of the Inkworks Press Collective, connecting it to a tradition of worker-owned collectives formed by workers in press and print shops. Sassafras and Design Action are just two examples of a growing movement of worker-owned design and tech cooperatives that are organizing and building collective resources around the world.

In the United Kingdom, multiple tech cooperatives have organized to form CoTech, the Co-operative Technologists network. CoTech's goal is to "create a better technology sector in the UK that focuses primarily on worker, customer and end-user needs rather than on generating private profit."[72] Each member organization is a tech company—sometimes quite a small one—that is "owned and democratically run by its workers (and, in some cases, customers). There are no private shareholders who take money out of the company and/or direct the company's activities."[73] This collective of cooperatives is governed through consensus processes using various digital platforms.

In high-value and highly competitive sectors such as the design and tech industries, workforce participation in governance presents an opportunity for broader participation in the decisions that may affect millions or billions of users. This form of democratization may also provide an antidote to some of the toxic workplace cultures for which both the design and tech industries are notorious.

Noncapitalist design practices create spaces, networks, and systems outside the direct reach of capitalism. As isolated practices, they evince an "escape" model, according to Wright's notion of *strategic logics*, even as they "resist" some of the negative impacts of capitalist models of design. Through collective and organized effort, the spread of these types of practices might "erode" the capitalist structures of civil society more broadly.

Postcapitalist Design

> Industrial transformation turned out to be a bubble of promise followed by lost livelihoods and damaged landscapes. And yet: such documents are not enough. If we

end the story with decay, we abandon all hope—or turn our attention to other sites of promise and ruin, promise and ruin.

—Ana Lowenhaupt Tsing, *The Mushroom at the End of* the World

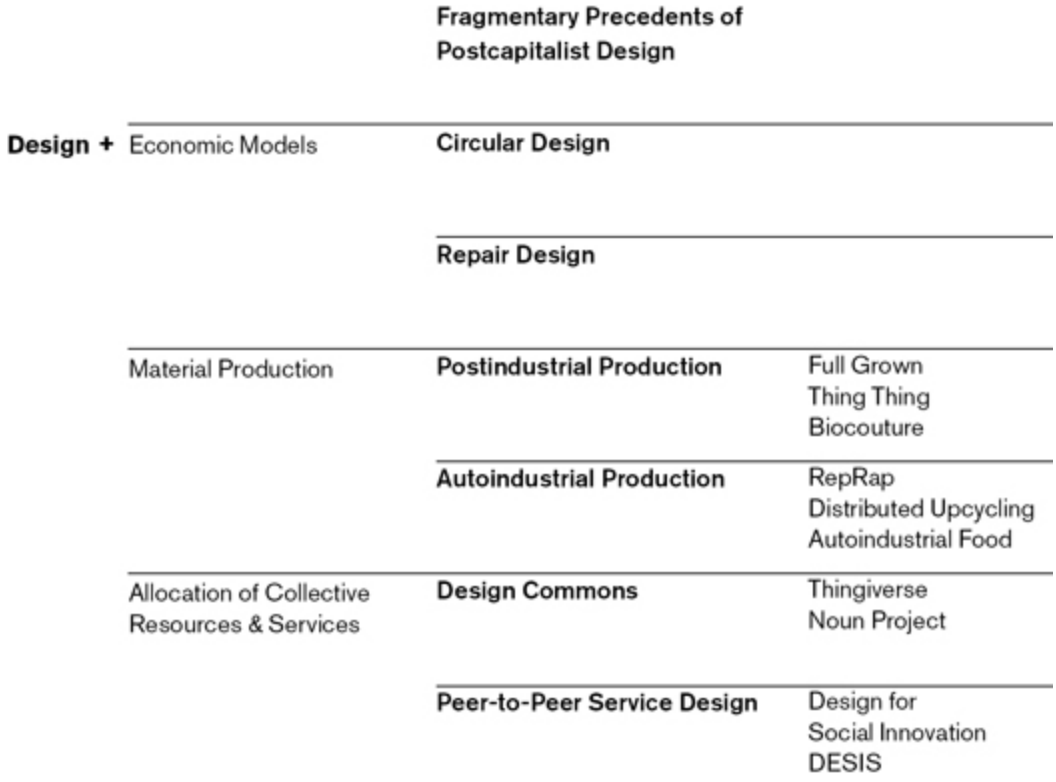


Figure 6.3
Fragmentary precedents of postcapitalist design.

The call for a new paradigm of postcapitalist design (PCD) is the central ambition of this book. Yet we should notice that the territory is already rich with precedent. Each of the following case studies offers a perspective emblematic of postcapitalist possibilities based on various approaches to the activity and outputs of design or the methods of sourcing materials, labor, and means of production. They illuminate hybrid forms of social organization among designers, producers, and users—in many cases, transforming "users" into active producers and/or designers. These precedents are fragmentary, but they point toward models that could catalyze robust disciplinary patterns of knowing and doing PCD.

The examples are organized by the different ways they reframe design's orientation to specific economic models (circular design and repair design), material production (postindustrial design and autoindustrial design), and

the production and allocation of collective resources, including labor and social organization (design commons and peer-to-peer service design). Taken together, they offer potent and vigorous views of a diverse but coherent new discipline of PCD, building new ways of knowing, doing, and being from capitalist ruin.

Circular Design: Design in a Circular Economy

The ecological impact of design has been a concern in different ways and under different names for several decades. Victor Papanek's work in the 1970s is generally credited with initiating a *green design* movement focused on "product design and development with a focus on single issues such as selecting recyclable materials." In the 1980s, *ecodesign* described product design development "with incremental add-on environmental considerations at each stage of the process." The 1990s saw broader adoption of approaches to *sustainable design*, "which includes ecological, economic and in addition, social considerations, such as the impact of product development on factory operators, and people living within the local environment." In the early 2000s, *design for sustainability* came to identify an even broader, systemic view of design, including "the radical redesign of products and services striving towards a sustainable future."[74]

From the environmental crises of the late twentieth century, an alternative economic concept contrasting to the "traditional linear business model of production of take-make-use-dispose and an industrial system largely reliant on fossil fuels" began to develop.[75] In a *circular economy*, "economic activity builds and rebuilds overall system health. The concept recognises the importance of the economy needing to work effectively at all scales—for large and small businesses, for organisations and individuals, globally and locally." Importantly, this shift to a circular economy is not about reducing the negative impacts of a *linear economy* based on the production and disposal of goods but about implementing an entire systemic shift that "builds long-term resilience, generates business and economic opportunities, and provides environmental and societal benefits."[76]

Circular design names a model for designing within a circular economy. It is shorthand for all the ways design can be employed to apply circular-

economy principles at every stage of the design of anything. Circular design replaces "the end-of-life concept with restoration, shifts towards the use of renewable energy, eliminates the use of toxic chemicals that impair reuse and aims for the elimination of waste through the superior design of materials, products, systems and business models."[77] In many ways, circular design seems as though it should be the default for what is considered "good" design. In fact, according to the former president of the International Council of Design, David Grossman, "If something is not designed in accordance with circular economy principles, it is not professional, responsible design—it is malpractice."[78] But this type of designing is not happening at the speed and scale required to curtail the parallel crises of climate change and economic equity.

Research on the circular economy has taken different directions. Most has focused primarily on business models and structures to capture value out of circular flows of resources, while some has focused on material possibilities for regeneration through technical and biological cycles. This variety means there is very little in the way of explicit and practical design strategies, "providing little guidance to designers and a lack of clarity on how designers are to design for new circular business models."[79]

In addition, "terminology around the 'circular economy' has been diverging rather than converging,"[80] creating a fractured discourse difficult to decipher for practical applications in design. Out of disparate scholarship and studies, scholars in the United Kingdom have identified five core strategies for circular design:

- Design for circular supplies: This strategy focuses mainly on the biological cycles and refers to thinking of "waste equals food" in which resources are captured and returned to their natural cycle without harming the environment.
- Design for resource conservation: This strategy focuses on both the technical and biological cycles and uses a preventative approach in which products are designed with the minimum of resources in mind.
- Design for multiple cycles: This strategy focuses on both the technical and biological cycle and refers to design aimed at

enabling the longer circulation of materials and resources in multiple cycles.

- Design for long-life use of products: This strategy focuses on the technical cycle and refers to extending the utilisation of a product during its use through extending its life and offering services for reuse, repair, maintenance and upgrade, or by enhancing longer-lasting relationships between products and users through "emotional durable design." Furthermore, changing the ownership of products through services could enhance longer utilisation of products and, therefore, move to a sharing system.
- Design for systems change: This strategy covers the whole spectrum of value creation for both biological and technical cycles and refers to design thinking in complex systems as a whole and between its parts to target problems and find innovative solutions.[81]

While the first four points refer to technical, material, and supply-chain concerns, the last point indicates that circular design requires the transition to a political economy beyond capitalism—despite the fact that this requirement is seldom explicitly stated in the literature. How exactly new political economies and new models of design will adopt the concept of circularity will most likely vary widely. However, such models are likely to be *postcapitalist* in the sense that they will leverage gains in economic and technological productivity ushered in by capitalism but radically transform the social and material processes of production—thus, "revolutionising the world."[82]

Repair Design: Design in a Repair Economy

A key element of circularity is the maintenance and longevity of material goods, which includes repair.

> Just a few decades ago, manufacturers packaged everyday appliances with instructions on how to repair them. Now they come with danger warnings and threats that doing so will void the warranty. Repair is

> discouraged by unavailable replacement parts, glued assemblies and tamper-proof cases that are difficult to open. So we discard things rather than fix them.[83]

The environmental case for an economy built on the repair and longevity of material goods is probably obvious: natural resources, energy for production, and waste are reduced. However, there is also evidence that a repair economy is economically and socially beneficial. The incentives for repair can create more high-skill jobs, connect people to the work of repair, and reduce the cost of living. A growing literature also shows that "repair economies can make people happier and more humane."[84] Many people of the Global North have lost any cultural context for fixing things, unlike in the culturally embedded practices of *jugaad* in India. Behaviors of hyperconsumption mixed with things deliberately designed for short-term use and quick replacement have contributed to a deskilling of repair capabilities. Several initiatives and organizations are attempting to change that.

For example, iFixit is a wiki-based site that "teaches people how to fix almost anything. Anyone can create a repair manual for a device, and anyone can also edit the existing set of manuals to improve them."[85] The project intends to empower individuals to share technical knowledge in a global communal effort to maintain the functional lives of objects, save money, save the environment, and "ditch the throwaway economy." It's also part of a movement against corporate and legal policies, restrictions, or other maneuvers that prevent people from repairing their personal property.

Repair typically signifies returning something to an original state of function, but designers are also finding creative ways to use repair as a model of innovation—making something new out of something old or damaged. Reknit Revolution provides one example of design's role in a repair economy. This ongoing initiative encourages *reknitting*: "the reworking of existing knitted items using knit-based skills, techniques and knowledge."[86] Knit garments that have become damaged, stained, or simply out of fashion are reimagined and re-created by opening, altering, unraveling, and reknitting the garment into something both different and totally unique. With an explicit goal of contributing to the "domestic circular economy," Reknit Revolution offers technical advice, organizes

workshops, exhibits striking new creations from the process of reknitting, but most of all encourages others to take up the challenge of doing it themselves. The fashion industry is one of the most environmentally destructive on the planet, and garment workers are notoriously exploited. This challenge to the knitted textile industry generates new territory for fashion designers to operate within a repair economy.

Postindustrial Design: New Models of Material Production

Almost 50 years after Enzo Mari gave instructions for building your own furniture, the British design and agriculture team Alice Munro and Gavin Munro are demonstrating that we can also *grow* furniture. Their organization Full Grown UK is researching, testing, prototyping, and putting into practice agricultural design methods to shape trees directly into furniture, "like an organic 3D printing that uses air, soil and sunshine as its source materials."[87] Although growing a chair may take seven to nine years, the end product requires no assembly, adhesives, hardware, or major machine processes. No global supply chains of materials or associated energy consumption and pollution are involved; all materials are inherently local to their production.

This novel approach remakes the design process by applying agricultural knowledge, some of it quite ancient. In designing chairs, end tables, lamp shades, and other domestic furnishings to be grown from various species of trees, Full Grown UK uses the ancient art of coppicing as one of its design methods. Coppicing is a practice of cutting down trees to encourage regrowth. Full Grown UK designers have discovered that certain species regrow new branches in patterns matching the Fibonacci sequence. Based on this knowledge, furniture can be designed specifically to incorporate natural patterns into structural components of the finished pieces. This rethinking of the means of production has the added benefit of enormously decreasing the environmental impact of producing home goods. Although labor and land are required, new models of collective ownership and production are currently being tested, including a crowdfunding platform to support future harvests of furniture.

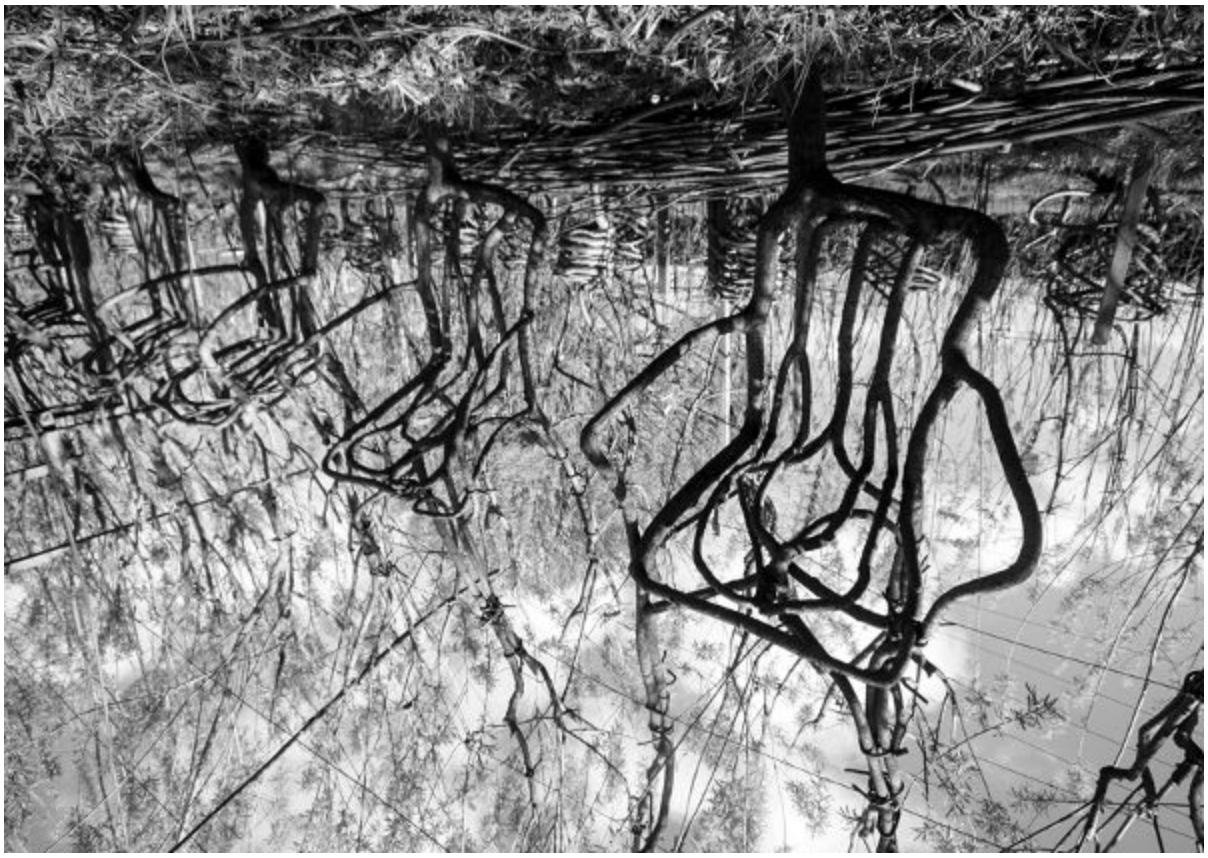


Figure 6.4

The “chair orchard” of Full Grown, Derbyshire, United Kingdom. (Photograph by Chris Robinson. Courtesy of Full Grown.)



Figure 6.5

An evolving collection of three “generations” of chairs grown by Full Grown. *Left to right*: the Dietel Chair, four years old, harvested in 2016; the Edwardes Chair, five years old, harvested in 2017; the Gatti Chair, six years old, harvested in 2018. (Photograph by Chris Robinson. Courtesy of Full Grown.)

The Detroit studio Thing Thing creates new processes for transforming discarded industrial and postconsumer materials into household goods. Established in 2012, the studio engineers and fabricates its own versions of industrial manufacturing machines “to experiment with materials typically unavailable to designers.” The team makes and modifies “plastic extruders, injection molders, rotational molding, compression molds, and more—developing an expertise and authorship in the entire making process.”[88] They create clocks, mirrors, lamps, vessels, and various furniture pieces in a process that involves transforming postconsumer, hand-recycled, high-density polyethylene plastic into objects that are some combination of sittable and sculptural. Their “Future Foam” project transforms scrap foam from the production of transportation interiors into benches. “Pan Fried Stools” are produced from plastic waste found all over Detroit—automotive

waste, product rejects, recycling centers, and even dumpster finds.[89] These repurposed materials are melted in a frying pan and then hardwood legs are attached. The team's deceptively simple open-source "recipe" for the stool belies their material expertise yet espouses an ethos of experimentally crafted DIY goods: "Bake plastic at 380 for 90 min or until gooey. Remove pan and insert legs. Polish all surfaces and serve."[90] An accompanying video tutorial shows you how to make your own stool.

The fashion designer Suzanne Lee's venture Biocouture is just one example of a rapidly expanding network of laboratories and studios exploring the use of *biotextiles* for textile manufacturing. In the Biocouture laboratory, bacterial cellulose is grown in a solution of sweetened tea, and during the fermentation process these cells can be modified to create textiles. Sample products from this process include the "bio-biker jacket," a biotextile send-up of the classic leather biker jacket.[91] Lee:

> There is a burgeoning global movement of makers and innovators who want to 'hack' materials in the same way we've seen open source software and hardware. Biocouture is building an open innovation resource to enable collaboration within the global biological materials community in order to rapidly advance innovations never previously imagined. Starting with microbial cellulose, which can be grown in a bucket and used to create a wide variety of biodegradable homewares and fashion accessories, Biocouture will provide recipes, methods, documentation and educational tools to enable widespread use and knowledge sharing globally. The big idea is to precipitate the development of many different products according to local innovation and need.[92]

The potential for organically produced materials that can be locally sourced, grown, and harvested indicates potential for a radical break from the ecologically disastrous linear model of textile and garment production within globalized supply chains.

Taken together, these postindustrial models of production demonstrate there are many different approaches that break with the take-make-use-dispose model initiated by industrial models of production. Although the latter model may sometimes seem inescapable for those of us living in the

overdeveloped, overindustrialized world, it is in fact only the tip of the iceberg when it comes to material production.

Autoindustrial Design: New Models of Production and Labor

In the 1940s, the Hungarian American mathematician and engineer John von Neumann was working on the theory for a "universal constructor." This idea was for a machine that could produce other machines—including machines that, like itself, were machine producing. These machines would be designed to pass information on from one generation to the next so that they could evolve in complexity, as in the evolution of biological organisms. What this idea holds at its core is the possibility of fully automated production—machines making the machines to make anything else people might need, creating productive abundance on their own.

One of the direct inheritors of this theory is the British engineer and mathematician Adrian Bowyer. In 2005, Bowyer set out on an ambitious project that now includes hundreds of collaborators from around the world. The RepRap project intends to produce a pure self-replicating device. Its name short for "replicating rapid prototype," RepRap is an attempt to prove that "rapid prototyping and direct writing technologies are sufficiently versatile to allow them to be used to make a von Neumann universal constructor."[93] The practical implications for this would be that people all over the globe could set up their own self-sustaining nanomanufacturing systems to produce many of the objects used in everyday life. By 2008, RepRap 1.0 (nicknamed "Darwin") made more than half of the parts for a new version of itself. The project continues in an effort to achieve the broader vision of a truly distributed, democratic, peer-to-peer model of material production.

RepRap is an open-source hardware project: the designs are made publicly available so that anyone can make their own device, including modifying and distributing new hardware based on that design. The open-source hardware concept encourages *peer-to-peer* collaboration so that the development of technology can take place in socially coordinated rather than corporate-controlled ways, with many people sharing knowledge in the process of exchange. "The picture that is beginning to emerge, then, is that the benisons [or blessings] of digital production can be enjoyed most widely

and most equitably when fabrication engines of a few different kinds are deployed in neighborhood-scale shared workshops."[94] Many of the challenges for scaling up this vision are, of course, posed by the commercialization of 3D printing. For many of the early enthusiasts, the vision of an autoindustrial, postscarcity world was their motivation, not narrow commercial success. This early foray into digital and autoindustrial production has generated a variety of interrelated projects that aim to create a robust ecosystem of technologies and materials necessary to realize this vision.

The research-and-development company Filabot is creating "cutting edge miniaturization of industrial polymer processes" in order to recycle plastics into filament to be used by 3D printers. The Filabot grinds up household plastic items—DVDs, sunglasses, water bottles, milk jugs, and so on—and melts them into pellets that can be fused into filament. If the process can be made entirely efficient, this would create the model for a closed loop of resources between the ongoing recycling of materials and the autoindustrial production by self-replicating machines. Filabot lists as one of its values: "To turn a throwaway society into a society that reclaims and creates. With a large focus on education, we strive to learn, teach, and share information so that everyone can make a positive impact."[95]

Meanwhile, The Protoprint Project is attempting to transform "informal recycling" into a systematic and circular model for recycling plastic into new forms, including 3D filament and injection-molded products. Based in Pune, India, the project collaborates with urban waste pickers and industrial partners to create an ecosystem of sustainable development and economic equity. Started in 2012, the Protoprint Project had by 2020 embarked on the construction of 50 plastic-processing facilities in collaboration with the government of Pune.[96]

The Dutch designer Chloé Rutzerveld is a self-proclaimed "food designer and food futurist." Her work confronts challenges in food production and consumption with designed interventions for rethinking the social, ecological, economic, and health of food and eating. Trained as an industrial designer, she brings methods from designing for small-batch production—in particular the use of 3D printing—to the making, modifying, and processing of new food types. What her experimentation and various publicly engaged collaborations propose is to remake our

common understandings of food by imagining new ingredients, structures, habits of making and eating, and, of course, new ways of "cooking" that use digital fabrication techniques.

Rutzerveld hosts "future food dinners," in which she engages a direct and participatory model for developing her ideas while bridging research, production, and consumption. The collaborative nature of these events encourages lively discussion among participants, particularly seeking the limits of adoption for some of the more radical ideas. Breaking down barriers to experimentation with one of the most culturally defined categories of everyday life—food—is facilitated through collaboration, cooperation, education, and self-production.

These projects and many other global efforts like them begin to frame a vision for fabrication and production that fundamentally debunks most economic thinking on material scarcity. The American designer and theorist Adam Greenfield clarifies how the economy of this vision contrasts with other types of economy: "A command economy seeks to plan ahead of time what things will be needed, and how many of each will be required; a market economy uses prices as signals to determine what should be made and when. An economy based on personal fabrication would undermine the assumptions common to both, by allowing end consumers to fulfill emergent demand more or less directly." The challenge for cultivating an *emergent demand economy* rests in combining the right social, political, and organizational configurations so that they can "effectively resist capture by existing logics of accumulation and exploitation, and bind them into processes that are generative of lasting and significant shared value."[97] These are the challenges of building an equitable and egalitarian postcapitalist society.

These disparate examples of autoindustrial production indicate alternative models of production, particularly pertaining to the use of technology. In these socially embedded approaches, automated methods of production don't erase human labor. Rather, new forms of decentralized, localized, and user-generated modes of production emerge—and more can be imagined—that put control of technical, economic, and labor concerns into the hands of individuals and communities. This shift has the subsequent effect of creating new subjectivities as users become more directly engaged in

designing and making—rather than only consuming—the material culture of their lives.

Design Commons & Platform Commons

It has been argued that any attempt to recuperate social and economic power from neoliberal capitalist exploitation will require a concerted effort to recapture the creativity, information, intellectual flourishing, opportunities for exchange, and value creation generated through digital platforms. Many platforms are already corporate enclosures, and those that have thus far resisted capitalist exploitation are typically at risk of some form or another of capitalist intervention—through buyouts, competition, and/or new platform positions that require technical dependence on capitalist firms.

The cultural and media theorist Joss Hands argues that any proposal for an egalitarian future is compelled to integrate the political spaces of digital culture, given "the centrality of digital communication to contemporary formations and conceptualisations of identity, self-awareness, social life and activism." Because "platforms are simply where the people are . . . platform politics requires more than just the taxonomic analysis of platforms; it should also include looking at alternative practices and the pragmatics of antagonism and collective modes of production."[98]

Two strategies are outlined in Joss's argument. The first has to do with an "exodus" from or decreasing reliance on capitalist platforms in the production of social and material life. The challenge, of course, is not minor, as indicated by Facebook's 2 billion users. A concerted effort toward "subtraction of capital" would be a major social and cultural project. To aid this effort, a second strategy is needed to create and maintain "common" platforms that offer alternative—and better—forms of social and material fulfillment. Indeed, some of this work is already underway.

Thingiverse is an open-source platform for sharing primarily free design files for objects to be produced on 3D printers, laser cutters, milling machines, and other digital manufacturing technologies. The platform is widely used by participants in maker communities and by those working in and with DIY technology, including the RepRap project. The platform is a shared repository for dissemination and innovation, including many object

files specifically designed for the purposes of repair or modification of products. Thingiverse is the clearest contemporary signal of the potential for a commons connecting designers to users and users-as-makers for distributed, local, small-scale, and fully social mode of production—what Greenfield has called an *emergent deman*d economy.

Founded in 2011, the Noun Project is a platform for designers to share, download, and use high-quality icons. These icons can be used in any context but are commonly adapted for use in user-interface design. Designed by contributors from around the world, they are free for download and unlimited use. The "free" account model allows users access to the full collection of more than 1 million vector icons but with the caveat that users must give credit to the icon's designer each time it is used. With a paid subscription, a member gets "perpetual, non-exclusive, worldwide right to use each icon you download as many times as you wish and in any medium, without having to credit the icon's creator."[99]

Previously mentioned work such as the *Autoprogetazzione* manual by Enzo Mari, the open-source instructions for "Pan Fried Stools" by Thing Thing, and Biocouture's vision for an open-source movement in biotextiles indicate the vast potential for a burgeoning design commons to radically remake both design and production as shared, open, local, and participatory processes. What becomes of branding for corporate interests when individual "subjects" in a postcapitalist society can directly modify, customize, and produce their own material goods?

Peer-to-Peer Service Design

One of the more promising precedents for postcapitalist practices can be found in *design for social innovation*. This term has been applied to a broad category of projects that typically bring together aspects of community-engaged research and practice, participatory design, and service design. The central theme of much of this work is the centering of citizen-participants in design processes so that individuals and groups work together on community-scale processes of creative problem solving to move toward new social and economic capabilities. The designer may be an organizer, a facilitator, a mediator, a technician, or any other guiding figure, yet the agency of the people involved is the driver of the work. In this sense, it is a

demonstration of designers committing to the notion that everyone can be a designer. Although this work is not always—or typically—explicitly labeled as anti- or postcapitalist, it is almost always exemplary of civil society working toward the *erosion* of capitalism through an intentional reorientation of social organization that transforms participants into subjects of communal sufficiency rather than turning them into consumers.

The combination of capitalism's division of labor and design's provision of easy-to-use commodities, packaged as "congealed labor," has gone a long way toward deskilling individuals from basic capabilities for sustaining their own material well-being. As J. K. Gibson-Graham put it, "A political project of economic becoming will require an imaginary in which economic possibility is plural and diverse. By speaking a language of economic diversity, we may be able to provide a context in which fleeing energies can be organized and amplified within alternative enactments of economy."[100] These projects emphasize long-term connections and commitments that foreground the importance (and challenge) of the effort to cultivate subjects for active participation in social organization and social —rather than purely economic—labor.

In 2009, the global network Design for Social Innovation towards Sustainability (DESIS) was formally organized to leverage and connect design labs based in universities and schools of design with local, regional, and global partners in working toward sustainable social change. The DESIS network was founded by the Italian design researcher, educator, and author Ezio Manzini, whose work has expanded the notion of what it means to design for sustainability and social innovation. Manzini criticizes most current *sustainability* efforts in design as generally focused on a similar idea of the commercialization of all social and political desires; the idea that you can "purchase" a sustainable lifestyle is at odds with the kind of sustainability that centers social and political life in an economic world of finite resources and deteriorating public and social infrastructures. Manzini envisions a sustainable world as pluralistic, occupied by many different scenarios that are "small, local, open, and connected" (the SLOC framework).[101]

As demonstrated by his efforts at global network building, Manzini is clear about the necessity of working at an institutional level to sustain this work.[102] Design for social innovation is not a project-scale endeavor.

Rather, it requires systemic and long-term efforts, first at the scale of practice and then at the scale of the design discipline. Through the DESIS network, researchers, practitioners, students, and community-citizen participants in DESIS-facilitated projects share their experiences, methods, best practices, results, and frustrations to build knowledge, enhance the disciplinary catalog of methods and outcomes, and connect designers and constituents around the globe.

From 2018 to 2020, DESIS organized the Design × Social Innovation and Cities initiative, which organized design schools around the world in direct participation with their communities. According to a report on the initiative published in 2019, "The possible intersections between city-making, social innovation and contemporary design are many, and highly diverse."[103] The report organized the wide-ranging activities under four subthemes: design for social cohesion, design for regenerating commons, design for urban production, and design for urban infrastructure.

One such collaboration was the Latham St. Commons project in Pittsburgh, Pennsylvania. Working on the site of vacant hundred-year-old garages, a team of students, faculty, researchers, and community participants put into motion a novel approach for improving community health and cooperation. "We are co-creating an inclusive and engaged community, connecting neighbors by collecting and sharing information. . . . We have worked extensively with the community over several years to learn about their needs and wants. This process allowed the project to grow out of the specificity of the place, our local economy and culture. . . . [W]e serve as facilitators to help leverage the community's ability to create healthy environments by providing resources to confidently pursue life changes."[104]

The Latham St. Commons (LSC) generated multiple small but interrelated thematic initiatives. Addressing challenges of health, the site operated work on delivering "important health information about HPV [human papillomavirus] and then offering HPV vaccinations on-site at LSC. Exceeding expectations, 297 young adults were vaccinated, preventing them from getting an HPV-related cancer." Addressing challenges of climate and energy, the site now redirects an estimated 137,916 gallons of waste water annually from the city's sewer system. "We built a continuous, 20 vessel rainwater collection and dispersion system." Addressing

challenges of food access, “for two years we have worked alongside the community, experimenting with growing food on rooftops and vertical walls and giving away significant amounts of produce.” Addressing challenges of underserved youth, the site has “workshops focused on 21st century literacies, strengthening identity, and the importance of self-care.”[105]

The Latham St. Commons project demonstrates *peer-to-peer urbanism*, a form of decentralized, participatory design in the urban environment. In such projects, knowledge about methods are freely exchanged so that they can be adapted to the unique local conditions of other places—an organizational idea borrowed from the open-source software movement.

Social innovation design, as a broad and expanded model of design practice, makes a great horizon for designers engaged in PCD practices. This horizon promotes and clearly articulates the need for new types of designers who may not be yet fully realized through education or practice. Many design-for-social-innovation projects are currently organized by academics, such as those in the DESIS network, and are often financed by grants or other external support. To truly realize the vision of design for social innovation would be to push these practices into fully self-sufficient states, including accommodating the labor of the designers and participants involved. This would represent one of multiple new disciplinary models of design after capitalism.

Conclusion

What becomes evident across these highly divergent snapshots is that making any kind of design truly transformative requires cultivating new subjects. This means not just new kinds of designers but also new kinds of users, audiences, and/or constituents of design as well as new kinds of clients, patrons, and/or collaborators of design—all of whom will become new collaborative agents in the making of new design disciplines. The designers and subjects of anticapitalist design practices have generally operated in a collective resistance to an oppressive or dominant force, such as colonial powers or racist institutions. The designers and subjects of noncapitalist design have generally operated outside of the reach of

capitalist modes of production, either as a deliberate escape or in some form of shelter, such as practices connected to academic research. The designers and subjects of emerging postcapitalist practices of design do so in somewhat isolated or incipient modes, even as attempts to network, connect, and bolster these practices are on the rise.

In gathering up these practices and their collective knowledge—their methods, mindsets, postures, and frameworks—we can see that they could benefit from additional explicit, deliberate, intentional, and collective action to identify them as oriented toward some common goals. *Design after capitalism* names one such direction. It identifies the need not only for new ways of "doing design" but a directional approach to the ways design cultivates its subjects on all sides of the designer's translational role between people and matter, and between culture and economy.

Just as the diagnosis and critique of capitalism are not meant to be totalizing—requiring the admission that a *capitolocentric* view of the world is an illusion, that in reality there is economic hybridity—these design practices also show a broad horizon of possibilities. Examples of anti-, non-, and postcapitalist forms of design practice illuminate the fact that a reorientation of various design practices could make for a very different design, a different material culture, and a different world.

7

Design after Capitalism, in Theory

> What I propose, therefore, is very simple: it is nothing more than to think what we are doing.
>
> —Hannah Arendt, The Human Condition

Institutional Framework for Postcapitalist Design

The goal of articulating postcapitalist design, or PCD—in theory and in principles for practice—is to mobilize the move away from the "competitive individualism" of business-as-usual design (BaUD) and toward establishing and maintaining a design discipline that systematically remakes social, political, and economic relations through its practice. It is an attempt to mobilize actors into new identities.

There may be two distinct ways to interpret the argument for a PCD.

1. Capitalism is a historical system that seems to have run its course. It is no longer fit for a global community of information-driven economies in which scarcity is often invented for the sake of maintaining the system.
2. Although capitalism has harnessed an immense productivity and transformed the world, it has always been a flawed system with dangerous and destructive core logics. It has operated past any reasonable claims of producing rational, distributed returns and is now in decline, making its negative impacts worse.

Following the first interpretation, designers may take it as a practical necessity to get ahead of the decline and disequilibrium of capitalism and prepare a postcapitalist model of design practice. Following the second, designers may take it as a moral necessity to create PCD practices to erode the destructive tendencies of capitalism. Perhaps either is equally valid for

taking up the challenge of transforming design practices. Either way, PCD itself can contribute to transforming both design as a discipline and the broader political economy.

PCD requires new theory to inform new mindsets and methods. To answer the question “Why design?,” PCD would respond: to bring about an equitable social world through the production of the objects and symbols of material culture. How this work is done is as meaningful as its outputs.

The term *postcapitalist design practice* refers to design practices that intentionally expand the purview of their activities to account for the social and economic relations produced by *designing*. As has been described, in postindustrial economies a higher degree of economic value is created through the production of information, services, and novel cultural forms—the products of McKenzie Wark’s *hacker class*—than through the manufacture of physical goods. However, this is not to say that the production of physical goods is reduced. It is not. The value of manufactured goods is increased through their integration with information, services, and novel cultural forms—including symbolic values. At the same time, immaterial (often digital) goods and services produce forms of value that are separate or added onto the manufacture of material goods. This is design’s currency in contemporary economies.

Although any individual economy may be, as a whole, predominantly capitalist, it also contains aspects that are noncapitalist. Erik Olin Wright contrasts the typical metaphor of an economy as a “body”—if part of the body gets sick, the whole body is sick—to the metaphor of an economy as a “pond.” A pond is a complex ecosystem in which many different species flourish, suffer, or adapt depending on changing dynamics in the whole ecosystem.

J. K. Gibson-Graham make a similar case in their articulation of *community economies*. In their examination of different kinds of transactions, different types of labor, and different forms of economic enterprise, they use the metaphor of an iceberg, in which “wage labor produced for a market in a capitalist firm” is just the “tip of the iceberg” of economic hybridity. Underneath the waterline, many other forms of (often unpaid) social labor are at work in the broader economy: within families, in neighborhoods, in schools, through cooperative production, in volunteering,

between friends, through informal lending, as well as through under-the-table services, self-employment, and even illegal economic activity.[1]

A simple example is the burgeoning phenomenon of community gardens in the United States. By producing a common space of shared land for local residents to grow their own fruits, herbs, and vegetables, community gardens also generate a new dynamic of social relations in which people work together, share knowledge, and often also share food in ways that are totally outside the economic logics of capitalism. Ezio Manzini has referred to community gardens, simple and commonplace as they have become, as "disruptive normality."[2] The social, economic, and productive activities that take place through participation in community gardens "disrupt" the typical consumer mindset and actions of buying food from a corporate retailer, which in turn purchases its goods from giant agricultural corporations. Simple acts "normalize" this disruptive activity. The act of growing food *builds* social power by connecting people, creating the conditions to share knowledge about agriculture, and sharing food itself. It undermines the capitalist model while creating new social desires: to join the community, participate, and enjoy the rewards.

Meanwhile, the proliferation of technologies and platforms for sharing information via decentralized peer-to-peer platforms represents an opportunity for other new forms of economic and social relations. Networks create a medium of exchange that can allow for open, collaborative sharing among geographically disparate people with the goal of "exchang[ing] gifts of intangible value" or working together to build new forms of value from their merger. Platforms provide the ground-plane upon which information can move freely if unchained from artificial scarcity, such as through intellectual-property law and protections. The potential forms of value made possible by freely shared or exchanged information demonstrates that the very things of most value today are not manageable by a market. "The proliferation of these non-market economic activities is making it possible for a cooperative, socially just society to emerge."[3]

In connection with Wright's two-by-two model for "strategic logics of anticapitalism," the *erosion* of capitalism within civil society is made possible by generating robust patterns of social empowerment within the existing capitalist state and business apparatus. Over time, these new patterns of doing and knowing—practice and discourse—become

institutionalized so that growing numbers of people will no longer need or desire to rely on capitalist modes of labor and consumption as their primary *ways of being* in society. What this means for PCD practices is that they can exist in the hybrid conditions of an economy that is predominantly capitalist, while building stronger and more robust institutional patterns that wither dependency on capitalism and transform the discipline of design. In this way, designers can play a role in transformational change beyond individual projects or practices alone. The theory of design after capitalism as a discipline rests on the capacity for design practices to actively engage in this process of *erosion*. In other words: play the long game.

Social Empowerment

Wright describes several "pathways to social empowerment," each of which exists at particular intersections among economic power, state power, civil society, and the economy. Those pathways situated entirely within civil society—not changing, attacking, or resisting state governmental systems—are described as "social empowerment directly over economic activity."[4] This is the focal point of PCD practice.

Although tactics of *erosion* based solely in civil society will inherently be limited by capitalist state powers and institutional grips on how designers operate (including legal and competitive obstacles), it is within civil society that design already wields power for transformative change. By transcending capitalist structures in civil society, people are socially empowered and taken care of in ways that also neutralize the harms of capitalist state and civil structures. Design practices affect political economy outside the world of lobbying, party politics, or direct action on policy (which is not say that kind of activity is not also valuable). The next two subsections outline two of Wright's "pathways to social empowerment" and describe their application to design practices.

It's important to note that these two pathways are not mutually exclusive. In fact, they may operate simultaneously, reinforcing each other, and the prolonged practice of the first pathway may create more fertile conditions for practice of the second.

Design in a Cooperative Market Economy

Wright describes how in a *cooperative market economy* the proliferation of diverse worker-owned cooperative enterprises produce the breadth of social and material needs for society in a model of collaboration rather than competition. Worker-members of a cooperative lend both labor and governance to the enterprise, including democratic forms of decision making on matters of how to invest, what to do with surplus, and more day-to-day matters such as logistical operations. According to Wright, "A stand-alone fully worker-owned cooperative firm in a capitalist economy is a form of social capitalism: the egalitarian principle of one-person-one-vote of all members of the business means that the power relations within the firm are based on voluntary cooperation and persuasion, not the relative economic power of different people."[5]

In the United States, worker-owned co-ops are probably most familiar in the narrow form of grocery suppliers. However, Wright's analysis of diverse cooperative enterprises and cooperative associations shows that far more is possible. The Mondragón Cooperative Corporation of worker-owned cooperatives in Spain's Basque Country is a prime example.[6] What began in the middle of the twentieth century as a small manufacturing cooperative making paraffin heaters has expanded to include worker-owned businesses in capital goods such as construction and machine equipment, banking and finance services, engineering, architecture, and insurance. The cooperative corporation even operates its own university. As a federation of independent cooperatives, the Mondragón Corporation is the tenth-largest company in Spain and the leading business group in the Basque Country. In 2015, the corporation's various cooperative enterprises employed more than 74,000 people. Mondragón cooperatives are run on the guidelines laid out in the "Statement on the Cooperative Identity," which defines cooperative principles and provides guidance to direct their behaviors and operations. These guidelines are produced and maintained by the International Cooperative Alliance, a cooperative union "representing cooperatives, which are estimated to be around 3 million worldwide, providing a global voice and forum for knowledge, expertise and coordinated action for and about cooperatives."[7]

Wright outlines the complex details of the governance structure of the Mondragón Cooperative Corporation in four broad points. First, there is a

dual structure of governance within the individual cooperatives. One is concerned with governance of the sociopolitical structure, which is responsible for the direct democratic involvement of worker-owners. The other concerns governance of the technostructure, which is "responsible for the managerial and technical functions of the cooperative, [and] is formally under the control of the sociopolitical structure." Second, each cooperative has a General Assembly, which is formally "the sovereign body of the cooperative . . . responsible for appointing the managing director and, in principle . . . to determine the broad strategies of the cooperative." Third, various Representative Councils operate the decision-making rights of individual cooperatives and bring their concerns to various standing committees that govern "the higher organizational levels of the [Mondragón Cooperative]. These governing bodies . . . coordinate activities across individual cooperatives . . . and formulate long-term strategic plans for Mondragón as a whole." Fourth, individual cooperatives are conjoined through a voluntary structure and retain withdrawal rights should they decide to leave the corporation.[8]

Of course, there are many tensions, pressures, and risks for such a collective of worker-owned cooperatives—some based on external competition, others from a political economy otherwise organized to support global capitalist interests, and yet others from the inherent difficulty of cohesively and effectively operating an organization of tiered forms of democratic governance.

Proponents of worker-owned cooperatives are sometimes cast as "naive" regarding the tough "real world" of business competition. It is thought that groups of workers without clear, hierarchical leadership are not capable of making decisions that can maintain a business and create the necessary conditions for organizational survival and thus the well-being of its workers. This is the myth of the need for a powerful, singular leader such as the contemporary CEO. However, the results of several long-term empirical studies of worker-owned enterprises debunk that myth. Over the past 40 years, several different studies by economists and organizations have yielded significant evidence that "worker participation usually enhances productivity in the short run, sometimes in the long run, and rarely has a negative effect. Moreover, participation is most conducive to enhancing productivity when combined with profit sharing, guaranteed long-range

employment, relatively narrow wage differentials, and guaranteed worker rights."[9]

Several worker-owned design enterprises already operate in this way, including the previously mentioned Sassafras Tech Collective and Design Action Collective in the United States as well as the member organizations of the CoTech network in the United Kingdom. However, many cooperatives still tend to exist as islands of social and economic productivity surrounded by capitalist competition. They are also typically limited to a single domain or industry, such as graphic design, web development, or grocery supply. Even if they do not work for capitalist interests, they compete with capitalist firms and do business with capitalist firms, such as professional service providers (production, development), equipment suppliers, and financial service providers (banks, insurance companies).

Wright argues, however, that "the situation could potentially be quite different if worker-owned cooperatives were embedded within what might be called a cooperative market economy. A cooperative market economy is one in which individual cooperative firms join together in larger associations of cooperatives—a cooperative of cooperatives—which collectively provide finance, training, problem-solving services, and other kinds of support for each other."[10] Worker-owned design cooperatives in federated cooperation with other cooperative enterprises—such as fabrication firms, software-development companies, printshops, and professional or technical service providers—would greatly expand the reach of this model and reduce the reliance on capitalist relations. Cooperative design enterprises *doing work for and with cooperative enterprises* could help build up the infrastructure for a decisively crafted social economy.

Design in a Social Economy

Wright describes a social economy as "an alternative way of directly organizing economic activity that is distinct from capitalist market competition, state-organized production, and household production. Its hallmark is production organized by collectivities directly to satisfy human needs not subject to the discipline of profit-maximization or state-technocratic rationality." Different variations of the term *social economy*

have been used to describe a wide range of "coalitions and solidarities across the full spectrum of activities that fall outside the ordinary capitalist practices."[11] This includes nonprofits and nongovernmental organizations, which are sometimes called the "third sector" (after business and government). Wright observes that the primary way that the social economy is financed in capitalist societies is through charitable donations. The goal of a postcapitalist transformation would be to decouple the construction and maintenance of the social economy from financial dependencies on capitalist relations—something that could be achieved through financial support of cooperatives and collective efforts.

Wright points to *Wikipedia* as a "striking example of almost pure social economy production." The immense scale of knowledge production and dissemination by *Wikipedia* is remarkable considering that it operates entirely outside of market or state support. Most of the effort that goes into producing *Wikipedia*'s entries involves the voluntary collaboration—as well as the occasional feuds, spats, and disagreements—of its large pool of contributors and editors. The infrastructural costs of running it are supported "largely from donations from participants and supporters of the Wiki foundation."[12] The underlying form of *Wikipedia* is very similar to social media, file sharing, and other social networks of voluntary association, yet many of those networks are tied to private interests for purposes of profit—from advertising, data collection, tertiary billable services, or all of the above.

One general role for the designer in a social economy would be to design and maintain nonmarket, voluntary networks through peer-to-peer service design. This role is akin to Manzini's vision of design for social innovation, in which designers become facilitators of community efforts at local problem solving and resource allocation. Participatory design and service design methods are applied within, with, and for the communities who will use them and sustain them. UX design and interaction design are oriented toward community-defined needs and result in digital platforms and physical touchpoints that serve as interfaces among people, information, and exchanges of goods.

A categorical distinction is worth noting here. Social innovation design makes a great horizon for designers engaged in PCD practices. Yet the world will still need material goods, and this need will still be part of

design's purview, which leads to another distinct role for the designer in a social economy. With the proliferation of technologies such as 3D printing, small-scale manufacturing, additive manufacturing, peer-to-peer production networks, and community-access fabrication labs, the production of material goods could be fully decoupled from a market economy and controlled at the community level. Theorists such as Murray Bookchin argued for this type of social economic system long before such autoindustrial technology was available at a community scale.[13] That technology is now largely here. Decentralized application of these technologies is in line with the vision of the RepRap project. One hindrance is that many people might not fathom what is possible with all this technology even if they know how to use it. Enter the designer. In this role, design practices could become ultracustomized to emergent local demand while expanding social forms of production—collectively determining what individuals and/or communities can support materially, technologically, and so on. This view of design's role aligns with endogenous growth theory's emphasis on broad distribution of both knowledge of and experience with technology as well as Greenfield's notion of an emergent demand economy. This approach to technology use and development could allow for the massive diffusion of intimate, even one-to-one designer–user relations. One can imagine *designing with* individuals anywhere in the world, while production can also happen at a microscale anywhere in the world.

Finally, all social endeavors take on symbolic forms. A third general role for designers in the social economy is to create cultural forms of value—building the symbols of association and coded systems of information and communication exchange that make life not just functional but also joyous, purposeful, and robust.

This vision of the social economy emphasizes the capacity for technological development and energy production to become decentralized, decoupled from market forces, and instead controlled, developed, and maintained by communities independently.[14] It puts not only more responsibility but also more opportunity for customization and variability in the hands of participants and creates new spaces where designers might find multiple outlets for their contributions to social and economic production.

Wright cautions that "there is no guarantee that a society within which power rooted in civil society predominates would be one that upholds

democratic egalitarian ideals. This, however, is not some unique problem[;] . . . it is a characteristic of democratic institutions in general."[15] In other words, this economic and social organization would also require political effort to ensure democratic processes of control over decisions related to resources, access, and production.

Community Economies

In challenging the *capitolocentric* view, J. K. Gibson-Graham invite us to look beyond the "tip of the iceberg" of wage labor and profit. Hybridity is found in the many ways that communities organize and exchange knowledge, services, goods, and social experiences. Gibson-Graham map out what an "ethical praxis of being-in-common" would look like by pointing toward issues that a community could determine based on its own values:

- What is *necessary* to personal and social survival
- How social *surplus* is appropriated and distributed
- Whether and how social surplus is to be produced and *consumed*
- How a *commons* is produced and sustained[16]

Asking "What is 'necessary'?" directly critiques the endless pursuit of growth for its own sake. Anything beyond "basic needs" might be considered surplus from a rational economic (capitalist) view, which has been the ongoing justification for surplus to be transferred to profit for owners and investors. Gibson-Graham argue in response that the concept of "basic needs" is in no way an objective, rational, or biological reality. Instead, it is always a "socially embedded ethical decision." The starting point for a *community economy** is to address what is considered socially "necessary" so that a recognition of what is "surplus"—that which is produced beyond necessity—can be identified and allocated.

It's important to stress that what is *necessary* does not end at some measure of "bare life" existence. It may be more akin to a common economic measure known as the *living wage* plus whatever additional

social experiences a community deems necessary. In a community economy, participants would determine what constitutes that living wage. Based on that determination and the needs for maintaining the economy, additional social and economic production would be considered surplus.

Asking how social *surplus* is appropriated, distributed, produced, and consumed brings light to questions about the forms of *social surplus*, including how they will be collected and how they will be used by the community. Taxes are the most adjacent example from a centralized government model. Yet taxpayers typically do not get to decide how much they will chip in, when and how they'll contribute, or how exactly those taxes will be used.[17] In a community economy, more than money would constitute both the necessary and the surplus. Cooking, cleaning, childcare, gifts or hand-me-downs, volunteer time, picking up or beautifying the neighborhood—these and many other "noneconomic" forms of social production enter the balance sheet of a community's needs and assets. How they are produced, how they are "consumed," and by whom create a terrain of social and ethical discussions that could take place outside the accounting of a market, in which different people have different dollar amounts attached to their time and activities—for example, high dollar amounts for superspecialized or competitive capitalist labor versus zero dollars for the equally if not more critical labor of maintaining the home, raising children, and other forms of so-called unproductive labor. As Gibson-Graham explain, "The boundary between necessary and surplus labor is an accounting device . . . and the desire to move this boundary can be seen to have motivated political struggles historically and to this day."[18] Surplus from social production—which may take endless forms, including symbolic and material-cultural forms such as those produced by designers—could contribute to the building and maintenance of a commons.

Commons

The *commons* can be most simply understood as any form of shared material or immaterial wealth or value, including the provision of services. *Commoning* is a form of social labor that manages the resources of a commons through participatory, nonhierarchical methods.

In precapitalist England, the “commons” were open lands that could be equally accessed and used by peasants for their own subsistence farming and gathering of resources. Since the birth of the internet, a whole new concept of commons has emerged, focused on the opportunities for sharing value through digital platforms and digital assets as well as in other ways that make use of global networks such as the World Wide Web. *Wikipedia* has been cited as a “commons” that has managed to make the 200-year-old encyclopedia industry irrelevant by translating the social labor of millions of knowledge contributors into a permanently free and open resource for the planet.[19]

The open-source movement marks another trajectory of commoning. The popular code-sharing platform github is an example. Users share the code for completed projects or projects in the making so that others may extend, modify, or otherwise make use of their work. Thus, an open-source platform produces new social relations. Individuals widely distributed by geography, occupation, interest, and skill level are found working together to solve immediate problems while also thinking up where their combined interests and skill sets might lead to new collaborations.

Designers have historically contributed in various ways to commons of design knowledge, even occasionally commoning their work. With the *Autoprogettazione* manual, Enzo Mari shared the design drawings and construction instructions for a suite of domestic furniture that could be built by amateurs with “off-the-shelf” materials and minimal carpentry skills. The plans for these furniture pieces were shared through a book that could be passed around within a community, where social empowerment could be further built from working together on furniture projects or sharing carpentry knowledge and skills.

In a digital context, the Noun Project has become a well-known and much-used resource for graphic designers. Despite some marketing-speak and capitalist maneuvers such as brokerage fees, it operates as a commons at least in concept.

Other smaller-scale examples of commons that have proliferated—some more recently than others—include peer-to-peer networks for file sharing, community gardens, Little Free Libraries (community book sharing), cooperative housing, neighborhood organizations, and, even more generally, the shared work of maintaining a household. Manzini’s

description of community gardens as “disruptive normality,” disrupting the consumer mindset of buying food from a corporate retailer while sharing knowledge and food itself, indicates the potential for a broader economic model built on similar principles. By transforming the role of “consumers,” new postcapitalist subjectivities are created in which motivations for participation in community economies are predicated on social power.

The work of commoning is generally conceived of in two ways: *communal labor* and *reciprocal labor*. *Communal labor** is the collective effort that a community puts into particular goals of the commons following its initiation. Commons projects that make use of communal labor may operate with loose expectations of participation by its community, or the community may organize itself more specifically with a charter that outlines commitments and responsibilities for participation. An example can be found in cooperative housing, where there may be shared resources for residents to use as well as organized responsibilities, such as scheduled days for members to cook, clean, wash dishes, and so on.

*Reciprocal labor** is oriented toward individual social labor, in which community members “gift” their time or effort with the expectation of reciprocity. Carpools are an example, in which drivers take turns driving each other or small groups to work, to run errands, and so on. In some senior communities, members voluntarily drive others who can no longer drive to meet with a doctor, pick up medicine, or do grocery shopping, perhaps anticipating the time when they are no longer able to drive and will need the community to provide these services for them.

One of the challenges for commoning to become a more broadly integrated model of economic and social life is the risk of commons being depleted, plundered, or privatized. Indeed, one of the strongest criticisms of the neoliberal variant of capitalism has been its seizure and privatization of public assets. However, if the expanding potential of commoning—both in local, physical examples such as community gardens and in digital, platform examples such as peer-to-peer sharing—can be made to fulfill a greater portion of the needs in people’s lives, then a true transformation toward a postcapitalist economy could be in the works. It may be the case that the ongoing enclosures, exploitations, and economic crises of capitalism, which are pushing many people to the margins of economic survival, may actually help trigger the transition toward commoning as a

necessity more than a choice. Here again, we can see novel opportunities for design to create symbolic values and symbolic forms of individual or associational identity to encourage acts of postcapitalist citizenry and render them more desirable. If designers can transform combinations of plastic and metal into objects considered "sexy," can the same be done to make communal living sexy?

Degrowth

The concept of *degrowth* is broadly applied to efforts at minimizing the extraction of natural resources and combating the capitalist logic of permanent growth. The Turkish economic scholar Bengi Akbulut sees degrowth not only as an issue of quantitative reduction but also as a social issue aimed at changing mindsets about the nature and goals of society. Akbulut describes degrowth as a "call for a radical break with the growth imperative and de-throning of economic growth as a societal goal," an agenda that she believes can "deconstruct and oppose the inevitability of the growth imperative." At its core, degrowth is a "radical questioning of the goals of efficiency and profit (or utility) maximization."[20] The Australian design studies scholar Cameron Tonkinwise similarly argues that one of the primary challenges facing designers in the Anthropocene is to "design away"—that is, to design so that fewer things get produced rather than to design existing things into future garbage through product obsolescence.[21]

There may be some obvious yet not trivial opportunities for designers working in material production—physical products, garments, printed materials, packaging design—to work according to a *degrowth* mindset by reducing the material needs of their work to the bare minimum. This impetus toward decreased material production is important, yet it represents only one opportunity for degrowth. Designers creating digital products and services, including web apps, mobile apps, interactive features for the web, video content, and so on, are often pushing the requirements of users' devices, operating systems, browsers or other software, and internet bandwidth. For example, by streaming video with steadily increasing resolution, streaming services create demand for the most recent hardware and/or operating systems on smartphones, tablets, laptops, and other

devices,[22] which in turn accelerates the need for upgrades. Let's not forget that the physical infrastructure and energy consumption of the internet are substantial. It has been estimated that the internet accounts for more than 9 percent of total electricity use globally.[23] Moreover, most of that electricity is used very inefficiently. Designing for degrowth means designing "down," not "up." Fortunately, working within constraints is exactly what designers should be good at.

While degrowth strategies at the level of individual projects will have a compound impact, design practices themselves can also be oriented toward *degrowth*. The legal standard of a 40-hour work week was the result of bloody social struggle in the early twentieth century, but it has largely disappeared, particularly for those doing cognitive, creative, and cultural labor (such as designers). Designers may naturally struggle "turning off" the creative brain once it's fixed on something, yet it is also possible to limit the entanglement of work and personal life that seems to be demanded by the communication channels that keep us constantly connected.

In 2017, France instituted a new labor law protecting employees' "right to disconnect" from email, smartphones, and other electronic communications on the weekend and after the end of the working day.[24] Adapting to this law, some French companies established rules to bar employees from using email or devices after hours, while others have automated email servers to shut down email systems outside of working hours. This new law adds to the labor law France passed in 2000 that sets the working week at 35 hours. Similar measures would be technically easy to adopt for any design practice and would significantly affect the social project of degrowth, particularly at the intersection of *labor* and output.

At a different scale, the Austrian American graphic designer Stefan Sagmeister has famously adopted a work pattern that includes self-induced *sabbaticals*, in which he takes one year off for every seven years of paid studio work. According to Sagmeister, he doesn't imagine he'll retire at the standard age of 65, so he's essentially borrowing time from his future retirement. Sabbaticals are typically found in academia—where they can provide the mental space and time to focus on new research or novel thinking—but they are becoming more commonplace elsewhere. Particularly for creative careers, they can be a buffer against burnout and an opportunity to invest the time to do the work of building up exactly the

kinds of social power that a postcapitalist politics demands (e.g., organizing community groups, distributing social surplus, establishing participatory budgeting in community economies, maintaining a commons, enabling decentralized forms of decision making). Google now even offers some employees sabbaticals that they can use to do volunteer work.[25] Obviously, having a sabbatical requires a certain level of financial security, and many of the examples given here are tightly bound up in otherwise capitalist relations. Yet structuring work life around designers practicing and embodying a degrowth mindset could shift expectations in the discipline and contribute to a new designer subject that works in contrast to stressful and exploitative existing practices. For example, when unpaid internships at well-known studios are oriented toward profit—for the studios' owners and/or clients—it is naked exploitation.

Key Economic Ideas

Additional terms and concepts from economic theory will help clarify the principles of PCD and their application.

Rivalry & Excludability

In economic terms, a good or service is considered *rivalrous* if consumption by one person makes it unavailable for simultaneous consumption by another person. This is the case for most physical commodities. For example, if I eat a slice of pizza, you can't eat that same slice of pizza. Making more slices of pizza will require an increase in *marginal production cost*,* or the cost required to produce one additional unit of the same thing. In contrast, a good or service is considered *nonrivalrous* or *nonrival* if it can be consumed by multiple people simultaneously without increasing the marginal production cost.

Many immaterial goods and services are *nonrival*—for example, information or media, music and movies; public goods such as roads or parks; and even natural goods such as sunshine and wind. If a good or service actually *increases in value* (in any sense of the term) when more people use it, then it is considered *antirival*. This is the case for social

media platforms, where more people using the platform makes it more valuable. *Nonrivalry* implies that the marginal production costs are zero, not that the total production cost is zero. This means there may be costs associated with the original production, such as recording a song, but no additional costs to produce more units, such as duplicating an audio file.

*Rivalry** can be seen as a continuum, not a binary. Many services are typically on the *nonrival* or *less-rivalrous* end of the spectrum, but in varying positions. For example, public roads and the internet can be used simultaneously by many people; however, there is a limit to capacity before congestion makes them slower or less usable. Services typically become *more rivalrous* when human interaction is involved but less when some aspects of the service are automated. A doctor can typically see only one patient at a time; an automated medical information and self-diagnostic service might handle many patients at a time.

Rivalry can also be thought of in terms of benefit of investment. For a *rival* good, such as a slice of pizza, the benefit of one more unit (another slice) is received only by the individual owner of that slice. However, for *nonrival* goods or services, the benefit from an additional unit or added capacity goes to everyone who uses the good or service. Investment in nonrival goods and services benefits a larger pool of people.

The second concept is *excludability*.* Goods or services that are available only to people who pay for ownership of them are considered *excludable*. They are privately owned. This is the case typically for the majority of the goods and services consumed in a market economy. Whether the good or service is rivalrous or not, excludability means that “producers can prevent some people from consuming the good or service based on their ability or willingness to pay.”[26]

Nonexcludable goods are those available to anyone and don’t require purchase of their use. This is the case for common resources such as forests and fishing stocks, public resources such as parks and roads, and even things such as knowledge, art, and scientific and technological information (not constrained by IP). In some cases, excludability is not inherent but artificially created. Compare public parks, pools, and playgrounds (*nonexcludable*) to gated communities with private parks, pools, and playgrounds (*excludable*). The goods are the same, but in the latter case

their use is artificially limited to certain people based on an ability and willingness to pay for them.

According to John Heskett, in "capitalist societies preaching the virtues of 'free markets,' of which the United States of America is the leading example, excludability is at the heart of the economic system and there is a constant struggle to extend and protect its boundaries. . . . There is a growing problem at present in enforcing excludability. . . . Attempts are constantly being made to strengthen laws and regulations to assert the rights of *intellectual property*. Excludability is the key characteristic of the concept of property, which hinges on something being sold or rented for profit and restricted only to those who can pay for it."[27]

Digital technology's capacity to duplicate, copy, and share so many different kinds of nonrival goods and services with no marginal production cost and without any inherent excludability is one of the key pressures on capitalism today. Put even more starkly by Paul Mason, "*An information economy may not be compatible with a market economy.* . . . [I]t is increasingly evident that information goods conflict fundamentally with market mechanisms."[28]

If we map these terms on a two-by-two diagram, we come up with four different designations for goods and services: private goods/services (rival, excludable), club goods/services (nonrival, excludable), common goods/services (rival, nonexcludable), and public goods/services (nonrival, nonexcludable).

One goal of PCD is to design goods, services, and experiences, along with their infrastructures and interfaces of access, toward nonrivalry and/or nonexcludability. To do so requires establishing design criteria such that products, services, systems, and symbols are designed as club goods, commons goods, or public goods, while designing private goods toward *nonrivalry* or *nonexcl*udability.

For private goods and services, this means designing into those goods and services the ways in which they can become either *nonrivalrous* (club goods) or *nonexcludable* (common goods). Excludability is typically defined by the material nature of the good, but this doesn't need to limit its capacity to be designed for shareability, which could make it a club or public good. For example, a bicycle might typically seem to be a private good, but a bike-share system makes a bicycle either common or public,

depending on the system of access. This designation doesn't pertain only to the material design of a good—the bike may be the same—but can be determined by the design of the object's experience or interaction as well as its "aura" (to use the language of branding) or symbolic value as something common, club, or public. Symbolic value that promotes the sharing or passing of goods around a community could move them along the excludability spectrum in the direction of nonexcludability. Can designers imbue symbolic values into shared goods, making their shareability an asset?

Club goods and services (nonrival, excludable) are almost always so because they are made to be artificially scarce. This means that it is the economic model of their access that limits their potential to be *nonrivalrous* or *nonexcludable*. PCD practices would work within design criteria that explicitly limit or abolish excludability or devise models for expanded access.

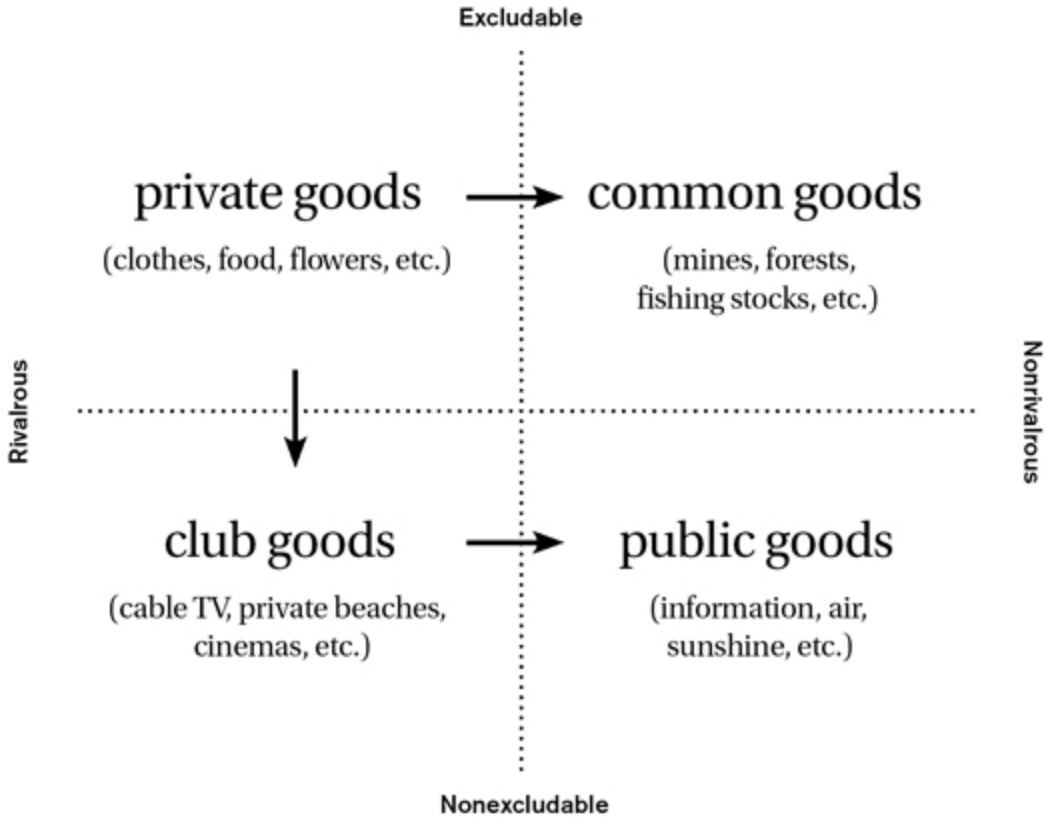


Figure 7.1

Four designations for goods and services, indicating PCD movement toward nonexcludable and nonrivalrous forms.

For common goods and services (rival, nonexcludable), the goal of PCD would be to design to maximize shareability. This maximization can be materially and symbolically achieved in many ways, but the point is that the design of common goods recalibrates the design criteria, as in designing for optimal benefit and endurance without exhausting the good or service's value. In other words, the work of designers specifically aims to prevent exhaustion of the commons from its use. Beyond designing for durability, PCD could include designing use-case scenarios that require and/or reward replenishing the commons. In conjunction with Tony Fry's claim that capitalist products are not really *consumed* but mostly wasted, the design goal would aim toward actual and full consumption, thus maximizing use and symbolic value for many.

For public goods and services (nonrival, nonexcludable), PCD practices would orient design criteria toward "resilience" through user-operated customization, updatable components, adaptability to new purposes or context, programs of maintenance, ease of repair, and so on. There would also be a focus on "universal design" to ensure that goods or services are accessible to as many people as possible regardless of their physical or cognitive ability. Another significant dimension to designing for public goods and services lies in creating the symbolic value for their *publicness*. In other words, designers can contribute to making shared, public experiences with goods and services desirable and *more attractive* than private experiences, using methods similar to branding but applying them to goals antithetical to the private ownership and profit extraction of branded experiences.

This economic framework reframes the nature of ownership and exclusion by asking who benefits from rivalry and excludability. PCD practices would seek ways to limit or transcend rivalry and excludability *throu*gh design.

Endogenous Growth Theory & Intellectual "Property"

The fundamental concept of endogenous growth theory is that "an increase in the share of people working in the knowledge sector will increase economic growth."[29] Because this line of thinking is rooted in a free-market context, in which the orientation of growth is intended for private

ownership and profit, it is commonly believed that intellectual-property law is required to generate the incentive for innovation. However, this assertion has been challenged by longitudinal research globally, which suggests that innovation is encouraged by the *belief in economic incentives* rather than by their reality. In a culture that valorizes risk-taking entrepreneurship, nine out of ten start-ups still fail. Surely, more than property laws must be driving these repeated attempts at failure (yet it's worth noting again the lack of rationality in this system).

Is it possible to do more with less, including broader and more diverse forms of technological innovation, without intellectual-property law? For example, could innovation be incentivized in different ways, through opportunities for collaboration and mutual exchange rather than through private profit? Would collective rather than competitive efforts prove more effective in technological innovation to minimize the use of natural resources for material production and other innovations to minimize or repair the climate catastrophe? Endogenous growth theory claims that focus on knowledge development and human capital—by expanding and distributing knowledge, particularly about technology, to the broadest possible audience—can lead to innovations in efficiency, better use of limited resources, and smarter ways for living with less. However, accomplishing these goals will require investment in those "internal" resources that can benefit the system. This means investment in people through broadly accessible forms of education, social services, and human and health support. Unfortunately, these are exactly the areas that have been massively undercut by neoliberal capitalism. In its neoliberal form, capitalism does not, in fact cannot, support these things because they require higher tax rates and offer no immediate return.

In the sphere of design, design practice must make its own forms of investment in human capital. Designers can take it as an expanded "professional obligation" to grow human capital *through* the practice of design by educating their constituents on design research, methods, and processes as they practice them. This aim to grow human capital aligns with the externalization of processes already taking place in many design practices—for example, as designers facilitate collaborative processes through participatory design.[30]

Conclusion

In summary, an outline of the theoretical framework of PCD looks something like this:

- Design is a social, political, economic, and cultural activity that takes place in civil society; design practices can be oriented toward expanding **social power** by eroding the structures of capitalism—rather than perpetuating capitalism and rather than attacking, evading, or transforming state power that supports capitalism. Two primary directions by which design practice can build social power are by working to create a *cooperative economy* and a *social economy*.[31]
- Design practices can actively participate in and produce **community economies** by recognizing the hybrid potential of community assets, including currently unpaid forms of labor, and by developing the networks, systems, interfaces, and symbolic values that generate participatory opportunities for the decision making that affects communities.[32]
- Design is often materially productive, but it can also work toward **degrowth**—that is, doing more with less—in part by investing in human capital, expanding and decentralizing knowledge about production as well as knowledge about technology and its applications, decreasing labor inputs, and decreasing material outputs of design activity.[33]
- Design can construct **postcapitalist subjectivities** by reimagining "users" as "constituents," designing products, services, and experiences toward increased social, political, and economic agency, and by becoming themselves more politically and economically committed to building the infrastructures needed for autonomous practices focused on community economies.[34]

The primary principles of PCD are to create **social power**, to build and maintain **community economies**, to enact **degrowth**, and to construct **postcapitalist subjectivities**. These principles can be applied toward

withering design for the private, profit-oriented interests of BaUD through the *institution* of design practice—merging design entrepreneurship with social empowerment. As PCD practices expand, a network effect can modify design as a discipline by creating new and more opportunities for designers to work in the PCD model and by expanding disciplinary knowledge and support. At the scale of the *project*, this effort can be implemented by shifting products and services increasingly toward nonrivalry and nonexcludability; at the scale of running a *practice* and sustaining a *discipline*, this can be implemented by efforts toward endogenous growth, investing in and distributing knowledge, access, and experience with and about design and technology. All of this will take concerted and deliberate efforts toward transformation.

III
Transformation

Another world is possible.

—many people, past and present

8
Postcapitalist Horizons

> For capitalism to end, then, it must provide for its own destruction—which, I would argue, is exactly what we are witnessing today.
>
> —Wolfgang Streeck, *How Will Capit*alism End?

Foresight

Strategic foresight is the study of the future for purposes of preparedness and change. In practice, the discipline makes use of *scenarios* to assess, evaluate, plan for, or work toward *possible futures* within a particular domain. The goal is most adamantly *not* prediction. "The core principle of strategic foresight is that the future is multiple, not singular."[1] Designers use similar methods when they cast yet-to-exist new products or services into a narrative account of the lives of potential users (personae), typically proposing multiple possibilities.

The foresight process involves determining key *drivers* affecting the domain of interest. For example, what social, technical, economic, ecological, or political transitions will affect the future of design? Drivers are thematically clustered bundles of smaller *trends*. Whereas trends are directional—for example, barriers to entry in the profession are going down, the number of college degrees are going up—drivers indicate broader movements of change whose directions are not yet determined or may be multidirectional. Drivers push and pull a domain in ways that can result in multiple possible futures. Generally, the idea is to map out how the drivers affecting a domain might lead in different directions over the course of several time horizons. The concept of *horizons* may be most familiar to designers from the "three-horizons model" for innovation strategy. In this model, Horizon One presents the space of opportunities for improved performance or enhancements of existing product or service offerings—for example, determining the new features to include in the next version of product X. Horizon Two presents the space of emerging opportunities, new

ventures that require investment to generate new capacities—for example, connecting product X to the yet-to-exist product Y by creating a new digital platform. Horizon Three is that space farther out where both opportunities and competition are unknown; however, it may be beneficial to invest in research and development to plan for a pivot into this space, even if only a preventive maneuver against competition.[2]

Foresight similarly sees the future as both multiple and evolving. The iconic diagram for this kind of thinking is the *cone of possibilities*: the farther we look into the future, the broader the range of possibilities appears. The method used to consider what's possible, probable, and preferable based on a particular horizon is the *scenario*. Descriptive scenarios of a domain's possible futures are useful for organizations and enterprises to think about "discontinuous change" that could come from external forces. Discontinuous change refers to a situation in which the systems, structures, and operating logics of an entire domain are transformed—such as an overhaul that dramatically remakes an industry (as is the goal of "disruptive innovation"). An iconic example is the disruption of video rental (a retail model) by on-demand streaming services (a subscription model).

There are several scenario-building methods, and their applications to organizational planning are numerous. One approach is to distinguish and articulate four *archetypes*: generic scenario types defined, in part, by their distinctions from each other and from the present. Archetypes operate as a scenario-building model that is separate from the specific content of the scenarios. The relationships and distinctions between the scenarios help give its audience a better understanding of the values and goals underlying each proposition.

The *baseline* scenario is one in which the current system—with its existing structures, logics, and patterns of relations between components—continues to grow.[3] Continued growth is the central assumption of capitalist society. It is both the official and unofficial (or assumed and unnamed) view of the future of most modern governments, businesses, educational systems, and organizations. "The purpose of government, education, and all aspects of life in the present and recent past, is to build a vibrant economy, and to develop the people, institutions, and technologies to keep the economy growing and changing, forever."[4] Baseline scenarios can be useful as

cautionary tales. As we think through the long-term implications of a system and its assumptions, charting out how continued growth could bloat and create conflict among components of the system, the problems and contradictions underlying the system become more apparent. This greater clarity creates a perspective from which future problems might be avoided or minimized through reform, or it may become apparent that the system itself requires transformation.

After articulating the baseline scenario, the foresight commitment to multiple possible futures calls for defining "alternatives." The *collapse* archetypal scenario examines a state of dysfunction in which the system in question falls into disrepair and the ways of doing things no longer work. A *new equilibrium* archetypal scenario examines how the system might adapt to confront a major challenge to its operation or existence, struggling but reforming and ultimately overcoming collapse. For capitalism, the collapse scenario would represent a rather bleak "falling apart" of the social fabric of democratic societies. The new equilibrium scenario might be equivalent to a reformist approach of "fixing" capitalism in order to save it from itself, if only temporarily. Both scenarios are interesting thought experiments. However, for the purposes of crafting a vision of postcapitalist horizons for design, we will compare the *baseline* to a *transformation* scenario.

A *transformation* scenario "entails fundamental change to the system/topic. The rules of the game are 'scrapped' and new ways of doing things emerge."[5] The pattern of change in transformation scenarios is represented by a slow adoption of a new system that coincides with the decline of the old system. At some point, the new system demonstrates its value and potential and begins to spread, and the old system withers. As the old way of doing things erodes, a new set of rules, relations, and roles are adopted within the new system ("erosion"). It is this aspect of transformation that is described by the notion of "discontinuous change." Discontinuous change contrasts with incremental growth and the latter's subsequent consequences and partial or temporary remedies.

Discontinuous change is sometimes expressed in terms of technology-driven transformation. However, as the history of capitalism has shown us, technological change alone tends to favor the already economically or politically powerful. Holistic social transformation requires a restructuring of social, economic, political, *and* cultural life, in which technology can

play a role but is not the primary agent of change. A transformative future is most likely to come about by concerted, collective, and intentional efforts toward discontinuous change. Transformation scenarios can be useful even when deliberately aspirational. They can set agendas by pointing toward a "preferred state" or preferable future. Preferable futures operate as horizons toward which an organization or enterprise can set its sights and direct its efforts, even when the details are a bit fuzzy. A transformation scenario reveals the gap between a preferred future on the horizon and the current situation. That gap expresses the scale and nature of the work that needs to be done to bring about intentional change.

Drivers

Based on the Framework Foresight model developed at the University of Houston, a *driver* is a "thematic cluster of related trends/inputs that is driving or shaping change." It's not surprising that the foresight world, as a discipline concerned with the study of the future, is rich with studies, scenarios, and images of the future of political economy, including variations on "postcapitalism" or "after capitalism" scenarios. Meanwhile, a diverse philosophical, cultural, economic, political, and sociological literature on the possible contours of postcapitalism has resulted in surprising agreement on the forces that today appear to be driving a new economic-political order. Peter Frase, Paul Mason, Nick Srnicek and Alex Williams, Erik Olin Wright, Peter Fleming, and Wolfgang Streeck all generally point toward climate change and automation of labor as the primary forces shaping *possible* postcapitalist futures. They do so with varying amounts of optimism about the likely outcome. Frase even uses a two-by-two framework to create four futures out of these two axes: the response to climate change can lead from scarcity to abundance on one axis; new forms of social organization can focus on hierarchy or equality on the other axis.[6]

Not only is the future multiple, but also, most importantly, it is dynamic. *How* large groups of people respond to social, political, and economic challenges determines the variety of possible futures. The possibility of an egalitarian, democratic postcapitalism is only one of those possibilities. It's

typically the only desirable one, and it's also the one that will take the most effort. Possibility precedes choice, which precedes action.[7]

The first driver is **climate change**. We're deep into an ecological crisis, and even the best prognosis from the scientific community is quite dismal. Destructive weather patterns will submerge, flood, burn, or blow away cities. The destruction of arable, livable lands will lead to increased migration. Water and food shortages will lead to increased human conflict. It is a planetary-scale challenge. "The biggest danger . . . is not that we simply fall off the climate cliff together. It is that human civilization does adjust to the climate catastrophe, but in a way that only carves out a comfortable existence for a tiny ruling class, cocooned in their bubbles of wealth strewn around a wider world of deprivation."[8] Mitigating this dystopia may depend on widespread cooperation, which could come in the form of voluntary cooperation through open or free sharing of information, resources, labor, or other efforts at collective sacrifice to reduce or deal with the effects of global warming locally and globally. Alternatively, change could come in the form of strict governmental regulations, ensuring and enforcing necessary changes. Either path is generally antithetical to capitalism. For design, the challenge concerns not only the production of physical products—and the impossibility of continuing apace with business as usual—but also the cultivation of subjectivities oriented toward social and material transformation.

The second driver is **automation of material and cognitive labor**. It includes robotic automation and artificial intelligence. The German economist and engineer Klaus Schawb, who founded the World Economic Forum, has popularized the term "Fourth Industrial Revolution": "We stand on the brink of a technological revolution that will fundamentally alter the way we live, work, and relate to one another. In its scale, scope, and complexity, the transformation will be unlike anything humankind has experienced before."[9] Estimates vary widely regarding just how many human jobs will be replaced by automation, but it is clear that most forms of work will be affected and modified. Many others will be rendered obsolete. It has been argued that previous waves of automation created new jobs even as some were destroyed (remember "creative destruction"?). However, it is often the case with automation that large numbers of lower-skilled jobs are destroyed while small numbers of higher-skilled jobs are

created. This process most hurts those with low levels of education. Heavy investment in education could help offset this negative effect (among other social issues). However, neoliberal economic policies head in exactly the opposite direction, defunding and privatizing public education and research. There are two extreme ends on the spectrum of possibilities ahead. On one end: billions of people around the world will be permanently unemployed, while the tiny ruling elite who own the technologies of production live comfortably off their yields—isolated from (and heavily protected against) the enormous population of dispossessed humans scratching out a meager survival. In this scenario, the vast majority of people are "renting" their existence.[10] On the other end: the abundance of automated material production will free people from most of the work required today. Most tedious, physical labor will be eradicated. Governments or other political forces will ensure that the yields of this automated efficiency are distributed widely so that people will have more discretion to choose what kind of and how much work they would like to do. This is the postscarcity, postwork version of the scenario.

The third driver is the proliferation of **the platform economy as social organization**. The economic structures and logics of the platform economy are already *embedded* in social life. However, as network effects reinforce these social and economic mergers, new social organizations proliferate. For example, both the Arab Spring and the rise of the "alt-right" political movement in the United States used decentralized methods of organization and platform connectedness to generate political power. The platform is the plane upon which most material production and information exchange already takes place, and it is through platforms—directly or indirectly—that most designed experiences reach their audiences. There may be obvious implications for the design of digital products and services, but physical experiences and material production also take place via platforms and the specific kinds of connectivity and exchange that platforms make possible. Multiple opportunities for design to generate new models of digital value unfold, dependent on the economic, social, political, and legal directions—for example, scarcity versus abundance mindsets, intellectual property versus open source, and so on.

The platform as social organization also shapes subjectivities. We can see ourselves as active participants in platforms, as passive consumers of platforms, or as objects trapped in and actively exploited by platforms. How

we see ourselves shapes the actions we take and thus permits us to shift our positions within the platform economy. As powerful as Facebook, Amazon, and Google may seem, they could come to an end through concerted effort motivated by the will of people around the world (a "political will" larger than the politics of specific nations or states). In the example of Facebook, all it would take is a mass exodus from the platform, and then *poof*, it's gone (unless Facebook were to hold our data for ransom or use it to blackmail us until we return, a scenario that actually doesn't seem *too* far-fetched). This exodus would have to start with a massive change in the subjectivity of Facebook's 2 billion users and be followed by those users' actions. No small order! In short, the work of cultivating new subjects for a postcapitalist society may be an even bigger challenge than the material-technical ones. Yet this is a territory in which design is uniquely situated to assist.

Baseline: Neofeudalism

Driver Features

Climate Change: Retreat and fortify; migration and nationalism

Technology: Neofeudalism; intellectual property, privately owned automation; privacy as luxury

Platform Economy: Rentism and exclusive access

Design Features

Paltry attempts at "sustainability"

Branded interfaces and mixed realities

UX design as rent collection

Baseline Scenario: Neofeudalism		
Drivers	Scenario Features	Design Implications
Climate Change	Retreat and Fortify; Migration and Nationalism	**Paltry "Sustainability"** **Designing Nationalism**
Automation	Neofeudalism; Intellectual Property; Privately Owned Automation; Privacy as Luxury	**Branded Mixed Realities (XR)** **Branded "Public" Spectacles** **Relentless XR Advertising** **Data Extraction**
Platform Economy	Rentism; Exclusivity of Access	**UX Design as Rent Collection** **Dark Patterns**

Figure 8.1
Baseline-scenario drivers, features, and design implications.

The *baseline* scenario for postcapitalism goes something like this: Technological innovation produces ongoing increases in material productivity, continually forcing down the costs of material production. However, the potential for accessing the abundance of this productivity is "stymied by ossified class structures and the state powers that defend them." Not only are the means of this productivity privately owned, but so too is the knowledge to create it. "Intellectual property and the rents that flow to it are the central categories" of this scenario.[11]

Robotics, machine learning, and various forms of artificial intelligence become new tools in the restructuring of work. Professional disciplines as diverse as law, medicine, and finance as well as labor fields as diverse as retail, truck driving, and factory work see massive cutbacks in human labor. These cutbacks create ever larger pools of labor for gig work, such as providing assistive, personal, or creative services for others in wealthier areas of the world, or for piecework, such as small batches of manufacturing, assembly, or physical labor.

Microtransactions of money and data are enacted in nearly every encounter. Tiny fractions of time and social or economic advantage are calculated and embedded into the fabric of everyday interactions with businesses, public services, and public spaces as well as between individual

people. The platforms that operate, monitor, and collect behavioral data from transactions reach their tendrils deeper into new territories, such as various augmented reality (AR), virtual reality (VR), and extended reality (XR) experiences. These experiences include a multitude of intersections with physical environments and objects (XR, IoT). Wearable technologies are deployed to assist in goal-oriented tasks for work, play, and personal improvement, giving platform owners deeper access to the psychological dimensions of everyday life.

The history of capitalism "has been about pushing market forces into sectors where they did not exist before."[12] New digitally enabled services appear in the last remaining cracks of yet untapped physical experience and flood into virtual spaces. These services begin to splinter. If you're willing to be constantly barraged with XR advertising, in which it becomes very difficult to distinguish commercial space from anything else, many services can be used freely at first—except for the cost of data extraction.

"In the end, it appears that platform capitalism has inbuilt tendencies to move towards extracting rents by providing services."[13] Accessing services, information, media, and cultural forms becomes dependent on the ability to pay rent in the form of subscription services, pay-as-you-go services, or "premium" access memberships that combine access through multiple platforms. "Those who control the most copyrights and patents become the new ruling class."[14]

There becomes scant difference between self-directed personal "improvement" and explicit coercion or behavioral modification. This is most evident in the context of the new forms of knowledge work, which demand constant flexibility, availability, and reinvention—to be always ready to do whatever is needed for the combined luxury of having cognitive work to do and the satisfaction of crushing the competition (even when there is none).

Because this system is built on the extraction of "rents" rather than on production, it is no longer capitalism as capitalism has been traditionally understood. This postcapitalist horizon looks like a tech-enabled return to something like feudalism. In fact, some have named this emerging phenomenon *neofeudalism*. "Monopoly concentration, intensified inequality, and the subjection of the state to the market have transformed

accumulation such that it now occurs as much through rent, debt, and force as it does through commodity production. . . . Capitalism is turning itself into neofeudalism."[15]

Platforms and their attendant services become more interdependent and more expensive over time. Once users are enclosed in platforms, the rent goes up. To remain in the system, you pay more, or you are left out with increased difficulty in accessing material prosperity due to a lack of connectivity and information. This tendency, based on the monopoly tendencies inherent in capitalism, suggests that the "existing inequalities in income and wealth would come to be replicated in access inequalities," in which the high price of entry becomes a barrier separating the wealthy from the rest.[16] Privacy becomes a luxury (as Apple's tightly sealed, luxury hardware and commitment to privacy already demonstrates). To opt out of data collection and other modes of surveillance, those with the means can purchase privacy services. This is one more way that growing levels of inequality are exacerbated by technology. The digital divide becomes the "access divide," deepening social inequalities.

Economic, social, and political asymmetry continue to grow. Because "the techniques to produce abundance are monopolized by a small elite," ownership and control over information—and thus access to material prosperity—are highly centralized through platforms owned by the *vectoralist* class.[17] This accumulated power continues to undermine any semblance of control by democratic processes, and the basic "contract" of democracy proves increasingly ineffectual. Attempts at regulation are continually rebuffed, increasing predatory practices, particularly by *vectoralist* interests.

Meanwhile, climate change leads to an acceleration in extinction, radical weather patterns, and challenges to survival for the world's poorest. These effects lead to increased climate migration and to the ratcheting up of nationalism and far-right populist political movements in response. Authoritarian tendencies in previously democratic societies not only increase but also become more and more popular, giving cover of legitimacy to autocratic rule. These movements seize on the impotence of democratic governments in the face of corporate power. See the example of Singapore: it is often touted as a leader in "smart city" development, showing that technologies can create a more efficient urban experience. Yet

this description belies the authoritarian political conditions behind that version of the “smart city.”

The divestment of citizens from political control leads to desire for symbols of strength and prosperity in the place of actual political and economic power for the many. Governments “deal with” growing crises of economic and ecological catastrophe by taking a harder-line approach to austerity, precarity, and international corporate control.

Neofeudalist Design

In this scenario, there is a continued instrumentalization of design through research, user-centered practices, and methods for the subjectification of users into consumers and passive data producers. Design practices are key tools in surveillance, extraction, and manipulation—all while normalizing these shifts in purpose under new and clever guises of efficient and rationalized living, working, and being amused. The use of artificial intelligence creates a technocratic form of design in which data collected from billions of users is applied toward generative methods of producing physical and informational products and services. The rents charged by platforms are collected both explicitly through various models of subscription or membership and implicitly through hidden forms of data and monetary extraction. These moves are aided and abetted by UX design: the design of experiences for the purpose of extraction.

Multiple internets become splintered with a sprawling mishmash of partially functional “free” content locations and layers of pay walls, pay-as-you-go services, memberships, partial subscriptions, and silver, gold, platinum, VIP, and Cloud-Ultra-Executive plans. The remaining free space of the internet is mostly a minefield of bait-and-switch lures to paid content—a particularly large and somewhat lucrative space for UX designers to operate dark patterns of manipulation and deception.

Designers work with artificial intelligence to create brands as highly nuanced, customized, and immediately responsive entities. Brands talk directly to individual user-consumers through conversational interfaces that attempt to build deeply involved, long-term relationships. Brands take on roles such as friend, assistant, and personal shopper—communicating or

making purchases in *anticipation of desires* rather than in response to explicit requests.

Brands are further detached from specific physical products and take highly differentiated approaches to engagement with their users. Brands become “lifestyle interfaces.” They offer different ways in which users can interact with an enormous suite of services, communications, and catalogs of physical products. In other words, the products and services don’t vary so much as the ways in which you access them: Alexa, with all her charms, is only one disembodied friend to choose from among many.[18]

Brands are both literally and figuratively *platforms*. The various virtual and mixed-reality chatbots that people use to help navigate their daily lives —for work, material goods, services, entertainment, and even companionship—are extensions of brands. Behind the scenes, the complex arrangement of licensing and contract agreements between product and service providers, manufacturers, and the brand platforms is far too baffling for users to ever know where their goods, services, or information are really coming from—let alone where, how many, or which humans are involved in the process.

Branded “experiences” become more pervasive and more diffuse. Walking the stretch of one city block, you might encounter what feels like endless branded experiences, ranging from physical forms of marketing embedded in the environment to a digital panorama of opt-out, members-only, or aggressively promoted AR, VR, XR, and mixed-reality experiences. Just as the neoliberal city comprises surfaces bought and sold for advertising, the “smart city” comprises privately owned vectors for embodied forms of advertising, immediate transactions, and data collection —all performed through various hybrids of mixed and extended reality. Nearly everywhere you go, you have the opportunity to virtually transport into some form of experiential marketing stunt. You’re never fully sure who else is “there” with you in a hybrid physical-virtual space—or of their intentions. Endless opportunities for user-to-user interactions blur distinctions between any stable notions of “here” and “now.” Because the economic incentive for most of these experiences is a hybrid of marketing and data extraction, the option to be left undisturbed or to mute these experiences requires purchasing “opt-out subscriptions” (which are also

available through brand memberships). Everyday life in public urban spaces begins to resemble a giant virtual casino.

Designers conceptualize, model user experiences, and produce hyperlocal third-party marketing and advertising that proliferate within mixed-reality interfaces. This process leads to a recursive nesting of interfaces within interfaces, even in those seemingly closed systems such as banking and personal finance. “It’s turtles all the way down.” Microtransactions attached to microtransactions abound, even when consent for purchases is dubious at best.

Designing interfaces for brand interactions requires constantly negotiating what the product or service is claiming to offer and the actual motivation of the brand or enterprise producing it. Branded interfaces connecting users to media content, local news, and navigation or accessing amenities in public space are actually designed to gather hypertargeted data on users’ movements, behaviors, and microinteractions. As the contours of this “surveillance dataism”[19] become more apparent but also more common, many people become resigned to the fact that accessing the digital overlay of the world comes at the price of their every step, breath, and decision being tracked.

Under the rubric of the IoT, designers create seemingly endless low-cost, sometimes even free IoT-connected products for the kitchen, bath, home, and work. These devices require service subscriptions in order to function. They are frustratingly *unusable* when the Wi-Fi goes out, which is often because demand on the limited bandwidth services available to low-income users stretches the services to their limits.

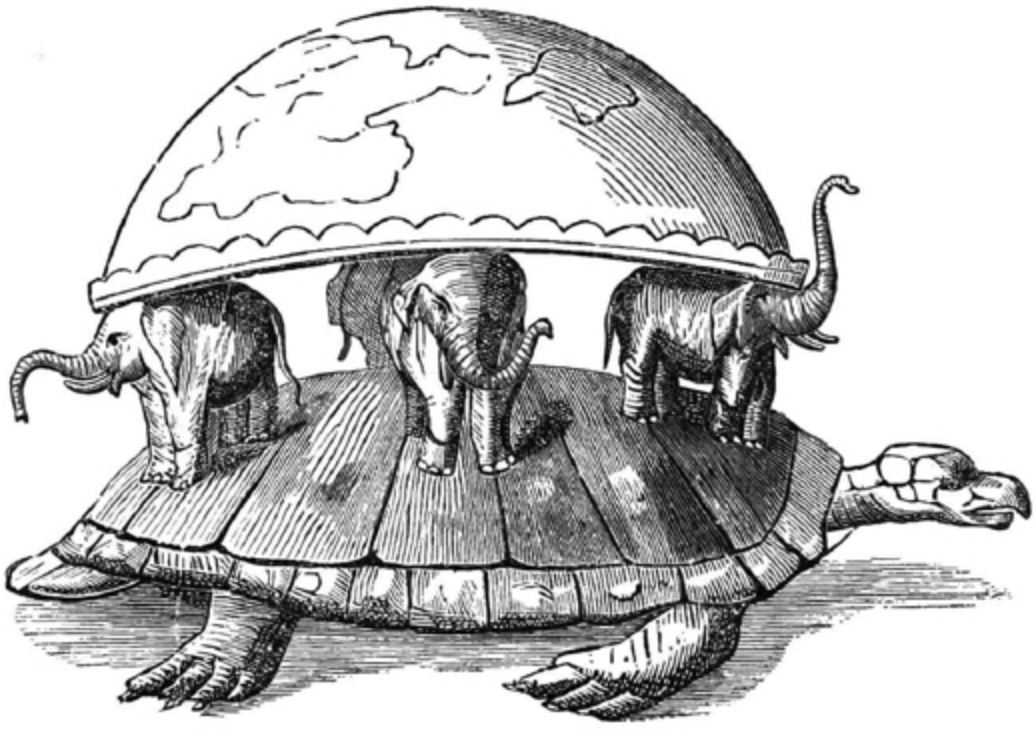


Figure 8.2
Mental models shape perception. “The Hindoos [*sic*] held the earth to be hemispherical, and to be supported like a boat turned upside down upon the heads of four elephants, which stood on the back of an immense tortoise. . . . [T]hese animals were merely symbolical, the four elephants meaning the four directions of the compass, and the tortoise meaning eternity.” This is possibly the origin of the phrase “it’s turtles all the way down,” which refers to an obscured view of the world in which the forces behind the immediate phenomenon are hidden, making things appear simpler than they are. (*Source*: “The Hindoo Earth,” figure 4 in “How the Earth Was Regarded in Old Times,” *Popular Science Monthly*, March 1877.)

The disciplinary language of design recognizes that its purpose has become more particular than its claims to “solve business problems” or “create value and growth.” It becomes explicitly clear that designers are hired to do the work of generating rent for platform owners (the *vectoralist* landlords). UX design evolves to become more explicitly focused on designers’ abilities to create novel forms of extraction. This is no longer work done by people who have an education steeped in humanist concerns. The field of design, in particular UX design, is populated by a labor force that brings a combination of business, psychological, and social scientific methods to the explicit tasks of dispossession. Dark patterns are no longer shameful offenses but essential tools in the designer’s skill set. Even the term *user-experience design* becomes dissociated from the notion of *the user* being at the center of the designer’s concern. It is abundantly clear—

explicit even—that *users* are actually targets for resources. UX *designer* is a professional title for one who models and produces new forms of experience that are in fact targeted at extraction. UX designers are the rent collectors for the ruling class.

One of the most anticipated, hyped, and disruptive modifications of everyday life in the near future is the proliferation of autonomous vehicles. In a project from 2013, speculative designers Anthony Dunne and Fiona Raby[20] gave a powerful account of possible futures for different political economies through the lens of *mobility*. The project United Micro Kingdoms illustrates multiple possible futures by producing four scenarios. Each scenario is animated by a design proposal and prototype that uses human transit as a lens through which to view the political, social, and economic values of the society. Of the four, the proposal titled "The Digitarians" is best aligned with the baseline future scenario just described. The digitarians "depend on digital technology and all its implicit totalitarianism-tagging, metrics, total surveillance, tracking, data logging. . . . Their society is organized entirely by market forces; citizen and consumer are the same." Digitarians get about in their "digicars"—electric, autonomous vehicles such as those currently being explored by Apple and Google. The digicars are more like media devices than vehicles. They are interfaces for movement in the physical world but also for "navigating tariffs and markets. Every square meter of road surface and every millisecond of access, at any moment, is monetized and optimized." Dunne and Raby summarize their goal in this work as critical rhetoric for designers to consider: "The digicar stands for all that is wrong with today's services. It is the ideal solution for a society that promotes freedom of choice and entitlement above all else, even in the face of ever diminishing resources, such as road space."[21] This is the neoliberal smart city, with all of its authoritarian politics and surveillance, brought to life through the design of sleek, seductive, "connected" products and services.

The everyday life of the world described in the baseline scenario is hardly far-fetched. Daily life today already "sounds like nothing so much as the conspiracy theory of a paranoid schizophrenic":[22] constant monitoring, manipulation, and surveillance by shadowy entities and the interests supporting them, acting upon our minute daily acts, words, and thoughts.

Designers today work hard to hack cultural “difference” out of the cycles and methods of extraction. The baseline scenario is just a lot more of that.

Transformation: Egalitarian Postcapitalism

Driver Features

Climate Change: Tech-enabled abundance; degrowth of material use; local technology

Tech: Automation of least-desirable labor; work, postwork, and working for each other

Platform Economy: New commons

Design Features

Remote, distributed, personal material design and production

Platform community economies, social and material exchange

Constructing postcapitalist subjectivities

A *transformation* scenario portrays a new system that is fundamentally different in its logics and structures from the existing one. The transformation brings into being—however quickly or slowly it comes about—new rules, norms, logics, social relations, and ways of doing things. Many scenarios posited for a transformative postcapitalist future are predicated on the notion that the Fourth Industrial Revolution will create automated, robotic, and/or intelligent technologies ample enough to create material *abundance* with little or no human labor. “What happens if production requires very little human labor or none at all?”[23] What appears today as the “threat” of eradicating jobs and creating a permanent underclass of the unemployable doesn’t have to be inevitably threatening, though. If automation can truly transform the productive requirements for the material necessities of human life into a nearly fully automated enterprise, it is possible for the benefits of this system to be distributed to reduce the dangerous and tedious labor required by humans around the

world while increasing the quality of living. Conversely, if the owners of automated production dramatically reduce the world's labor force, they will run into one of capitalism's inherent risks: the downward spiral of plummeting demand from a growing population who can't afford the products that fuel production and thus profit.

	Transformation Scenario: Egalitarian Postcapitalism	
Drivers	Scenario Features	Design Implications
Climate Change	Degrowth; Decentralizing, Growing, and Repairing Production	**Remote, Distributed, Personal Material Design & Production**
Automation	Automation of Tedious Labor; Work and Postwork; Working for Each Other	**Platformed Community Economies; Social & Material Exchange**
Platform Economy	New Commons	**Constructing Postcapitalist Subjectivities**

Figure 8.3
Transformation-scenario drivers, features, and design implications.

This means that automation is largely a social and political issue. There are many different political projects to consider inside a postcapitalist scenario. One that comes up repeatedly—and has received an uptick in mainstream attention recently—is the universal basic income (UBI). A UBI is a governmental public program for a periodic payment delivered to all citizens of a given population without any requirements. It is a guaranteed living wage. The idea has surfaced many times in US political history. It's closest moment of adoption was under President Richard Nixon.[24] In a bid for the presidential election in 2020, Andrew Yang ran on a platform centering a UBI, or what he called the "freedom dividend." It shouldn't be too surprising that some of the loudest support for UBI today comes from Silicon Valley—indicating that many in the tech industry recognize the risks that the technologies they're working on pose for current social structures (and their own enterprises).[25]

In the transformation scenario, technological innovation produces ongoing increases in material productivity, continually forcing down the costs of material production—as in the baseline scenario. However, rather than submitting to state-supported class structures that keep abundance limited to a small set of owners, the network of cultural and material producers modifies social relations *through their means of production*. This work is difficult in the face of both the political structures that enforce private ownership as well as the competitive capitalist landscape that surrounds it. However, the steady building up of social economies through concerted, collaborative effort—including in the creative fields Wark identifies as the *hacker class*—creates new institutions, networks for exchange, models of practice, disciplinary knowledge, and social relations that erode capitalism from within. As the hacker class shifts away from subservience to the vectoral class, it finds mutual compatibility, creative fulfillment, intellectual rigor, social satisfaction, and a sustainable level of material exchange within its own communities, both local and global.

Strengthening and growing the independence of robust social structures for the economy does not require capitalism to disappear altogether. Reflecting Wright's "pond" metaphor and J. K. Gibson-Graham's "tip of the iceberg" metaphor, the project is aimed at growing capacity for alternatives to capitalism over multiple generations—a project that can (and, in fact, probably *must*) take place within a hybrid economy that includes capitalism.

The End of Design (as We Knew It): Goals & Principles

Paul Mason lays out the goals and principles for what he calls "Project Zero": "a zero-carbon energy system; the production of machines, products and services with zero marginal costs; and the reduction of necessary labour time as close as possible to zero."[26] With a focus on eliminating unnecessary forms of labor, Mason is describing a particular postcapitalist scenario that emphasizes local control over the determination of needs and desires, small-scale production processes, and collective forms of technological development. These activities make active use of globalized commons made possible through platform-based exchanges of information, knowledge, and value. In this context, the designer has multiple new roles to play in directing the flows of material resources, labor, and

communications as well as in the formation of new subjectivities. Because design contributes to the construction of subjectivities—individual and collective perceptions of identity, purpose, role, and relations—it also contributes to the cultural dimension missing from Mason's economic and political agenda.[27] In other words, designers have roles to play in constructing the material, informational, economic, and *cultural* structures, infrastructures, and systems for an *egalitarian* postcapitalist society.

Adam Greenfield combats the corporate (and authoritarian) vision of the "smart city" as a singular, top-down approach to "highly technologised urban environments" by subverting some of its own "lean development" strategies.[28] He modifies the concept of the *minimum viable product*—a product version with just enough features to satisfy early customers and provide feedback for future development—to describe the making of *minimum viable utopias*: "an experimental, participatory, iterative and, above all, multiple approach to the making of urban place—one that sensitively leverages advanced technology, but is not driven by it." Greenfield articulates four distinct ways that communities—that is, regular people, not experts or government officials—can localize control of the determination of needs, development, and use of technologies in their daily lives: people making things, people making data, people making networks, and people making places.[29] Through this lens, we can see four distinct types of activities—all interlinked—in which designers take on active and specific roles in developing and maintaining social power through the "making of." These four areas offer a framework to describe different roles for postcapitalist design in this transformation scenario.

People Making Things: Decentralizing, Growing, & Repairing In the sphere of material production, designers remain intimately involved in determining specifications for the production and use of material goods. However, designers use this process to work toward *degrowth*, aimed particularly at hyperconsumption in the overdeveloped world. For the achievement of this goal, true participatory design in the cooperative model —rather than exploitative codesign practices—becomes common. Some designers' material expertise will be targeted toward ultracustomized garments, objects, goods—many upcycled from older versions—or toward communication and information infrastructures for personalized peer-to-peer services. These activities draw on many different design skills and take

many different forms because they are situated within small scales of emergent social demand. These practices operate at different proximities to their constituents. This ranges from intimate designer–user relationships, including one to one, at the local scale to designing with and for individuals anywhere in the world.

Rather than technology being harnessed for competitive advantage, new possibilities for the automation of material production are oriented toward equitable fulfillment of material and social needs. The existing network of community-based maker spaces becomes a socially significant workshop for learning, making, critiquing, and basic communal joy. Designers organize, facilitate, and participate in workshops rather than specify production in sweatshops. Using distributed digital manufacturing technologies, designers work with communities to make their own material goods, specified to their unique conditions and desires and on an as-needed basis. Digital files, 3D printable models, and other forms of drawings or plans can be parametrically modified by inputting data specific to users or content.

Building on commons-based networks such as Thingiverse and github, files and production knowledge are shared and continually modified. Each new design innovation is a contribution to the commons. Designers share their “hacks” and continue to culturally modify known forms of objects and things even as they operate as producer-technicians-constituents through peer-to-peer networked collaboration.

Designers contribute their future-oriented research and thinking to regularly organize needs and desires assessments—a socially embedded evolution of what was formerly known as “market research.” They are looking for “opportunity spaces” to continually update or improve communal sufficiency and flourishing. Design “problems” are still economic problems, meaning that resource allocation is a central concern. However, they are not economic in the limited sense of “business problems,” which frames problems through a production model directed at profit. Instead, the economic problems of design require “just enough” resources of knowledge, material, and labor—a *minimum viab*le utopia.

Community gardens give way to a variety of new ways that people adopt agricultural methods for producing natural materials that intersect with digital production technologies. In turn, local innovations for making

everything from novel food types to furniture emerge. As Full Grown UK has demonstrated, with the right combination of knowledge, labor, and patience the industrial production of furniture can be replaced by the growing of durable goods. Imagine the community garden that sets aside room just past the herbs for growing chairs and end tables. The ongoing need for education provides ample opportunities for innovation: new object types, new furniture designs based on what has been learned about growth patterns of different species of plants and trees, novel processes, features, and so on. In combination with the makerspace as a hub of productive, educational, and social activity, these primarily "natural" objects can be integrated with low- or high-tech components: lamps with motion sensors, chairs with touch capacitance to adjust based on the body's weight distribution, and so on. That these novel approaches can be produced at the nanoscale in highly nuanced ways is indicative of the potential for removing these decisions from the corporate profit model.

Community-scale or even in-home, kitchen-scaled labs become the site for growing biotextiles and other material experiments. Knowledge exchange for these material innovations and experiments is again predicated on the commons. Some designers operate like "tutors"—visiting people in their labs to help set up, provide instruction, offer feedback, and so on.

Beyond the design and production of new things, a whole category of designers is directly involved in novel methods of repair and repurposing existing material goods. Clothing that has become damaged, worn, or simply no longer desirable is dynamically altered by *refashion designers* (e.g., Reknit Revolution). Industrially manufactured products that are damaged or have decreased performance are modified, retrofitted, or tuned up, particularly through the replacement of parts via 3D printing, rapid prototyping, and/or micromanufacturing. *Postindustrial designers* take up the challenge of both repairing and augmenting the enormous supply of consumer products already in the world.

This is a postcapitalist model of material production in that it builds on the productive capacities and innovation that capitalism has pushed, but it does so in a socially distributed and decentralized way, removing the enormous weight of private control over production.

People Making Data: Building & Replenishing the Commons In the transformation scenario, renewable energy production becomes a

decentralized and locally managed common resource. Combined with small-scale manufacturing, tailored to social and material contexts, micromanufacturing satisfies many of the material needs of communities. These peer-production models extend the reach of the commons with the aid of design's capacity to culturally modify—that is, functionally and symbolically adjust—common stock designs, materials, and techniques for hypernuanced uses.

"It could be said that every Commons scheme basically has four interlinked components: a resource (material and/or immaterial; replenishable and/or depletable); the community which shares it (the users, administrators, producers and/or providers); the use value created through the social reproduction or preservation of these common goods; and the rules and the participatory property regimes that govern people's access to it."[30] Designers have a role to play in each of these four components. Based on this loose schematic, each commons also becomes a new community of social relations as well as a new political and cultural space.

Designers will be important contributors to the work of building and maintaining the various forms of commons that link social, productive, and knowledge-based resources. Designers work in a variety of roles related to facilitating networks for exchange of things; creating digital or informational services, systems for material production, and systems for resource management; and developing tools, systems, and processes for organizational structures (including participatory forms of governance) and communication structures (including the symbols and subjectivities of new social identities).

As everybody's "work" has become much broader and less alienating than under the capitalist division of labor, all participants in platform models of exchange and production are represented by a multiplicity of interests. Designers employ participatory design methods to shape, deploy, and use such systems. Like all democratic processes, this one is discussion intensive. Participatory design leads to its own form of degrowth: increased deliberation and decreased consumption.

People Making Networks: Peer-to-Peer Social Services Participatory design regularly identifies needs, desires, or possibilities that are entirely removed from the economy and can be provided socially. Current design and business practices that seek "innovative solutions" for dealing with

social challenges such as isolation of the elderly, caring for the homebound, and providing companionship for those at risk of or dealing with mental health disorders typically look toward "technological solutions." What these solutions really represent are extractions of social value into business value. "Taking care of each other, overcoming our isolation and loneliness, is at the essence of being human. But is what we want a world where we are all *paid* for that activity?"[31] UX design often claims to require a deep capacity for "empathy." Yet it's hard to make a convincing case for this if design researchers who are studying the intricate needs, desires, and barriers for a "user" to achieve human flourishing only proceed to develop those particular services that are easily converted into something to be purchased. Reclaiming "empathy," UX designers seek social solutions for social problems. Through community-economy-mapping techniques, local resources are identified and matched to needs through the design of novel service-exchange configurations, customized to the context and needs of multiple parties. Many of the approaches, methods, and contexts being developed by the design-for-social-innovation movement already demonstrate a model for this kind of participatory development of peer-to-peer service design.[32]

People Making Selves: Subjectivity Designers no longer spend the greater part of their lives creating cultural distinctions—*hacking* novelty out of the cultural milieu—for proprietary ownership of "brands." Instead, they create novel distinctions specific to the social contexts of their work, building up the material, structural, and infrastructural environment of community economies, global and local. This is also central to the work of building the commons. In doing so, designers also build symbolic patterns of social and individual identities and generate new forms of subjectivity. Building the human connections to generate and maintain a commons isn't simply another "rational" model of production; it would include all the irrationality that makes material culture pleasurable and joyful, including unique aesthetic experiences produced by design. This work would also include opportunities for surprise, weirdness, and delight.

"Under this new system, the goal of design is not just to build an artifact to fulfill some genuine social need 'out there' but also to make us reflect on how that need has emerged, how it has become a project worth pursuing, and how, all things considered, it may actually not be worth pursuing at all.

Designers shouldn't force these answers on users, but they should make it easier for users to ask questions that may or may not lead to such answers."[33] In transformative design, designers no longer work for "users" alone. What is desirable for any single constituent is balanced with the desires of the social body.

In this way, participants in a project that will affect their daily lives weigh in on the potential value of what is being designed, with the always-present option to simply *not build it*. Subjects of this postcapitalist economy exercise democratic processes in matters of design, production, and resource allocation. In this context, Greenfield's "people making places" can be recast as "people making *identities*"—replacing the social making of "places" with the social making of "selves."

Capitalism produces subjectivities through labor and consumption. "In order to imagine a postscarcity world as a utopia, then, it is necessary to imagine the sources of meaning and purpose in a world where we are not defined by our paid work."[34] Although many fear the loss of identity that would come with a postwork society or even one in which the work is far less "disciplinary" or less defined, new subjectivities will emerge with new multidimensional roles to be played.

Generally, in participatory design more people are involved in the activities typically set aside for professional designers. This brings us close to Manzini's vision of social innovation design as a continual process of community empowerment through creative processes, in which designers facilitate research and creative activities, negotiating, working toward consensus, and determining plans for the development of new material or social experiences. This role will require designers to also work toward social connectedness as a *primary need* in this new order. "Neoliberalism has established a set of incentives that compel us to act and identify ourselves as competitive subjects. Orbiting around this subject is a constellation of images related to self-reliance and independence that necessarily conflict with the programme of a post-work society. The fact that so many people find it impossible to imagine a meaningful life outside of work demonstrates the extent to which the work ethic has infected our minds."[35] By establishing projects, processes, and activities for communal production, designers can help give people roles in their own ongoing social and economic development. Through participatory processes, designers

facilitate the self-making of subjectivities predicated on active collaboration and belonging—something built into and yet always simmering in the design process itself.

Conclusion

Designers today are accustomed to calling themselves (sometimes albeit smugly) “agents of change.” But what is the direction of that change? This is often left unsaid. The limits of design’s possibility for change are created in part by beliefs in those limits—beliefs defined, in part, by the political economy in which it operates. The *egalitarian* postcapitalist scenario laid out here attempts to expand those limits of possibility. Achieving this scenario will require a political subject who can truly be an agent of change. A “political subject ‘emerges’ by the act of naming, that is by revealing.”[36] Clearly, designers alone can’t bring about this scenario. However, the social, material, and human-centered expertise of design can serve as a catalyst to bring into being such a society. The first step is naming the political subject: the postcapitalist designer. To achieve the goals of PCD, the postcapitalist designer will require clear principles for action and strategies to apply them.

9
Design after Capitalism, in Practice

How to Do Postcapitalist Design

The diagnosis and critique of capitalism and business-as-usual design (BaUD) laid out in the first part of this book attempted to pose an argument for *why* design should transform itself to transcend the structures and logics of capitalism. The foresight scenario of an egalitarian postcapitalism combined with the theoretical framework laid out in chapter 8 outline *what* postcapitalist design, or PCD, could be. This chapter indicates *how* that transformation could come about.

In discussing transformation, Wright describes three different approaches: ruptural, symbiotic, and interstitial. In alignment with the objective of *erosion*, the interstitial model of transformation is adopted in the PCD proposal described as "the development of alternative institutions within the framework of capitalist society, such as cooperatives or peer-to-peer collaboration in the digital world."[1] To achieve a preferred state of design within an egalitarian postcapitalist society, new structures, logics, methods, mindsets, and institutions need to be built within the current framework. This begins by naming a core set of principles that define PCD. These principles are then applied at three layers—the project, the practice, and the discipline—resulting in a set of distinct strategies.

This layering of the application of PCD principles is akin to "pace layering" as identified by the American environmentalist Stewart Brand and described by the American science fiction author Bruce Sterling: "The core idea of pace layering is that there is not just one system in the world. The world is multisystemic; it's onion-like. The upper layers are light, airy, and swift-moving, while the deeper layers beneath them are dense, seismic, and slow."[2]

Design projects, the upper layer, move quickly. Practices, the middle layer, move more slowly but tend to have more lasting effects. Design

practices adapt and evolve over time, including how designers work with each other, with collaborators, and with clients or constituents. Design as a discipline, the bottom layer, is dense and slow. However, as a critical mass of practices transform, new disciplinary knowledge and standards emerge (including new methods, vocabulary, intentions, and leaders). Design education is also a reflection and catalyst of the discipline. Postcapitalist ways of doing projects and organizing practices will create institutionalized patterns of thinking and doing design—theory and practice—and these patterns will penetrate into those deeper layers, eventually transforming the discipline.

At the intersection of the principles and the layers, there are strategies (Principle × Layer = Strategy). These strategies describe *how* PCD can reorient projects, practices, and the discipline for the kinds of design referred to in this book.

Many of the world's contemporary problems are problems of scale. While the practice-based transformations described here are focused primarily on local, community-network operations to build solidarity and community economies, achieving concerted social power at the disciplinary level is critical for transformation.

A few points also should be made about what PCD *does* or *does not* mean or need to do.

PCD *does not* mean a withdrawal from all forms of markets. Markets for exchange have been with human beings since the earliest civilizations. However, it was the "market society" ushered in by capitalism that marketized things such as people (labor), nature (land), and purely symbolic value forms (money, risk). Instead, PCD can use markets for the exchange of ideas, information, goods and services, and even labor—both with and without the involvement of money—so long as those efforts maintain social empowerment at their core rather than to build hierarchies.

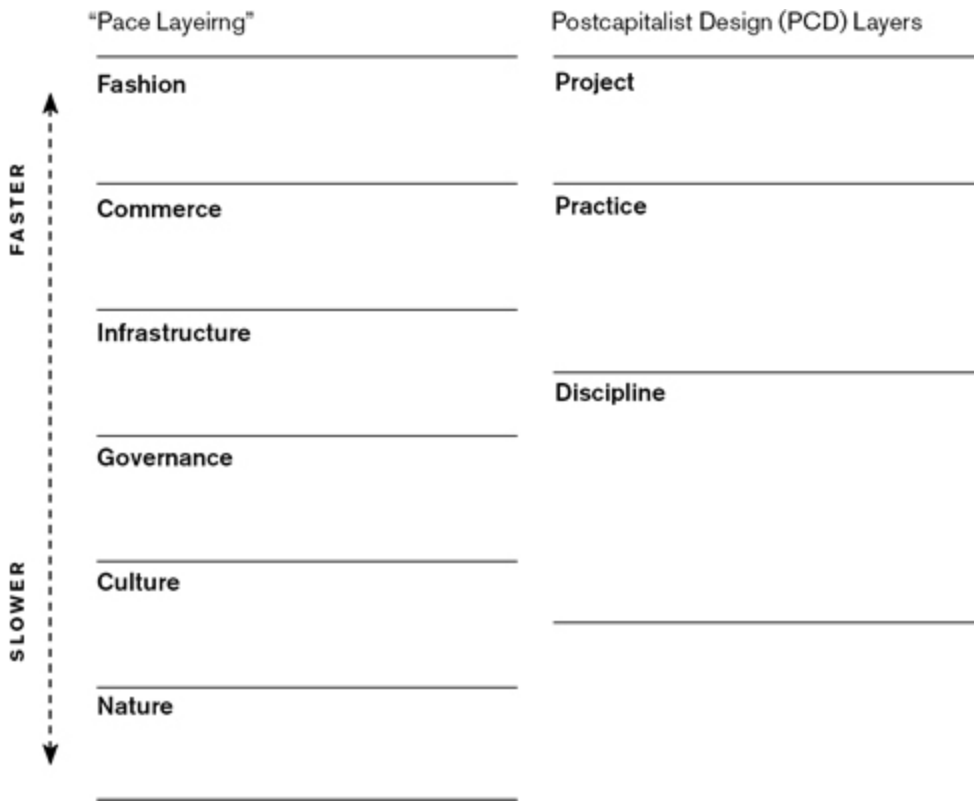


Figure 9.1

PCD layers in relation to Stewart Brand's pace layering. (*Source*: Image based on Stewart Brand, *The Clock of the Long Now* [New York: Basic Books, 1999], 37.)

PCD *does not* mean ending free enterprise as such. The egalitarian social economic vision laid out in the transformation scenario emphasizes decentralized methods for material production, technological development, and the resulting construction of new subjectivities between designers and citizen-constituents. In other words, this is *not* an argument for a centralized command economy, such as state-run socialist economies, that instructs producers what to make and designers how to do it. Instead, in the context of PCD, designers look to the communities in which they live, work, and serve to ask, discuss, and evaluate what is needed and what resources exist (per the methods of community economies).

PCD *does* necessitate a heightened level of professional and social responsibility for designers. This is what is meant by a transformation of the discipline—it is immanently responsible for its role in developing sustainable social and economic relations. This heightened level of professional and social responsibility includes the demand to grow capacities for the intellectual, technical, and material production of life

through the practice of design. Designers teach as they do, sharing process, tools, methods, and techniques.

PCD *does not* produce "commodities"—at least not in the historical sense of the term—including commodified services, information or communications, and experiences. PCD instead builds social relations through the production of material, digital, and experiential culture to move toward their decommodification. This building may begin by working within capitalist structures, logics, and subjectivities, but every project and each practice will be taken as an opportunity to expand social power, sustain community economies, build and maintain commons, and transform subjectivities *as much as possible*. This is the work of *erosion*, and it is iterative.

Although it may sound as if these demands require a utopian level of good will, consider how designers *value* their own work—and the work and worth of other designers—outside the economic system of value. Profit-seeking clients don't typically seek "great" design from the same perspective that most designers would evaluate "great" design. Profit-seeking clients want work that gets the job done efficiently and effectively to turn a profit for them so that they can move on to the next venture. Designers want the work to satisfy the client, but they often strive even harder in the knowledge that their work will be evaluated by their peers according to criteria different from those of their clients. In order to build peer regard, designers put in extra work—extra iterations, extra testing and evaluation, extra peer input, extra detailing, *extra time*—to achieve aesthetic or conceptual excellence that will go above and beyond the economic and functional needs of their clients or corporate teams. And they don't do this extra work just out of vanity. Work done beyond bare functionality means that design has its own disciplinary criteria for excellence, *as it should!* Designers reading this book may know the satisfaction of sharing with a client that work produced for them has won a design award and thus achieved recognition in a sphere of influence beyond the client's domain. So, too, will it be with PCD. New measures of impact in terms of social empowerment and well-being will be adopted to evaluate and circulate disciplinary success.

Principles & Strategies

Designed products, services, and experiences are like tentacles of congealed labor, politics, and subjectivity. They are distributed into the world, where they create subjects whenever and wherever they are used (consumed) by others. The work of creating PCD can start here, with the project. Designers today, whatever their current practices or professional obligations, can begin this effort now.

PCD principles are applicable in a variety of institutional forms of practice, from independent designers (freelance or solo practitioners) to large consulting studios or agencies to in-house design teams or designers working within capitalist business enterprises. Obviously, the context will shift how or which strategies are most effective, particularly at different layers.

In a design consulting/studio model, designers will have a great deal of control over the organizational structure of the practice and potentially more control over project selection and development. In the postcapitalization of such practices, tensions may arise from dealing with external competition and keeping afloat in a capitalist context. The community economy assessment of needs, surplus, and assets described in the practice guidelines can be useful to keep the enterprise mission driven.

In an organizational design/in-house model, designers will likely have less control over organizational structure and management as well as project selection and development. Attempts to postcapitalize design practices from within a capitalist business model will likely face tensions from the enterprise and its institutional practices. However, there are potentially larger-scale transformations to be made. Even small moves can make an impact so long as the participating designers are strategic and relentless.

Principles applied to projects will push them to the next level, practice. Framing organizational needs within a postcapitalist agenda shifts the frame so that the demands of postcapitalism are not an antagonism to be overcome with each new project or organizational challenge. Instead, each project, organizational change, distribution of surplus, or building of networks and infrastructures within the discipline represents a challenge framed in an opportunity to *postcapitalize*. Being strategic begins with understanding the ideology behind what you do. Acting strategically shifts the coordinates of

reality. What seems limited by the “real world” is itself a discourse of ideology. “The limits of possibility are always in part created by belief in those limits.”[3]

Many of the practical principles described here are already being put into effect by some design practitioners or can be found in adjacent disciplines. Others will require systematic and collaborative effort. The goal is to gather up both ends to articulate strategies that can be both immediately adopted and oriented toward long-term goals.

PCD Strategies
Principles × Layers ~ Strategies

	Social Power	Community Economies	Degrowth	Postcapitalist Subjectivities
Project	**Facilitating Community Design**	**Local Production & Local Control**	**Minimum Viable Utopias & Circular Design**	**Postcapitalist Constituents**
Practice	**Socializing the Enterprise**	**Commoning the Surplus**	**Organizational Degrowth**	**Postcapitalist Designers**
Discipline	**Socializing Disciplinary Resources**	**Confederated Cooperative Design**	**Disciplinary Degrowth**	**Postcapitalist Citizens**

Figure 9.2
PCD strategies. Principles × Layers = Strategies.

As in the theoretical framework for PCD described in chapter 7, there are four core PCD principles. These principles operate at three layers.

As each principle is applied at each layer, a distinct strategy is defined.

Principles: PCD Practices

- Create social power
- Build and sustain community economies

- Enact degrowth
- Construct postcapitalist subjectivities

Layers

- Project
- Practice
- Discipline

Strategies

- Facilitating community design (*social power* × *project*)
- Local production and local control (*community economies* × *project*)
- Minimum viable utopias (*degrowth* × *project*)
- Postcapitalist constituents (*postcapitalist subjectivities* × *project*)
- Socializing the organization/workplace (*social power* × *practice*)
- Commoning the surplus (*community economies* × *practice*)
- Organizational/practical degrowth (*degrowth* × *practice*)
- Postcapitalist designers (*postcapitalist subjectivities* × *practice*)
- Socializing disciplinary resources (*social power* × *discipline*)
- Confederated cooperative design (*community economies* × *discipline*)
- Disciplinary degrowth (*degrowth* × *discipline*)
- Postcapitalist citizens (*postcapitalist subjectivities* × *discipline*)

In the subsequent sections, the strategies are described first, followed by the guidelines, which can be used to prompt designers or design teams

through a process of postcapitalizing their projects, practices, and disciplines.

Project Strategies

Facilitating Community Design (Social Power × Project) Working toward a vision of peer-to-peer service design, PCD designers practice community design in the spirit of the Scandinavian tradition of *cooperative design*. Rather than working with potential users or consumers to create better commodities, the cooperative method truly designs *with the people for whom the work is intended*. What this means is that the process must directly engage a broader group of those who will be *affected* by the work, not just those who choose to purchase or use the commodity produced by the work. For example, designing for a local business would necessitate designing for more than just potential customers. It would involve designing for the physical community of the business; the local supply chain of labor and resources; underemployed, homeless, and otherwise underserved people in the area; transportation systems; functions of public space in the area; and so on. This process would seek every opportunity for the economic functions of the specific project at hand to generate noneconomic social relations, while decreasing the material impacts of the enterprise.

Most generally, the orientation of the participatory project expands well beyond and in some cases even shifts away from a focus on aligning a consumer's experience with a client's economic gains. To achieve this shift, primary questions the designer adds to the challenge would include: How can designing this project increase quality of life, build and maintain noneconomic social relations, and decrease the material and attention demands on those who will use it? What new "forms" of material things or immaterial experiences can be crafted from designing in this way? How can this project contribute to a commons of material or cultural resources, general knowledge, or practical skills?

Reducing Externalities. Part of working in the expanded field of designing for communities includes explicit attempts to reduce negative "externalities" or, rather, to turn what a purely economic orientation would deem "external" into internal opportunities for endogenous growth.

Designers working on mobile medical apps may have only the best intentions for helping their users access accurate and useful medical information by tracking, monitoring, and/or analyzing their own health data to make informed health decisions. However, designers working on such projects should ask themselves: How could these data be used to hurt, physically or otherwise, our users? Could our users be extorted by insurance companies, financial institutions, or others if this data were to end up being extracted, consensually or otherwise? What can we do to make sure that never happens? If we can't guarantee that it will never happen, how can we ethically proceed with this project? Combining speculative design, to determine what the negative implications might be, with cooperative design, to find hidden forms of noneconomic value, PCD community designers expand the threshold for "due diligence" by rigorously mapping out the potential long-term impacts of a project and presenting them for evaluation and assessment to the communities that the project may affect. Development and innovation of new technological systems would include educational experiences with and for participants (actual "users") of the project, building new literacies to ensure competent and informed constituents of design.

Nonrivalrous, Nonexcludable. Although current practitioners will likely be working with profit-seeking, capitalist clients, they can still make the effort to move the production of private goods into nonrivalrous and nonexcludable economic models. What is it that makes a private good or service rivalrous and/or excludable? Are those criteria artificially established? How can the designing of the good or service move toward a less- or nonrivalrous model? How can the good or service shift from excludable toward less- or nonexcludable? How can it be moved into broader circulation to create noneconomic value for a broader group of people?

Local Production, Local Control (Community Economies × Project) In striving to build community economies that boost noneconomic relations even while engaging economic activity, PCD designers work as tightly as possible within the existing assets of the community for production, fabrication, software development, and other services required for the actual "making of" designed products, services, and experiences. The "community" starts with the local community in which the design practice is embedded but may also include social relations in remote geographic

areas that are directly connected to that community—for example, distant family or friends or communities of mutual and reciprocal support anywhere in the world. By regularly mapping the community economy of assets (see “Practice Guidelines”), PCD designers will be aware of their communities’ vast, often invisible resources. They will activate this knowledge of local potential and design within those opportunities and those constraints. Localizing production is not intended to create regional or nationalist tendencies of exclusion; it may be necessary for designers to actively work against these risks by clearly articulating the agenda of a PCD model for building diverse, multicultural community economies. The goals are to produce a localized capacity for community control over production, development services and experiences, and innovation in design and development of technology to increase economic self-sufficiency, including through noneconomic practices of social exchange. PCD is a practice of collaboration and inclusion that brings all related systems of resource allocation and social relations into design’s purview.

Minimum Viable Utopias & Circular Design (Degrowth × Project) Adapting Greenfield’s concept of *minimum viable utopias*, designers work toward adaptable, customizable prototypes that can be circulated to engage constituents in making things, making data, making networks, and making places (or making identities through making places)—achieving these goals while contributing to the project’s specifications. PCD designers create small, local, open, and adaptable objects and experiences that are intended to evolve continually through use in a constant state of becoming. Designers’ constituents make their own worlds by modifying *minimal viable prototypes*, a practice that creates a feedback loop in which the designer intervenes in modifications, customizing the prototype for specific needs or use-case scenarios unintended by the designers. In these iterative encounters in which the designer returns to a project to maintain or adjust the designed outcomes, a dialogue forms that can facilitate future projects.

This endeavor combines degrowth and the building of postcapitalist subjectivities among the constituents. Objects and experiences are produced at smaller scales: less of them are made, but they are made more useful, more durable, and more meaningful. This creates circularity in the design process and improves the circularity of the designed goods and services—reducing material and energy demands through decreased scale and increased longevity. The first step is to reduce material and energy needs,

which includes limiting use of nonrenewable materials (particularly fossil fuels) and designing for production methods that limit nonrenewable-energy use. Degrowth strategies also apply to digital services and experiences: reducing the use of streaming video and animations, reducing page weight on web and mobile services through attention to image sizes and font selection, and reducing other processing-intensive media to reduce both energy consumption and the pressure to continually update devices to accommodate newer operating systems.[4] The second step is to design for durability and maintenance. This requires consideration of the maintenance and future forms of the good or service, including interchangeability of parts, durability and circularity of materials, labor processes for repair and/or disassembly, and the labor and materials supply chains involved.

Design Constituents (Postcapitalist Subjectivities x Project) Most design currently creates capitalist subjectivities, and this will be one of the more difficult and ingrained mindsets for PCD designers to overcome. A starting point will be to recalibrate the language and mindset of design. Over time, as PCD practices erode the model in which designers are hired (for wage labor) by clients to create design work to be purchased by customers, these assumptions and "norms" will wither such that designers are increasingly working directly for the people who use their services or outputs. The term *constituents* may be the most accurate generalized term for those who "use" design. A constituent is an empowered member of a community, one who takes an active role in the community's activities—not a passive consumer. A constituent is a part of a whole. In the language of PCD, "users" become *constituents* or in some cases specific people with specific names. Designers work for and with their *con*stituents.

The diffusion of user-centered design methods has proliferated the use of personae in the framing of design *problems* and *opportunities*. These fictional composites attempt to find useful averages of people and everyday contexts studied through design or ethnographic research. This practice tends to flatten out significant nuances of individual humans, asking people to accommodate themselves to the average users imagined by the designer. Meanwhile, casting the humans studied in exploratory research as "users" or "consumers" already foregrounds certain roles. Thus, subjectivities are made.

The practical everyday challenges of the human experience insist that the fleeting moments one spends with something such as a digital product, service, or experience are probably distracted moments in which there are multiple competing goals—many of which are also likely more important. By conceptualizing humans as *users*—presumably dedicated to the functional and symbolic values associated with the product, service, or experience—designers act on the premise that they are granting agency or capability. Which is true to an extent. Yet it is the constituent's *latent agency** that designers are tapping into, joining forces with, or collaborating with to achieve new or different ends. By directing design research and development toward socially productive goods, services, and experiences, they are designing with a human-as-social-agent rather than a human-as-consumer directive in mind, regardless of the project's domain.

Decentering the User. An adjacent strategy is to decenter the individual user as the locus of design concern. This strategy is aimed at combating the competitive individualism of capitalist subjectivities and foregrounding social collaboration instead. Designing objects, experiences, and services that require or reward collaboration, draw distinct connections to local contexts, resources, or needs, and keep users cognizant of the consequences of their individual actions, including resource limitations, would be central to this method of working.

Identities (Not Brands). As their horizon, PCD designers aim to work outside of capitalist economic and social relations. As a transitionary pathway, the private ownership of "branded" products, services, or experiences can be shifted (even slowly) away from capitalist social and economic relationships by nudging the frame toward experiences *supported by* capitalist interests. This move requires reconsidering what brands are, what they do, and whom they serve. Brands have become rather malleable through their multitude of forms of expression, such as personalized verbal tones on social media or simulated models of direct conversation with customers. The opportunity is to find footholds in which emphasis on identities of experiences create relationships with the identities of those who experience them without directly commodifying (granting "ownership") to a privately owned interest. There are endless examples of corporate-sponsored experiences that are thematically named or "branded" on their own, "brought to you by" a corporate interest. For example, museum exhibitions are designed within a specific identity for the

exhibition while also typically sponsored by a corporation (and potentially fitting within the brand of the museum). The goal here is to erode continually the "branded" experience toward experiences with distinct identities. These experiences may still be *capitalist sponsored* in a transition toward other models of sponsorship. This may seem like a trivial distinction, but it is part of PCD designers' effort to shift both their constituents' and their own subjectivities. The shift might just shake loose the contemporary hypnosis of some designers under the spell of claiming themselves "brand warriors." Perhaps by making novel social experiences without ownership, they could become "human interest warriors."

Confederated Support/Sponsorship. A similar strategy is to conceptualize and execute the design of products, services, or experiences through collaborative consortia of multiple "sponsors." A consortium of private sponsors engaged in developing new public-facing (rather than consumer-facing) experiences can merge resources while transforming the logics of competition into those of collaboration.[5] In this way, corporate interests may underwrite socially beneficial services or experiences. What needs to be avoided in this case is designing toward exclusivity for the sake of increased pricing (and profit). Designers are in a position to navigate these waters—including the refusal to work toward artificially exclusive outcomes—instead shifting toward nonrivalry and nonexcludability through design.

Prototypes as Minimum Viable Utopias. PCD designers circulating minimal viable prototypes insist that constituents make their own worlds by modifying, refining, and repurposing. This direct engagement with designing or redesigning, which is an "experience" to be designed by the PCD designer, insists on a shift in subjectivity away from that of passive consumer. Creating the channels for dialogue and discourse, prompting opportunities for refinement or alteration, and engaging in the ongoing life cycle of a product or experience may occupy much of the PCD designer's time (which is also a strategy of degrowth, maintaining what already is).

Project Scenario

Here's an example of how a design team might implement these strategies to operate PCD principles on the project layer. The scenario is intentionally

common (although not banal) to make clear that these principles can immediately expand the scope of any design opportunity.

A new restaurant is opening, and Design Studio X has been commissioned to develop its entire brand, including visual identity system, interior design, environmental graphic design, website, and promotional materials.

The designers would begin in a familiar way: by conducting an audit of restaurants in the area with the goal of determining how this new offering fits into the existing landscape. This audit may contribute to establishing, to some degree, pricing and menu offerings. The designers would also conduct research with community residents on culinary and dining-experience preferences to aid in determining what the restaurant might offer not only in terms of the experience of dining but also in terms how the restaurant might complement other leisure and food-related activities in the area. So far these same steps might be done through the lens of determining a competitive advantage for the business. However, through the PCD lens these same methods are used to conceive the primary function of the enterprise—making and serving food—as an integral component of a social, cultural, and economic context. The PCD designers would work with local residents to map the *community economy*—assessing unserved needs and overlooked assets in the community. This will help determine what varieties of social surplus the restaurant may generate and how it will distribute that social surplus, including to new or existing commons. Working with individuals and organizations such as community gardens, homeless populations, unemployed or underemployed area residents, home care or other unpaid domestic workers, young people, and those working in the food-service industry, designers construct a working model of the community economy to gain insights on what the new enterprise can do to contribute responsibly rather than only to extract. Designers might recommend operational supports for the restaurant, how food waste should be handled, how employment opportunities can be crafted and promoted, how the “culture” of the restaurant can be developed to build the social power of its employees, or how shared, nonrivalrous, or nonexclusive use-case opportunities can be created for the restaurant’s interior or exterior spaces. “The details are not details,” as Charles and Ray Eames famously said. That ethos and commitment to thorough rigor applies also to seemingly simple possibilities of how a “brand” can build social identity within its

neighborhood while reciprocally engaging its community economy. Designers would continue to work in a cooperative design model with the enterprise and community to develop operational protocols for systems, interfaces, or communications along with the identity design program. For example, software systems for management would be deployed or developed in the cooperative methods, through consultation with servers and cooks and dishwashers. As good stewards of the project, the PCD designers would iterate continually within this expansive participatory model to monitor the project's impact and adjust as needed. The designers share a stake in the ongoing well-being of both the enterprise and the community that supports it.

Practice Strategies

Socializing the Organization/Workplace (Social Power × Practice) All design practices institutionalize social, economic, and power relations through patterns of operation—*how* they do the work they do within networks of clients, constituents, and collaborators. The institution of design practice includes many actors and actions. PCD practices consider each of these relations, patterns, and social elements an opportunity for rethinking *wage labor for profit* and reorienting it toward *increased noneconomic social relations* and *distribution of social surplus*. These opportunities include:

- How a practice engages, contracts, and exchanges money or other services for work with clients or constituents
- How designers conduct research into the terrain of their projects or interventions, including how research is conducted to understand the social and environmental contexts of their work
- How designers conceptualize, present, and evaluate design proposals with their clients or constituents, including how the decision-making process is formulated
- How designers, technical experts, professional services, and supporting staff are hired, incentivized, supervised, and/or managed

- How practices contract, employ, direct, or otherwise engage third parties in development or fabrication processes for prototyping or other productive processes during the work of designing
- How technical, material, manufacturing, or other production-related specifications are established to determine how the products or services of their design will be produced by others, even products or services contracted and managed by their clients and constituents
- How designers determine the various life-cycle states of goods, services, or experiences they design on behalf of clients or constituents, including efforts toward a circular economy
- How designers forecast, plan, staff, or otherwise provide planning and resources for maintenance of goods, services, or experiences throughout their life cycles
- How involvement and representation of diverse cultural, social, and economic perspectives in the designing process are incorporated
- How surplus from payments or exchanges received from designing are distributed
- How contracts, agreements, exchanges, or service provisions are made with other service providers related to the infrastructural needs of the designing work space, such as landlords, utilities, equipment purchases or leasing, technical services, communication services, and daily needs, such as food, transportation, and janitorial services

This partial list of social and economic commitments already demonstrates the vast web of relations established in the institutional operations of design practice. *Socializing the workplace* means privileging social power over economic gain in every one of these exchanges.

Cooperative Design Practice. Perhaps an obvious starting point to socializing the workplace would be for independent design practices to operate as worker-owned cooperatives. There are plenty of existing models from which to build, including guidance from the International Cooperative Alliance.[6] To increase social power, each practice has the responsibility to

design its own working methods and structures unique to its participants and context. This is a great starting point for PCD designers to practice the work necessary for cultivating and maintaining community economies.

Commoning the Surplus (Community Economies × Practice) *Tacit Knowledge*. Designers learn a great deal in the process of designing. This is the *tacit knowledge* of design. It may include generating new knowledge about people as potential *constituents* (formerly *users*) gained through ethnographic research; knowledge about a particular industry, domain, topic, or trends gained through exploratory research; or knowledge about technical processes for achieving a particular goal in the design and production process.

One step toward *commoning the surplus* of design practice is to share this knowledge: to *make the act of designing teachable*. Today, open sharing of research insights, data, methodological innovations, process modifications, new research tools, and technical skills for design is largely the domain of design research and education. However, there is also a growing movement of open sharing of results of the designing process from studios or agencies doing mission-oriented work—for example, those working with nonprofit or humanitarian groups. Designers share free versions of their customized research tools, share stories of novel approaches to doing research, write articles summarizing their research insights, and post tutorials on something they learned while working in a new technology or modifying a process of production.[7] These practices set in motion one basic ingredient for PCD practice: *commoning the surplus of tacit* knowledge.

One hurdle to this effort is the nondisclosure agreement, which has become common for designers in an era of heightened intellectual-property protections. Strategies for jumping this barrier include negotiating better agreements rather than simply accepting them. For example, if the research must remain confidential, designers can argue for publishing extensible insights without disclosing the industry-relevant details, thus sharing new directions or patterns of work. This approach is akin to academic research practices in which anonymity can be protected when doing ethical research with human subjects.

Building a common platform and maintaining its operations for the open exchange or practical knowledge represents a significant project for PCD as a discipline. It will require an infrastructure for the dissemination,

organization, navigation, and ease of searching or discovering content of value by and for other practitioners. (See the section “Confederated Cooperative Design [Community Economies × Discipline].”)

Distribution of Surplus. In articulating the issue of *surplus* in postcapitalist politics, J. K. Gibson-Graham provide a framework of questions to be considered: “Who is to be included in decision making over the rate of appropriation of surplus and its distribution? Is there a maximum size of enterprise or social grouping in which such hands-on involvement is not feasible? Under what conditions might surplus appropriation be seen as nonexploitative? How might nonproducers of social surplus have a say in how surplus is generated, appropriated, distributed? And, of course, what are the social destinations to which surplus is to be distributed, and those to which it will not?”[8] These questions are not easy to answer, nor are there universal answers. Economic relations are, after all, socially embedded. Yet these questions should be addressed, and the responses that a design practice implements will go a long way toward defining its institutional political economy. Recall also that *social surplus* does not mean “money”; however, the *economic surplus* of practice may also be considered in these questions.

The first step is to establish measures of the social surplus generated by the practice. What kinds and how much of knowledge, skills, social experiences, networks, communication patterns, material exchanges, or other social patterns of being and doing emerge over the course of a single year of design practice? *A lot*. Mapping the social surplus and taking measure of it presents an image of existing nonmarket *value* being created by design practice. Communicating that surplus and sharing authority in its distribution establishes a “postcapitalist politics” of practice. Let’s look at these questions in more detail.

“Who is to be included in decision making over the rate of appropriation of surplus and its distribution?” Obviously, employees of the practice should be considered, but what about all of those other contributors to the enterprise listed in the network of operations—third-party providers of professional and support services, vendors, producers, developers, and others engaged in related production processes? They also produce both economic and social surplus for the practice, so they should be involved in the decision making. This is not to suggest they directly *receive* the surplus.

Through annual (or more frequent) conferences, the practice would disclose social surplus with its organizational network, including clients, constituents, and both paid and nonpaid contributors. Disclosure leads to discussion of distribution, addressing the needs and assets of the community economy as a whole (see the section “Practice Scenario”).

“How might nonproducers of social surplus have a say in how surplus is generated, appropriated, distributed?” Community input on how social surplus is used might similarly include annual conferences or more regular events, meetings, surveys, and tools to share what is produced and how it can be generated, appropriated, and distributed. Who is the “community” beyond the organizational network of clients, constituents, and contributors? Local communities where offices are located and where designers live, online communities of knowledge sharing in which designers engage, professional networks or disciplinary communities, as well as educational communities for next-generation designers—all represent opportunities to distribute social surplus into community and social power. Sharing knowledge and social authority with these communities further builds a discipline of postcapitalist practice that maintains and sustains its own development.

Is there a maximum size of enterprise or social grouping in which such hands-on involvement is not feasible? Under what conditions might surplus appropriation be seen as nonexploitative? There are no static answers to these questions, but addressing them directly, transparently, democratically, and regularly would be central to the operations of PCD practice.

Organizational/Practical Degrowth (Degrowth × Practice) Beyond minimizing material and energy inputs for individual projects, PCD practices work to help constituents modify, adjust, or otherwise work with what they have—to “design away” rather than reflexively produce something new. The same effort will be needed for PCD practice. As PCD practices are embedded in community economies, what is considered economically *necessary* should decline. Designers may seek methods for reducing economic needs, which will also build new networks of exchange—for example, decreasing dependency on proprietary software or tools and instead embracing open-source or self-generated equivalents.

PCD practice is also oriented toward degrowth of labor. As mentioned in chapter 7, the legal standard of a 40-hour work week was hard won in the

early twentieth century, only to largely disappear in the early twenty-first century. It is still possible to structure a practice that limits the entanglement of work and personal life even as digital communications keep people constantly connected to work.

Postcapitalist Designers (Postcapitalist Subjectivities × Practice) Working in community economies, designers are directly embedded in the social life of communities and are able to witness the long-term impacts of their work. They see themselves neither as isolated creative laborers toiling away in anonymity nor as creative heroes valorizing their individual talents. This altered perspective does not diminish the rigor, talent, or skills required to work as a designer. There can still be (moderate) vanity, too. Rather, it means seeing a designer's work as intimately connected to the lives and well-being of others—something that can't be done alone or removed from the social lives of others.

Of course, this does not suggest there is a singular *postcapitalist design subject*. There will need to be many, as diverse at least as the variety of constituencies for design. Similarly, new forms of practice will emerge, specializing in different aspects of designing for and with constituents and in different forms: material, experiential, and service productions of different kinds and in nearly endless sociocultural and socionatural contexts.

Practice Scenario

Participatory budgeting is a "democratic process in which community members decide how to spend part of a public budget. It gives people real power over real money."[9] In 1989, participatory budgeting was implemented in Porto Alegre, Brazil, as an antipoverty measure, and it helped reduce child mortality by nearly 20 percent. Since then, the practice has spread to more than 7,000 cities around the world "to decide budgets from states, counties, cities, housing authorities, schools, and other institutions."[10] Although this process varies, it could benefit from design research and practice. This work is not always officially requested or sanctioned by a governing body that spends tax revenue. The activity itself generates an act of political organization as well as economic literacy about the community's needs and desires. It builds social relations and solidarity

through a political process. Particularly with the involvement of designers, whose persuasive capabilities have long been part of their value to capitalist enterprise, community proposals could put pressure on local governments to help address a community's economic vision.

Design and design research could be instrumental collaborators in this work. Designers collaborating on a participatory budgeting process, including cooperative design and speculative design to create plans, visions, and tools (digital and/or analog), would situate a PCD practice directly into a community economy. This process would lead to opportunities for identifying additional community needs and resources for ongoing design interventions by the PCD practice. Organizing and facilitating the process, doing the research, and producing digital and analog working tools would be a collaborative endeavor between designers and community stakeholders —including those already engaged in community organizing, providing social services, and doing policy work. Developing digital tools or services for the participatory budgeting project presents opportunities to work with local software developers, students, and others interested in learning technical skills to produce a *technical commons* while localizing technological innovation and development.

Although this undertaking would be massive, it is an ideal starting point for a new PCD practice within a community economy. It would also be an excellent opportunity for an in-house design team for an otherwise capitalist enterprise to engage in expanding its company's PCD orientation—an act of *erosion* in the name of "community outreach."

Discipline Strategies

Socializing Disciplinary Resources (Social Power × Discipline) PCD practices will need infrastructures and platforms to generate and maintain this new disciplinary model—creating its own community economy of assets to sustain itself. One strategy is to aggregate resources among practicing PCD designers to share resources while maintaining the autonomy of local and regional varieties of independent practices. The primary disciplinary principle is *cooperation over competition*: decreasing individual dependencies on market-based resources while reducing the risks

of economic failure within a competitive landscape. Shared resources among PCD practices might include:

- *Health care plans*: Bargaining power increases with scale; practices combining resources to participate in health plans (particularly worker-owned or -governed providers) can achieve economies of scale to address one of the most expensive and vital challenges of creating livable working conditions.
- *Operational services union*: Networked resources can be set up for shared employment-management services, such as pay roll.
- *Legal and financial services*: Credit union(s) can be opened for PCD designers and a broad cooperative of professional services, such as accounting, legal, insurances, and tax services, can be provided.
- *Interagency consortia*: Multiple PCD enterprises create cross-team consortia and cross-team hires that work together to address large-scale challenges or projects and generate employment opportunities where designers work across multiple practices in a consortium in order to decrease the precarity of “freelance” or gig work, offer a diverse working life for designers without job hopping, and circulate design skills among practices.
- *Work spaces*: In place of the prevalent landlord/rentism model, coworking spaces can be established as a commons with the requisite communal and/or reciprocal labor to maintain them. These spaces can become conduits for cross-disciplinary team building and interagency consortia.
- *Collaborative work systems*: The social and productive infrastructure for independent, small-team, and remotely working PCD designers can be created, as well as shared sources for contacts, research data, project-impact data, and many of the other knowledge-based resources competitors would typically hide from each other. Together, these shared infrastructure and resources would create a platform of work services (calendar, documentation, workflow) that can be combined with a knowledge commons as a collection of open-source repositories.

All of these practices already exist in some form. Yet many of them are currently structured on a profit model of competition and extraction rather than on a cooperative model of mutual and reciprocal coordination.

A major challenge for PCD as a discipline is to transform PCD practices and design teams within BaUD, or capitalist, enterprise. Yet BaUD teams exist in positions of social and economic power. They can become PCD allies through shared disciplinary infrastructures, postcapitalist social pressures, and solidarities from outside the enterprise.

- PCD practices put pressure on what are currently attractive and high-paying design careers inside a BaUD enterprise (in particular the tech industry) by demonstrating what a meaningful, collaborative, and beneficial alternative practice (and life) is like.
- PCD disciplinary training, resources, and methods or tools for postcapitalizing design practice can be collaboratively produced and shared between independent (PCD) and corporate (BaUD) design teams.
- Disciplinary certifications for PCD skills or qualifications would create opportunities for capitalist support of postcapitalist education.
- PCD leadership and practice influencers steer the course of the discipline, encouraging new disciplinary forms to take root in diverse contexts, including BaUD enterprises.
- PCD education bakes postcapitalist logics and subjectivities into the long-term methods and practices of PCD regardless of the variety of working contexts designers take on during their careers. (See the section “Design Education” later in this chapter.)
- Disciplinary press, publications, conferences, events, and recognition focused on PCD circulate case studies, research findings, best practices, and inspirational projects that cross boundaries of practice types and create new sites for PCD disciplinary knowledge and discussion.
- Peer-to-peer education, training, and evaluation channels are part of the cross-boundary effort to build disciplinary knowledge through practice. Perhaps most critical is the work of mentorship,

using peer-to-peer and platform models of exchange so that best practices by recognized designers can be understood, not just praised.

- Based on recent polling, the majority of young people in the United States, regardless of career or discipline, have lost faith in the capitalist system.[11] This means the majority of young Americans working in diverse fields within capitalist enterprise would prefer to live and work in a different way. PCD practices can provide thoughtful leadership, influence, and a replicable model of the ways that a discipline can reorient itself toward building social relations away from the logics, structures, and limitations of capitalism and its many "bullshit jobs." PCD leadership is measured through respectful influence, not authority.

Nearly half a trillion dollars is spent globally each year on enterprise soft-ware—building the networks, platforms, and digital tools for businesses to do their work. This industry has tremendously benefited from the labor of UX and other designers; yet this arena also represents a great deal of money and effort dedicated to building new tools to do (often more) work. Transferring even a fraction of this effort toward building the infrastructures of a collaborative, cooperative working world would make a tremendous impact.

Clearly, none of the efforts outlined here will bring about an explicit end to BaUD on its own. Yet together these efforts would erode the disciplinary logics and structures of capitalism within design. By practicing PCD, designers can produce positive *spillover effects** into the many domains and communities in which they already work.

Confederated Cooperative Design (Community Economies × Discipline) PCD practices generate their own discipline-scaled community economy when unified in a shared effort of *commoning*. By directly confederating or indirectly networking or combining resources, cooperative PCD enterprises further erode their individual and thus collective dependencies on capitalist relations with clients and/or business partners (as in the Mondragón case study). As confederation scales up, resources like those listed earlier become integrated into the community of cooperatives to

create a conglomeration with its own financial, legal, professional, and even educational resources built into its postcapitalist structure and ethos.

Building the infrastructure and platforms to maintain confederations and networks of cooperative PCD practices will be critical to sustain this kind of work. As a contribution to the support of the discipline, PCD platforms serve to connect designers with opportunities; share common resources among the broader design community and its constituents; disseminate best practices; set ethical guidelines; offer legal, financial, and regulatory support; connect designers and their constituents to knowledge commons and technical information; and generate and sustain the variety of commons central to this new disciplinary model. Professional organizations for designers, such as AIGA (the largest professional membership organization for design), the Industrial Designers Society of America (IDSA), the Society for Experiential Graphic Design (SEGD), and the Interaction Design Foundation, although primarily directed toward predominantly capitalist models of design practice as it exists today (BaUD), also include various models of discipline-scale knowledge commons, support, and social interactions. For example, AIGA offers its members health benefits and legal services, and offers anyone free access to the Design Teaching Resource, "a peer-populated platform for educators to share assignments, teaching materials, outcomes, and project reflections."[12] A PCD infrastructure—postcapitalist in both origin and operations—for commoning, connecting, and collaborating would provide further resources for in-house design teams or for those working directly in BaUD enterprise to initiate postcapitalist strategies internally. (Remember that Amazon built up AWS to handle its own logistics, only to discover these platform services were valuable to outside interests, a discovery that put into motion one of the largest platform revenue streams in the global economy.) This PCD platform, or series of networked platforms, would also provide invaluable resources for other fields of work, producing an influential model for disciplinary strategies toward social change.

Disciplinary Degrowth (Degrowth × Discipline) To accomplish the scaling up of cooperative design practices, PCD designers and practices will commit a percentage of their intellectual and practical labor power—a reasonable percentage of working hours—to the efforts of building the cooperatives, confederations, platforms, and infrastructures required of a PCD discipline. This is not "volunteer work," nor should it be considered

personal or pet projects. It is the shared labor and shared governance of building democratic and egalitarian institutions. You don't get paid to vote, participate in, or inform yourself of issues in a democracy; you do so to participate in shaping the infrastructure of society. Design practices—studios, consultancies, agencies—commit a named percentage of the time their teams are dedicated to this work, akin to models for determining vacation or paid sick leave. Design teams working within BaUD enterprises—in-house designers, product or service designers within businesses—may have a tougher sell in getting these enterprises to commit paid time to building the infrastructures of PCD. Nonetheless, designers in these positions can leverage corporate benefits such as "professional development," time offered for humanitarian projects, and even sabbaticals as opportunities to incorporate the building of PCD disciplinary structures into BaUD enterprises.

If this sounds like "more work" in contrast to "degrowth," that may be true in the short term. However, a large, distributed, and diverse cohort of designers could use PCD principles to build the infrastructures of a PCD discipline, all the while constructing their own postcapitalist subjectivities. These hours of labor power will be diverted from doing the work that currently stuffs landfills with its products, burns up the atmosphere in order to cool off massive data centers, and perpetuates or even increases social and economic inequities around the globe and in nearly every neighborhood. Remember Adolfo Natalini's words: "Until all design activities are aimed towards meeting primary needs . . . design must disappear."[13] All the while, this work will pave the way for a transformed discipline, building the institutions required for it to thrive. (We'll probably also need powerful political representation to combat the imminent legal obstacles, but that's the subject for another book. Meanwhile, see David Schweickart's book *After Capitalism* for concrete proposals on how to remake state institutions for a postcapitalist political economy.)

Postcapitalist Citizens (Postcapitalist Subjectivities × Discipline) One of the long-term projects of PCD is to construct a *semiotics of postcapitalist society*. What this means is designing the material and symbolic culture of that society, including images of new social-cultural-political subjects. This project works at one of the deepest and slowest "pace layers," *culture*. It will take time to construct such a semiotics. Yet PCD's powerful role is rooted in design's capacity for creating desire. Can PCD, as a discipline,

make the hard work, negotiation, and compromise of *actual* democratic processes desirable and even normal? Design can make the physical and interactive processes more manageable, more efficient, and more satisfying through known methods of user-centered design. PCD can expand these capabilities within the context of degrowth, decreasing time constraints for excessive work. Going even further, PCD can make politically and socially engaged lifestyles *symbolically* valuable, desirable, and sexy, just as it has done for many brands.

PCD Guidelines

These PCD guidelines are intended to practically guide the work of transforming projects, practices, and disciplines—including and beginning with design currently done for capitalist enterprise. As a group, the guidelines are designed to function uniquely at each of these layers, representing different contexts, scales, and speeds of transformation. Each set of guidelines comprises a series of questions for designers to consider as they navigate the unique contexts of their work. In these guidelines, the strategies take a form directly applicable to designerly approaches for synthesizing disparate needs into design solutions. Each guideline is posed as a "*How might we . . .*" question.

Project Guidelines: PCD Strategies × Project

The project guidelines are intended for designers and design teams to ask how any given design project—including for capitalist clients or within businesses—can be used as an opportunity to install and enact postcapitalist principles. They operate like a worksheet or checklist: each question offers a discrete opportunity for postcapitalizing *some* aspect of the project at hand, organized by the four principles of PCD. Designers and design teams are encouraged to use these questions throughout the design process—starting with early research to define the strategy and scope and concluding with finalizing specifications for production, maintenance, and circularity. It's best to begin a project's orientation through this framework. For example, a project brief could be expanded or reframed by asking these

questions at the start. In addition, new opportunities for postcapitalizing the project may turn up anywhere in the process. This process initiates *interstitial* erosion of capitalism at its most frequent point of contact with designing: the project.

How might we design this project to . . .

Create Social Power?

- replace economic (wage) labor with social exchange?
- increase and sustain social connectedness beyond economic relations?
- support economic independence and well-being?
- support economically disadvantaged communities?
- build opportunities for education of related skills (e.g., programming and development) for constituents or others who are outside the project's immediate reach?
- deliver value that can't be commodified—is nonexclusive, nonrivalrous?
- fulfill at least one additional nonexclusive, nonrivalrous need or desire for each one discovered through research to fit the client's economic concerns?
- engage constituents in learning valuable skills related to design processes?
- increase human capital by distributing knowledge of and experience with applications of new technologies for the project's constituents?
- ensure that any knowledge, skills, or innovative techniques are shared and explained within the work itself?
- support participants/constituents in becoming self-sufficient by allowing them to increasingly design and produce for themselves?

Build and Sustain Community Economies?

- determine what the constituent *community*—not just the client or "user"—actually needs, socially and materially?
- determine what the constituent community deems the appropriate social surplus to gather and distribute and how it can do so?
- build or sustain a commons?
- engage the fabrication, production, distribution, or other third-party work to be developed within the community economy, using local assets and resources?
- shift designed outcomes from rivalry and excludability to nonrivalry and nonexcludability, from ownership to sharing, from copyright to open source, from intellectual property to shared knowledge?

Enact Degrowth?

- minimize or reduce material inputs?
- work with community partners to innovate more efficient processes?
- plan for efficient maintenance and repair, particularly with local resources, including knowledge and labor?
- integrate design outcomes into to a circular and repair economy?
- remain durable, usable, and meaningful?
- be shared, reused, or circulated within a community?

Construct Postcapitalist Subjectivities?

- address the needs and desires of constituents as citizens and/or specific groups of individuals (not "users")?
- decenter the individual and individualist competition?
- prioritize communal processes of making, defining, and distributing value?
- require or reward collaboration, draw connections to local contexts, resources, and needs?

- ensure that constituents are cognizant of resource allocations and the global and local implications of those allocations?
- encourage positive ecological actions, such as replenishing, restoring, and nurturing the ecological environment?
- incentivize collaboration, participation, negotiation, and compromise?
- make political participation in acts of everyday life attractive and desirable?

Practice Guidelines: PCD Strategies × Practice

The *practice guidelines* are to be used by design practices—the entire staff of independent design consultancies or entire design teams within businesses—to address how the functional operations of designing can be transformed toward postcapitalist principles both in the long term and on a day-to-day basis. The goal is to erode capitalist structures and logics by formalizing and instituting postcapitalist models in the structures and daily operations of practice. The questions that make up the guidelines should be considered a starting point. They should be addressed and updated through ongoing discussions with a practice's participants and constituents (in today's terms, "clients" and "users") as well as with the broader communities—local geographic and global associations—in which a practice is socially embedded.

It is recommended to begin and sustain this process through regular *workplace-shared-governance* meetings and *community-of-practice* meetings. These meetings should be held at least annually, if not more often. The goal of these meetings is to map challenges and opportunities in determining the future work to be done by the practice. Workplace-shared-governance meetings include everyone *directly* involved in the design practice, even administrative, technical, facilities maintenance, and support staff. Community-of-practice meetings include the entire "community," which comprises all employees, partners, vendors, clients, representatives of the constituent community (people who use the designed outcomes), and neighbors in the physical community. Community-assets and community-

needs maps should be revised at these meetings as living documents that can continually guide the PCD practice.

The purpose of these discussions is to ask the four questions posed by J. K. Gibson-Graham:

1. What is necessary for each participant's personal and social survival?
2. How will social surplus be appropriated and distributed?
3. Will social surplus be produced and consumed? If so, how?
4. How shall the commons be produced and sustained?

Through workplace-shared-governance and community-of-practice meetings, specific objectives and strategies can be set to address opportunities for postcapitalizing the practice. The following thematically grouped questions can be used to inform organizational and operational decision making in these meetings or more regularly whenever an organizational change or decision needs to be made.

How might we . . .

Create Social Power?

- distribute and institutionalize social, economic, and political relations through patterns of operation?
- build a socially empowered organizational network of constituents, collaborators, and contributors?
- socialize the workplace, including everyone directly involved in the design practice?
- socialize the community, including all employees, partners, vendors, clients, and neighbors?
- operate cooperatively, sharing ownership and governance with those whose efforts, knowledge, and labor support the enterprise?

Build and Sustain Community Economies?

- determine and achieve what is *necessary* for the practice's and community's personal and social survival?
- determine whether and how social *surplus* will be produced and consumed with and for the practice and its community?
- appropriate and distribute social *surplus* collectively, with and for the practice and its community?
- build and sustain a (or some) *commons*—materials, equipment, knowledge, information, space to be used for diverse activities, file sharing, peer-to-peer instruction or guidance, and so on?

Enact Degrowth?

- reduce material and energy needs in operating a practice?
- reduce or eliminate the use of fossil fuels in operating a practice?
- strategize the ongoing reduction of fossil fuels and other nonrenewable resources in all that the practice produces and consumes, including energy needs of digital services and production of designed outcomes?
- localize production of material goods to minimize shipping and fuel use?
- ensure participants the "right to unplug," support non-work-related interests, fulfill educational interests and personal flourishing of participants, collaborators, and constituents?
- design so that our collaborators and constituents can flourish and live comfortably while consuming less material, energy, and cognitive or attention resources?

Construct Postcapitalist Subjectivities?

- normalize PCD practice by sharing processes, methods, outcomes, and challenges?
- demonstrate successful models of design and community flourishing through our Practice, sharing our best practices and nurturing new participants?
- generate design-driven desire for PCD ways of living, making, knowing, and working?
- share, disseminate, and promote PCD models of practice for adoption by others?

Disciplinary Guidelines: PCD Strategies × Discipline

The *disciplinary guidelines* are intended to be used to formalize, codify, circulate, and normalize patterns of doing and knowing PCD into disciplinary infrastructures and institutions. These guidelines are also framed as questions, which are a starting point to be regularly modified and made more specific as they are addressed toward what needs to be done, who can do it, and how. These questions should be asked by individual designers seeking deeper engagement in the transformation of design, by design practices committed to and engaged in transformation, and by the members, constituents, and participants of disciplinary institutions. The details are the fuzziest here because the specific challenges, needs, and resources required for disciplinary transformation will necessarily emerge from those engaged in it through practice and theory, and through research and education. Nonetheless, the basic contours are laid out for taking the first steps.

How might we . . .

Create Social Power?

- participate in, contribute to, and maintain shared disciplinary resources, such as health care, political representation, legal and professional advising?
- create and maintain commons of practical, material, and knowledge-based assets within the discipline and among

its many constituents?

- contribute PCD methods, processes, and resources toward in-house, corporate design practices to support internal erosion of capitalist design enterprise?
- continually provide and seek mentorship to build long-term and evolving forms of disciplinary patterns of postcapitalist action and knowledge?
- contribute to publications, media, and events that share case studies, research, and best practices and that generate new sites for PCD knowledge, discussion, and association?

Build and Sustain Community Economies?

- work to cooperate, confederate, or network PCD practices to build broader social power within the discipline?
- work to collaboratively achieve greater political power for PCD practitioners?
- contribute to building and maintaining the infrastructure and platforms of a distributed PCD discipline?

Enact Degrowth?

- commit a dedicated quantity of working time to the intellectual and productive labor required for building PCD infrastructures, networks, commons, and other shared resources?
- commit efforts to build cross-boundary initiatives between independent (PCD) design practices and corporate, in-house (capitalist) practices?

Construct Postcapitalist Subjectivities?

- support postcapitalist designers through ongoing education and participation in extra- or interdisciplinary programs to extend the reach of PCD and create positive spillovers in other communities of practice?

- support postcapitalist citizens in their efforts to postcapitalize their lives by promoting a PCD that is truly participatory and cooperative?
- work in ways that transform constituents of design from passive consumers to active agents who participate in and facilitate the design of their own lives?
- normalize PCD as a disciplinary form and the generally perceived notion of what Design is and *should be*?
- make PCD *the* desirable model for design, internal both to the discipline and to the general public?

Design Education

One of the most significant sites for the transformation of design's various disciplines will be design education. Currently, design education, along with much of higher education in general, is in a moment of crisis—and that crisis provides an excellent opportunity for discontinuous change. However, change can take many paths.

Design became a modern profession, particularly in the United States, during the early twentieth century, largely outside of the university—thus outside of theory. However, design started to become an academic discipline around the 1970s by asserting the need for "designerly approaches to research and action."[14] This move established design within the disciplinary needs and confines of the academy, which were quite different from the commercial needs of design practice.

For the past two decades (at least), there have been various calls for the remaking, remodeling, or complete overhauling of design education. The kinds of changes to be imposed vary widely based on the critic's position. Some have called for design education to place stronger emphasis on empirical research, research methods, and processes that can generate more objectively evidence-based practices in design. These voices often solicit design education to emphasize design-specific research methods and standards, many of which are generally adaptations from the social sciences. The resulting social *scientificity** of design practices would

accentuate methods for studying human behaviors, social interactions, human interactions with environments and objects, and varying aspects of perception, cognition, and human factors. From this perspective, the master–apprentice model of “studio” education from which many design schools emerged (or are still embedded) lacks the ability to cover necessary topics and their application to designing. The claim here is that the human-contextual nature of design requires a rational, evidence-based approach to interventions in social contexts (sometimes admitting the largely irrational behaviors of individuals). Yet how or toward what ends this evidence-based, research-driven design should be oriented remains a relatively open question.

Others have called for design education to engender greater technical knowledge, participation in and familiarity with material sciences as well as new or emerging manufacturing processes, or other forms of integrating design directly with the physical or technical sciences. The proponents of this view typically assert design’s need for integration with engineering, computer science, HCI, and other fields of material and technological innovation. The claim here is that the material-technical nature of designing objects, manufacturing processes, communication systems, and the multitude of human–computer interactions that make up much of contemporary life requires a rational, evidence-based approach to innovate new possibilities for material or experiential interventions. Again, the specific orientation of such efforts typically remains unstated.

Others emphasize that designers need to understand how things get done, sometimes referring to the “real world” of productivity as dictated by the practices of business. This knowledge is seen to be particularly necessary as designers take on responsibilities in working with or overseeing complex research and production systems with diverse stakeholders, experts from multiple disciplines, large and varied teams, and operations spread around the globe. Some argue that these expanded roles require designers to possess knowledge of the methods and applications of management, organizational leadership, administration, and strategic allocation of resources. Here, the ideology is clear even when unnamed—design’s function is to innovate for competitive advantage and profit. In other words: designers need to be business savvy.

Perhaps in reaction to these various empiricist, materialist, and business realist calls for change in design education, still others claim that design practices are and always will be centered in the traditions of the humanities. Designers are not scientists or technicians or businesspeople. They work in the space of culture, studying and producing cultural knowledge and cultural productions, building on history, engaging diverse forms of knowing and awareness, generating ever new material and symbolic vocabularies of what the earthly experience of being human means (Arendt's "human condition" comprises both the particular confines of planet Earth and the "artificial world" of human "work"). In this view, it is the humanities that define the values, ethics, aesthetics, and cultural norms that should inform and be informed by design.

So we essentially have demands for designers to be versed in most of the disciplines that make up a comprehensive university, while specializing in an already highly diverse discipline called "design." That's a tall order! Yet these demands are generally appropriate. Design was born into nationally scaled, industrial, manufacturing-based societies but now operates (or aspires to operate) in a global postindustrial economy and multicultural society.

Earlier in this book, the statement "objects disappeared into systems" was used as a metaphor because, of course, both objects and systems are still with us and have become more complex and interdependent. Design in its broadest sense needs to be capable of working at a huge range of scales and in a huge range of contexts. This is the reason why many of those calling for reform in design education, regardless of the preferred direction, tend to insist on the need for increased specialization. It has already been argued here that the postindustrialization of advanced capitalist economies has blurred many of the industrial-era distinctions still commonly used in design education (perhaps more so than in practice), such as graphic design, industrial design, and fashion design.

Many of the most vocal critics of design education today direct their ire at design education rooted in stand-alone art and design schools. They might even concede that programs based within research universities are better adapted and may even be fairly well suited to producing the breadth of knowledge and methodological acumen designers require for working in the contemporary world.[15] However, the latter institutions are also often

under attack by the same ideology that this book has sought to identify and to encourage design disciplines to work against—contemporary neoliberal capitalism.

Design has indeed grown out of the contexts of its origins and has at least somewhat modified its methods to adapt to the postindustrial world. Yet what is still often lacking in discourse on design education is a framework to address perhaps the most important question: What is the purpose of design? To answer that question requires adopting some ideological position or another. The theoretical frameworks that build from this initial perspective can lead to highly divergent forms of design.

Across these and other competing demands, two salient arguments about the needs that contemporary design education must now satisfy stand out:

1. Design students today need to become familiar with methods necessary to engage, think, and operate at the level of systems—moving beyond the skills of crafting objects alone.
2. Design students today need methods, frameworks, and approaches to think more critically about the impacts of their work and to evaluate risks, negative impacts, and potential for social harms. This is particularly important as designers engage new, emerging, and unprecedented technologies.[16]

Where shall the responses to these challenges be oriented? The answer this book offers is that design should be reoriented, integrated into, and operated toward deliberately remaking the political economy. No doubt, this goal is an ambitious one. Yet it is strategically situated at the intersection of design's varied capabilities and at the heart of this historical moment. In place of "Facebook classes," perhaps we should have postcapitalism classes.

PCD offers a general theory of change around which new pedagogical methods and models can follow. PCD education would require specialization *in addition to* visual, material, and formal design competencies. Additional specialization *within* PCD education would scaffold around four general areas. In keeping with the nature of disciplinary formations, the goal would be to immerse students in the breadth of PCD design discourse, methods, and practice while providing

pathways for depth in one of the areas described. This is a cursory and tentative outline of a significant project in its own: constructing a PCD educational experience that can be appropriately modified and customized to distinct contexts.

Communal Social Sciences. How can designers study people in communal settings, shifting from the user-centered mindset of *the individual* to the mindset of the constituent—an individual part of a whole, *always embedded* in a social fabric? How shall designers study the social, psychological, and cultural conditions of people working together to participate in the making of their own material, symbolic, and social lives? Although designers and design researchers have often borrowed research methods from the social sciences, they also must regularly adapt or modify these methods for their unique circumstances. (Here, applied social psychology may be more useful than applied cognitive psychology, which today can look rather exploitative.) What needs to be taught and formalized in PCD design education is a community-centered set of methods for research and practice. The rapidly expanding field of design anthropology is beginning to chart this territory. As in PCD practices themselves, there are many existing examples, fragments, and case studies to draw upon. They need to be codified, organized into curricular frameworks and structures, and implemented in a concerted manner. The knowledge and methods designers gain from this territory would be applied in doing the highly social activities required for facilitating participatory community design. In the broader field of PCD, this means providing specialized knowledge in the work of communal social research as a model of design research and practice.

Open, Distributed Technical Studies. How can both design experts and nonexpert communities make use of peer-to-peer networks; digital fabrication; extant and emerging 3D printing for making physical objects as well as foods and pharmaceuticals; material knowledge for maximizing inputs for micromanufacturing; processes for closed-loop manufacturing; circular economies; and application of agricultural or biological materials? Designers will be needed not just to *use* these tools but also to help set up community-scaled production systems, specifying the right mix of tools, processes, and installation instructions as well as creating and maintaining commons for file sharing, material supplies, and process knowledge. In another category within this area, designers will need to be versed in

information sciences to understand how structure, storage, distribution, and interaction with information and communications create conditions for cooperative human flourishing, in part because they all relate to networks of material production. Clearly, designers can't do all of these things alone, but they should be central actors in working with communities to make productive and self-sufficient use of material and informational resources, specific to the community economy. The use of specialized, proprietary software would be dislocated as a central component of design education. Instead, implementing knowledge of and experience in working with open-source technologies and digital commons should be the dominant norm for PCD education. From this groundwork, design students can take up the ongoing challenges and opportunities of making open, custom tools specific to their practical needs throughout their evolving careers.

Postcapitalist Enterprise and Organizing. How can designers organize and participate in the ongoing operational needs of shared governance, processes of stakeholder decision making, workflow management, resource management, and the generation of context-specific models and approaches to postcapitalist enterprise? This combination of political science, network theory, and business administration would be reoriented toward nonhierarchical forms of "management" and nonexploitative productivity. In this area, PCD designers can reclaim the term *leadership* from its common misuse in business and academic settings (where it is typically used to describe some combination of management and authority). They will promote instead a form of "design leadership" predicated on setting out the process for crafting a vision of the future and enlisting the voluntary participation of others to work toward its realization. Borrowing from methods and tactics of community organizing, this area of specialization provides the tools for designers to participate in collective action. Designers can also bring to these tasks the ability to create symbols and experiences of association that contribute to the affective, emotional, and even aesthetic dimensions of organizing and facilitating shared governance of an enterprise.

Humanities of the Commons. How do designers make ethical decisions? How do they understand and practice value judgments? How do they know what they know, how do they pursue it, and what do they do when they "make worlds" *with* and *for* others? By studying political economy, political science, philosophy, history, ethics, cultural studies,

critical theory, and other fields in the humanities, designers gain perspectives and frameworks to understand what they do and how to act within what they do. The intellectual and philosophical trajectory of the Enlightenment has largely informed the world we inhabit today (including the notion of "the individual" as the primary social, political, and economic unit). Yet designers need a critical bearing on the limits and problems of this conception, among others. As a focus area, a *humanities of the commons* would emphasize those critical methods that allow designers to conceptualize and scaffold their understanding of what it means to participate in *commoning* worlds through design. Today, emerging technologies are rapidly producing new ethical challenges for designers (among others). Yet the charge to "act ethically" is impossible without a framework that would allow a designer to make ethical judgments in practice, particularly when faced with unprecedented scenarios. "Do no harm" and "don't be evil" are too simplistic, as are many of the "manifestos" spilling out of design studios and schools. Meanwhile, the growing interest in and discussions on *posthumanism* might play a role in the study of what design *could* be doing. This area of specialization and its extensive potential for inter- and transdisciplinary knowledge could inform what PCD is doing and why—particularly as it provides many opportunities to confront what *should not be don*e and why.

This too is a tall order because it's essentially asking for a "universal" education as one subset of a discipline's educational framework. Yet for a PCD model of design education, a cross-section of these subjects can frame the theory of change and provide the intellectual apparatus for designers to continually ask the important ontological and epistemological questions—to continue to "think what they are doing."[17]

Is the university the best place for this design education? Yes and no. Generally speaking, the stand-alone, studio model of design education cannot produce PCD designers. Designers will need to maintain material expertise, visual acumen, and formal sensitivity, but these abilities need to be developed within a context where other competencies can inform and be informed by the traditional competencies of design in order to give those skills an orientation. A research university can provide this kind of comprehensive education so long as the particular university, school, or program is not wholly indoctrinated in the model of "real-world" practice as

education, which can only reproduce the structures and logics of capitalist BaUD design. The DESIS network offers a promising model by engaging design education programs and research labs in direct community-engaged social innovation. However, PCD design will need other bridges between direct-action community contexts and the "designing of" things, systems, and symbols that can bring about new structures, logics, and subjectivities of postcapitalism explicitly (not only immediate "problem solving").

The research university can probably play only a partial role in the future education of postcapitalist designers. It may be the place to develop fundamental design competencies and intellectual grounding described in the four focus areas described earlier. However, this education must be supplemented with community-based practice—with, through, or totally separate from the university. One possibility would be a resident model akin to medical education, with design students working "in residence" with communities. This would build points of contact for design education in the community: community education programs, workshops, peer mentorship, group advising, and similar arrangements—all become sites for varying levels of design education to be distributed to nonexperts who will participate with designers to make their worlds.

A particular point of distinction is that PCD models of education must dismantle the myth and ethos of competitive advantage and the claims of its social benefit. One simple way to do this is to immediately halt all practices of intellectual-property protection in the realm of higher education and research—particularly in publicly funded universities. Of course, the logics of neoliberal policy have infected the education "sector," and, of course, those same logics have insisted that private, individual gain is the primary incentive for innovation. It only stands to reason, then, that the university (whether public or private), as a great source of innovation through publicly funded research, has become a critical site of the privatization of ideas (which has been followed by the privatization of student loan debt and the heavy debt burdens of many universities themselves). Public universities focused on commercializing research are typically doing so to stay financially solvent (and competitive) in a scenario made difficult by policies that have relentlessly defunded public support for education. Yet this situation only demonstrates that the neoliberal ideology has been highly effective in its persistent privatization of the public sphere, not that it was ever a consensual arrangement or is particularly productive for society at

large. Remember that publicly funded research gave us the internet, space travel, satellite communications, and life-saving pharmaceuticals; these forms of knowledge just happened to be paid for and sold back to us as consumers through this model of privatization.

The PCD educational approach would be to *teach others* how to design and work with open-source tools, how to embed design practices into communities, and how to work with and work for *constituents*. Design education should foster sociopolitical awareness to generate sustainable new *ways of being in the world*. This approach may sound utopian, but creating a sense of possibility and nurturing the capabilities to transform the world through knowledge should be the function of education—not chasing a career of endless financial, social, and personal competition.

10
Conclusion: Another Design Is Possible

> While recognizing the important capacity of horizontalist tactics to provide small-scale support to communities and to temporarily disrupt certain exploitative practices, the commitment to fetishized versions of consensus, direct action, and particularly prefigurative politics, constrains the possibilities of expanding and overtaking existing social systems.
>
> —Nick Srnicek and Alex Williams, *Inventing* the Future

Will the strategies laid out here to transform design be enough to erode and eventually overtake existing social and economic systems?

We end here with clear-eyed recognition that social change takes time; takes robust, ongoing, and iterative efforts; and requires the proper historical conditions. However, when social change does come about, it's when a clearly articulated alternative is on hand with people ready to work for it at the right moment. As Erik Olin Wright says in his book *Envisioning Real Utopias*, "Capitalism will survive, for the foreseeable future anyway. . . . [S]o long as a viable alternative to capitalism is not actively on the historical agenda—and with broad popular support linked to a political movement able to translate that support into political power—capitalism will remain the dominant structure of economic organization. This book has tried to contribute to the task of placing alternatives on the historical agenda."[1]

Design after Capitalism has attempted to do the same for design—to contribute to the task of placing an alternative design on the historical agenda. The transformation of design from a discipline largely at the disposal of capitalist interests into a postcapitalist reconfiguration will not happen quickly. It will require dedicated and long-term committed actions by many actors, and those actions will need to take many forms. It offers only the first step, which is to spell out a diagnosis and critique of current conditions, lay out alternative possibilities, and highlight the practical

contours of transformation. Meanwhile, we have to remember that the future is not singular; it is plural.

Design practices are undergoing rapid change. New methods, intentions, and specializations crop up almost daily. The goal of this book has been to name a specific orientation for transformative practices. Some of the ideas come from speculation on what design could be, based only on fragmentary precedents of what has been. Others come from contemporary practices already well documented. Ezio Manzini's vision of design for social innovation is one trajectory this proposal builds upon—one focused particularly on communally embedded service design. However, even the term *social innovation* has been wielded for purposes in contrast to those laid out here. With the naming of postcapitalism as the ideological foothold for a transformative model of design, a navigational direction is set.

Efforts to "decolonize" design are embedded in this project. To decolonize design disciplines would be to decenter them from their capitalist frameworks and move them to operate from positions outside or counter to patriarchal Western modernity. Of course, this is already happening now, today, in many places. Yet many of the structures and infrastructures of design disciplines continue under the hegemonic sway of capitalism. Postcapitalism and decolonization are intertwined.

Business-as-usual design (BaUD) is highly compromising of its own goals—typically adjusting itself to the demands of a paying client in one form or another. PCD brings an ideology and an agenda of its own to the table. This postcapitalist orientation is a strategic approach toward naming possibilities beyond the capitalist paradigm in order to transcend its multiple grips on the practices, discourse, structures, and logics of design—particularly as that paradigm is in a downward spiral of decline and disequilibrium.

The effort to create a coherent argument for postcapitalism can appear totalizing. However, to critique the structural and systemic problems in capitalism does not mean everything about it is bad or that its end will make everything good. It is abundantly clear that design, as significant as its impact is on the nature of everyday life, is only a tiny piece of the puzzle.

And yet design is a place to start. It is in the nature of hyperobjects, just like "wicked problems," that you never have enough information, you can't see the entirety of the situation, you can't see the full picture of the

“preferred state” toward which you’re striving, and you might not even see your own individual impacts. In that case, you can start anywhere. As a discipline, PCD can adopt measures to transform the practices that shape material and economic flows as well as mindsets and social commitments. Designers can also be involved in changing political power through various nondesign social and political activities.

“Another world is possible.” So say generations of activists and critics of capitalism. This is also the basic and most essential claim of design.

You can never relax.

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Glossary

ascriptive status: A mechanism by which people are stratified into social classes based on qualities not in their control—for example, gender or race

capitalism: An economic system based on private ownership of means of production, wage labor, and the accumulation and investment of capital through profits; a system of social classes organized by access to ownershipof capital

class: A group that shares economic, political, and/or social status

commoning: Social labor that manages the resources of a commons through participatory, nonhierarchical methods

commons: Any form of shared wealth or value, including the provision of services

communal labor: Labor to support a commons based on expectations of participation by its community, whether explicit or implicit

community economy: An economic system that encourages local forms of production, resource allocation, and social forms of labor and exchange

conspicuous consumption: Consuming goods for their symbolism of status

constituent: An empowered member of a community; one part of a whole

consumerism: Social, economic, and political organization based on the theory that increased consumption of goods is desirable; preoccupation with buying of consumer goods

contradictory class locations: Professions or individuals occupying multiple class positions between capital and labor, based on access to authority and/or particular skills or expertise

critical theory: Critique of culture and society through the lens of power structures; engaging philosophy to help overcome social structures that oppress or dominate people

degrowth: Political and economic theories and movements emphasizing minimal production, ecological and social sustainability, and cooperation and working against paradigms of economic growth

discourse: The use of language about a subject that shapes how that subject is thought about, how it is communicated, and (in disciplinary terms) how it is practiced

endogenous growth theory: Theory arguing that economic growth is primarily the result of increasing knowledge of and about technology, distributing access to knowledge and thus growing potential for innovation; "economics of ideas" rather than of objects

erosion: The action of transcending the structures and neutralizing the harms of capitalism in civil society while also neutralizing the harms inflicted by capitalist government to create new forms of economic and social functioning

excludability: Whether goods or services are available only to people who pay for their use or ownership (*excludable*) or are freely available (*nonexcludable*)

Fordism: Assembly-line methods for increased production efficiency; division of labor and specialization of production operations for mass production

functionalism: Modernist design ethos that "form follows function"; technical, material, and aesthetic unity under the demands of instrumental purpose

hacker class: McKenzie Wark's term for the class of creative producers of cultural forms of value in an "information political economy"

hegemony: Pervasiveness of a system of thinking or doing that asserts itself over other possibilities, often without being chosen or without even being noticed

innovation pessimism: Belief or concern that technological achievements are of decreasing significance for humanity

Keynesianism: Governmental policies and programs to increase employment as well as consumer demand and spending; named for the British economist John Maynard Keynes

labor theory of value: Theory arguing that the economic value of a good or service is determined by the amount of "socially necessary labor" required to produce it

latent agency: A person's or group's unused or not (yet) enabled capacity to act in a certain way

marginal production cost: The cost required to produce one additional unit of a good or service

Marxism: Socioeconomic analysis that evaluates modes of material production to understand class relations and views class conflicts as the primary driver of historical change; named for the German political philosopher Karl Marx

mass culture: Capitalist consumer society in which public opinion, values, and political views are shaped by mass (broadcast) media and the mass production of consumable goods

military Keynesianism: Economic policy that advocates government spending on the military to boost economic growth

neoliberalism: Twentieth-century resurgence of free-market liberalism; associated with privatization, globalization, deregulation, free trade, austerity of public spending, and minimal government for protections of private rights and ownership

planned obsolescence: Design or production of a product with an artificially limited lifespan through deliberately decreased functionality

postmodernism: Broad social, philosophical, and cultural movement emerging in the late twentieth century that challenged claims of universality and "grand narratives" of modernism; the ambiguous historical era after modernity

product (or design) obsolescence: Design of a product so that it becomes less usable and/or less desirable or appears out of date before reaching its functional lifespan

proletarianization: The social process of people becoming wage laborers rather than working for their own subsistence

reciprocal labor: Labor to support a commons in which community members contribute time or effort with the expectation of reciprocity

rivalry: Whether a good's consumption by one consumer prevents consumption by others (*rivalrous*) or simultaneous consumption by multiple consumers is possible (*nonrivalrous*)

scientificity: The quality of being or seeming scientific

social relations: Interactions between two or more people or groups that can be shaped by the cultural, political, and economic conditions of society

spillover effects: Economic events in one context that occur because of events in a seemingly unrelated context

stagflation: An economic condition of high inflation and high unemployment

subjective theory of value: Theory arguing that the value of goods and services is determined by the individual choices regarding the perceived benefit or pleasure provided, not by the amount of labor or materials costs

subjectivity: Cultural understandings of one's self and the world based on external phenomena

surplus value: The difference between the value of the inputs of production (labor, materials, tools, knowledge) and the value they produce

Taylorism: Scientific approach to production to increase factory efficiency; evaluating each step in a manufacturing process to break it down into specialized, repetitive tasks with measurable inputs and outputs, named after the American engineer Frederick Winslow Taylor (1856–1915)

toil glamour: The presentation of intense cultural or cognitive work as attractive or exciting

vectoralist class: McKenzie Wark's term for the class of owners who own, operate, and control information networks and platforms in an "information political economy"

Notes

Preface

1. Paul Verhoeven, dir., *Robocop* (Los Angeles: Orion Pictures, 1987).
2. Ivica Mitrović, "Western Melancholy/How to Imagine Different Futures in the 'Real World,'" *Interakcije*, August 27, 2018, http://interakcije.net/en/2018/08/27/western-melancholy-how-to-imagine-different-futures-in-the-real-world/.
3. "Putting technology at the center of anything is profoundly conservative" (Cameron Tonkinwise, "Just Design," *Medium*, August 21, 2015, https://medium.com/@camerontw/just-design-b1f97cb3996f).
4. Tsing, *The Mushroom at the End of the World*, 18.

Introduction

1. There are many references loaded in this language. Martin Heidegger articulates "being in the world" and the concept of "world" as signifying "the totality of those entities which can be present-at-hand in the world" (Heidegger, *Being and Time*, 93). Hannah Arendt's articulation of "the human condition" situates humanity within the contexts of both a particular planet and a human-made "artificial world" (Arendt, *The Human Condition*). The design historian Tony Fry regards design's fundamental purpose to be "a remaking of 'worlds,' not however by a literal material reconstruction (although material change has to concur) but by the creation of a perceptual shift so that the meaning of one's world and actions alters" (Fry, *Design as Politics*, 16).
2. Despite offering the most thorough and expert translation of the relevance of various strands of economic theory to design practice—which has provided a strong foundation for this book—Heskett still falls back on statements such as "The main arena in which design is practiced is business" (Heskett, *Design and the Creation of Value*, 55). Clive Dilnot notes this issue in his introduction to Heskett's book: "The dominance of the market as is (and even of market society) becomes the effective horizon" (13). Design scholars who directly analyze the relationship between design and capitalism typically emerge with faith in the potential of the "virtuous cycle" and "dynamism" of capitalism, thus neglecting the broader social and ecological crises.
3. For example, David Schweickart's model for "economic democracy" in his book *After Capitalism*.
4. Abdullah et al., "A Manifesto for Decolonising Design."
5. Wright, *How to Be an Anti-capitalist*, 56.
6. Michael Meyer and Don Norman state that *only* 2 to 4 percent of Fortune 500 companies have chief design officers, which seems a very odd criterion to measure the success of a discipline's educational apparatus. Why not measure how many (or few) designers are political leaders or community organizers? See their article "Changing Design Education for the 21st Century."
7. Kari Paul and Dominic Rushe, "Tech Giants' Shares Soar as Companies Benefit from Covid-19 Pandemic," *Guardian*, July 30, 2020, https://www.theguardian.com/business/2020/jul/30/amazon-

apple-facebook-google-profits-earnings; Robinhood Foundation, “Poverty Tracker: Spotlight on Hunger,” November 2020, https://www.robinhood.org/poverty-tracker/2020_PovertyTracker_Food.pdf.

8. Simon, *The Sciences of the Artificial*, 111.
9. Haraway, “Anthropocene, Capitalocene, Plantationocene, Chthulucene.”
10. “Externality,” in *Lexico by Oxford*, https://www.lexico.com/en/definition/externality.
11. Morton, *Hyperobjects*, 1.
12. Morton, *Hyperobjects*, 15.
13. Morton, *Hyperobjects*, 27.
14. Morton, *Hyperobjects*, 115.
15. Drazen, *Political Economy in Macroeconomics*, 3.
16. Wright, *Envisioning Real Utopias*, 10.
17. Graeber, “On the Phenomenon of Bull-shit Jobs.”

Chapter 1

1. Arendt, *The Human Condition*, 1.
2. Arendt, *The Human Condition*, 1.
3. Fukuyama, “The End of History?”
4. Jameson, “Future City,” 76.
5. Graeber, “On the Phenomenon of Bull-shit Jobs.”
6. Kennedy School Institute of Politics, Harvard University, “Harvard IOP Spring 2016 Poll,” 2016, https://iop.harvard.edu/youth-poll/past/harvard-iop-spring-2016-poll; Frank Newport, “Democrats More Positive About Socialism Than Capitalism,” *Gallup*, August 13, 2018, https://news.gallup.com/poll/240725/democrats-positive-socialism-capitalism.aspx.
7. Max Ehrenfreund, “A Majority of Millennials Now Reject Capitalism, Poll Shows,” *Washington Post*, April 26, 2016, https://www.washingtonpost.com/news/wonk/wp/2016/04/26/a-majority-of-millennials-now-reject-capitalism-poll-shows/.
8. Tormey, *Anti-capitalism*.
9. Mason, *Postcapitalism*, xix, 25–26, original emphasis.
10. “Capitalism,” Merriam-Webster.com, 2021, https://www.merriam-webster.com/dictionary/capitalism?utm_campaign=sd&utm_medium=serp&utm_source=jsonld.
11. Chin, *My Life with Things*, 45.
12. Matt Simon, “Companion Robots Are Here. Just Don’t Fall in Love with Them,” *Wired*, August 2, 2017, https://www.wired.com/story/companion-robots-are-here/; Kevin Stacey, “Brown / Hasbro Team to Design Smart Robotic Companions to Assist Seniors,” *News from Brown*, Brown University, last modified January 11, 2017, https://www.brown.edu/news/2017-11-01/aries.
13. Streeck, *How Will Capitalism End?*, 44.
14. Tsing, *The Mushroom at the End of the World*, 118, quoting from Korzeniewicz, “Commodity Chains and Marketing Strategies,” 252.
15. “Billions of dollars dedicated to researching and developing complex weapons systems generated the first transistors that found their way into computers; hard plastics eventually used in automobiles and appliances; optical fibers that came to be the ‘information highways’ of the Internet; lasers that could eventually reshape eyeballs; jet engines that allowed commercial aircraft to travel ten thousand miles without refueling; and a vast array of precision gauges, sensing devices, and electronic gadgets that found their way into hundreds of thousands of commercial products and services” (Reich, *Supercapitalism*, 43).
16. “David Laurence Painter (b.1913) was among the early students in the Industrial Design program at The School of The Art Institute of Chicago, enrolling in 1933 and receiving his diploma in 1936. From 1935 to 1950, Painter was vice-president in charge of design at the Chicago design firm of Barnes & Reinecke, later becoming co-head of the product styling division. In 1950, Painter and two colleagues bought the company’s design interests and formed their independent partnership, Painter/Teague/Petertil, Industrial Designers” (Cooper-Hewitt, “Painter/Teague/Petertil,” accessed February 4, 2021, http://cprhw.tt/p/2Lpjmr/).
17. Sony Design, “TR-55,” accessed February 4, 2021, https://www.sony.net/Fun/design/history/product/1950/tr-55.html.
18. V&A Museum, “Radio in a Bag,” accessed February 4, 2021, https://collections.vam.ac.uk/item/O85208/radio-in-a-bag-radio-weil-daniel/.
19. Tharp and Tharp, *Discursive Design*.

20. Polanyi, *The Great Transformation*; Wright, *Envisioning Real Utopias*; Gibson-Graham, *The End of Capitalism*; Gibson-Graham, *A Postcapitalist Politics*.
21. Feola, "Capitalism in Sustainability Transitions Research," 243.
22. Gibson-Graham, *A Postcapitalist Politics*, 6.
23. Land, *Fanged Noumena*, back cover.
24. Tsing, *The Mushroom at the End of the World*, 128, 65.
25. Wright, *Envisioning Real Utopias*.
26. Wright, *How to Be an Anti-capitalist*, video.
27. Wright, *How to Be an Anti-capitalist*, video.
28. Jodi Dean, "Neofeudalism," LA *Review of Books*, May 2, 2020, https://lareviewofbooks.org/article/neofeudalism-the-end-of-capitalism.
29. Wark, *Capital Is Dead*.

Chapter 2

1. Interestingly, the word *capital* from the Latin term *capitalis*, "of the head" and the medieval Latin term *capitale*, "stock, property," was first used to describe a person's wealth in the early 1600s. From the 1640s, it began to describe "the wealth employed in carrying on a particular business" ("Capital," *Etymology Online*, 2021, https://www.etymonline.com/word/Capital).
2. The term *trade-mark* or *trademark* was not used with this meaning until the nineteenth century ("Trademark," *Etymology Online*, 2021, https://www.etymonline.com/search?q=trademark).
3. One of Polanyi's chief arguments and lasting contributions to the discourse of political economy is that all economic practices are *embedded* in the particular practices of a social world. He considers any distinction or separation between the social and the economic to be an oversimplification (Polanyi, *The Great Transformation*).
4. Polanyi, *The Great Transformation*.
5. Wood, *Origin of Capitalism*, 72.
6. Wood, *Origin of Capitalism*, 2–3.
7. Wood, *Origin of Capitalism*, 15.
8. Wood, *Origin of Capitalism*, 95.
9. Wood, *Origin of Capitalism*, 101, 102.
10. Wood, *Origin of Capitalism*, 102.
11. Wood, *Origin of Capitalism*, 104.
12. Heilbroner and Thurow, *Economics Explained*, 15.
13. Heilbroner and Thurow, *Economics Explained*, 14.
14. Heilbroner and Thurow, *Economics Explained*, 16. See also Braudel, *The Structures of Everyday Life*.
15. Meggs, *History of Graphic Design*, 73.
16. Meggs, *History of Graphic Design*, 85.
17. Drucker, *History of the Book*, chap. 5 passim, last updated January 2018, https://hob.gseis.ucla.edu/index.html.
18. Adam Smith's work *The Wealth of Nations* (1776) was the second-most-cited publication in the social sciences prior to 1950, behind only Karl Marx's *Das Kapital* (1867).
19. Heskett, *Design and the Creation of Value*, 65, emphasis in original.
20. Heskett, *Design and the Creation of Value*, 66.
21. Heskett, *Design and the Creation of Value*, 69–70, 76.
22. Polanyi, *The Great Transformation*, 87.
23. Arendt, *The Human Condition*, 4.
24. Fulcher, *Capitalism*, 38–39.
25. Heilbroner and Thurow, *Economics Explained*, 33.
26. US Small Business Administration, Office of Advocacy, "Frequently Asked Questions," last updated August 2018, https://www.sba.gov/sites/default/files/advocacy/Frequently-Asked-Questions-Small-Business-2018.pdf.
27. Chad Otar, "What Percentage of Small Businesses Fail—and How Can You Avoid Being One of Them?," *Forbes*, October 25, 2018, https://www.forbes.com/sites/forbesfinancecouncil/2018/10/25/what-percentage-of-small-businesses-fail-and-how-can-you-avoid-being-one-of-them/#18bdc63d43b5.

28. Karl Marx, “A Contribution to the Critique of Political Economy” (orig. pub. in German in 1859), trans. S. W. Ryazanskaya, Marxists Internet Archive, accessed February 4, 2021, https://www.marxists.org/archive/marx/works/1859/critique-pol-economy/ch01.htm#.

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31. Aimee Picchi, “40% of Americans Only One Missed Paycheck away from Poverty,” CBS *News*, January 29, 2019, https://www.cbsnews.com/news/40-of-americans-one-step-from-poverty-if-they-miss-a-paycheck/.

32. Veblen, *Theory of the Leisure Class*, 25.

33. Veblen, *Theory of the Leisure Class*, 19.

34. Fulcher, *Capitalism*, 41.

35. Fulcher, *Capitalism*, 43.

36. Meggs, *History of Graphic Design*, 258.

37. Meggs, *History of Graphic Design*, 259.

38. Louis Kalff Instituut, “Louis Kalff,”accessed February 4, 2021, https://www.louiskalffinstituut.nl/en/louis-c-kalff/.

39. Philips, *Celebrating 90 Years of Design at Philips*, 2015, 6, https://www.newscenter.philips.com/main/design/resources/pdf/Philips-Design-90-years-backgrounder.pdf.

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41. Gantz, *The Industrialization of Design*, 3.

42. Gantz, *The Industrialization of Design*, 4.

43. Hunter Oatman-Stanford, “Let There Be Light Bulbs: How Incandescents Became the Icons of Innovation,” *Collectors Weekly*, July 16, 2015, https://www.collectorsweekly.com/articles/let-there-be-light-bulbs/.

44. In a fire station in the northern California town of Livermore, there is a light bulb that was manufactured in the 1890s by Shelby Electric Company and that has been burning since 1901. In 2015, the “Centennial Light Bulb” reached an incredible milestone of 1 million hours of operation. You can follow along or occasionally check in on the BulbCam—a webcam focused on the bulb for the whole world to see (“Bulbcam,” Centennial Bulb, accessed February 4, 2021, https://www.centennialbulb.org/cam.htm).

45. *Automotive News*, “Annual Model Change Was the Result of Affluence, Technology, Advertising,” September 14, 2008, https://www.autonews.com/article/20080914/OEM02/309149950/annual-model-change-was-the-result-of-affluence-technology-advertising.

46. Jordan Grant, “When Tail Fins Made Waves,” National Museum of American History blog, June 28, 2016, https://americanhistory.si.edu/blog/when-tail-fins-made-waves.

47. The very first Beetle convertible, built in 1938, was given to none other than Adolf Hitler.

48. Jerry Hirsch, “First Volkswagen Beetle Arrived in a U.S. Showroom 65 Years Ago,” *Los Angeles Times*, January 31, 2014, https://www.latimes.com/business/la-xpm-2014-jan-31-la-fi-hy-volkswagen-beetle-65-years-20140130-story.html.

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50. Samuel Gibbs, “Apple and Samsung Fined for Deliberately Slowing Down Phones,” *Guardian*, October 24, 2018, https://www.theguardian.com/technology/2018/oct/24/apple-samsung-fined-for-slowing-down-phones; Janet Burns, “Italy Fines Apple, Samsung a Few Mil for ‘Planned Obsolescence’ in Phones,” *Forbes*, October 24, 2018, https://www.forbes.com/sites/janetwburns/2018/10/24/italy-fines-apple-samsung-a-few-million-for-planned-obsolescence-in-phones/?sh=32e7eccd5afb.

51. The reality may be a bit more complex because paying a “living wage” was also an investment in decreasing the turnover rate of employees, various forms of absenteeism, and the lack of productivity. Regardless, the results were similar, if not the same.

52. Heilbroner and Thurow, *Economics Explained*, 42.

53. Heilbroner and Thurow, *Economics Explained*, 40.

54. Heilbroner and Thurow, *Economics Explained*, 42, emphasis in original.

55. Menger, *Principles of Economics*, trans. James Dingwall and Bert F. Hoselitz (Auburn, AL: Ludwig von Mises Institute, 2007; originally published in German in 1871), 146, 121, quoted in Heskett, *Design and the Creation of Value*, 81, 80.

56. Drucker, *Innovation and Entrepreneurship: Practices and Principles* (New York: Harper and Row, 1986), 228, quoted in Heskett, *Design and the Creation of Value*, 86–87.

57. Heskett, *Design and the Creation of Value*, 77, emphasis in original.

58. Heskett, *Design and the Creation of Value*, 77.

59. Joseph A. Schumpeter, *Capitalism, Socialism, and Democracy* (New York: Harper, 1942), 83, quoted in Heskett, *Design and the Creation of Value*, 104.

60. “First, defense spending is not an effective stimulus package. Investments in the civil sector, as several studies have demonstrated, produce more jobs than investments in the military sector. Second, in order to secure sufficient funds to address economic recession and climate change—without saddling future generations with crippling debt—governments must begin to shift military spending toward human needs” (Feffer, “The Risk of Military Keynesianism”).

61. Raizman, *History of Modern Design*, 251.

62. Reich, *Supercapitalism*, 15.

63. Michael W. Clune, “What Was Neoliberalism?,” LA *Review of Books*, February 26, 2013, https://lareviewofbooks.org/article/what-was-neoliberalism/.

64. Reich, *Supercapitalism*, 47–48, emphasis in original.

65. Raizman, *History of Modern Design*, 306–307.

66. Jared Goss, “Design, 1950–75,” Metropolitan Museum of Art, October 2004, https://www.metmuseum.org/toah/hd/dsgn3/hd_dsgn3.htm.

67. Raizman, *History of Modern Design*, 257.

68. Raizman, *History of Modern Design*, 256.

69. Raizman, *History of Modern Design*, 257.

70. Raizman, *History of Modern Design*, 307.

71. Raizman, *History of Modern Design*, 307.

72. Raizman, *History of Modern Design*, 323.

73. Raizman, *History of Modern Design*, 322–323.

74. Raizman, *History of Modern Design*, 322.

75. Raizman, *History of Modern Design*, 322.

76. Raizman, *History of Modern Design*, 305.

77. “Today consumption—if this term has a meaning other than that given it by vulgar economics—defines precisely *the stage where the commodity is immediately produced as a sign, as sign value, and where signs (culture) are produced as commodities*” (Baudrillard, “Toward a Critique of the Political Economy of the Sign,” 150–151, emphasis in original).

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Chapter 3

1. Bell, *The Coming of Post-industrial Society*, 297–298; see also Cohen, *Three Lectures on Post-industrial Society.*
2. Reich, *Supercapitalism*, 50.
3. Streeck, *How Will Capitalism End?*, 17.
4. Streeck, *How Will Capitalism End?*, 15–17.
5. Reich, *Supercapitalism*, 51.
6. Raizman, *History of Modern Design*, 369.
7. Streeck, *How Will Capitalism End?*, 99.
8. Streeck, *How Will Capitalism End?*, 99.
9. "As widely used today, the term late capitalism . . . is not merely an emphasis on the emergence of new forms of business organization (multinationals, transnationals) beyond the monopoly stage but, above all, the vision of a world capitalist system fundamentally distinct from the older imperialism, which was little more than a rivalry between the various colonial powers" (Jameson, *Postmodernism*, xiv).
10. Jameson, *Postmodernism*, xiv.
11. Fuller, "Jean Baudrillard, Design Theorist."
12. Baudrillard, *The Consumer Society*, 61.
13. Levitt, "Marketing Success through Differentiation—of Anything."
14. Meggs, *History of Graphic Design*, 491.
15. Raizman, *History of Modern Design*, 369.
16. Raizman, *History of Modern Design*, 369.
17. John M. Carroll, "Human Computer Interaction—Brief Intro," Interaction Design Foundation, accessed February 4, 2021, https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/human-computer-interaction-brief-intro.
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20. Daniele Catalanotto, "A Tiny History of Service Design," *Service Design Magazine*, August 19, 2018, https://service-design.co/book-a-tiny-history-of-service-design-368ed603797c.
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23. Kimbell and Blomberg, "The Object of Service Design," 85, 90.
24. Kimbell and Blomberg, "The Object of Service Design," 85, 91.
25. Kimbell and Blomberg, "The Object of Service Design," 86, 91, 90.
26. Kimbell and Blomberg, "The Object of Service Design," 89–90.
27. Prahalad and Ramaswamy, "Co-creation Experiences," 8.
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29. John Chisholm, "What Is Co-design?," Design for Europe, accessed January 29, 2021, http://designforeurope.eu/what-co-design.
30. Harvey, *Seventeen Contradictions*, 23.
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37. Nussbaum, "Design Thinking Is a Failed Experiment."
38. Nussbaum, "Design Thinking Is a Failed Experiment."
39. Srnicek, *Platform Capitalism*, 20–21.
40. Srnicek, *Platform Capitalism*, 41, 43.
41. Srnicek, *Platform Capitalism*, 52, 53.
42. Srnicek, *Platform Capitalism*, 49–50.
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46. Heskett, *Design and the Creation of Value*.
47. Heskett, *Design and the Creation of Value*, 106.
48. Heskett, *Design and the Creation of Value*, 107.
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50. Morley, "What Is Endogenous Growth Theory?"
51. Heskett, *Design and the Creation of Value*, 178.

Chapter 4

1. Wright, *Envisioning Real Utopias*, 45.
2. Streeck, *How Will Capitalism End?*, 56.
3. Fry, *Design as Politics*, 65.
4. Nicole Aschoff, "Why Capitalism and Feminism Can't Coexist," *Jacobin*, September 2019, https://jacobinmag.com/2019/09/capitalism-socialist-feminism-inequality-sexism.
5. The primary sources for this section include the work of Erik Olin Wright, Wolfgang Streeck, Karl Polanyi, Herbert Marcuse, Robert Reich, J. K Gibson-Graham, Paul Mason, Nick Srnicek, and Alex Williams.
6. Srnicek, *Platform Capitalism*, 23.
7. Mason, *Postcapitalism*, 104.
8. Mason, *Postcapitalism*, 3.
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10. Streeck, *How Will Capitalism End?*, 26.
11. Robert J. Samuelson, "The Rise of Downward Mobility," *Washington Post*, August 5, 2018, https://www.washingtonpost.com/opinions/upward-mobility-is-a-myth/2018/08/05/bb960ce4-972c-11e8-80e1-00e80e1fdf43_story.html.
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16. Streeck, *How Will Capitalism End?*, 29.
17. Nicholas Confessore, Sarah Cohen, and Karen Yourish, "Small Pool of Rich Donors Dominates Election Giving," *New York Times*, August 1, 2015, https://www.nytimes.com/2015/08/02/us/small-pool-of-rich-donors-dominates-election-giving.html.
18. Bell, *The Coming of Post-industrial Society*.
19. Reich, *Supercapitalism*, 108.
20. *Medium*, "Peter Thiel on Stagnation, Innovation, and What Not to Call Your Company," April 6, 2015, https://medium.com/conversations-with-tyler/peter-thiel-on-the-future-of-innovation-77628a43c0dd. Thiel seems particularly critical of government regulation or at least inconsistencies and unevenness in regulation between "bits and atoms," meaning that the truly revolutionary potential of biotech is limited by the intense regulation of human bodies, while the world of software innovation has a much lower barrier to entry but less to offer in return.
21. Srnicek and Williams, *Inventing the Future*, 178–179.
22. The term *stack* is used to describe the various software components or subsystems needed to create a platform. Benjamin Bratton has extended this concept as a metaphor to describe the political regime of platforms. In his analysis, "jurisdiction" is not drawn "only according to the

horizontal subdivision of physical sites by and for states, but also according to the vertical stacking of interdependent layers on top of one another." The power and control of platforms (often by multinational corporate enterprises) operate in both tension and coordination with the power and control of sovereign nations. "In my analysis, there are six layers to this Stack: Earth, Cloud, City, Address, Interface, and User" (Bratton, "The Black Stack").

23. Srnicek, *Platform Capitalism*, 105.

24. Srnicek, *Platform Capitalism*, 106.

25. United Nations, Department of Economic and Social Affairs, *World Mortality 2019 Data Booklet* (2019), https://www.un.org/en/development/desa/population/publications/pdf/mortality/WMR2019/WorldMortality2019DataBooklet.pdf.

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28. Kolata and Tavernise, "It's Not Just Poor White People."

29. Rosenthal, "Insurance Policy."

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32. "Defuturing . . . seeks to disclose the bias and direction of that which is designed and how it is totally implicated in the world we conceptually constitute, materially produce, waste (rather than consume), occupy and use as an available material environment" (Fry, *A New Design Philosophy*, 11).

33. The French philosopher, anthropologist, and sociologist Bruno Latour defines the "critical zone" as that narrow habitat that contains all known forms of life: "The idea of the 'critical zone' is useful because it gets you out of nature. Nature is very big. It covers everything from the big bang to microbes. Conceptually, that makes it a complete mess. The critical zone is limited. It is just a few kilometres thick—above and below the surface of the Earth. But all discovered life is within it. This brings us inside in a way that nature does not. It is very different from the way of thinking that makes people such as Elon Musk think they should go on a mission to Mars. That is escapist. But when you think in terms of a critical zone, you are locked in, you cannot escape. What does it mean for politics if we are locked in and not in the infinite cosmology opened by Galileo? It means we cannot behave in the same way. It means we cannot just endlessly extract resources and discard our waste. In the critical zone, we must maintain what we have because it is finite, it's local, it's at risk and it's the object of conflict" (quoted in Jonathan Watts, "Bruno Latour: 'This Is a Global Catastrophe That Has Come from Within,'" *Guardian*, June 6, 2020, https://www.theguardian.com/world/2020/jun/06/bruno-latour-coronavirus-gaia-hypothesis-climate-crisis). See also Latour, *Down to Earth*, 77–80.

34. Kolbert, *The Sixth Extinction*.

35. Earth Observatory, NASA, "Global Temperatures," accessed February 4, 2021, https://earthobservatory.nasa.gov/world-of-change/decadaltemp.php.

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37. Matt Simon, “Capitalism Made This Mess, and This Mess Will Ruin Capitalism,” *Wired*, September 20, 2019, https://www.wired.com/story/capitalocene/. See also Harraway, “Anthropocene, Capitalocene, Plantationocene, Chthulucene.”

38. Wright, *Envisioning Real Utopias*, 47.

39. It’s worth noting that the English word *individual* was not used to refer to humans until the seventeenth century and did not become colloquial for “a person” until the eighteenth century—a usage that is contemporaneous with the Enlightenment and the birth of capitalism. “Meaning ‘a single human being’ (as opposed to a group, etc.) is from [the] 1640s. Colloquial sense of ‘person’ is attested from 1742” (“Individual,” *Online Etymology Dictionary*, 2021, https://www.etymonline.com/word/individual).

40. Adorno, *Minima Moralia*.

41. Marcuse, *One-Dimensional Man*, 9.

42. Wright, *Class Counts*, 13–14.

43. Wright, *Class Counts*, 15–16.

44. Wright, *How to Be an Anti-capitalist*, 7.

45. Wright, *Class Counts*, 16.

46. Wright, *Class Counts*, 16.

47. Wright, *Class Counts*, 18.

48. Reed, “Marx, Race, and Neoliberalism,” 49.

49. Wallerstein, *Historical Capitalism*, 25.

50. The ratification of the Nineteenth Amendment to the US Constitution gave women the right to vote in the United States in 1920.

51. Nancy Fraser, “How Feminism Became Capitalism’s Handmaiden—and How to Reclaim It,” *Guardian*, October 14, 2013, https://www.theguardian.com/commentisfree/2013/oct/14/feminism-capitalist-handmaiden-neoliberal.

52. Aschoff, “Why Capitalism and Feminism Can’t Coexist.”

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54. Fraser, “How Feminism Became Capitalism’s Handmaiden.”

55. Wallerstein, *Historical Capitalism*, 153.

56. “Expansion of Rights and Liberties—the Right of Suffrage,” Charters of Freedom, National Archives, Washington, DC, last updated July 6, 2016, https://web.archive.org/web/20160706144856/http://www.archives.gov/exhibits/charters/charters_of_freedom_13.html.

57. US Constitution, article 1, section 2, clause 3: “Representatives and direct Taxes shall be apportioned among the several States which may be included within this Union, according to their respective Numbers, which shall be determined by adding to the whole Number of free Persons, including those bound to Service for a Term of Years, and excluding Indians not taxed, three fifths of all other Persons.”

58. Reed, “Marx, Race, and Neoliberalism,” 51.

59. Reed, “Marx, Race, and Neoliberalism,” 50.

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Chapter 5

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Chapter 7

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7. International Cooperative Alliance, "About Us," accessed February 4, 2021, https://www.ica.coop/en/about-us/international-cooperative-alliance.
8. Wright, *Envisioning Real Utopias*, 242.
9. Schweickart, *After Capitalism*, 60.
10. Wright, *Envisioning Real Utopias*, 139–140.
11. Wright, *Envisioning Real Utopias*, 140–141.
12. Wright, *Envisioning Real Utopias*, 142.
13. Bookchin, *Post-scarcity Anarchism*.
14. See Stewart Brand's *Whole Earth Catalog* (published 1968–1972). See also Hallé and Ladue, *Karl Hess*, video.
15. Wright, *Envisioning Real Utopias*, 147.
16. Gibson-Graham, *A Postcapitalist Pol-itics*, 88.
17. A notable exception is the practice of participatory budgeting. See Wright, *Envisioning Real Utopias*, 155–160, and Participatory Budgeting Project, https://www.participatorybudgeting.org/.
18. Gibson-Graham, *A Postcapitalist Pol-itcs*, 89.
19. Wright, *Envisioning Real Utopias, 194–203*.
20. Akbulut, "Anti-capitalist Commons as a Degrowth Strategy."
21. Tonkinwise, "Design Away."
22. "Eroom's law [an inversion of Moore's Law of Growth] exemplifies a growing awareness across the sciences that something is deeply and widely wrong with scientific research. The number of new results is not only falling, but those results are becoming less trustworthy" (Bridle, *New Dark Age*, 86).
23. Srnicek, *Platform Capitalism*, 39.
24. Alanna Petroff and Océane Cornevin, "France Gives Workers 'Right to Disconnect' from Office Email," CNN *Business*, January 2, 2017, https://money.cnn.com/2017/01/02/technology/france-office-email-workers-law/index.html.

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27. Heskett, *Design and the Creation of Value*, 72.
28. Mason, *Postcapitalism*, 25–26, emphasis in original.
29. Morley, “What Is Endogenous Growth Theory?”
30. Schuler and Namioka, *Participatory Design*.
31. Wright, *Envisioning Real Utopias.*
32. Gibson-Graham, *A Postcapitalist Politics.*
33. Akbulut, “Anti-capitalist Commons as a Degrowth Strategy.”
34. Gibson-Graham, *A Postcapitalist Politics.*

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2. *McKinsey Quarterly*, “Enduring Ideas: The Three Horizons of Growth,” December 1, 2009, https://www.mckinsey.com/business-functions/strategy-and-corporate-finance/our-insights/enduring-ideas-the-three-horizons-of-growth.
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4. Andy Hines, “Fun with Scenario Archetypes,” *Hinesight*, October 17, 2014, https://www.andyhinesight.com/forecasting/fun-with-scenario-archetypes/.
5. Hines, “Evolution of Framework Foresight, Part 5.”
6. Frase, *Four Futures*, 29.
7. “Here, possibility precedes choice: the choice is a choice among possibilities” (Žižek, *First as Tragedy*, 150).
8. Frase, *Four Futures*, 100.
9. Klaus Schawb, “The Fourth Industrial Revolution: What It Means, How to Respond,” World Economic Forum, January 2016, https://www.weforum.org/agenda/2016/01/the-fourth-industrialrevolution-what-it-means-and-how-to-respond/.
10. Frase, *Four Futures*, 69–90.
11. Frase, *Four Futures*, 70.
12. Mason, *Postcapitalism*, 175.
13. Srnicek, *Platform Capitalism*, 126.
14. Frase, *Four Futures*, 71.
15. Dean, “Neofeudalism.”
16. Srnicek, *Platform Capitalism*, 126.
17. Frase, *Four Futures*, 71.
18. “‘We carried out research and found that a woman’s voice is more “sympathetic” and better received,’ said head of Amazon’s ‘Smart Home’ division, Daniel Rausch, in a conversation with Business Insider” (Hannah Schwär and Qayyah Moynihan, “Companies Like Amazon May Give Devices Like Alexa Female Voices to Make Them Seem ‘Caring,’” *Business Insider Deutschland*, April 5, 2020, https://www.businessinsider.com/theres-psychological-reason-why-amazon-gave-alexa-a-female-voice-2018-9?op=1).
19. Morozov, “Capitalism’s New Clothes.”
20. From 2001 to 2015, Dunne and Raby ran the Design Interactions program at the Royal College of Art in London. Their teaching, writing, and creative works have informed much of the territory of speculative design.
21. Dunne and Raby, *Speculative Everything*, 175–176.
22. Greenfield, *Radical Technologies*, 243–244.
23. Frase, *Four Futures*, 48.
24. Rutger Bregman, “Nixon’s Basic Income Plan,” *Jacobin*, May 5, 2016, https://www.jacobinmag.com/2016/05/richard-nixon-ubi-basic-income-welfare/.
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basic-income-y-combinator.

26. Mason, *Postcapitalism*, 266.
27. See also Schweickart, *After Capitalism*.
28. Greenfield, “Practices of the Minimum Viable Utopia.”
29. Greenfield, “Practices of the Minimum Viable Utopia,” 18.
30. Kostakis and Bauwens, *Network Society and Future Scenarios*, 40.
31. Frase, *Four Futures*, 47.
32. Manzini, *Design, When Everybody Designs*.
33. Morozov, *To Save Everything, Click Here*, 329.
34. Frase, *Four Futures*, 38.
35. Srnicek and Williams, *Inventing the Future*, 124.
36. Tarizzo, “What Is a Political Subject?”

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1. Wright, *How to Be an Anti-capitalist*, 153.
2. Sterling, “Pace Layers,” 216.
3. Wright, *How to Be an Anti-capitalist*, 49.
4. Mightybytes, “Web Fonts and Sustainability,” last modified March 16, 2021, https://www.mightybytes.com/blog/sustainability-web-fonts/.
5. Under neoliberal logic, this would insist that privately funded enterprises take ownership of public resources. The proposal here is that this must be inverted. For example, efforts at gentrification in many American cities bring together private and public interests, resulting in private real estate developers taking partial or total ownership of public assets, such as parks, buildings, and even residential complexes, which allows them to extract rents and set market prices through controlling the housing supply. This is a danger for any public–private merger. Such scenarios must be diligently avoided and control transferred to democratically governed rather than private interests.
6. International Cooperative Alliance, “About Us.”
7. As just one example, the nonprofit social innovation firm Design Impact regularly shares free versions of its research tools and publishes extensible research insights. See its website at https://d-impact.org/.
8. Gibson-Graham, *A Postcapitalist Politics*, 93.
9. Participatory Budgeting Project, “What Is PB?,” accessed February 4, 2021, https://www.participatorybudgeting.org/what-is-pb/.
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11. Kennedy School Institute of Politics, “Harvard IOP Spring 2016 Poll.”
12. American Institute of Graphic Arts, “Design Teaching Resource,” accessed February 4, 2021, https://teachingresource.aiga.org/.
13. Quoted in Didero, *SuperDesign*, 210.
14. Tonkinwise, “Post-normal Design Research,” 32.
15. Meyer and Norman, “Changing Design Education for the 21st Century.”
16. AIGA Design Educators Community, “AIGA Designer 2025,” August 22, 2017, https://educators.aiga.org/aiga-designer-2025.
17. A fascinating story from the *autonomia* movement in Italy in the 1970s: As a concession to their strike demands, factory workers were granted free access to study at local universities. The organizing groups “steered workers using their 150 study hours away from cultural courses (through which they might transcend the world of work) into courses such as economics, which they expected workers to find interesting. If anyone had bothered to ask, they would have discovered that many of the workers were often more interested in youth culture than in studying the dynamics of production, and many women gravitated toward feminism rather than traditional leftist theory” (Katsiaficas, *The Subversion of Politics*, 38).

Chapter 10

1. Wright, *Envisioning Real Utopias*, 366.

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